

RATIONAL GERSTEN KERNELS IN RAMIFIED REGULAR LOCAL RINGS

ABSTRACT. We construct infinite-order elements in the kernels of rational generic restriction maps for ramified regular local rings of mixed characteristic. The basic examples have dimensions two and three, with residue fields $\mathbf{F}_5(T)$ and $\mathbf{F}_3$, respectively. In both cases an actual K_3 -class remains of infinite order after completion and vanishes integrally in the fraction field. Explicit products of three units are given in each dimension. The proofs combine supported coefficient classes, arithmetic divisor calculations, and two cohomological detectors. Analytic Deligne regulators survive every finite localization needed for the arithmetic local rings; relative continuous cyclic characters detect the completed classes in ordinary continuous de Rham quotients. We extend the constructions to every residue prime and obtain arbitrarily large kernel ranks in fixed local dimension. Cyclotomic coefficients and continuous Laurent residues give classes in higher degrees. At every prime, fixed three-dimensional rings, including complete rings, have infinite rational generic-kernel rank in every degree at least three. A fixed complete two-dimensional ring in residue characteristic five has the same property. We also construct explicit Milnor, Beilinson-motivic, and higher-Chow classes with zero generic restriction, keeping actual classes distinct from their images in completed spectra.

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1. INTRODUCTION

For a regular local domain R with fraction field F , consider the rational restriction map

$$K_j(R)_{\mathbf{Q}} \longrightarrow K_j(F)_{\mathbf{Q}}, \quad K_j(R)_{\mathbf{Q}} = K_j(R) \otimes_{\mathbf{Z}} \mathbf{Q}.$$

Its injectivity is the first assertion of the augmented rational Gersten conjecture. We construct classes in its kernel for ramified regular local rings of mixed characteristic. The classes are represented in actual algebraic K -groups: they have infinite order, and their restrictions to the fraction field vanish before tensoring with $\mathbf{Q}$.

The two basic rings are

$$(1.1) \quad \begin{aligned} A_2 &= \left(\mathbf{Z}_{(5)}[T, x, y] / (\Phi_5(1 + y) + x^4 + Txy^3) \right)_{(5, x, y)}, \\ A_3 &= \left(\mathbf{Z}_{(3)}[x, y, z] / (3 + xy + z^2) \right)_{(x, y, z)}, \end{aligned}$$

where $\Phi_5(t) = 1 + t + t^2 + t^3 + t^4$. The ring A_2 has dimension two, maximal ideal (x, y) , and residue field $\mathbf{F}_5(T)$; the ring A_3 has dimension three, maximal ideal (x, y, z) , and residue field $\mathbf{F}_3$. Write $\hat{A}_i$ for completion at the maximal ideal. Theorem 3.2 and Theorem 2.1 construct classes $c_i \in K_3(\hat{A}_i)$ whose images in $K_3(\hat{A}_i)$ also have infinite order. In each of the four rings the corresponding class vanishes after inverting x , and hence in the fraction field, integrally.

In dimension three the class is the product of three explicit units:

$$(1.2) \quad c_3 = 2 \left\{ \frac{1+z}{2}, 1+x, 1 + \frac{(1+x)y}{4} \right\}.$$

The braces denote the product of unit classes in Quillen K_1 . Dennis–Stein identities prove its integral vanishing after inverting x . Its infinite order is detected on a smooth formal quadric by a convergent three-adic endpoint series. The actual vanishing of the special-fiber group K_4 makes the relative-to-absolute map injective, so this argument detects an actual K_3 -class. The two-dimensional construction starts with a coefficient in the K_3 of a cyclotomic integer ring and pushes it forward from the regular divisor $x = 0$. A second construction identifies the explicit unit product

$$\left\{ 2 + y, 1 - x, 1 + (1 - x)(x^3 + Ty^3)y((1 + y)^5 + 1) \right\} \in K_3(A_2)$$

as an infinite-order generic-kernel class in both A_2 and $\hat{A}_2$; see Proposition 27.2.

The same methods give families with larger rank, higher K -degree, and one fixed ring supporting infinitely many independent classes. Table 1 locates these extensions. Every rank assertion concerns the rational generic kernel; the witnesses are actual integral classes vanishing on the indicated principal open. The parameters introduced in residue fields preserve the local Krull dimension.

The geometric and completed proofs detect the supported classes by different cohomology theories. On an arithmetic quadric, a small resolution shows that the ruling has infinite order in the local class group. This calculation forces each geometric boundary component removed by an allowed denominator to have zero divisor class. Consequently the ruling’s Betti class survives every finite localization. Multiplying by a number-field coefficient with nonzero regulator gives a nonzero analytic Deligne class on every such open. Filtered continuity then proves

Dim.	Result	Reference
2	At every residue prime, tame families have unbounded kernel rank in degree three and nonzero kernel in every odd degree at least three.	28.1, 28.3
3	Eisenstein families have explicit rank formulas before completion and arbitrarily large lower bounds after completion.	14.4, 17.1
3	Explicit cyclotomic triples at every prime retain their independence after henselization and completion.	17.4
3	Products of j specified units give infinite-order kernel classes for every $j \geq 3$.	21.5
2,3	At each prime, a fixed ring has infinite kernel rank in every odd degree at least three; in dimension three this also holds for its completion.	29.1, 23.6
3	At every prime, one fixed ring and one fixed complete ring, with residue field $\overline{\mathbf{F}}_p(T)$, have infinite kernel rank in every $j \geq 3$.	24.2, 24.4
2	One complete ring with residue field $\overline{\mathbf{F}}_5(T, S)$ has infinite kernel rank in every $j \geq 3$.	31.2
2,3	Explicit degree-three classes yield Milnor kernels; the three-dimensional class also gives Beilinson-motivic and higher-Chow kernels.	27.3, 25.1

TABLE 1. Families and consequences of the basic constructions. The cited statements give the coefficient rings and all rank bounds.

nonvanishing on the local ring. For the quartic surface, the corresponding divisor calculation uses the exceptional quartic curve and its Jacobian.

Completion requires a separate detector. The Beilinson fiber square of Antieau–Mathew–Morrow–Nikolaus identifies the relevant relative p -completed K -theory with continuous cyclic homology [2]. A finite mixed Hochschild–Kostant–Rosenberg comparison retains the ordinary continuous de Rham quotients. On a quadric, an explicit residue operator detects the ruling class. On the quartic family, the elliptic quotients of its Jacobian and a bounded slope-one functional detect the weighted root class after passage to the generic Cohen completion. In each case the proof separately controls the relative-to-absolute map and localization of the integral test ring. This is the step that returns a cohomological nonvanishing statement to the actual algebraic K -group.

For the rank constructions, cyclotomic classes supply finite families of independent coefficient values. Their Chern classes factor through spectral completion, so independence passes to the continuous cyclic point coordinates without requiring surjectivity from actual K -theory. A continuous Laurent constant-term operator preserves the ruling detector after adjoining units. Products with these units raise the K -degree, while unramified coefficient extensions increase the rank. Finite descent handles each possible relation before completion; a separate comparison for completed essentially smooth models handles the fixed complete rings.

We use $K(R)$ for the algebraic K -theory spectrum and $K(R; \mathbf{Q}_p) = K(R)_p^\wedge[1/p]$ for its rationalized spectral p -completion. This notation is distinct from tensoring each actual K -group with $\mathbf{Q}_p$. Throughout the continuous de Rham arguments, cohomology means the quotient by the image of the differential, with no closure of that image. The final mathematical section gives an explicit obstruction to the strict simplicial-functor construction in [37, Lemma 13(III)]. The scope of the accompanying computational and formal checks is recorded separately at the end of the article.

2. AN EXPLICIT THREE-ADIC KERNEL CLASS

2.1. The ring and the class. Generic restriction. Products of units give explicit classes with which to study restriction from a regular local ring to its fraction field. Dennis–Stein

identities can establish the vanishing of such a class after localization [50, III, Definition 5.11 and Theorem 5.11.1]. For nonvanishing, we use the continuous fiber comparison [2, Proposition 4.10], together with multiplicative traces and the module structure on cyclic homology [9, 38, 28, 10]. The example below requires both an integral calculation in algebraic K-theory and an ordinary, possibly non-Hausdorff, differential quotient.

An explicit class. For units $b_1, \dots, b_r$ in a commutative ring R , write $\{b_1, \dots, b_r\} = [b_1] \cdots [b_r] \in K_r(R)$ for the product of their classes in actual Quillen algebraic K-theory.

Theorem 2.1. *Let*

$$A_3 = \left(\mathbf{Z}_{(3)}[x, y, z] / (3 + xy + z^2) \right)_{(x, y, z)}, \quad \widehat{A}_3 = \mathbf{Z}_3[[x, y, z]] / (3 + xy + z^2).$$

Then $\widehat{A}_3$ is the maximal-ideal completion of A_3 , and both rings are ramified regular local domains of dimension three. For either $R = A_3$ or $R = \widehat{A}_3$, the class

$$c_R = 2 \left\{ \frac{1+z}{2}, 1+x, 1 + \frac{(1+x)y}{4} \right\} \in K_3(R)$$

has infinite order and maps to zero in $K_3(R[1/x])$ integrally. Consequently,

$$K_3(R) \otimes_{\mathbf{Z}} \mathbf{Q} \longrightarrow K_3(\text{Frac}(R)) \otimes_{\mathbf{Z}} \mathbf{Q}$$

is not injective.

The same unit formula therefore gives a nonzero rational class in both the algebraic local ring and its completion. Its generic vanishing already holds before rationalization.

Method. The integral calculation uses a Dennis–Stein symbol whose product with twice a unit class is c_R . For nonvanishing, we map to a smooth formal quadric over an Eisenstein quadratic extension of $\mathbf{Z}_3$. Proposition 2.6 constructs the relative character with its K-module action and fixes its value on relative units. Lemma 2.7 then detects its top component by two endpoint values, without constructing a convergent primitive for a closed one-form. The endpoint difference is nonzero by an estimate valid for every term of its defining series. Vanishing of the actual fourth K-group of the special fiber finally passes this detection from relative to absolute K-theory.

Section 2.2 fixes the conventions and the continuous comparison. Section 2.3 proves integral vanishing. Section 2.4 constructs and normalizes the character. Section 2.5 proves the residue criterion. Section 2.6 completes the proof.

2.2. Preliminaries. We use Quillen algebraic K-theory and its unit products with the conventions of [40, 50]. For continuous cyclic constructions we follow [2, Definition 4.1 and Proposition 4.2]. We write HH, HC, HC^- , and HP for Hochschild, cyclic, negative cyclic, and periodic cyclic homology, respectively. The symbols THH and TC denote topological Hochschild and topological cyclic homology.

Notation 2.2. An uncompleted group $K_i(D)$ always means an actual Quillen K-group. For an ideal $J \subseteq D$, the relative spectrum is

$$K(D, J) = \text{fib}(K(D) \longrightarrow K(D/J)), \quad K_i(D, J) = \pi_i K(D, J).$$

Thus relative K-theory is defined by a homotopy fiber, rather than by the kernel of a map on homotopy groups.

For a spectrum-valued invariant, the notation $;\mathbf{Q}_p$ means p -completion of the spectrum followed by inversion of p . It does not mean tensoring its actual homotopy groups with $\mathbf{Q}_p$. The superscript cts denotes the homotopy inverse limit over the p -power reductions.

Unit products are natural under ring maps, multilinear, and graded commutative. They satisfy $\{b, 1-b\} = 0$ whenever both entries are units; see [50, III, Lemma 5.10.2, Definition 5.11, and the unit-product discussion following Theorem 5.11.1]. The distinction in Notation 2.2 will be retained when these products are mapped to continuous invariants.

Setup 2.3. Put

$$W = \mathbf{Z}_3[u]/(u^2 + 3), \quad E = W[1/3].$$

The Eisenstein equation makes W a complete discrete valuation ring with uniformizer u and residue field $\mathbf{F}_3$. Its u -adic and 3-adic topologies agree. For a smooth W -algebra D_0 of relative dimension at most two, let D denote its 3-adic completion, and put

$$\Omega_D^j = \left(\varprojlim_n \Omega_{D_0/W}^j / 3^n \Omega_{D_0/W}^j \right) [1/3].$$

All differentials on these modules are relative to W . The notation $W\langle T_1, \dots, T_r \rangle$ denotes the restricted power series algebra, namely the 3-adic completion of $W[T_1, \dots, T_r]$.

The two cases used below are $D = W\langle T \rangle$ and

$$B = W\langle X, Y, Z \rangle / (XY + Z^2 - 1).$$

The algebra $W[X, Y, Z]/(XY + Z^2 - 1)$ is smooth of relative dimension two: its partial derivatives $Y, X, 2Z$ have no simultaneous zero on the quadric.

The continuous comparison will be used in its formal-scheme form.

Proposition 2.4 (Continuous fiber comparison [2, Proposition 4.10]). *Let $\mathcal{O}$ be the integer ring of a complete discretely valued characteristic-zero field with perfect residue field k of characteristic p . Let $\mathfrak{X}$ be a smooth p -adic formal $\mathcal{O}$ -scheme. There is a natural diagram*

$$(2.1) \quad \begin{array}{ccc} K^{\text{cts}}(\mathfrak{X}; \mathbf{Q}_p) & \longrightarrow & K(\mathfrak{X}_k; \mathbf{Q}_p) \\ \downarrow & & \downarrow \\ \text{TC}^{\text{cts}}(\mathfrak{X}; \mathbf{Q}_p) & \longrightarrow & \text{TC}(\mathfrak{X}_k; \mathbf{Q}_p) \\ \downarrow & & \downarrow \\ \text{HC}^{-, \text{cts}}(\mathfrak{X}/\mathcal{O}; \mathbf{Q}_p) & \longrightarrow & \text{HP}^{\text{cts}}(\mathfrak{X}/\mathcal{O}; \mathbf{Q}_p) \end{array}$$

in which both squares are cartesian.

The proof of [2, Proposition 4.10] identifies the horizontal fibers after projection from absolute Hochschild homology to Hochschild homology relative to $\mathcal{O}$. For an algebraization with bounded p -power torsion, [2, Proposition 4.2] identifies continuous Hochschild and cyclic constructions with their p -completed counterparts. The relative version is stated immediately before [2, Proposition 4.10]. These assertions apply to the smooth, W -flat algebraizations in Setup 2.3.

2.3. Integral vanishing. In this section, we prove the regularity and integral vanishing assertions of Theorem 2.1.

Proposition 2.5. *The rings A_3 and $\hat{A}_3$ of Theorem 2.1 are ramified regular local domains of dimension three, with $\hat{A}_3$ the maximal-ideal completion of A_3 . For either ring R , all entries defining c_R are units, and*

$$c_R|_{R[1/x]} = 0 \quad \text{in } K_3(R[1/x]).$$

Proof. The ambient local ring $\mathbf{Z}_{(3)}[x, y, z]_{(3, x, y, z)}$ is regular of dimension four. The image of $3 + xy + z^2$ in its maximal ideal modulo its square is the nonzero image of 3. Its quotient is therefore regular of dimension three, with maximal ideal (x, y, z) , and is a domain. Completion gives the displayed presentation of $\hat{A}_3$. The same parameter calculation in $\mathbf{Z}_3[[x, y, z]]$ proves that this completion is regular of dimension three. In both rings, $3 = -xy - z^2$ belongs to the square of the maximal ideal, so the rings are ramified. Each entry in the definition of c_R has nonzero image in $\mathbf{F}_3$.

We use Dennis–Stein symbols with the convention that $\langle r, s \rangle$ is defined when $1 - rs$ is invertible. In additive notation their relations are

$$\begin{aligned}\langle r, s \rangle &= -\langle s, r \rangle, \\ \langle r, s \rangle + \langle r, t \rangle &= \langle r, s + t - rst \rangle, \\ \langle r, st \rangle &= \langle rs, t \rangle + \langle tr, s \rangle\end{aligned}$$

whenever the symbols involved are defined [50, III, Definition 5.11, relations (D1)–(D3)]. If r is a unit, then $\langle r, s \rangle = \{r, 1 - rs\}$. Skew-symmetry gives $\langle r, s \rangle = \{1 - rs, s\}$ when s is a unit. In particular, $\langle r, 1 \rangle = 0$ whenever it is defined. These identities hold in actual K-groups.

In R , put

$$a = \frac{1+z}{2}, \quad t = \frac{y}{4}, \quad \lambda = 1+x, \quad \mu = 1+\lambda t, \quad q = a(1-a).$$

The defining equation of R gives

$$q = \frac{1-z^2}{4} = 1+xt.$$

The elements $a, 1-a, \lambda, \mu, q$, and λq are units. The second Dennis–Stein relation gives

$$\delta := \langle -x, t \rangle = \langle -x, t \rangle + \langle -x, 1 \rangle = \langle -x, t + 1 + xt \rangle = \langle -x, \mu \rangle = \{\lambda q, \mu\},$$

where the last equality uses $1 + x\mu = \lambda q$.

Graded commutativity gives $2\{a, a\} = 0$, and the Steinberg relation gives $\{a, 1-a\} = 0$. Hence

$$2\{a, q\} = 2\{a, a\} + 2\{a, 1-a\} = 0.$$

Multiplying δ by $2[a]$ now yields

$$2[a]\delta = 2\{a, \lambda, \mu\} + 2\{a, q, \mu\} = c_R.$$

After inverting x , the first-entry formula for the Dennis–Stein symbol identifies δ with $\{-x, q\}$. Consequently,

$$c_R|_{R[1/x]} = 2\{a, -x, q\} = -2\{a, q, -x\} = 0.$$

No completion or rationalization of a K-group enters this calculation. $\square$

2.4. The relative character and its normalization. The goal of this section is to construct the character used to detect c_R and to determine its value on a relative unit multiplied by two absolute unit classes.

Proposition 2.6. *In Setup 2.3, the continuous comparison defines a natural map from the actual relative spectrum*

$$\chi_D: K(D, uD) \longrightarrow \Sigma \mathrm{HC}^{\mathrm{cts}}(D/W; \mathbf{Q}_3).$$

This map is $K(D)$ -linear, where the target action is obtained by restriction along

$$K(D) \longrightarrow \mathrm{HC}^{-, \mathrm{cts}}(D/W; \mathbf{Q}_3).$$

For $v \in 1 + uD$, let $[v]_{\mathrm{rel}} \in K_1(D, uD)$ be the relative unit class specified by the scalar automorphism v and its identity reduction modulo u . Then

$$\chi_D([v]_{\mathrm{rel}}) = \log v \quad \text{in } \mathrm{HC}_0^{\mathrm{cts}}(D/W; \mathbf{Q}_3) = \Omega_D^0.$$

For $b_1, b_2 \in D^\times$, its product character satisfies

$$(2.2) \quad \chi_D([v]_{\mathrm{rel}}[b_1][b_2])_{\mathrm{top}} = \pm \log(v) \, d\log b_1 \wedge d\log b_2 \quad \text{in } \Omega_D^2/d\Omega_D^1.$$

Here the top component is the first summand of

$$(2.3) \quad \mathrm{HC}_2^{\mathrm{cts}}(D/W; \mathbf{Q}_3) = (\Omega_D^2/d\Omega_D^1) \oplus \ker(d: \Omega_D^0 \longrightarrow \Omega_D^1).$$

The sign in (2.2) is the universal suspension sign. Equation (2.3) uses the ordinary differential image.

Proof. The character is obtained from the horizontal fibers of (2.1) for $\mathrm{Spf} D$. Compatible maps from actual to continuous K-theory, followed by completion of spectra and rationalization, give a map from $K(D, uD)$ to the fiber of the top row. The two squares carry this fiber to the fiber of $\mathrm{HC}^{-, \mathrm{cts}} \rightarrow \mathrm{HP}^{\mathrm{cts}}$. For a circle object M , the norm sequence is

$$\Sigma M_{hS^1} \longrightarrow M^{hS^1} \longrightarrow M^{tS^1};$$

see [38, §I.3 and Corollary I.4.3]. By [28, Theorem 2.1, Lemma 2.2, and Theorem 2.3], its cyclic interpretation identifies the last fiber with

$$\Sigma \mathrm{HC}^{\mathrm{cts}}(D/W; \mathbf{Q}_3).$$

This defines a map out of actual relative K-theory.

The required module action can be followed through the construction of the comparison. Put $H = \mathrm{THH}(D)$, regarded as a commutative algebra in cyclotomic spectra. For a bounded-below cyclotomic spectrum H , the construction in [2, Proposition 2.8] tensors the cyclotomic maps $\mathbf{Z}^{\mathrm{triv}} \rightarrow \mathrm{THH}(\mathbf{F}_p)$ with H . It gives a cartesian square with TC in the top row and TC^- in the bottom row. Its horizontal fiber is the p -completion of $\Sigma(H \otimes \mathbf{Z}_{hC_p})_{hS^1}$. The comparison in [2, Corollary 2.10] replaces the lower-right term by a Tate construction, using the equivalence induced by $\mathbf{Z}^{\mathrm{triv}} \rightarrow \mathrm{THH}(\mathbf{F}_p)$, and then inverts p . Specializing to $H = \mathrm{THH}(D)$ and using $\mathrm{THH}(D) \otimes \mathbf{Z} \rightarrow \mathrm{HH}(D)$ gives the square with rows $\mathrm{TC}(D) \rightarrow \mathrm{TC}(D \otimes_{\mathbb{S}} \mathbf{F}_p)$ and $\mathrm{HC}^-(D) \rightarrow \mathrm{HP}(D)$ in [2, Theorem 2.12]. The reduction construction of [2, Definition 2.14] replaces derived reduction by ordinary reduction through their rational TC-equivalence. Finally, [2, Proposition 4.10] replaces reduction modulo p by reduction modulo a uniformizer and projects to relative Hochschild homology.

Every object $H \otimes T$ in the first two constructions is an H -module through its first factor, and every map $\mathrm{id}_H \otimes f$ is H -linear. The lax monoidal structure of TC therefore makes their TC images modules over $\mathrm{TC}(H) = \mathrm{TC}(D)$. The homotopy-fixed-point and Tate objects carry actions of $\mathrm{TC}^-(H)$, and hence of $\mathrm{TC}(H)$ through its canonical map to $\mathrm{TC}^-(H)$. Their norm and comparison maps preserve these actions. This compatibility follows either from the lax monoidal constructions in [38, §I.3 and Corollary I.4.3], or by forming the norm inside the category of modules over the circle-equivariant algebra.

In particular, the Tate equivalence inverted in [2, Corollary 2.10] is a module map. Its inverse is taken in the module category. The map $\mathrm{TC}(H) \rightarrow \mathrm{TC}(H \otimes \mathbf{Z}^{\mathrm{triv}})$, which becomes an equivalence after rationalization in [2, Theorem 2.12], is a module map induced by the unit of $\mathbf{Z}^{\mathrm{triv}}$. After forgetting to circle objects, the maps

$$H \otimes \mathbf{Z} \longrightarrow \mathrm{HH}(D) \longrightarrow \mathrm{HH}(D/W)$$

are maps of H -algebras. The reduction maps $D \otimes_{\mathbb{S}} \mathbf{F}_3 \rightarrow D/3 \rightarrow D/u$ are maps of D -algebras. Thus the rational TC-equivalences used in [2, Definition 2.14 and Proposition 4.10] are $\mathrm{TC}(D)$ -linear as well.

Completion, rationalization, and passage to fibers preserve these module constructions. The resulting action on the cyclic fiber is restriction along

$$\mathrm{TC}(D) \longrightarrow \mathrm{TC}^-(D) \longrightarrow \mathrm{HC}^{-, \mathrm{cts}}(D/W; \mathbf{Q}_3).$$

The multiplicative cyclotomic trace, whose composite with THH is the topological Dennis trace [9, Theorems 7.3 and 7.4], and the compatible actual-to-continuous maps make χ_D a $K(D)$ -module map. The suspension contributes the usual sign for a suspended module. Only the negative cyclic action on cyclic homology is used here; no ring structure on HC is asserted.

For the calculation of the target, the relative dimension bound gives a finite mixed-complex model. On the algebraization D_0 , the normalized Hochschild map is

$$(2.4) \quad b_0 \otimes \cdots \otimes b_n \longmapsto \frac{1}{n!} b_0 db_1 \wedge \cdots \wedge db_n \quad (0 \leq n \leq 2),$$

and is zero in higher degrees. The factorials occurring here are units in W . The Hochschild differential maps to zero, while the Connes operator maps to d ; beyond degree two its target is zero. Smooth Hochschild–Kostant–Rosenberg identifies the underlying Hochschild homology

with $\Omega_{D_0/W}^n$, vanishing for $n > 2$. This identification follows from the Koszul resolution of the smooth diagonal and is realized by (2.4). Hence (2.4) is a quasi-isomorphism of mixed complexes to $(\Omega_{D_0/W}^*, 0, d)$. Only this finite mixed complex is needed, rather than an infinite rational splitting.

The cyclic, negative cyclic, and periodic total complexes of this model contain only finitely many form modules in each total degree. Those modules are W -flat, and their reduction towers modulo 3^n are termwise surjective. Derived 3-completion is therefore given by termwise completion of these total complexes. The relative continuity assertion of [2, Proposition 4.2 and the paragraph preceding Proposition 4.10] identifies them with the continuous constructions. Inverting 3 gives $\mathrm{HC}_0^{\mathrm{cts}}(D/W; \mathbf{Q}_3) = \Omega_D^0$ and (2.3). Indeed, cyclic degree two has Ω_D^2 in column zero and Ω_D^0 in column minus one. The differential out of the second term is d , and boundaries into the first term form $d\Omega_D^1$. These are homology groups in the ordinary derived category, with no closure taken on a differential image.

A unit $v \in 1 + uD$ determines a particular relative class: its scalar automorphism is a loop whose reduction is the identity automorphism. This construction is natural in D and maps to $[v] \in K_1(D)$. To determine its character, first take $D = W\langle T \rangle$ and $v = 1 + uT$, and write

$$f = \chi_D([1 + uT]_{\mathrm{rel}}) \in E\langle T \rangle.$$

Choose the circle orientation for which the norm is Connes' operator, as in [28, Lemma 2.2 and Theorem 2.3]. The norm followed by projection to HH_1 then sends f to df . On the K-theory side, this composite is the Dennis trace. The scalar-unit calculation for that trace represents the class of b by $b^{-1} \otimes b$ in HH_1 , which (2.4) sends to $d\log b$. It follows that

$$df = d\log(1 + uT).$$

Evaluation at $T = 0$ sends the specified relative unit to zero, so $f(0) = 0$. The kernel of differentiation on $E\langle T \rangle$ consists of constants: for each coefficient c_n with $n \geq 1$, the equation $nc_n = 0$ forces $c_n = 0$. The convergent restricted series

$$\log(1 + uT) = \sum_{n \geq 1} \frac{(-1)^{n+1} u^n T^n}{n}$$

has the required derivative and constant term. Thus $f = \log(1 + uT)$. For general $v \in 1 + uD$, substitution $T = (v-1)/u \in D$ defines a continuous W -algebra map from $W\langle T \rangle$ to D . Naturality gives $\chi_D([v]_{\mathrm{rel}}) = \log v$.

It remains to specify the coefficient in the product formula. Over any commutative base ring, negative cyclic homology acts on cyclic homology by ordinary shuffle plus the cyclic variable times cyclic shuffle [10, §2, Definition 2.1 and Remark 2.2]. The cyclic, negative cyclic, and periodic long exact sequences are module sequences, with the usual sign on connecting homomorphisms [10, Proposition 2.3]. These statements precede, and do not require, the characteristic-two hypotheses imposed in later sections of that reference.

Write the cyclic complex, with h of degree -2 , as

$$C^+ = (E[h, h^{-1}]/hE[h]) \otimes_E C_*,$$

where C_* is the rationalized Hochschild chain complex. The function $\log v$ is represented in column zero. A negative cyclic representative of a unit has leading Hochschild class $b^{-1} \otimes b$; its remaining terms contain positive powers of h . Multiplication on column zero kills all those terms, including the cyclic-shuffle correction $h \, \mathrm{sh}'$. The first multiplication remains in column zero, so the second multiplication has the same property. The degree-two shuffle has two terms. After antisymmetrization both give the same wedge, canceling the factor $2!$ in (2.4). Changing a leading Hochschild representative by a boundary does not change its resulting cyclic class. The coefficient in (2.2) is therefore one, apart from the suspension sign already specified. $\square$

2.5. An ordinary residue criterion. In this section, we detect differential classes on the formal quadric of Setup 2.3 by a residue on the overlap of two affine-plane charts.

Lemma 2.7. *For $F \in E\langle Z \rangle$, the form initially defined where X is invertible by*

$$\omega_F = \frac{dX}{X} \wedge dF$$

extends to a global element of Ω_B^2 . If ω_F belongs to the ordinary image $d\Omega_B^1$, then $F(1) = F(-1)$.

Proof. Differentiating the equation of the quadric gives $XdY + YdX + 2ZdZ = 0$. On the locus where Z is invertible,

$$\frac{dX}{X} \wedge dZ = -\frac{dX \wedge dY}{2Z}.$$

The loci where X or Z is invertible cover the quadric, since $X = Z = 0$ is incompatible with $XY + Z^2 = 1$. This identity extends ω_F across its potential pole.

Before completion, let U_+ be the complement of the line $X = 0, Z = -1$, and let U_- be the complement of the line $X = 0, Z = 1$. Put $a_+ = 1$ and $a_- = -1$. Each U_i is an affine plane with coordinates X, t_i , under

$$(2.5) \quad Z = a_i + Xt_i, \quad Y = -t_i(2a_i + Xt_i).$$

To see this, write $U_i = D(X) \cup D(Z + a_i)$. The inverse coordinate is $(Z - a_i)/X$ on $D(X)$ and $-Y/(Z + a_i)$ on $D(Z + a_i)$. The equation $(Z - a_i)(Z + a_i) = -XY$ identifies these functions on the overlap. Their preimages cover the (X, t_i) -plane, since $2a_i$ is a unit. Thus (2.5) is an isomorphism. The intersection $U_+ \cap U_-$ is $D(X)$. After completion these charts are formal bidiscs, with overlap $W\langle X, X^{-1}, Z \rangle$.

On the i -th bidisc, define

$$q_i = -(F(Z) - F(a_i)) \frac{dX}{X}.$$

Substitution from (2.5) makes $F(Z) - F(a_i)$ divisible by X in the restricted series algebra, so q_i is regular. Differentiation gives $dq_i = \omega_F$. On the overlap,

$$(2.6) \quad q_+ - q_- = (F(1) - F(-1)) \frac{dX}{X}.$$

We use a coefficient calculation for closed one-forms on a bidisc. Let

$$A(X, t)dX + D(X, t)dt$$

be closed over $E\langle X, t \rangle$, and write $A = \sum_{m,n \geq 0} a_{m,n} X^m t^n$ and $D = \sum_{m,n \geq 0} b_{m,n} X^m t^n$. The coefficients tend to zero. For $c \in W$, substitution $t = c/X$ is defined on $E\langle X, X^{-1} \rangle$. The coefficient of $X^{-1}dX$ in the resulting form is the convergent sum

$$\sum_{m \geq 0} (a_{m,m+1} - b_{m+1,m}) c^{m+1}.$$

Closedness means $\partial_t A = \partial_X D$, hence

$$(m+1)a_{m,m+1} = (m+1)b_{m+1,m} \quad (m \geq 0).$$

Since E has characteristic zero, each summand in the residue is zero. This calculation does not divide an infinite series to construct a primitive.

Assume now that $\omega_F = dq$ for a global $q \in \Omega_B^1$. The forms $q_i - q$ are closed on the two bidiscs. Restrict them to the common slice $Z = 0$. In chart i , this restriction is the substitution $t_i = -a_i/X$, with $-a_i \in W$. Both restricted forms have residue zero by the coefficient calculation. Their difference, by (2.6), has residue $F(1) - F(-1)$. Therefore $F(1) = F(-1)$. The argument concerns exactness in the ordinary quotient $\Omega_B^2/d\Omega_B^1$. $\square$

2.6. Nonvanishing of the explicit class. The goal of this section is to complete Theorem 2.1 by detecting an actual relative class on B and passing to its absolute image.

Proof of Theorem 2.1. Proposition 2.5 proves regularity and integral vanishing. It remains to prove that c_{A_3} and $c_{\widehat{A}_3}$ have infinite order.

The substitutions

$$x = uX, \quad y = uY, \quad z = uZ$$

define a continuous homomorphism $\widehat{A}_3 \rightarrow B$. Indeed, every formal series in x, y, z converges u -adically after substitution, and the relation maps to $3 + u^2(XY + Z^2) = 0$. In B , put

$$a = \frac{1 + uZ}{2}, \quad \lambda = 1 + uX, \quad \mu = 1 + \frac{(1 + uX)uY}{4}, \quad q = a(1 - a) = \frac{1 + 3Z^2}{4}.$$

These elements are units, and a^2 reduces to 1 modulo u . Consequently, the specified relative unit of Proposition 2.6 gives an actual class

$$(2.7) \quad \gamma = [a^2]_{\text{rel}}[\lambda][\mu] \in K_3(B, uB).$$

Its image in $K_3(B)$ is $[a^2][\lambda][\mu] = 2[a][\lambda][\mu] = c_B$, where c_B is the image of either source class.

The map from relative to absolute K-theory is injective in this degree. To prove this using actual groups, put

$$C = B/uB = \mathbf{F}_3[X, Y, Z]/(XY + Z^2 - 1).$$

A second pair of affine-plane charts describes $\text{Spec } C$ as an affine-line torsor over $\mathbf{P}_{\mathbf{F}_3}^1$. The first chart has coordinates X, t and equations

$$Z = 1 + Xt, \quad Y = -t(2 + Xt).$$

It is the complement of the line $X = 0, Z = -1$. The second has coordinates Y, s and equations

$$Z = -1 - sY, \quad X = -s(2 + sY).$$

It is the complement of the line $Y = 0, Z = 1$. These charts are affine planes by the inverse-coordinate calculation used in (2.5). Their omitted lines are disjoint, so the charts cover $\text{Spec } C$. On their intersection,

$$(2.8) \quad s = t^{-1}, \quad Y = -t^2X - 2t.$$

Thus projection to t on the first chart and to s on the second glues to a morphism $\text{Spec } C \rightarrow \mathbf{P}_{\mathbf{F}_3}^1$. It is locally the affine-line projection, with affine-linear transition of invertible slope $-t^2$. In particular, it is a torsor under a line bundle.

Homotopy invariance on the two charts and their overlap, followed by the two-open Mayer-Vietoris sequence, shows that pullback gives an equivalence on K-theory [40, §7, Remark 3.5 and Proposition 4.1]. The projective bundle formula [40, §8, Theorem 2.1] identifies $K(\mathbf{P}_{\mathbf{F}_3}^1)$ with $K(\mathbf{F}_3) \oplus K(\mathbf{F}_3)$. For a finite field, the actual-group computation is

$$K_{2i}(\mathbf{F}_q) = 0, \quad K_{2i-1}(\mathbf{F}_q) = \mathbf{Z}/(q^i - 1) \quad (i \geq 1);$$

see [41] and [50, IV, Theorem 1.12 and Corollary 1.13]. Only its consequence $K_4(\mathbf{F}_3) = 0$ is needed here. It follows that

$$(2.9) \quad K_4(C) = K_4(\mathbf{F}_3) \oplus K_4(\mathbf{F}_3) = 0.$$

The long exact sequence of the actual relative spectrum contains

$$K_4(C) \longrightarrow K_3(B, uB) \longrightarrow K_3(B).$$

Hence (2.9) makes $K_3(B, uB) \rightarrow K_3(B)$ injective. This argument uses the actual fourth K-group.

We detect γ by its character. Define the 3-adic logarithm of 2 by

$$\log 2 = \frac{1}{2} \log 4 = \frac{1}{2} \log(1 + 3), \quad L(Z) = -\log 2 + \log(1 + uZ).$$

The logarithm series are additive on $1 + uB$ by the formal power series identity and their u -adic convergence. Since $4 \in 1 + uB$ and $a^2 = (1 + uZ)^2/4$, this gives $\log(a^2) = 2L$. This use of the

logarithm remains valid at ramification index two. Proposition 2.6 identifies the top component of $\chi_B(\gamma)$ with $\pm\tau$, where

$$(2.10) \quad \tau = 2L \, \mathrm{dlog} \, \lambda \wedge \mathrm{dlog} \, \mu.$$

For the differential calculation, put $t = uY/4$. Then $q = 1 + uXt$, $\mu = 1 + \lambda t$, and $\lambda q = 1 + uX\mu$. Differentiating the last identity and wedging with $\mathrm{dlog} \, \mu$ gives, where X is invertible,

$$(2.11) \quad \mathrm{dlog} \, \lambda \wedge \mathrm{dlog} \, \mu = \frac{\mathrm{d}X}{X} \wedge \mathrm{dlog} \, q - \mathrm{dlog} \, q \wedge \mathrm{dlog} \, \mu.$$

More explicitly, the left-hand side plus $\mathrm{dlog} \, q \wedge \mathrm{dlog} \, \mu$ is

$$\frac{u \, \mathrm{d}X \wedge \mathrm{d}\mu}{\lambda q} = \frac{u \, \mathrm{d}X \wedge \mathrm{d}t}{q} = \frac{\mathrm{d}X}{X} \wedge \frac{\mathrm{d}q}{q}.$$

The form involving $\mathrm{d}X/X$ is global by the extension calculation in Lemma 2.7, since q depends only on Z . Thus (2.11) is an identity of global regular two-forms.

Choose $F \in E\langle Z \rangle$ satisfying

$$(2.12) \quad F'(Z) = \frac{6ZL(Z)}{1 + 3Z^2}, \quad \text{so that} \quad \mathrm{d}F = L \, \mathrm{dlog} \, q.$$

Such an F exists in the restricted series algebra. With the absolute value normalized by $|3| = 1/3$, both L and $(1 + 3Z^2)^{-1}$ converge for $|Z| < \sqrt{3}$. Termwise integration divides each coefficient by a positive integer, whose 3-adic valuation grows at most logarithmically. The integrated series therefore still has radius of convergence at least $\sqrt{3}$, which is greater than one. Equations (2.10) and (2.11) now give the global identity

$$(2.13) \quad \tau = 2 \frac{\mathrm{d}X}{X} \wedge \mathrm{d}F - 2\mathrm{d}(F \, \mathrm{dlog} \, \mu).$$

By Lemma 2.7, a nonzero difference $F(1) - F(-1)$ makes the class of τ nonzero in the ordinary quotient $\Omega_B^2/\mathrm{d}\Omega_B^1$.

Only the odd part of L contributes to this endpoint difference. Since $u^2 = -3$, that odd part is

$$u \sum_{k \geq 0} \frac{(-3)^k Z^{2k+1}}{2k+1}.$$

Multiply this series by $6Z/(1 + 3Z^2)$, integrate, and subtract the values at 1 and -1 . All series converge on a disc of radius greater than one, which justifies termwise operations and regrouping by $j = k + l$. The resulting formula is

$$(2.14) \quad \frac{F(1) - F(-1)}{u} = 12 \sum_{j \geq 0} \frac{(-3)^j S_j}{2j+3}, \quad S_j = \sum_{k=0}^j \frac{1}{2k+1}.$$

Its $j = 0$ term is 4.

For completeness, an estimate for all indices controls the tail. Normalize $v_3(3) = 1$. Every denominator in S_j is at most $2j+1$, so

$$v_3(S_j) \geq -\lfloor \log_3(2j+1) \rfloor.$$

For $j \geq 1$, the valuation of the j -th summand in (2.14) is therefore at least

$$A_j = 1 + j - \lfloor \log_3(2j+1) \rfloor - v_3(2j+3).$$

Direct substitution gives $A_1 = 1$, $A_2 = 2$, and $A_3 = 1$. For $j \geq 4$, we have

$$(2.15) \quad \lfloor \log_3(2j+3) \rfloor \leq \lfloor j/2 \rfloor.$$

Equivalently, $2j+3 < 3^{\lfloor j/2 \rfloor + 1}$. This holds for $j = 4$ and $j = 5$. Advancing j by two preserves the inequality because $2j+7 < 3(2j+3)$. Both $\lfloor \log_3(2j+1) \rfloor$ and $v_3(2j+3)$ are bounded by $\lfloor \log_3(2j+3) \rfloor$. Thus (2.15) gives

$$A_j \geq 1 + j - 2\lfloor j/2 \rfloor \geq 1.$$

Moreover,

$$A_j \geq 1 + j - 2 \log_3(2j + 3) \longrightarrow +\infty.$$

The entire tail of (2.14) consequently converges in $3\mathbf{Z}_3$, and

$$(2.16) \quad \frac{F(1) - F(-1)}{u} \in 4 + 3\mathbf{Z}_3.$$

In particular, the endpoint difference is nonzero.

Equations (2.13) and (2.16), together with Lemma 2.7, show that $\chi_B(\gamma)$ has nonzero top component. Its target is a $\mathbf{Q}_3$ -vector space, so a torsion element cannot have this image. Hence γ has infinite order in the actual group $K_3(B, uB)$. The injection proved using (2.9) shows that c_B has infinite order in $K_3(B)$. As c_B is the image of both $c_{\widehat{A}_3}$ and c_{A_3} , both source classes have infinite order.

For either source ring R , Proposition 2.5 gives integral vanishing in $K_3(R[1/x])$. Since R is a domain, $R[1/x]$ embeds in $\text{Frac}(R)$, so the class also vanishes in $K_3(\text{Frac}(R))$. An element of an abelian group becomes zero after tensoring with $\mathbf{Q}$ exactly when it is torsion. Thus c_R remains nonzero in $K_3(R) \otimes_{\mathbf{Z}} \mathbf{Q}$, proving the asserted failure of injectivity. $\square$

Remark 2.8 (Ordinary differential homology). Every differential quotient used above is ordinary algebraic homology after derived completion and rationalization. No differential image is replaced by its closure. The argument makes no assertion about the order of the class in an individual finite Artin quotient, and uses no identification

$$K_i(D) \otimes_{\mathbf{Z}} \mathbf{Q}_3 = \pi_i(K(D)_3^{\wedge}[1/3]).$$

It also requires no injection from actual K-theory into completed K-theory.

The distinction between an ordinary image and its closure is visible on this quadric. The global form

$$\omega_0 = \frac{dX}{X} \wedge dZ$$

is nonexact by Lemma 2.7, applied to $F(Z) = Z$, whose endpoint difference is 2. For every integer $j \geq 0$, however, the one-form

$$\beta_j = -(Z - Z^{3^j}) \frac{dX}{X}$$

is global and regular. Indeed, $Z - Z^{3^j}$ vanishes at both 1 and -1 , and is divisible by $Z^2 - 1 = -XY$. Its derivative is

$$d\beta_j = \omega_0 - 3^j Z^{3^j-1} \omega_0 \longrightarrow \omega_0.$$

Thus ω_0 lies in the closure of the differential image while remaining outside that image. Passing to the Hausdorff quotient would discard the obstruction used in the proof.

3. THE QUARTIC CONSTRUCTION

Supported classes and generic restriction. A class pushed forward from a Cartier divisor vanishes after restriction to the complement of that divisor. For a regular local ring, the question is whether such a class can retain infinite order in algebraic K-theory. The localization sequence and the projection formula express this question in terms of supported classes and their products; we use these results in the forms of [50, V.3–V.6]. Our examples allow two different regulators to detect one such product. The first uses the analytic regulator and its comparison with the regulator of a number ring [12, 14]. The second uses the relative fiber comparison with cyclic homology [2, 9].

The ring is defined by a quartic equation whose constant term is the residue characteristic. Its cyclotomic divisor splits into four components after a flat coefficient extension. The coefficients of these components are conjugates of one actual class in degree three. The resulting divisor calculation is the common starting point of both proofs.

3.1. A rational generic kernel. The geometric argument detects the class on every finite open that occurs in the localization defining the local ring.

Theorem 3.1. *Let*

$$A = \left(\mathbf{Z}_{(5)}[T, x, y] / (\Phi_5(1 + y) + x^4 + Txy^3) \right)_{(5, x, y)}, \quad \Phi_5(z) = z^4 + z^3 + z^2 + z + 1.$$

Then A is a regular local ring of dimension two with residue field $\mathbf{F}_5(T)$. There is an actual infinite-order class $c \in K_3(A)$ whose image in $K_3(\text{Frac } A)$ is zero. Here K_3 denotes Quillen algebraic K-theory.

The class is the pushforward of a cyclotomic coefficient from $x = 0$. After the divisor splits, its analytic regulator is a linear combination of four first Chern classes. A blowup calculation shows that the only relation among these classes is the diagonal relation. It also controls every geometric component of every divisor removed by an allowed localization.

3.2. Persistence after completion. The completed argument requires a coefficient with prescribed nonzero components at both odd characters.

Theorem 3.2. *Let A be the ring in Theorem 3.1, and let $\hat{A}$ be its (x, y) -adic completion. Both rings are regular local domains of dimension two, with maximal ideal (x, y) and residue field $\mathbf{F}_5(T)$. There is an actual class $c \in K_3(A)$ such that its image $\hat{c} \in K_3(\hat{A})$ has infinite order and*

$$c|_{\text{Frac } A} = 0, \quad \hat{c}|_{\text{Frac } \hat{A}} = 0$$

integrally. Consequently both maps

$$K_3(A) \otimes \mathbf{Q} \longrightarrow K_3(\text{Frac } A) \otimes \mathbf{Q}, \quad K_3(\hat{A}) \otimes \mathbf{Q} \longrightarrow K_3(\text{Frac } \hat{A}) \otimes \mathbf{Q}$$

have nonzero kernel.

The class in Theorem 3.2 can also be used in the geometric proof of Theorem 3.1. We establish this identification after constructing its arithmetic coefficient. In particular, the passage to completion does not require choosing a second supported class.

Proof architecture. The completed detector is a continuous two-form relative to a ramified coefficient ring. Its parameter differential is retained. A tame trace to a constant ordinary elliptic curve, followed by a bounded contraction, takes the weighted root-divisor class to $4e_3(b)$ times a nonzero one-form class on a generic Cohen completion. The exact linear identity is proved in Proposition 9.1; it holds for every anti-invariant coefficient b . Nonvanishing of the one-form rests on scalar Frobenius pole removal in Lemma 8.1 and on the rigid-to-convergent results of [15].

Proposition 7.1 identifies the relevant relative K-theory fiber with ordinary cyclic homology after spectral completion and inversion of 5. Its product calculation detects an actual relative class. The smooth-surface finite-coefficient bound of [25, Theorem 8.4] then applies twice: first to the fiber of reduction, and then to the support term in localization. These are distinct fiber sequences.

Section 4 fixes the K-theory conventions and the supported construction. Sections 5 and 6 prove the geometric route. Section 7 establishes the relative cyclic comparison. Section 8 develops the Cohen and elliptic arguments. Section 9 computes the weighted quartic detector. Section 10 constructs its actual arithmetic coefficient. Section 11 proves Theorem 3.2 and identifies the coefficient used in both routes.

4. PRELIMINARIES

We use algebraic K-theory, transfers, localization, and finite coefficients with the conventions of [50, Chapters IV and V].

Notation 4.1. For a ring H and an ideal I , put

$$K(H, I) = \text{fib}(K(H) \longrightarrow K(H/I)), \quad \mathcal{K}(H) = K(H)_5^\wedge[1/5], \quad \mathcal{K}(H, I) = K(H, I)_5^\wedge[1/5].$$

The completion is the derived inverse limit of the spectra $K(H)/5^n$, and likewise for a relative spectrum. It is not defined by tensoring the actual K-groups with $\mathbf{Z}_5$. Completion and inversion of 5 preserve fiber sequences.

On adically completed rings, differential forms are continuous forms. Every de Rham quotient in this article uses the ordinary image of the differential, unless a closure is explicitly introduced in the construction of a bounded functional.

For regular noetherian rings of finite dimension, projective-module K-theory agrees with coherent-module G-theory. Finite closed immersions have G-theory pushforwards by restriction of scalars. These pushforwards commute with flat base change and satisfy the projection formula

$$i_*(i^*\alpha \cdot z) = \alpha \cdot i_*z.$$

For an effective Cartier divisor with invertible ideal I , its two-term resolution gives $i_*[\mathcal{O}] = 1 - [I]$. We use [50, V.3.3, V.3.3.2, V.3.7.2, V.3.12] for these statements. In each application, the immersions are finite between affine schemes and the base change in question is flat. Thus the required properness and Tor-independence hold.

For a regular noetherian ring H , a nonzero divisor s , and a regular quotient H/sH , localization and dévissage give

$$K(H/sH) \longrightarrow K(H) \longrightarrow K(H[1/s]),$$

where the first map is support pushforward [50, V.4.1, V.6.1]. K-groups commute with filtered colimits of rings [50, IV, Section 6.4]. The same holds with finite coefficients, by the cofiber construction. In particular, an equality in a filtered localization holds after some finite further localization.

Setup 4.2. Write

$$\begin{aligned} P(y) &= \Phi_5(1+y) = 5 + 10y + 10y^2 + 5y^3 + y^4, \\ F &= P(y) + x^4 + Txy^3, \\ R_0 &= \mathbf{Z}_{(5)}[T, x, y]/(F), \quad \mathfrak{m} = (5, x, y), \quad S = R_0 \setminus \mathfrak{m}, \quad A = S^{-1}R_0. \end{aligned}$$

Fix a primitive fifth root ρ , and put

$$E_{\text{ar}} = \mathbf{Q}(\rho), \quad W_{\text{ar}} = \mathbf{Z}_{(5)}[\rho], \quad \pi = \rho - 1, \quad G = (\mathbf{Z}/5\mathbf{Z})^\times.$$

For $g \in G$, let $\sigma_g(\rho) = \rho^g$.

The two coefficients xy^3 and $P(y) + x^4$ of the linear polynomial F in T have no common irreducible factor in $\mathbf{Z}_{(5)}[x, y]$. Neither x nor y divides $P(y) + x^4$. Gauss's lemma therefore shows that R_0 is a domain. Solving for T gives $\text{Frac } A = \mathbf{Q}(x, y)$.

The ambient local ring $\mathbf{Z}_{(5)}[T, x, y]_{(5, x, y)}$ is regular of dimension three, with residue field $k_0 = \mathbf{F}_5(T)$. The class of F in its cotangent space is the nonzero class of 5. Thus A is regular local of dimension two. The identity

$$(4.1) \quad 5(1 + 2y + 2y^2 + y^3) = -(x^4 + y^4 + Txy^3)$$

shows that its maximal ideal is (x, y) . Completion preserves the dimension and cotangent space of a noetherian local ring, so $\hat{A}$ is also regular local of dimension two, with the same residue field. Both rings are domains.

The polynomial P is Eisenstein at 5. Consequently W_{ar} is finite free of rank four over $\mathbf{Z}_{(5)}$. Its reduction modulo 5 is $\mathbf{F}_5[\pi]/(\pi^4)$, so it is local. Moreover,

$$(4.2) \quad 5 = \pi^4 \epsilon, \quad \epsilon = -\frac{1}{1 + 2\pi + 2\pi^2 + \pi^3}, \quad \epsilon \bmod \pi = -1.$$

Thus W_{ar} is a discrete valuation ring with uniformizer π and residue field $\mathbf{F}_5$. Taking the quotient by x gives

$$(4.3) \quad A/xA = W_{\text{ar}}[T]_{(\pi)}, \quad y \longmapsto \pi.$$

This is a regular discrete valuation ring. Its completion $\widehat{A}/x\widehat{A}$ is regular as well.

Definition 4.3. Let $i: \operatorname{Spec}(A/xA) \rightarrow \operatorname{Spec} A$ be the closed immersion. For an actual class $\beta \in K_3(W_{\text{ar}})$, define

$$c(\beta) = i_*(\beta|_{A/xA}) \in K_3(A).$$

Restriction to $A[1/x]$ kills $c(\beta)$, because the modules in its pushforward are supported on $x = 0$. Hence its generic image is zero integrally. The map $A \rightarrow \widehat{A}$ is flat. Flat base change identifies its completed image with the same pushforward on $\operatorname{Spec} \widehat{A}$, so that image also vanishes over $\operatorname{Frac} \widehat{A}$. The remaining issue in both theorems is infinite order.

5. THE DIVISOR LATTICE AND ITS FINITE OPENS

In this section, we prove the Picard and Betti persistence used in Theorem 3.1.

Proposition 5.1. *In Setup 4.2, put*

$$B_{\text{ar}} = A \otimes_{\mathbf{Z}_{(5)}} W_{\text{ar}}, \quad \mathcal{T} = \operatorname{Spec} B_{\text{ar}}[1/\pi, 1/y],$$

and

$$\mathcal{Q} = \operatorname{Spec} E_{\text{ar}}[T, x, y, y^{-1}]/(F), \quad D_g = V(x, y - (\rho^g - 1)) \subset \mathcal{Q}.$$

For $s \in S$, write $\mathcal{Q}_s = D(s) \subset \mathcal{Q}$. Then $\mathcal{Q}$ is a smooth geometrically integral surface, $\mathcal{T} = \operatorname{Spec} S^{-1}\mathcal{O}(\mathcal{Q})$, and

$$\operatorname{Pic}(\mathcal{Q}) = \mathbf{Z}^4/\mathbf{Z}(1, 1, 1, 1),$$

where the standard basis maps to $\mathcal{O}(D_g)$. This calculation persists under every extension of E_{ar} , and $\operatorname{Pic}(\mathcal{Q}) \rightarrow \operatorname{Pic}(\mathcal{T})$ is injective.

For every embedding $E_{\text{ar}} \rightarrow \mathbf{C}$, the classes

$$e_g = c_1^{\text{top}}(\mathcal{O}(D_g)) \in H^2(\mathcal{Q}(\mathbf{C}), \mathbf{R})$$

have exactly the real diagonal relation. For every $s \in S$, restriction induces an injection

$$H^2(\mathcal{Q}(\mathbf{C}), \mathbf{R}) \longrightarrow H^2(\mathcal{Q}_s(\mathbf{C}), \mathbf{R}).$$

Proof. The ring B_{ar} is finite free over A . Its closed fiber is $k_0[\pi]/(\pi^4)$, so it has the single maximal ideal $\mathfrak{n} = (\pi, x, y)$. Tensoring $A \hookrightarrow \mathbf{Q}(x, y)$ with the free $\mathbf{Z}_{(5)}$ -module W_{ar} embeds B_{ar} in $E_{\text{ar}}(x, y)$. Thus B_{ar} is a domain. With $V_{\text{ar}} = W_{\text{ar}}[T]_{(\pi)}$, comparison of localizations gives

$$B_{\text{ar}} = (V_{\text{ar}}[x, y]/(F))_{(\pi, x, y)}.$$

Here every polynomial whose reduction is nonzero modulo π becomes a unit. Inverting π and y gives $S^{-1}E_{\text{ar}}[T, x, y, y^{-1}]/(F)$, as asserted.

Consider the blowup $\mathcal{Y} = \operatorname{Bl}_{\mathfrak{n}}(\operatorname{Spec} B_{\text{ar}})$. Its tangent coordinates $(U : X : Y)$ correspond to (π, x, y) . The associated graded ring is

$$k_0[U, X, Y]/(X^4 + Y^4 + TXY^3 - U^4).$$

Indeed F has order four in the ambient regular local ring, and its nonzero initial form generates the initial ideal of the principal ideal (F) . The exceptional scheme is the plane curve

$$C_{\text{exc}}: X^4 + Y^4 + TXY^3 = U^4.$$

A geometric singular point of this curve would satisfy $U = 0$. It cannot have $Y = 0$, since then $X = 0$. Putting $s = X/Y$, the remaining equations are

$$s^4 + Ts + 1 = 0, \quad 4s^3 + T = 0.$$

In characteristic five these imply $T = s^3$, $s^4 = 2$, and $T^4 = 3$, contrary to the transcendence of T . Thus C_{exc} is geometrically smooth. Distinct geometric plane components would intersect by Bézout and give a singular point; repeated components would also be singular. Therefore the curve is geometrically integral.

The strict-transform equations retain the coefficient $\epsilon = 5/\pi^4$ from (4.2):

$$\begin{aligned} h_\pi &= X^4 + Y^4 + TXY^3 + \epsilon(1 + 2\pi Y + 2\pi^2 Y^2 + \pi^3 Y^3), \\ h_x &= 1 + Y^4 + TY^3 + \epsilon U^4(1 + 2xY + 2x^2 Y^2 + x^3 Y^3), \\ h_y &= X^4 + 1 + TX + \epsilon U^4(1 + 2y + 2y^2 + y^3). \end{aligned}$$

The substitutions are respectively $(x, y) = (\pi X, \pi Y)$, $(\pi, y) = (xU, xY)$, and $(\pi, x) = (yU, yX)$. The latter two ambient charts retain the relations $\pi = xU$ and $\pi = yU$.

These ambient charts are regular. For example, on the x -chart one can eliminate x or U when the other is a unit. At a point where both vanish, $\pi - xU$ has the nonzero linear term π in a regular local polynomial ring over V_{ar} . The argument on the y -chart is identical with x replaced by y ; the π -chart is a localization of $V_{\text{ar}}[X, Y]$.

Let t denote the exceptional parameter on any one of these charts. It is a nonzero divisor, the ambient quotient modulo t is a polynomial domain over k_0 , and the corresponding h has nonzero reduction. If $ta = hb$, reduction modulo t forces $b = tb'$, and cancellation gives $a = hb'$. Thus the quotient by h has no t -torsion. The displayed equations describe the strict transform without an additional exceptional component.

At a point of C_{exc} , its smoothness gives a nonzero differential of the reduced equation in an exceptional-plane direction. Therefore the strict transform is regular there. Away from the exceptional divisor, where π is invertible, the characteristic-zero hypersurface is smooth: if $F_T = xy^3 = 0$, then either $x = 0$, where $P'(y) \neq 0$ at each root of P , or $y = 0$, where $x^4 = -5$ and $F_x = 4x^3 \neq 0$. At a nonclosed point over $\pi = 0$, the binary form $x^4 + y^4 + Txy^3$ is geometrically square-free by the repeated-root calculation above. Its affine zero locus is smooth away from $(x, y) = (0, 0)$, so its x, y Jacobian proves regularity there. Consequently $\mathcal{Y}$ is regular everywhere and is an isomorphism off $\mathfrak{n}$. The hypersurface B_{ar} is Cohen–Macaulay and regular in codimension one, hence normal.

The exceptional ideal satisfies

$$\mathfrak{n}\mathcal{O}_{\mathcal{Y}} = \mathcal{O}_{\mathcal{Y}}(-C_{\text{exc}}), \quad \mathcal{O}_{\mathcal{Y}}(C_{\text{exc}})|_{C_{\text{exc}}} = \mathcal{O}_{C_{\text{exc}}}(-1).$$

These identities follow from the exceptional parameters on the charts, or from [44, Lemmas 31.33.2 and 31.33.4, Tags 0804 and 02OS]: the charts are $R[I/a]$, the extended ideal is generated by a , and the degree-minus-one tautological line is the exceptional-divisor line.

Write $a_g = \rho^g - 1$ and $b_g = a_g/\pi$. Then b_g is a unit with reduction g . The strict transform $\tilde{D}_g$ of the prime $(x, y - a_g)$ meets the exceptional curve only at

$$P_g = [1 : 0 : g].$$

Its parameter is π , giving this tangent direction. On the π -chart, setting $X = 0$ gives

$$h_\pi(0, Y) = \prod_{g=1}^4 (Y - b_g),$$

because $P(\pi Y) = \pi^4 \prod_g (Y - b_g)$. The Y -derivative is a unit at $(\pi, X, Y - b_g)$. The maximal ideal there is generated by π, X , while $\tilde{D}_g$ is given by $X = 0$. Its intersection with C_{exc} , given by $\pi = 0$, has multiplicity one.

We next compute the lattice on $C_{\text{exc}}^* = C_{\text{exc}} \cap \{UY \neq 0\}$. Extend the constants to $k = \overline{\mathbf{F}}_5$, and consider

$$M = \text{Spec } k[X, Z, Y, Y^{-1}]/(XZ - (1 - Y^4)), \quad T = \frac{Z - X^3}{Y^3}.$$

This surface is smooth, since the derivative $4Y^3$ is a unit. The open $X \neq 0$ has coordinate ring $k[X^{\pm 1}, Y^{\pm 1}]$, which is factorial. The omitted prime divisors are $L_g = V(X, Y - g)$, each with multiplicity one in $\text{div}(X)$. They generate $\text{Cl}(M) = \text{Pic}(M)$ by divisor localization. A relation among them is the divisor of a function that becomes a unit on the Laurent open. Such a unit is $cX^r Y^s$. Since Y is already invertible on M , the only relation is

$$\text{Pic}(M) = \mathbf{Z}^4/\mathbf{Z}(1, 1, 1, 1).$$

For every $t \in k$, the fiber $T = t$ is

$$X^4 + Y^4 + tXY^3 = 1, \quad Y \neq 0.$$

With $s = X/Y$ and $w = 1/Y$, its coordinate ring is

$$k[s, w, w^{-1}]/(w^4 - (s^4 + ts + 1)).$$

The polynomial $s^4 + ts + 1$ is not a square in $k(s)$. A rational square that is polynomial is a polynomial square by unique factorization. Comparing it with $(s^2 + bs + c)^2$, after normalizing the leading coefficient, gives $b = c = 0$ from the cubic and quadratic coefficients, contradicting the constant term. Since k contains the fourth roots of unity and has characteristic different from two, adjoining a fourth root of this nonsquare has degree four. The extension is a splitting field with Galois group inside μ_4 ; a subgroup of order at most two would fix the square of the root and make the polynomial a square. Thus every fiber is integral and reduced. Its single vertical prime divisor is $\text{div}(T - t)$, with multiplicity one.

Localization to $k(T)$ removes exactly these vertical prime divisors. Their classes are zero, so

$$\text{Pic}(M) \xrightarrow{\sim} \text{Pic}(M_{k(T)}).$$

The generic fiber is $C_{\text{exc}, k(T)}^*$, and L_g restricts to P_g with multiplicity one. Hence a principal combination $\sum_g n_g P_g$ on C_{exc}^* must be diagonal: extend to $k(T)$ and use the lattice just computed.

The substitution $v = x^3 + Ty^3$ identifies

$$\mathcal{Q} = \text{Spec } E_{\text{ar}}[x, v, y, y^{-1}]/(xv + P(y)).$$

This surface is smooth because P has four distinct roots, all in E_{ar} . Its open $x \neq 0$ is the Laurent factorial surface with coordinates x, y . The removed prime divisors are exactly D_g , each occurring once in $\text{div}(x)$. Every relation is obtained from a Laurent unit $cx^r y^s$. It follows that $\text{Pic}(\mathcal{Q}) = \mathbf{Z}^4/\mathbf{Z}(1, 1, 1, 1)$. This also proves geometric integrality. The same argument works over every extension of E_{ar} .

Suppose that $\sum_g n_g D_g$ is principal on $\mathcal{T}$, with rational function f . On the regular surface $\mathcal{Y}$, its divisor differs from $\sum_g n_g \tilde{D}_g$ by

$$mC_{\text{exc}} + \sum_j m_j V_j,$$

where V_j are strict transforms of primes in $\pi = 0$ or $y = 0$. Indeed $\mathcal{Y}[1/\pi, 1/y] = \mathcal{T}$. The strict transforms over $\pi = 0$ meet the exceptional curve only in $U = 0$, and those over $y = 0$ meet it only in $Y = 0$, as the chart equations show. They therefore miss C_{exc}^* . The line $\mathcal{O}_{C_{\text{exc}}}(-1)$ is trivial where $U \neq 0$. Restriction of the principal-divisor identity gives

$$\sum_g n_g [P_g] = 0 \quad \text{in } \text{Pic}(C_{\text{exc}}^*).$$

The coefficients n_g are equal. The original combination is already a multiple of $\text{div}(x)$ on $\mathcal{Q}$, which proves Picard injectivity.

Fix an embedding into $\mathbf{C}$. The divisors $D_g(\mathbf{C})$ are disjoint affine lines, and their complement is $(\mathbf{C}^\times)^2$. The Thom isomorphism gives the exact segment

$$H^1((\mathbf{C}^\times)^2, \mathbf{R}) \longrightarrow \bigoplus_g \mathbf{R} \longrightarrow H^2(\mathcal{Q}(\mathbf{C}), \mathbf{R}).$$

The second map sends the standard vectors to e_g . The angular classes of x, y form a basis of the first group. Around each D_g , x winds once because its zero is simple, whereas y winds zero times because it is invertible. The image of the first map is precisely the diagonal line. This proves the asserted relation among the e_g .

Let Z be an irreducible E_{ar} -divisor in the zero locus of some $s \in S$. Its restriction to $\mathcal{T}$ is empty. Picard injectivity therefore gives $[Z] = 0$ on $\mathcal{Q}$. Over an algebraic closure, its geometric components $Z_1, \dots, Z_d$ form one Galois orbit and each has multiplicity one, since the characteristic is zero. The geometric Picard group is the torsion-free lattice computed above.

Galois acts trivially on it because its generators D_g are already defined over E_{ar} . Thus all $[Z_j]$ equal a class h , and $dh = 0$. It follows that every $[Z_j]$ vanishes. Geometric irreducibility persists under extension of algebraically closed fields, so every irreducible complex component of the boundary has zero Picard class and zero topological first Chern class.

Let H be the reduced complex divisor cut out by s . If it is empty, restriction is the identity. Otherwise remove the finite set Σ of singular points and component intersections. Local cohomology at a point of a smooth real four-manifold is concentrated in degree four, by the cohomology of a ball relative to its punctured ball. Therefore

$$H_{\Sigma}^2(\mathcal{Q}(\mathbf{C}), \mathbf{R}) = H_{\Sigma}^3(\mathcal{Q}(\mathbf{C}), \mathbf{R}) = 0.$$

Put $X = \mathcal{Q}(\mathbf{C})$, $X^\circ = X \setminus \Sigma$, and $D = H \setminus \Sigma$. The support exact sequences give

$$H_H^2(X, \mathbf{R}) \xrightarrow{\sim} H_D^2(X^\circ, \mathbf{R}), \quad H^2(X, \mathbf{R}) \xrightarrow{\sim} H^2(X^\circ, \mathbf{R}).$$

Each irreducible component H_i of H has connected smooth part $D_i = H_i \setminus \Sigma$: its normalization is a connected Riemann surface, and removing finitely many points preserves connectedness. The D_i are disjoint closed smooth curves in X° .

A tubular neighborhood is a disk neighborhood in their normal complex line bundles; a varying radius accommodates noncompact curves. Excision reduces cohomology with supports to the relative cohomology of this neighborhood and its complement of the zero section. Over a contractible trivializing open, the normal fiber pair $(B^2, B^2 \setminus \{0\})$ has relative cohomology $\mathbf{R}$ in degree two and zero otherwise. Changes of trivialization preserve its complex orientation and therefore its generator. Relative Mayer–Vietoris on a good trivializing cover gives

$$H_D^2(X^\circ, \mathbf{R}) = H^0(D, \mathbf{R}) = \bigoplus_i \mathbf{R}.$$

This uses ordinary cohomology of D , without a compactness assumption.

The canonical section of $\mathcal{O}_X(H_i)$ is transverse to zero along D_i on X° , with complex-linear normal derivative. Pulling back the oriented fiber Thom class gives the corresponding support generator. After forgetting supports, homotopy of this section to the zero section identifies its class with the Euler class of the underlying oriented real plane bundle. This equals $c_1(\mathcal{O}_X(H_i))|_{X^\circ}$: both classes are given by the winding-number cocycle of the circle-valued transition functions, with the complex orientation fixing the positive generator. The isomorphism on ambient H^2 above identifies its preimage with the global class $c_1(\mathcal{O}_X(H_i))$ on X . Every such class is zero by the Picard-group calculation. The restriction sequence

$$H_H^2(\mathcal{Q}(\mathbf{C}), \mathbf{R}) \longrightarrow H^2(\mathcal{Q}(\mathbf{C}), \mathbf{R}) \longrightarrow H^2(\mathcal{Q}_s(\mathbf{C}), \mathbf{R})$$

now proves injectivity for every s . □

6. THE MULTIPLICATIVE ANALYTIC REGULATOR

The goal of this section is to prove Theorem 3.1 from Proposition 5.1.

Proof of Theorem 3.1. Regularity and the supported construction were established in Section 4. Let $\mathcal{O}_{E_{\text{ar}}}$ be the integer ring of E_{ar} . It is contained in W_{ar} : an element integral over $\mathbf{Z}$ is integral over W_{ar} , which is integrally closed in E_{ar} .

The field E_{ar} has two complex-conjugate pairs of embeddings and no real embedding. By [12, Proposition 12.2],

$$\dim_{\mathbf{R}}(K_3(\mathcal{O}_{E_{\text{ar}}}) \otimes \mathbf{R}) = 2.$$

The analytic regulator is injective on this real vector space by [14, Section 6, especially (6.4)]. The comparison there with the Borel regulator is up to a nonzero factor 2, which does not affect injectivity. At a point, its weight-two target is $\mathbf{C}/\mathbf{R}(2)$, where $\mathbf{R}(p) = (2\pi i)^p \mathbf{R}$. Conjugate embeddings give conjugate values. The interval description has point representatives $i^{p+1} dt$; in weight two they are purely imaginary.

Choose $\gamma_0 \in K_3(\mathcal{O}_{E_{\text{ar}}})$ with nonzero regulator, and set

$$\gamma = \gamma_0 - \sigma_{-1}\gamma_0, \quad \beta = \gamma|_{W_{\text{ar}}}.$$

Since $\mathbf{R}(2)$ is the real subspace of $\mathbf{C}$, conjugation negates the unique imaginary representative of a point regulator. Thus the regulator of γ is twice that of γ_0 and remains nonzero after restriction to E_{ar} . The identity $\sigma_{-1}\beta = -\beta$ holds in actual K-theory. Put $c = c(\beta)$. Its generic restriction is zero by Definition 4.3.

On $\mathcal{Q}$, define

$$J_g = \mathcal{O}_{\mathcal{Q}}(-D_g), \quad \eta_g = 1 - [J_g], \quad \kappa = \sum_{g \in G} (\sigma_g \beta)|_{E_{\text{ar}}} \cdot \eta_g.$$

The resolution of the Cartier divisor gives $\eta_g = [\mathcal{O}_{D_g}]$ in $K_0(\mathcal{Q})$. We will detect κ on every $\mathcal{Q}_s$.

The regulator is first defined at a finite arithmetic stage. Choose N divisible by 5 and by the denominators needed for s . The roots $a_g = \rho^g - 1$ and their pairwise differences are units over $\mathcal{O}_{E_{\text{ar}}}[1/N]$. Indeed their norms have absolute value 5: for a_g this is $\Phi_5(1) = 5$, and a difference is $\rho^h(\rho^{g-h} - 1)$. The scheme

$$X_N = \text{Spec } \mathcal{O}_{E_{\text{ar}}}[1/N][x, v, y, y^{-1}]/(xv + P(y))$$

is smooth over $\mathcal{O}_{E_{\text{ar}}}[1/N]$. At a residue point where either x or v is nonzero, one of the corresponding partial derivatives is nonzero. Where both vanish, $P'(y) \neq 0$, since the four roots remain distinct. Thus X_N and $(X_N)_s$ are regular separated schemes of finite type over $\mathbf{Z}$. The root divisors extend as Cartier divisors: near $y = a_g$, the other factors of P are units and the equation expresses $y - a_g$ as a multiple of x . The coefficient classes $\sigma_g \gamma$ extend there as well.

For such schemes, [14, Theorem 2.31 and Remark 2.27] gives a natural regulator of ring spectra with homotopy maps

$$K_n(X) \longrightarrow \prod_{p \geq 0} H_{\mathcal{D}, \text{an}}^{2p-n}(X(\mathbf{C}), \mathbf{R}(p)).$$

The target satisfies the arithmetic conjugation invariance. After constructing this map, project to the component selected by a fixed embedding $\tau: E_{\text{ar}} \rightarrow \mathbf{C}$. Increasing N is compatible with this construction. Filtered-colimit continuity [50, IV, Section 6.4] therefore defines the regulator on $K_3(\mathcal{Q}_s)$; the complex component is always $U_s = \mathcal{Q}_s(\mathbf{C})$.

We use its multiplicative structure explicitly. By [14, Remark 2.6 and Definition 2.7], analytic Deligne cohomology is the shifted cone of

$$(2\pi i)^p A_{\mathbf{R}}^{\bullet}(U) \oplus F^p A_{\mathbf{C}}^{\bullet}(U) \longrightarrow A_{\mathbf{C}}^{\bullet}(U), \quad (a, b) \longmapsto a - b.$$

Here $A^{\bullet}$ denotes smooth forms, and F^p means at least p holomorphic differentials. There is no logarithmic boundary condition. Its interval model consists of forms on $[0, 1] \times U$ whose endpoint values are respectively real-twisted and in F^p . Multiplication is wedge product. The map to the cone is

$$\theta \longmapsto \left(\theta|_{0, \theta|_1, - \int_{[0,1]} \theta \right);$$

see [14, Proposition 2.8 and Corollary 2.14].

For a complex surface, $F^3 A_{\mathbf{C}}^{\bullet}(U) = 0$, because at most two holomorphic coordinate differentials can occur. The cone sequence consequently identifies

$$(6.1) \quad H_{\mathcal{D}, \text{an}}^3(U, \mathbf{R}(3)) = H^2(U, \mathbf{C})/H^2(U, \mathbf{R}(3)).$$

The next possible term is the kernel of $H^3(U, \mathbf{R}(3)) \rightarrow H^3(U, \mathbf{C})$, which is zero by extension of scalars. This argument does not require properness.

Let $r_g \in i\mathbf{R}$ be the imaginary representative of the point regulator of $\sigma_g \beta$ at τ . At least one r_g is nonzero, since the embeddings $\tau \sigma_g$ exhaust all embeddings of E_{ar} . Antisymmetry and commutativity of G give $r_{-g} = -r_g$. The vector (r_g) is therefore nonzero and not diagonal.

The interval representative of this point class is $-r_g dt$. Choose a Hermitian metric and Chern connection on $\mathcal{O}(D_g)$, and let ω_g be the resulting real closed representative of e_g . Its weight-one regulator is the interval-constant form $2\pi i \omega_g$. This follows from the unitary, type-(1, 1) curvature and the degree-two term of the unnormalized character $\text{Tr exp}(-\text{curvature})$ in [14,

Definitions 2.16–2.20]. The weight-one class of η_g is the same, by additivity of the first Chern class. Wedge multiplication followed by minus integration gives

$$(-r_g dt) \wedge (2\pi i \omega_g) \mapsto 2\pi i r_g \omega_g.$$

Thus the weight-three regulator of $\kappa|_{\mathcal{Q}_s}$ is

$$(6.2) \quad \left[2\pi i \sum_g r_g e_g|_{U_s} \right] \in H^2(U_s, \mathbf{C})/H^2(U_s, \mathbf{R}(3)).$$

No other point weight contributes: point Deligne cohomology is nonzero only in degrees zero and one, and for K_3 the equation $2p - 3 = 1$ forces $p = 2$.

Write $r_g = it_g$ with $t_g \in \mathbf{R}$. Proposition 5.1 shows that $\sum_g t_g e_g \neq 0$, since (t_g) is not diagonal. It stays nonzero on every U_s . The representative in (6.2) is the nonzero real class

$$-2\pi \sum_g t_g e_g|_{U_s}.$$

By contrast, $H^2(U_s, \mathbf{R}(3)) = iH^2(U_s, \mathbf{R})$ is purely imaginary. The real and purely imaginary subspaces intersect in zero. Hence (6.2) is nonzero. Its target is a real vector space, so $\kappa|_{\mathcal{Q}_s}$ has infinite order for every $s \in S$.

The map $A \rightarrow B_{\text{ar}}[1/\pi, 1/y]$ is flat. Over it, the divisor $x = 0$ splits into the disjoint components $D_g \cap \mathcal{T}$. On the g -th component, the coefficient map sends $\rho = 1 + y$ to ρ^g . Flat base change and the projection formula give the equality of actual classes

$$(6.3) \quad c|_{\mathcal{T}} = \sum_g (\sigma_g \beta)|_{E_{\text{ar}}} [\mathcal{O}_{D_g \cap \mathcal{T}}] = \kappa|_{\mathcal{T}}.$$

The ring of $\mathcal{T}$ is regular, being a localization of the smooth surface $\mathcal{Q}$. No identification of K-theory and G-theory on the possibly singular ring B_{ar} is involved.

Finally,

$$K_3(\mathcal{T}) = \varinjlim_{s \in S} K_3(\mathcal{Q}_s).$$

If a nonzero integer killed c , it would kill κ in this colimit by (6.3). The resulting equality would hold at a finite stage, contrary to (6.2). Thus c has infinite order. $\square$

7. RELATIVE CYCLIC DETECTION

In this section, we prove the comparison and product formula that will detect the completed supported class.

Proposition 7.1. *Let $k = \overline{\mathbf{F}}_5$, $W_0 = W(k)$, and let R be the integer ring of a finite totally ramified extension of $W_0[1/5]$. Write π_R for a uniformizer and $E_R = \text{Frac } R$. Let B_0 be a localization of a smooth R -algebra whose relative cotangent module is finite projective of constant rank two. Assume that $B = \varprojlim_n B_0/5^n B_0 \neq 0$. There is a natural equivalence*

$$(7.1) \quad \mathcal{K}(B, \pi_R B) \simeq \Sigma \text{HC}^{\text{cts}}(B/R; \mathbf{Q}_5)$$

compatible with the $K(B)$ -module action through the negative cyclic Chern character.

The degree-two cyclic group has the canonical summand

$$\Omega_{B/R}^2[1/5]/d\Omega_{B/R}^1[1/5].$$

For an actual $\beta \in K_3(R, \pi_R R)$, let $b \in E_R$ be its relative point coordinate, so its point character is $t^{-1}b$ in the mixed-HKR convention with $|t| = -2$. There is a scalar $\lambda_R \in E_R^\times$, independent of B , J , and β , such that for every invertible B -module J , the top-form component of $\beta_B(1 - [J])$ under (7.1) is

$$-\lambda_R b c_1^{\text{dR}}(J).$$

Nonvanishing of $-b c_1^{\text{dR}}(J)$, or of a finite sum of these forms, proves nonvanishing of the corresponding actual relative class after rationalized spectral completion: the whole sum is multiplied by the same nonzero λ_R . No injectivity from actual K-theory to its completion is asserted.

Proof. The fiber theorem [2, Theorem A] applies to a commutative ring henselian along (5). Its coefficients $\mathbf{Q}_5$ mean spectral completion followed by inversion of 5. It gives a cartesian square with rows

$$K(T; \mathbf{Q}_5) \longrightarrow K(T/5; \mathbf{Q}_5), \quad \mathrm{HC}^-(T; \mathbf{Q}_5) \longrightarrow \mathrm{HP}(T; \mathbf{Q}_5),$$

and hence identifies the relative K-theory fiber with $\Sigma \mathrm{HC}(T; \mathbf{Q}_5)$. The continuity result [2, Proposition 4.2] applies to bounded 5-power torsion and compares completed Hochschild and cyclic objects with their infinitesimal inverse limits. The paragraph preceding [2, Proposition 4.10] also treats relative Hochschild and cyclic theories. We extend that relative argument to B_0 by bounds in each finite degree range; no topological finite-presentation assertion about B is needed.

Both B_0 and B are 5-torsion-free, by flatness over the discrete valuation ring and flatness of noetherian adic completion. Their continuous relative forms are the completions of $\Omega_{B_0/R}^j$. Completion is exact on these finite projective modules, and they vanish for $j > 2$.

For every n , the ring $W_0/5^n$ is the filtered union of the finite étale Witt extensions of $\mathbf{Z}/5^n$. Its relative cotangent complex over $\mathbf{Z}/5^n$ is zero. Flat base change therefore gives $L_{W_0/\mathbf{Z}_5}^\wedge = 0$. Choose an Eisenstein presentation $R = W_0[\pi_R]$, with polynomial f . Then

$$L_{R/\mathbf{Z}_5}^\wedge \simeq [R \xrightarrow{f'(\pi_R)} R]$$

in cohomological degrees $-1, 0$. Separability in characteristic zero gives $f'(\pi_R) \neq 0$. Choose $a \geq 0$ with $5^a \in f'(\pi_R)R$. Multiplication by 5^a on this complex is null-homotopic.

Completed cotangent transitivity gives a triangle

$$M \longrightarrow \hat{L} \longrightarrow P, \quad M = [B \xrightarrow{f'(\pi_R)} B], \quad \hat{L} = L_{B_0/\mathbf{Z}_5}^\wedge, \quad P = (\Omega_{B_0/R}^1)_5^\wedge.$$

The module P is projective in degree zero and M lies in degrees $-1, 0$. The connecting map $P \rightarrow M[1]$ is zero, so the triangle splits as a triangle of B -complexes. The kernel terms in the i -th derived exterior power of the projection are sums of

$$(\Lambda^r M) \otimes \Lambda^{i-r} P, \quad 1 \leq r \leq i.$$

The r -th exterior-power functor takes the null-homotopy of $5^a \mathrm{id}_M$ to one of 5^{ar} . Thus each displayed term is killed as an object by 5^{ar} .

The Hochschild–Kostant–Rosenberg filtration has graded pieces $(\Lambda^i L)[i]$, and its tails become increasingly connective. Only finitely many pieces affect any fixed homotopy range. This remains true after derived completion. For a c -connective spectrum, the only possible additional bottom group is the derived inverse limit of the quotients of its c -th homotopy group; that quotient tower is surjective. Completion therefore preserves connectivity. Combining the exterior-power bounds along the finitely many extensions bounds the 5-power torsion in the comparison cone in each fixed range. Hence

$$(7.2) \quad \mathrm{HH}(B_0; \mathbf{Z}_5)[1/5] \xrightarrow{\sim} \mathrm{HH}(B_0/R; \mathbf{Z}_5)[1/5]$$

is an equivalence of spectra with circle action.

Ordinary cyclic homology is the circle homotopy-orbit construction. For a connective spectrum, the omitted part of the skeletal filtration of its bar construction becomes arbitrarily connective. Each finite stage uses finite colimits and tensors with finite cell spaces, which commute with spectral completion. The finite-range argument proving (7.2) thus also proves the ordinary cyclic comparison. Equivalently, the circle norm sequence makes the square comparing the absolute and relative maps $\mathrm{HC}^- \rightarrow \mathrm{HP}$ cartesian. We do not need the separate vertical maps on HC^- and HP to be equivalences; those constructions involve infinite limits.

The complete ring B is henselian along (5). Applying the fiber theorem to the ordinary ring B , and replacing the cyclic fiber by the relative one just computed, gives the desired comparison with $\mathcal{K}(B, 5B)$. The map $B/5 \rightarrow B/\pi_R$ has nilpotent kernel. By [2, Corollary 3.8], a nilpotent relative extension in a ring with nilpotent 5 has rationalized completed relative K-theory zero. Exactness therefore identifies $\mathcal{K}(B, 5B)$ with $\mathcal{K}(B, \pi_R B)$. This proves (7.1). Precomposing with the map from actual relative K-theory defines the actual character.

The module structure requires more than naturality. Put $H = \mathrm{THH}(B)$, a commutative algebra object in cyclotomic spectra. The coefficient comparisons in [2, Proposition 2.8 and Corollary 2.10] are tensored with H ; their maps are id_H tensored with the coefficient maps. They are consequently maps of H -modules. The maps from H , the linearization $H \otimes H\mathbf{Z} \rightarrow \mathrm{HH}(B)$ used in [2, Theorem 2.12], and projection to $\mathrm{HH}(B/R)$ are H -linear.

The functor TC is lax symmetric monoidal, so these diagrams become diagrams of $\mathrm{TC}(B)$ -modules. Circle invariants, orbits, and the norm fiber carry the corresponding actions. The norm projection formula identifies the fiber action with the usual HC^- -action on $\Sigma \mathrm{HC}$. The equivalences used above may therefore be inverted in the module category. The relative K -to- TC map is induced by the multiplicative cyclotomic trace, whose E_∞ refinement is [9, Theorem 7.4]. Restricting the module action along this trace proves the asserted $K(B)$ -linearity through the negative cyclic character.

It remains to compute the ordinary cyclic component. In relative dimension two the normalized HKR map is integral, because 2 is invertible:

$$a_0[a_1 | \cdots | a_j] \mapsto \frac{1}{j!} a_0 da_1 \wedge \cdots \wedge da_j \quad (0 \leq j \leq 2),$$

and higher chains map to zero. The Hochschild boundary maps to zero. The alternating sum defining the Connes operator contributes $j+1$ before normalization, so the factors $1/j!$ make it correspond to d . For $j=2$, the target of d is zero. The antisymmetrized smooth HKR representative maps to the original form, and higher relative Hochschild homology vanishes. Thus this is a quasi-isomorphism of mixed complexes. The completed comparison gives the finite continuous form modules already identified.

Give a bookkeeping variable t homological degree -2 , and give a j -form degree j . The mixed differential is td . The ordinary cyclic model retains the powers $1, t^{-1}, t^{-2}, \dots$ and discards positive powers. In degree two it has $\Omega^2 \oplus t^{-1}B$. The differential sends $t^{-1}b$ to db . The degree-three boundaries in the top-form term are exactly $d\Omega^1$. Consequently

$$(7.3) \quad \mathrm{HC}_2^{\mathrm{cts}}(B/R; \mathbf{Q}_5) = \frac{\Omega_{B/R}^2[1/5]}{d\Omega_{B/R}^1[1/5]} \oplus t^{-1} \ker(d: B[1/5] \rightarrow \Omega_{B/R}^1[1/5]).$$

In particular the image in the first summand is the ordinary differential image.

At the coefficient point, the degree-two group is $t^{-1}E_R$. The class β has representative $t^{-1}b$, and b is constant for differentiation relative to R . We retain this point coordinate without further normalization. The character comparison in [2, v2, Constructions 4.7–4.8 and 4.11, Propositions 4.10 and 4.12, pp. 21–23] gives a common nonzero scalar, whose image in E_R we denote by λ_R , for which the first Chern component of the periodic character is $\lambda_R c_1^{\mathrm{dR}}$. Here equation (28) of that reference gives the special-fiber character, equation (24) gives the crystalline/de Rham comparison, and equation (25) is the fiber square. The comparison preserves first Chern classes because their transition functions give the same $d \log$ cocycle. The scalar is common to all smooth formal finite-type models over the fixed coefficient ring; neither scalar one nor $\mathbf{Q}_5$ -rationality is required.

This formula applies to every invertible B -module J here. The finite presentation of $J/\pi_R J$, together with the finite equations witnessing rank-one projectivity, descends to the special fiber of a finite localization of a smooth relative rank-two R -algebra. Complete this finite model. An idempotent presenting the descended module lifts: starting with a matrix lift e , iterate $e \mapsto 3e^2 - 2e^3$. If $\delta = e^2 - e$, the new error is $\delta^2(4\delta - 3)$, so the errors tend to zero adically and the matrices converge to an idempotent. Its projective module has rank one since its reduction does and π_R lies in the Jacobson radical. The completed model maps naturally to B . After pullback, an isomorphism modulo π_R with $J/\pi_R J$ and its inverse lift between the projective modules. Their composites are invertible because they reduce to the identity modulo the Jacobson radical, so the pulled-back line is isomorphic to J . Naturality therefore carries the finite-model character formula to B .

In the rank-two mixed complex, both degree-zero negative and periodic cyclic homology are

$$\ker(d: B[1/5] \longrightarrow \Omega_{B/R}^1[1/5]) \oplus t(\Omega_{B/R}^2[1/5]/d\Omega_{B/R}^1[1/5]).$$

The comparison is the identity on these summands: total degree zero contains functions and t -times two-forms, total degree one supplies the boundaries $t d\Omega^1$, and higher forms vanish. Thus the negative-to-periodic map is injective in this degree. The commutative fiber square identifies the negative cyclic character of J with $1 + \lambda_R t c_1^{\text{dR}}(J)$. The character of $1 - [J]$ is consequently $-\lambda_R t c_1^{\text{dR}}(J)$, whose action on $t^{-1}b$ has top component $-\lambda_R b c_1^{\text{dR}}(J)$. At the normalized bar level, the shuffle term gives this wedge product, while cyclic-shuffle corrections insert the relative scalar b in a bar slot and vanish. This is the module action established above. The common nonzero scalar preserves nonvanishing of every finite sum.

A nonzero top-form component cannot be the image of a zero rationalized completed relative class. The same calculation is additive for finite sums. It makes no assertion that actual K-theory injects into completion. $\square$

8. COHEN COMPLETION AND ELLIPTIC CONTRACTION

We prove that the graph-divisor class survives Cohen completion.

8.1. Scalar Frobenius pole removal.

Lemma 8.1. *Let p be a prime, $k = \overline{\mathbf{F}}_p$, $W = W(k)$, and $K = W[1/p]$. Let D be a smooth affine W -algebra of relative dimension one with connected nonempty special fiber $U = \text{Spec}(D/pD)$. Put*

$$S_D = \{s \in D \mid s \bmod p \neq 0\}, \quad \widehat{D} = \varprojlim_n D/p^n D, \quad C = \varprojlim_n S_D^{-1} D/p^n D, \quad F_C = C[1/p].$$

Let $D^\dagger$ be the weak completion. Assume that an integral Frobenius lift $\phi: D^\dagger \rightarrow D^\dagger$ restricts to the d -th Witt Frobenius on W , where $d \geq 1$, and reduces to the $q = p^d$ power map. If $\omega \in \Omega_{D^\dagger/W}^1[1/p]$ has nonzero rigid cohomology class and

$$\phi^* \omega - a\omega = dh \quad (a \in W, h \in D^\dagger[1/p]),$$

then its image in

$$\Omega_{C/W}^1[1/p]/dF_C$$

is nonzero.

Proof. A smooth connected curve has disjoint irreducible components, since its local rings are domains. Therefore $D_0 = D/pD$ is an integral normal domain. Write $L = \text{Frac } D_0$. The localization $S_D^{-1} D = D_{pD}$ is a regular local ring of dimension one with parameter p and residue field L . Thus C is a complete discrete valuation ring with these same data.

For every n , the map $D/p^n D \rightarrow S_D^{-1} D/p^n D$ is injective. If $sa \in p^n D$ with $s \in S_D$, reduction modulo p gives $a \in pD$. Write $a = pa_1$, cancel p by flatness over W , and repeat. Taking limits embeds $\widehat{D}$ in C and gives

$$(8.1) \quad \widehat{D} \cap p^n C = p^n \widehat{D}.$$

The continuous forms on C are the completions of the forms on $S_D^{-1} D$.

We first determine its continuous constants:

$$(8.2) \quad \ker(d: F_C \rightarrow \Omega_{C/W}^1[1/p]) = K.$$

After inverting an element of S_D , smoothness supplies an étale coordinate t and a ring D' étale over $W[t]$. The residue of t is a separating parameter for L , and the generic-special completion of D' is still C . Formal étaleness extends $t \mapsto t + z$ to a Taylor map $D' \rightarrow C[[z]]$, successively modulo powers of z . Further denominators have unit constant coefficient, so this map extends to the localization and then to C , coefficient by coefficient. Write its coefficients as integral continuous W -linear operators $H_j: C \rightarrow C$.

Uniqueness of the étale lift identifies successive translations by z, w with translation by $z + w$. With $\partial = H_1$, this gives $j! H_j = \partial^j$ and $\partial(t) = 1$. The continuous cotangent module is free on

dt . If $f \in C$ satisfies $df = 0$, then $j!H_j(f) = 0$ for every $j > 0$. The ring C is torsion-free over $\mathbf{Z}$, so $H_j(f) = 0$.

On reduction modulo p , all positive Hasse operators annihilate $\bar{f} \in L$. For each $r \geq 1$,

$$1, t, \dots, t^{p^r-1}$$

is a basis of L over L^{p^r} . Indeed finite separability of $L/k(t)$ gives $[L : L^p] = p$, and $dt \neq 0$ implies $t \notin L^p$. Iteration gives $[L : L^{p^r}] = p^r$. A smaller purely inseparable degree for t over L^{p^r} would force t to be a p -th power in L .

Expand $\bar{f} = \sum_{i=0}^{p^r-1} b_i^{p^r} t^i$. The Taylor series of $b_i^{p^r}$ has no terms of degrees $1, \dots, p^r - 1$. Comparing the coefficients of $z^{p^r-1}, z^{p^r-2}, \dots, z$ successively gives $b_i = 0$ for every $i > 0$. Thus $\bar{f} \in \bigcap_r L^{p^r} = k$. For the last equality, the valuation of a nonzero element of the intersection at every point of the smooth proper model of L is divisible by all p^r , hence zero. Such an element is a global regular function on a proper connected curve and belongs to k ; conversely k is perfect.

Choose $w_0 \in W$ lifting $\bar{f}$. Then $(f - w_0)/p$ is integral and again has zero differential. Repeating gives $f = \sum_{i \geq 0} p^i w_i$, with $w_i \in W$. This converges in W and in C , so $f \in W$. Scaling by a power of p proves (8.2).

The integral map ϕ fixes p and extends continuously to $\hat{D}$. Its reduction preserves nonzero residues, so inversion of S_D and completion extend it to C . We claim that

$$(8.3) \quad f \in C, \quad \phi(f) - af \in \hat{D} \implies f \in \hat{D}.$$

Write $\phi(f) - af = r$. A pole of $\bar{f}$ on U of order $m > 0$ would give pole order qm in $\bar{f}^q$ and at most m in $\bar{a}\bar{f}$. These cannot cancel. Hence $\bar{f}$ has no poles on U . Normality gives $\bar{f} \in D_0$.

Lift $\bar{f}$ to $g_0 \in D$, and put

$$f_1 = (f - g_0)/p, \quad r_1 = (r - \phi(g_0) + ag_0)/p.$$

The numerator defining r_1 belongs to $\hat{D} \cap pC = p\hat{D}$ by (8.1). Thus $r_1 \in \hat{D}$ and $\phi(f_1) - af_1 = r_1$. Iteration yields

$$f = \sum_{i=0}^{n-1} p^i g_i + p^n f_n, \quad g_i \in D, \quad f_n \in C.$$

The partial sums converge in $\hat{D}$, and the remainder tends to zero in C . This proves (8.3). No W -linearity of ϕ was used.

Suppose now that $\omega = df$ in F_C . The Frobenius equation gives $d(\phi(f) - af - h) = 0$. By (8.2), $\phi(f) - af = h + c$ for $c \in K$. Multiply by one sufficiently large power of p to make f, h, c integral in $C, \hat{D}, W$, respectively. The equation is preserved because $\phi(p) = p$. Equation (8.3) then puts the scaled f in $\hat{D}$. Undoing the scaling makes ω exact in the rational completed de Rham complex.

For the smooth finite-type curve U and its constant overconvergent Frobenius coefficient, the map

$$H_{\text{rig}}^1(U/K) \longrightarrow H_{\text{conv}}^1(U/K)$$

is injective by [15, Theorem 3.2.1]. The weakly completed de Rham complex computes the first group, and the completed de Rham complex computes the second, with comparison induced by $D^\dagger \rightarrow \hat{D}$; see also [15, Theorem 2.1.2]. Both use ordinary differential images in degree one. The primitive just constructed contradicts this injectivity. Only the original finite-type curve U enters this external comparison. $\square$

8.2. A bounded elliptic contraction.

Setup 8.2. For the remainder of this section, let $k = \overline{\mathbf{F}}_5$, $W = W(k)$, and $K = W[1/5]$. Let

$$\mathcal{E}/W: v^2 = z^3 + z$$

be the smooth projective elliptic curve with origin O , and put

$$\mathcal{E}^\circ = \mathcal{E} \setminus \{O, (0, 0)\}, \quad E_0 = \mathcal{E} \otimes_W k, \quad A_{\text{ell}} = \Gamma(\mathcal{E}^\circ, \mathcal{O}).$$

Let D be a smooth affine W -algebra of relative dimension one with connected nonempty special fiber. Let C be its generic-special completion, and put

$$B_{\text{ell}} = (\hat{A}_{\text{ell}} \otimes_W C)_5^\wedge.$$

Proposition 8.3. *Choose rational dagger one-forms ω, η on $\mathcal{E}^\circ$ representing nonzero generators of the proper slope-one and slope-zero crystalline lines, respectively. There is a bounded K -linear functional*

$$\ell: \Omega_{\hat{A}_{\text{ell}}/W}^1[1/5] \longrightarrow K$$

such that

$$\ell(\omega) = 1, \quad \ell(\eta) = 0, \quad \ell(d\hat{A}_{\text{ell}}[1/5]) = 0.$$

It extends to a continuous degree-minus-one chain operator

$$L: \Omega_{B_{\text{ell}}/W}^\bullet[1/5] \longrightarrow \Omega_{C/W}^\bullet[1/5][-1]$$

with $L_2(\alpha \wedge \gamma) = \ell(\alpha)\gamma$, where the elliptic form is placed first. The map

$$J: H_{\text{dR}}^1(C/W)[1/5] \longrightarrow H_{\text{dR}}^2(B_{\text{ell}}/W)[1/5], \quad [\gamma] \longmapsto [\omega \wedge \gamma],$$

is split injective, with left inverse induced by L . In particular, a class represented by $\omega \wedge \gamma + \eta \wedge \delta$ is nonzero whenever $[\gamma] \neq 0$ in the ordinary one-form quotient of C .

Proof. The cubic has distinct roots in characteristic five, so E_0 is smooth. The coefficient of z^4 in $(z^3 + z)^2$ is 2, which is nonzero modulo five. The Hasse-invariant criterion [43, Theorem V.4.1(a), p. 148] gives ordinarity. Proper rational crystalline H^1 therefore has two lines of slopes zero and one. Restriction to E_0° is injective in rigid H^1 . The removed reduced finite set $Z = \{O, (0, 0)\}$ has codimension one in the smooth curve E_0 . The support-vanishing assertion of [39, Theorem 1.1, third assertion, p. 63] gives $H_{Z, \text{rig}}^1(E_0/K) = 0$, since $1 < 2$. The localization sequence [39, Proposition 1.3, p. 64] therefore gives

$$0 \longrightarrow H_{\text{rig}}^1(E_0/K) \longrightarrow H_{\text{rig}}^1(E_0^\circ/K).$$

This natural restriction respects Frobenius.

For unramified Witt coefficients, [15, Theorem 7.3.1] identifies slope-one rigid and convergent cohomology in degree one. For smooth affine schemes with unit-root coefficients, the slope-one convergent part meets the closure of zero trivially [15, Theorem 7.3.6]. The slopes in $[0, 1)$ lie in the closure of zero by [15, Theorem 7.2.1]. All hypotheses apply to E_0° , constant coefficients, and the unramified ring W .

Put $M_0 = \Omega_{\hat{A}_{\text{ell}}/W}^1$, $M = M_0[1/5]$, and let N be the closure of $d\hat{A}_{\text{ell}}[1/5]$ in M . All one-forms are closed, since the relative dimension is one. The topology and convergent-cohomology description in [15, Definition 7.1.1] identify the separated quotient with M/N . The class of ω in this quotient is nonzero, whereas $\eta \in N$. We use this separated quotient only to construct ℓ .

The finite projective, complete, separated, torsion-free module M_0 defines a Banach norm on M with discrete values. The quotient $V = M/N$ is Banach for the quotient norm. For a nonzero quotient vector, the infimum of the norms of its representatives is positive and attained, because the norm values are discrete and N is closed. Completeness follows by taking a Cauchy subsequence, lifting its successive differences with norms tending to zero, and summing those lifts in M .

Let V_0 be the unit ball, and lift a basis of the k -vector space $V_0/5V_0$ to elements $e_i \in V_0$. Every $v \in V_0$ has a unique expansion

$$v = \sum_i a_i e_i, \quad a_i \in W,$$

where the coefficients tend to zero off finite subsets. To obtain the expansion, reduce modulo five, subtract a finite lifted basis expansion, divide by five, and repeat. The resulting series converges by completeness. For uniqueness, a nonzero coefficient family has a largest absolute value. Scaling to make that value one and reducing modulo five contradicts independence of

the finitely many largest coefficients. Scaling also gives such expansions over K for all $v \in V$. The coordinate maps are consequently bounded and K -linear.

Some coordinate is nonzero on the image of ω . Divide that coordinate by its value on ω and compose with $M \rightarrow V$. This gives ℓ . It annihilates N , hence both η and exact forms. For some $e \geq 0$, boundedness gives $5^e \ell(M_0) \subset W$.

Set $\ell_0 = 5^e \ell$. Reduce this integral map modulo 5^n , tensor with $C/5^n C$, and pass to the inverse limit. This defines $M_0 \hat{\otimes}_W C \rightarrow C$. Tensoring instead with $\Omega_{C/W}^1/5^n$ defines $M_0 \hat{\otimes}_W \Omega_{C/W}^1 \rightarrow \Omega_{C/W}^1$. After rationalization divide by 5^e . These are limits of compatible operators, with no interchange between cohomology and inverse limits.

At each finite precision, the tensor-product formula for differentials gives

$$(8.4) \quad \Omega_{B_{\text{ell}}/W}^1 = (M_0 \hat{\otimes}_W C) \oplus (\hat{A}_{\text{ell}} \hat{\otimes}_W \Omega_{C/W}^1),$$

$$(8.5) \quad \Omega_{B_{\text{ell}}/W}^2 = M_0 \hat{\otimes}_W \Omega_{C/W}^1.$$

Higher forms vanish. The two factors have cotangent rank one; for C , this follows from completion of an étale-coordinate localization of D . The form modules are finite projective. Flatness over W ensures that completed tensor products reduce to the ordinary tensor products modulo 5^n , so the displayed limits are valid.

Define $L_0 = 0$. Define L_1 by the extended ℓ on the first summand of (8.4) and zero on the second. Define L_2 by the second tensor extension. On functions, $L_1 d = 0$. For an elliptic one-form α and a base function f ,

$$d(\alpha f) = -\alpha \wedge d_C f, \quad L_2 d(\alpha f) = -\ell(\alpha) d_C f = -d_C L_1(\alpha f).$$

For an elliptic function a and a base one-form γ , $d(a\gamma) = d_A a \wedge \gamma$, so its image under L_2 is zero, as is $d_C L_1(a\gamma)$. Finite sums of pure tensors are dense, and all operators are continuous. Thus $L_2 d = -d_C L_1$ on the completed complex, which is the chain identity for the shift $[-1]$.

The form $\omega \wedge \gamma$ is closed. If $\gamma = d_C f$, then $\omega \wedge \gamma = -d(f\omega)$. Hence J is defined on ordinary cohomology. Finally $L_2(\omega \wedge \gamma) = \gamma$. This proves the splitting, and $\ell(\eta) = 0$ proves the last assertion. $\square$

8.3. Graph divisors.

Proposition 8.4. *In Setup 8.2, assume that $\tilde{Q}: \text{Spec } D \rightarrow \mathcal{E}^\circ$ has nonconstant special fiber $Q: U \rightarrow E_0^\circ$. Let $\mathcal{L}_Q$ be the pullback to $\text{Spec } B_{\text{ell}}$ of*

$$\mathcal{O}(\Gamma_{\tilde{Q}} - \Gamma_{-\tilde{Q}})$$

on $\mathcal{E}^\circ \times_W \text{Spec } D$. Then

$$c_1^{\text{dR}}(\mathcal{L}_Q) \neq 0 \quad \text{in} \quad \Omega_{B_{\text{ell}}/W}^2[1/5]/d\Omega_{B_{\text{ell}}/W}^1[1/5].$$

Nonvanishing persists under every finite extension of K , with continuous relative forms over the extended integer ring.

Proof. Choose the proper slope-zero generator η and slope-one generator ω over $\mathbf{Q}_5$, and normalize $\text{Tr}(\eta \cup \omega) = 1$. The curve is defined over $\mathbf{F}_5$. Proper smooth crystalline/de Rham comparison gives corresponding algebraic de Rham classes on $\mathcal{E}_K$. Their restrictions to $\mathcal{E}_K^\circ$ have algebraic one-form representatives because the curve is affine. These also define dagger and continuous representatives, so Proposition 8.3 applies.

With the principal-polarization convention, the normalized Poincaré bundle is

$$\mathcal{P} = \mathcal{O}(\Delta - \mathcal{E} \times O - O \times \mathcal{E}).$$

Its restrictions to the origin slices are trivial, so its Chern class has only a mixed term in the proper Künneth decomposition. The diagonal acts as the identity on H^1 . With our pairing convention, the mixed tensor is

$$\omega \otimes \eta - \eta \otimes \omega.$$

Multiplying by η on the first factor and tracing returns η on the second. Multiplying by ω returns ω . Perfectness of the pairing determines the tensor. The trace, duality, Künneth, and

proper crystalline compatibilities used here are those of [4, §1.7(i), §2.3(i), Theorems 2.4 and 3.2, and §3.3(i)].

The graph line for a map q is the pullback of the diagonal line by $\text{id} \times q$. In the difference for $\tilde{Q}$ and $-\tilde{Q}$, the first-factor origin line cancels, and the second-factor terms cancel because $[-1]^* \mathcal{O}(O) \simeq \mathcal{O}(O)$. Inversion acts by -1 on proper H^1 . Thus

$$(8.6) \quad c_1^{\text{dR}}(\mathcal{L}_Q) = \left[2(\omega \wedge \tilde{Q}^* \eta - \eta \wedge \tilde{Q}^* \omega) \right].$$

This equality first holds on the finite-type affine product over K . The difference from another Chern representative is the differential of an algebraic one-form there. It remains exact under weak completion, generic localization, and continuous completion. No Künneth formula for the completed product is being used.

Compactify U to its smooth proper connected curve $\bar{U}$. Properness extends Q to $\bar{Q}: \bar{U} \rightarrow E_0$. The map $\bar{Q}$ is finite and flat of rank $m = [k(\bar{U}) : k(E_0)] > 0$: its finite direct image is torsion-free over the local discrete valuation rings of E_0 . This rank includes the inseparable degree. Write ξ_C for the rigid homological fundamental class of a smooth proper curve C , viewed as its top-degree trace functional. Proper cycle pushforward gives $\bar{Q}_*[\bar{U}] = m[E_0]$; compatibility of cycle classes with proper pushforward therefore gives $\bar{Q}_* \xi_{\bar{U}} = m \xi_{E_0}$ [39, Proposition 2.2, Section 2.1, and Proposition 6.4]. The homological pushforward is the transpose of cohomological pullback. Consequently, for every $u \in H_{\text{rig}}^2(E_0/K)$,

$$\text{Tr}_{\bar{U}}(\bar{Q}^* u) = m \text{Tr}_{E_0}(u).$$

Taking $u = a \cup b$ and using the naturality of cup products gives

$$\text{Tr}_{\bar{U}}(\bar{Q}^* a \cup \bar{Q}^* b) = m \text{Tr}_{E_0}(a \cup b).$$

The pairing is perfect and m is invertible in K . Consequently $\bar{Q}^*$ is injective on degree-one cohomology, without a separability assumption. Restriction from $\bar{U}$ to U is also injective. If the reduced finite complement $Z = \bar{U} \setminus U$ is nonempty, it has codimension one in the smooth curve $\bar{U}$. Support vanishing gives $H_{Z, \text{rig}}^1(\bar{U}/K) = 0$, and the localization sequence then makes $H_{\text{rig}}^1(\bar{U}/K) \rightarrow H_{\text{rig}}^1(U/K)$ injective [39, Theorem 1.1, third assertion, and Proposition 1.3, pp. 63–64]. For an empty complement restriction is the identity. Thus $Q^* \eta \neq 0$ in $H_{\text{rig}}^1(U/K)$, represented by $\tilde{Q}^* \eta$.

The special curve and map descend to some finite field $\mathbf{F}_{5^d}$. The four $\mathbf{F}_5$ -points of E_0 are O and the three points with $v = 0$, at $z = 0, 2, 3$. At $z = 1, 4$ the cubic is a nonsquare. The Frobenius polynomial is therefore $X^2 - 2X + 5$. Its reduction has simple roots $0, 2$, so its two roots lie in $\mathbf{Z}_5$, one a unit α . We choose η with Frobenius eigenvalue α .

The weak completion $D^\dagger$ admits an integral, Witt-semilinear Frobenius lift by the smooth affine lifting statement in [19, Introduction, p. 198]. Iterate it d times to obtain ϕ . Its action on rigid cohomology is canonical and independent of the lift. Descent of Q gives $\phi^*[Q^* \eta] = \alpha^d [Q^* \eta]$. On the affine dagger complex this means

$$\phi^*(\tilde{Q}^* \eta) - \alpha^d \tilde{Q}^* \eta = dh \quad (h \in D^\dagger[1/5]).$$

Lemma 8.1 applies, since $\alpha^d \in W$. Hence the image $[\tilde{Q}^* \eta]$ in the ordinary Cohen one-form quotient is nonzero.

Apply L from Proposition 8.3 to (8.6). The result is $2[\tilde{Q}^* \eta]$, which is nonzero. Because L takes exact forms to exact forms, the graph class cannot be exact.

For a finite extension K'/K , its integer ring W' is finite free over W and complete. Scalar extension therefore commutes with the adic limits defining the product and its finite projective form modules. Relative forms over W' are the old forms tensored with W' . After inversion of five, the complexes are tensored with K' . Field extension is exact and faithfully flat, so it commutes with kernels and ordinary images and preserves the nonzero class. $\square$

9. THE QUARTIC QUOTIENT AND ITS WEIGHT

In this section, we construct the elliptic quotient and compute the linear functional on the four root-divisor classes.

Proposition 9.1. *Let $k = \overline{\mathbf{F}}_5$, $W = W(k)$, $R = W[\rho]$, and*

$$V = \varprojlim_n W[T]_{(5)}/5^n.$$

There is a finite unramified extension C/V , obtained as the generic-special completion of a smooth affine W -curve étale over an open of the T -line, with the following property. Put $C' = C \otimes_W R$ and

$$B = C' \langle X, Y \rangle / \left(X^4 + \frac{\Phi_5(1 + \pi Y)}{\pi^4} + TXY^3 \right).$$

For $g \in G = \mathbf{F}_5^\times$, set

$$a_g = \frac{\rho^g - 1}{\pi}, \quad J_g = (X, Y - a_g), \quad \gamma_g = -c_1^{\text{dR}}(J_g).$$

The ideals J_g are invertible. Let $E = \mathbf{Q}_5(\rho)$, let $\omega_T: G \rightarrow \mathbf{Z}_5^\times$ be the Teichmüller character, and put

$$e_r = \frac{1}{4} \sum_{g \in G} \omega_T(g)^{-r} \sigma_g \quad (r = 1, 3)$$

on $\mathbf{Q}_5$ -vector-space targets. If $b \in E$ satisfies $\sigma_{-1}b = -b$ and $e_1b, e_3b \neq 0$, then

$$\sum_{g \in G} \sigma_g(b) \gamma_g \neq 0 \quad \text{in} \quad \Omega_{B/R}^2[1/5]/d\Omega_{B/R}^1[1/5].$$

The relative forms retain dT .

Proof. Over $k(T)$, consider

$$H: X^4 + Y^4 + TXY^3 = Z^4, \quad H^\circ = H \cap \{Z \neq 0\}, \quad P_g = [0 : g : 1].$$

The singular-point calculation in Proposition 5.1 applies with Z in place of U . It gives $T^4 = 3$ at a hypothetical singularity. Thus H is smooth and geometrically integral.

The same auxiliary surface

$$\text{Spec } k[X, U, Y, Y^{-1}]/(XU - (1 - Y^4)), \quad T = (U - X^3)/Y^3,$$

has root-divisor lattice $\mathbf{Z}^4/\mathbf{Z}(1, 1, 1, 1)$. For clarity, the mechanisms remain unchanged here: inversion of X gives the factorial Laurent ring; its units give only the diagonal relation; each fiber becomes $w^4 = s^4 + ts + 1$, whose right side is not a square by comparison with $(s^2 + bs + c)^2$; hence each vertical divisor is the reduced principal divisor $T - t$. Divisor localization therefore gives this same lattice on $H^\circ \cap \{Y \neq 0\}$. A torsion relation on $P_1 - P_{-1}$ in the proper Jacobian would restrict to a torsion relation in this lattice, which is impossible. Finite field extension preserves nontorsion: norm after pullback multiplies divisor classes by the extension degree.

We construct the quotients over an étale parameter cover. Set $\Delta = 256 - 27T^4$, and order the roots $b_1, \dots, b_4$ of $s^4 + Ts + 1$ over a finite étale cover of $W[T, 1/\Delta]$. Put

$$a = \frac{b_2 - b_4}{b_2 - b_1}, \quad x = a \frac{s - b_1}{s - b_4}, \quad \lambda = a \frac{b_3 - b_1}{b_3 - b_4}.$$

The elements $a, a - 1, a - \lambda, \lambda, \lambda - 1$ are units, as follows from the pairwise root differences. Adjoin c with $c^4 = a(a - 1)(a - \lambda)$. Replacing $w = Z/Y$ by $cw/(s - b_4)$ gives

$$(9.1) \quad w^4 = x(x - 1)(x - \lambda).$$

Indeed the product of the three transformed factors is $a(a - 1)(a - \lambda) \prod_{j=1}^3 (s - b_j)/(s - b_4)^3$, which is $c^4(s^4 + Ts + 1)/(s - b_4)^4$.

Adjoin r with $r^2 = \lambda$ and $\alpha_{\pm}$ with $\alpha_{\pm}^2 = r \pm 1$. All these extensions are étale, because the radicands and 2, 4 are units. Set

$$z = \frac{w^2}{x}, \quad v_{\pm} = \frac{w(x \pm r)}{x}.$$

Equation (9.1) gives

$$(9.2) \quad z^2 = x + \frac{r^2}{x} - 1 - r^2, \quad v_{\pm}^2 = z^3 + (r \pm 1)^2 z.$$

After $z = \alpha_{\pm}^2 \zeta$, $v_{\pm} = \alpha_{\pm}^3 v$, the target is the constant elliptic curve $v^2 = \zeta^3 + \zeta$.

Let $\tau(x, w) = (x, -w)$ and

$$\iota(x, w) = \left(\frac{r^2}{x}, \frac{rw}{x} \right).$$

The maps h_+, h_- are the quotients by ι and $\tau\iota$, respectively. Their degree is two: x satisfies a quadratic over the target functions by (9.2), w is recovered from $x, v_{\pm}$, and the displayed nontrivial involution fixes the target functions. They extend between the proper smooth curves.

The boundary $Z = 0$ consists of the four points $x = 0, 1, r^2, \infty$. At $0, \infty$, the orders of $z, v_{\pm}$ are $(-2, -3)$, so the image is O with local degree one. At $1, r^2$, these orders are $(2, 1)$, so the image is $(0, 0)$, again with local degree one. Everywhere else z is finite and nonzero. The exact inverse image of $\{O, (0, 0)\}$ is therefore the boundary. Both maps restrict to finite maps of all of H° to E_0° .

The equations are defined over W . Shrinking the parameter open by elements nonzero at its generic special point makes these finite flat quotient maps between smooth relative curves. Choose a connected component of the special fiber of the finite étale parameter cover. It can be isolated in a smooth affine W -model by inverting a lift of its characteristic idempotent. Write D for the resulting smooth W -algebra. Its generic-special completion C is the corresponding finite unramified extension of V ; subsequent open restrictions are units in C .

The involutions $\tau, \iota, \tau\iota$ form a Klein four group. Its total quotient is rational with coordinate $z = w^2/x$. Recovering x and then w gives degree at most four, and the four automorphisms fix z , so $k(T)(z)$ is the invariant field. If q is the quotient by τ , the pullback and norm identities on Jacobians give

$$(9.3) \quad q^* q_* + h_+^* h_{+*} + h_-^* h_{-*} = 2\text{id} + \sum_{g \in \{1, \tau, \iota, \tau\iota\}} g = 2\text{id}.$$

The full group norm is zero because it factors through the Jacobian of the rational total quotient. The class $P_1 - P_{-1}$ is killed by q_* . If its images under both h_+ and h_- were torsion, (9.3) would make it torsion. Choose a quotient h for which the image is nontorsion. Since $h\tau = [-1]h$, this image is $2Q$, where $Q = h(P_1)$. Every point in $E_0(k)$ is defined over a finite field and is torsion. Hence Q is nonconstant. The lifted root section gives $\tilde{Q}: \text{Spec } D \rightarrow \mathcal{E}^{\circ}$ with this reduction.

Let $i \in W$ be the fourth root of unity reducing to 2. The rotation $[X : Y : Z] \mapsto [iX : iY : Z]$ fixes s , sends the transformed w to $-iw$, sends z to $-z$, and sends $v_{\pm}$ to $-iv_{\pm}$. Thus it induces $[-i]$ on the elliptic target. It sends P_1 to P_i , so

$$(9.4) \quad h(P_g) = [g^{-1}]Q$$

for all fourth roots g , including the lifted Teichmüller sections.

We need an integral trace on total forms, including base differentials. For a tame double quotient of smooth relative curves, the trace is the étale trace away from ramification. At ramification, choose an anti-invariant fiber parameter t ; étale-locally the quotient parameter is $s = t^2$. Invariant fiber one-forms are multiples of $t dt = ds/2$. Invariant base forms and their wedges also descend. Thus invariant total forms are exactly the forms on the quotient. Summing $1 + \iota^*$ and descending defines a chain map

$$\text{Tr}_h h^* = 2, \quad h^* \text{Tr}_h = 1 + \iota^*.$$

It is integral and continuous, and extends through localization and completion, first relative to W and then relative to R . For an invertible module L , the Cartier-divisor norm identity

is $h^* \text{Nm}_h L = L \otimes \iota^* L$, including ramification multiplicities. Since h^* is injective on rational cohomology by $\text{Tr}_h h^* = 2$, it follows that

$$(9.5) \quad \text{Tr}_h(c_1^{\text{dR}}(L)) = c_1^{\text{dR}}(\text{Nm}_h L).$$

This is an identity on ordinary de Rham quotients, because the trace is an actual chain map.

Consider first

$$B_{\text{cyc}} = C' \langle X, Y \rangle / (X^4 + Y^4 + TXY^3 - 1), \quad \gamma_g^{\text{cyc}} = c_1^{\text{dR}}(\mathcal{O}(P_g)).$$

The norm of its root line under h is the graph line of $[g^{-1}] \tilde{Q}$. Choose η, ω with the polarization convention of (8.6). The unit-root character can be determined exactly. On the elliptic curve,

$$[i](z, v) = (-z, iv), \quad [i]^* \left(\frac{dz}{2v} \right) = i \frac{dz}{2v}.$$

By the perfect pairing, the character on $H^1(\mathcal{O})$ is i^{-1} . A primitive integral generator of the unit-root crystalline line has nonzero image in $H^1(\mathcal{O})$ modulo five. Otherwise its nonzero reduction would lie in the holomorphic subspace, which crystalline Frobenius kills modulo five, contradicting its unit Frobenius eigenvalue. The two order-four eigenvalues have distinct reductions. Consequently the character on the unit-root line is exactly ω_T^3 , rather than an unspecified odd character.

Let $\lambda_g = \sigma_g(b)$ for any anti-invariant b . Then $\lambda_{-g} = -\lambda_g$. Pair opposite roots and apply (9.5), followed by the contraction L of Proposition 8.3. The graph-difference formula (8.6) contributes twice the pullback of η . Equation (9.4) gives $[g^{-1}]^* \eta = \omega_T(g)^{-3} \eta$. Hence the exact linear identity is

$$(9.6) \quad L \text{Tr}_h \left(\sum_{g \in G} \sigma_g(b) \gamma_g^{\text{cyc}} \right) = \left(\sum_{g \in G} \sigma_g(b) \omega_T(g)^{-3} \right) [\tilde{Q}^* \eta] = 4e_3(b) [\tilde{Q}^* \eta].$$

The factor two from the graph difference is exactly the contribution of the two opposite roots in the full sum. The polarization convention fixes the common sign in this formula.

Proposition 8.4, using Lemma 8.1, shows that $[\tilde{Q}^* \eta]$ is nonzero in the ordinary Cohen one-form quotient. Its finite-constant-extension assertion preserves this class over R . Thus (9.6) is nonzero when $e_3(b) \neq 0$; in particular it is nonzero under the two nonvanishing assumptions of the proposition. Both operators preserve exactness, so the weighted class on B_{cyc} is nonzero. Formula (9.6) remains valid for all anti-invariant b , even if an odd component vanishes.

Finally we transport all four root lines to B . The cyclotomic identity gives $5/\pi^4 \equiv -1 \pmod{\pi}$, and all other coefficients below degree four in $\Phi_5(1 + \pi Y)/\pi^4$ are divisible by π . Thus B and B_{cyc} have the same special fiber

$$X^4 + Y^4 + TXY^3 = 1.$$

For $G_0 = X^4 + Y^4 + TXY^3$, Euler's identity $X(G_0)_X + Y(G_0)_Y = 4G_0$ proves smoothness on this fiber. The two equations are monic in Y , so the models are π -torsion-free. Their completions are formally smooth over C' .

Lift the identity modulo π successively modulo π^n . This gives a continuous C' -linear map between the two completed models. Successive approximation proves surjectivity. An element of its kernel lies in π times the source; torsion-freeness of the target permits division of the relation by π . Iteration and separatedness prove injectivity. Thus the map is an isomorphism.

On the support of J_g , the Y -derivative is $\prod_{h \neq g} (a_g - a_h)$, a unit. The equation expresses $Y - a_g$ locally as a multiple of X ; elsewhere J_g is the unit ideal. Hence J_g is invertible. Its reduction is $(X, Y - g)$, equal to the reduction of the cyclic root ideal. In a complete ring, π lies in the Jacobson radical. An isomorphism between the reductions of two projective rank-one modules lifts by projectivity; Nakayama at every maximal ideal makes the lift an isomorphism. Therefore the formal replacement identifies every root Picard class.

The replacement fixes C' , so it fixes all $\sigma_g(b)$. It induces an isomorphism of continuous forms relative to R , with dT retained. The weighted nonzero class, and the transported linear functional (9.6), therefore give the assertion for B . $\square$

10. AN ACTUAL ARITHMETIC COEFFICIENT

The goal of this section is to construct one actual coefficient whose two odd point-character components are nonzero.

Proposition 10.1. *Put $W_E = \mathbf{Z}_5[\rho]$, $E = \mathbf{Q}_5(\rho)$, and $\pi = \rho - 1$. Define*

$$r_E: K_3(W_E) \longrightarrow E$$

by passage to $\pi_3\mathcal{K}(W_E)$, the inverse of $\pi_3\mathcal{K}(W_E, (\pi)) \rightarrow \pi_3\mathcal{K}(W_E)$, and the relative point equivalence with $\mathrm{HC}_2^{\mathrm{cts}}(W_E/W_E; \mathbf{Q}_5) = E$. There is an actual class

$$\beta = (1 - \sigma_{-1})\beta_0 \in K_3(W_{\mathrm{ar}})$$

whose reduction in $K_3(\mathbf{F}_5)$ is zero, such that $\sigma_{-1}\beta = -\beta$ integrally and

$$b = r_E(\beta|_{W_E}) \quad \text{satisfies} \quad e_1 b \neq 0, \quad e_3 b \neq 0.$$

The projectors e_r act only on the indicated $\mathbf{Q}_5$ -vector-space targets.

Proof. For $r = 1, 3$, set

$$\tau_r = \sum_{g=1}^4 \omega_T(g)^{-r} \rho^g.$$

Its π -adic valuation is r . To see this, expand $\rho^g = (1 + \pi)^g$. For $j < r$, the coefficient $\sum_g \omega_T(g)^{-r} \binom{g}{j}$ vanishes modulo five: $\binom{g}{j}$ has degree j in g , and the power sums on $\mathbf{F}_5^\times$ vanish unless the exponent is divisible by four. For $j = r$, the coefficient is $4/r! \bmod 5$, which is nonzero. Since 5 is a unit times π^4 , terms with $j < r$ have valuation at least $4 + j > r$, while terms with $j > r$ have valuation at least $j > r$. The degree- r term is the unique one of minimal valuation.

Let Li_2 be the Coleman dilogarithm normalized by $\sum_{m \geq 1} z^m/m^2$ on the open unit disc. Define

$$S_j(z) = \left(z \frac{d}{dz}\right)^j \frac{1}{1-z}, \quad F_2(z) = \mathrm{Li}_2(z) - 5^{-2} \mathrm{Li}_2(z^5).$$

Partitioning the indices prime to five into $a + 5m$, and expanding $(a + 5m)^{-2}$, gives on that disc

$$(10.1) \quad F_2(z) = \sum_{j \geq 0} (-1)^j (j+1) 5^j S_j(z^5) \sum_{a=1}^4 \frac{z^a}{a^{j+2}}.$$

The right side converges uniformly on $|z| \leq 1$, $|1 - z^5| = 1$. Each S_j has integral coefficients and unit denominator there, so its j -th term has norm at most 5^{-j} . This affinoid is connected: its reduction is the integral curve $\mathrm{Spec} \mathbf{F}_5[z, (1-z)^{-1}]$, and its integral affinoid algebra has no nontrivial idempotent.

By [6, Theorem 2.7(3)], the depleted polylogarithm $\mathrm{Li}_n(z) - p^{-n} \mathrm{Li}_n(z^p)$ is analytic outside the closed disc $|z - 1| \leq p^{-1/(p-1)}$, with a convergent expression in $1/(1-z)$. For $p = 5$, this region contains the affinoid just used. Both sides of (10.1) are analytic there. Their equality on the open unit disc extends by the identity principle.

At $z = -\rho$, $z^5 = -1$. The value $\mathrm{Li}_2(-1)$ is fixed by G , so its odd projections vanish. Permuting the summation indices gives the exact identity

$$e_r(\rho^a) = \frac{\omega_T(a)^r}{4} \tau_r.$$

Applying it to (10.1) yields

$$(10.2) \quad \frac{4e_r \mathrm{Li}_2(-\rho)}{\tau_r} = \sum_{j \geq 0} (-1)^j (j+1) 5^j S_j(-1) \sum_{a=1}^4 \frac{(-1)^a \omega_T(a)^r}{a^{j+2}}.$$

All $S_j(-1)$ are 5-integral. Modulo five, only the $j = 0$ term remains:

$$\frac{1}{2} \sum_{a=1}^4 (-1)^a a^{r-2}.$$

For $r = 1$ it is $\frac{1}{2}(-1 + 3 - 2 + 4) = 2$ in $\mathbf{F}_5$. For $r = 3$ it is $\frac{1}{2}(-1 + 2 - 3 + 4) = 1$. Thus both odd projections of $\mathrm{Li}_2(-\rho)$ are nonzero. Division by τ_r is legitimate by its valuation, and the projection identity is exact, so no precision loss is involved.

The special-unit symbol construction [8, Theorems 1.6 and 1.10, Remark 1.7] maps the degree-one, weight-two symbol complex of a localized number ring to rational K-theory. Its syntomic regulator is $[x]_n \mapsto \pm(n-1)!L_{\mathrm{mod},n}(x)$; the sign is the single convention in the cited remark. In weight two there is no conjectural vanishing assumption.

Take $x = -\rho$ in W_{ar} . Both x and $1-x = 1+\rho$ are units, since the latter reduces to 2 modulo π . The symbol $[-\rho]_2$ is a rational degree-one cocycle: its differential is $(1+\rho) \otimes (-\rho)$, or the exterior version, and the root-of-unity factor is zero after rationalization. It gives $\beta_{\mathrm{rat}} \in K_3(W_{\mathrm{ar}}) \otimes \mathbf{Q}$. Its syntomic regulator is $\pm \mathrm{Li}_2(-\rho)$, because the modification terms contain $\log(-\rho) = 0$. Choose a positive integer N and an actual β_0 representing $N\beta_{\mathrm{rat}}$, and set $\beta = (1 - \sigma_{-1})\beta_0$. This is an operation on actual groups. The automorphism σ_{-1} induces the identity on the residue field, so β reduces to zero. Its integral antisymmetry follows by squaring σ_{-1} . Equivariance of the regulator and (10.2) show that its two odd syntomic components are nonzero.

For the possibly ramified field E , the syntomic-to-étale comparison in [29, Section 2] has no ramification restriction. The degree-one syntomic coordinate for weight $n > 1$ is E [29, Example 2.2.4]. It agrees with the definition used above [29, Proposition 2.2.7]. Its Chern-compatible map to $H^1(E, \mathbf{Q}_5(n))$ is the Bloch–Kato exponential [29, Propositions 2.2.9 and 2.3.4], which is an isomorphism for $n > 1$, as recalled in [29, Section 1.3]. These maps are natural under field automorphisms. Thus the actual étale Chern class $c_{2,1}(\beta) \in H^1(E, \mathbf{Q}_5(2))$ has both odd components nonzero.

There is a natural G -equivariant $\mathbf{Q}_5$ -linear map

$$(10.3) \quad \pi_3 \mathcal{K}(W_E) \longrightarrow H^1(E, \mathbf{Q}_5(2))$$

through which the Chern class of every actual element factors. To construct this map, first work before inverting five. Map from $\pi_3(K(W_E)^\wedge_5)$ to $\pi_3(K(E)^\wedge_5)$, and then to $\varprojlim_n K_3(E; \mathbf{Z}/5^n)$. Apply the finite-coefficient Chern classes

$$K_3(E; \mathbf{Z}/5^n) \longrightarrow H^1(E, \mu_{5^n}^{\otimes 2}).$$

Naturality, compatibility with reduction of integral classes, additivity for odd coefficients, and compatibility with coefficient transitions are respectively [49, Proposition 2.1.1, Lemma 2.3, Propositions 2.4 and 2.8]. The resulting compatible additive maps recover the actual Chern class. Each finite target is killed by 5^n , so the inverse-limit map is $\mathbf{Z}_5$ -linear. The groups $H^0(E, \mu_{5^n}^{\otimes 2})$ are finite, hence Mittag–Leffler. The continuous-cohomology inverse-limit sequence therefore identifies the target limit with $H^1(E, \mathbf{Z}_5(2))$. Inverting five gives (10.3). No derived inverse-limit vanishing on the K-theory side is needed.

The finite-field calculation [41][50, IV, Corollary 1.13] gives

$$K_{2j}(\mathbf{F}_5) = 0, \quad K_{2j-1}(\mathbf{F}_5) = \mathbf{Z}/(5^j - 1) \quad (j \geq 1).$$

Multiplication by five is invertible on these positive groups. The finite-coefficient exact sequences show that all positive homotopy groups of $K(\mathbf{F}_5)^\wedge_5$ vanish. In particular the groups in degrees three and four of $\mathcal{K}(\mathbf{F}_5)$ vanish. The relative fiber sequence identifies $\pi_3 \mathcal{K}(W_E, (\pi))$ with $\pi_3 \mathcal{K}(W_E)$.

The point case of [2, Theorem A, Proposition 4.10, Corollary 3.8] identifies this group with $\mathrm{HC}_2^{\mathrm{cts}}(W_E/W_E; \mathbf{Q}_5) = E$. Here W_E is a complete mixed-characteristic discrete valuation ring with perfect finite residue field, its formal spectrum is smooth over itself, and changing (5) to (π) changes the special fiber only by a nilpotent thickening. The comparison is natural for automorphisms of the coefficient point, hence G -equivariant.

Transporting (10.3) through this point identification gives a G -equivariant $\mathbf{Q}_5$ -linear map from E to $H^1(E, \mathbf{Q}_5(2))$, carrying b to $c_{2,1}(\beta)$. If $e_r b = 0$, the corresponding Chern component would vanish. This contradicts the syntomic calculation for $r = 1, 3$. Equivariance also gives $\sigma_{-1} b = -b$. No equality between a chosen syntomic normalization and the point coordinate was required. $\square$

11. ASSEMBLY OVER THE LOCAL RING AND ITS COMPLETION

In this section, we prove Theorem 3.2 and show that its actual coefficient also gives the geometric detector.

Proof of Theorem 3.2. Choose the actual β of Proposition 10.1, and set $c = c(\beta)$. Section 4 proves regularity of $A, \hat{A}$ and both integral generic vanishings. We prove that the image of this same c in $K_3(\hat{A})$ has infinite order.

Put $k = \overline{\mathbf{F}}_5$, $W = W(k)$, and $R = W[\rho]$. Choose C , arising from a smooth affine curve D , as in Proposition 9.1. Write L for the function field of $D/5D$. It is a finite separable extension of $k(T)$. The coefficient ring $C' = C \otimes_W R$ is a complete discrete valuation ring with uniformizer π and residue field L . Use the test algebra

$$(11.1) \quad B = C' \langle X, Y \rangle / \left(X^4 + \frac{\Phi_5(1 + \pi Y)}{\pi^4} + TXY^3 \right).$$

The polynomial is integral over R : every coefficient below degree four before division has π -valuation at least four. It is monic in Y , so the polynomial quotient and its completion are π -torsion-free.

Its special fiber is

$$(11.2) \quad B/\pi B = L[X, Y]/(X^4 + Y^4 + TXY^3 - 1).$$

For $G_0 = X^4 + Y^4 + TXY^3$, Euler's identity gives $X(G_0)_X + Y(G_0)_Y = 4G_0 = 4$ on this fiber. It is therefore smooth over L . Its projective closure is the smooth geometrically integral quartic of Proposition 9.1, so the fiber is a domain.

The ring B is noetherian by its monic presentation and adic completion. Since π lies in its Jacobson radical, all maximal ideals contain π . At each one, the quotient by π is regular and π is a nonzero divisor. Lifting a regular system of parameters from the quotient and adjoining π proves regularity of that local ring. Hence B is regular. Its dimension is two: the restricted two-variable ring over C' has dimension three, and the equation is a nonzero monic equation. Equivalently, the parameter argument gives dimension at most two at every maximal ideal and equality at closed points of the special curve. Separatedness, torsion-freeness, and the domain special fiber show that B is a domain: the first nonzero π -adic terms of a product of two nonzero elements have nonzero product.

We construct the essentially smooth rank-two model required in Proposition 7.1. Let $U_D = D_{5D}$, whose completion is C , and put

$$H = ((U_D \otimes_W R)[X, Y]) / \left(X^4 + \frac{\Phi_5(1 + \pi Y)}{\pi^4} + TXY^3 \right).$$

Its completion is B . The two relative partial derivatives generate the unit ideal modulo π by the Euler calculation. Choose an element j of their generated ideal with $j \equiv 1 \pmod{\pi}$. Then $H[1/j]$ is smooth of relative dimension one over $U_D \otimes_W R$. Inverting j does not change the completion.

The ring $U_D \otimes_W R$ is a localization of the smooth R -algebra $D \otimes_W R$, of relative dimension one. Transitivity of smoothness makes $H[1/j]$ a localization of a smooth R -algebra with relative cotangent rank two. The finite coefficients and the choice of j spread over a further localization of D . This gives the required B_0 . The 5-adic and π -adic completions agree because 5 is a unit times π^4 . In particular, relative forms over R retain the differential dT .

Write $b = r_E(\beta|_{W_E})$. Proposition 9.1 gives invertible ideals

$$J_g = \left(X, Y - \frac{\rho^g - 1}{\pi} \right)$$

and the nonzero class

$$(11.3) \quad \Xi = \sum_{g \in G} \sigma_g(b) (-c_1^{\text{dR}}(J_g)) \neq 0 \quad \text{in} \quad \Omega_{B/R}^2[1/5]/d\Omega_{B/R}^1[1/5].$$

The reduction of $\beta|_{W_E}$ in $K_3(\mathbf{F}_5)$ is zero. It lifts to an actual $\beta_{\text{rel}} \in K_3(W_E, (\pi))$. This lift is unique, since $K_4(\mathbf{F}_5) = 0$ by [41][50, IV, Corollary 1.13]. Naturality makes the lifts of all conjugates compatible. Pull them through $W_E \rightarrow R$. The relative point coordinates are $\sigma_g(b)$ in $R[1/5]$: the point comparison is natural for this coefficient extension, and the map on degree-two point cyclic homology is inclusion of the coefficient fields.

Form the actual relative class

$$\delta_{\text{rel}} = \sum_{g \in G} (\sigma_g \beta_{\text{rel}})|_B \cdot (1 - [J_g]) \in K_3(B, \pi B),$$

and let δ be its image in actual $K_3(B)$. Proposition 7.1, applied to the rank-two model above, identifies its top-form component with $\lambda_R \Xi$, where $\lambda_R \neq 0$. Thus the image of δ_{rel} in $\pi_3 \mathcal{K}(B, \pi B)$ is nonzero.

We establish the special-fiber vanishing before using either fiber sequence. The ring in (11.2) is the filtered colimit of

$$(D/5D)[1/s][X, Y]/(X^4 + Y^4 + TXY^3 - 1), \quad 0 \neq s \in D/5D.$$

These are smooth affine surfaces over the perfect field k . The base is a smooth affine curve, and the Euler identity makes the displayed curve over it smooth. For a smooth d -dimensional variety over a perfect field of characteristic p , the finite-coefficient theorem [25, Theorem 8.4] gives $K_j(-; \mathbf{Z}/p^n) = 0$ for $j > d$. Its groups are the homotopy groups of the cofiber spectrum, with their torsion contribution, rather than mere tensor products of actual K-groups. The theorem follows there from the motivic-to-K spectral sequence and the vanishing of logarithmic de Rham–Witt sheaves above the dimension. Applying its smooth-surface form and then filtered continuity gives

$$K_j(B/\pi B; \mathbf{Z}/5^n) = 0 \quad (j > 2, n \geq 1).$$

For $j \geq 3$, both the degree- j and degree- $(j+1)$ groups of the finite-coefficient tower vanish. The Milnor exact sequence therefore gives $\pi_j K(B/\pi B)_5^\wedge = 0$, and inversion of five preserves this vanishing. In particular,

$$(11.4) \quad \pi_3 \mathcal{K}(B/\pi B) = \pi_4 \mathcal{K}(B/\pi B) = 0.$$

First use the reduction fiber sequence

$$(11.5) \quad \mathcal{K}(B, \pi B) \longrightarrow \mathcal{K}(B) \longrightarrow \mathcal{K}(B/\pi B).$$

Equation (11.4) makes the first map an isomorphism on degree three. The image of δ in $\pi_3 \mathcal{K}(B)$ is consequently nonzero.

Next use regularity of B , regularity of $B/\pi B$, and the nonzero divisor π . Localization and dévissage [50, V.4.1, V.6.1] give the different fiber sequence

$$(11.6) \quad \mathcal{K}(B/\pi B) \longrightarrow \mathcal{K}(B) \longrightarrow \mathcal{K}(B[1/\pi]).$$

The left map is support pushforward, not reduction. Its source has zero degree-three group, so the second map is injective on degree three. Thus $\delta|_{B[1/\pi]}$ has nonzero image in $\pi_3 \mathcal{K}(B[1/\pi])$. That target is a $\mathbf{Q}_5$ -vector space, so the actual class $\delta|_{B[1/\pi]}$ has infinite order. Only the fact that torsion maps to zero in a characteristic-zero vector space is used here.

There is a ring map $A \rightarrow B$ defined by

$$x \mapsto \pi X, \quad y \mapsto \pi Y, \quad T \mapsto T.$$

The defining equation maps to π^4 times the equation of B . Every element inverted in forming A reduces to a nonzero polynomial in T modulo $(5, x, y)$. Its image modulo π is a nonzero element of L , hence a unit. Since π lies in the Jacobson radical, its image is a unit in B . This verifies the map on the localization. It takes (x, y) into πB , so completeness extends it continuously to $\hat{A} \rightarrow B$.

We match the actual supported class only after a flat splitting. Set

$$S_{\text{spl}} = (A \otimes_{\mathbf{Z}_{(5)}} W_{\text{ar}})[1/5].$$

This is flat over A , and regular because it is finite étale over $A[1/5]$. The polynomial $P(y)$ splits there with roots $\rho^g - 1$, whose differences are units. The divisor $x = 0$ is the disjoint union of the Cartier divisors with ideals

$$I_g = (x, y - (\rho^g - 1)).$$

Near a component, the other factors of P are units, and the equation expresses $y - (\rho^g - 1)$ as a multiple of x . Away from that component the ideal is the unit ideal. The element x remains a nonzero divisor by flatness, and the component rings are localizations of $E_{\text{ar}}[T]$, hence regular.

On the g -th component, the original coefficient map sends $\rho = 1 + y$ to ρ^g . Flat base change and the projection formula give the equality of actual classes

$$(11.7) \quad c|_{S_{\text{spl}}} = \sum_{g \in G} (\sigma_g \beta)|_{S_{\text{spl}}} \cdot (1 - [I_g]).$$

No scalar projector acts on an actual K-group in this identity.

The map $A \rightarrow B$, together with the coefficient embedding, gives $S_{\text{spl}} \rightarrow B[1/\pi]$. There

$$x = \pi X, \quad y - (\rho^g - 1) = \pi(Y - a_g).$$

The pulled-back invertible module I_g identifies with $J_g[1/\pi]$. Locally on its support it is generated by x , whose image πX is a nonzero divisor in the domain $B[1/\pi]$; off the support it is generated by a unit. Thus this identification is an isomorphism of invertible modules, not only an equality of generated ideals.

Pulling back (11.7) now gives

$$c|_{B[1/\pi]} = \sum_g (\sigma_g \beta)|_{B[1/\pi]} (1 - [J_g[1/\pi]]) = \delta|_{B[1/\pi]}.$$

The right side has infinite order. The map factors through $\hat{A}$, so it is also the image of $\hat{c}$. Hence $\hat{c}$, and therefore c , has infinite order.

An element of an abelian group maps to zero after tensoring with $\mathbf{Q}$ precisely when it is torsion. The two integral generic vanishings established at the start thus give the two asserted nonzero rational kernels.

We finish by identifying this coefficient with one allowed in the geometric proof. The inclusions $\mathbf{Z}[\rho] \subset \mathcal{O}_{E_{\text{ar}}} \subset W_{\text{ar}}$ show that W_{ar} is the localization of $\mathcal{O}_{E_{\text{ar}}}$ obtained by inverting integers prime to five. Localization, filtered continuity, and K_2 of finite fields being zero give a surjection

$$K_3(\mathcal{O}_{E_{\text{ar}}}) \longrightarrow K_3(W_{\text{ar}});$$

see [50, V.6.1], [50, IV, Section 6.4], and [41][50, IV, Corollary 1.13]. Lift the actual β_0 in Proposition 10.1 to $\gamma_0 \in K_3(\mathcal{O}_{E_{\text{ar}}})$. Then $\gamma = (1 - \sigma_{-1})\gamma_0$ maps exactly to β .

The nonzero point coordinate makes β nontorsion, so γ is nontorsion. The regulator injectivity of [14, Section 6], with the rank calculation of [12, Proposition 12.2], gives γ a nonzero complex regulator. All calculations in Section 6 therefore apply to this same β . Thus both detectors apply to the actual class c just constructed, and its image under completion is exactly $\hat{c}$. $\square$

12. ARITHMETIC QUADRICS AND GEOMETRIC DETECTION

The supported classes on an arithmetic quadric are detected by a ruling line bundle. The geometric argument has two steps: the ruling survives every finite localization permitted by the arithmetic local ring, and its regulator pairs nontrivially with a number-field coefficient.

12.1. The arithmetic quadric and its coefficient class. For an abelian group M , write $M_{\mathbf{Q}} = M \otimes_{\mathbf{Z}} \mathbf{Q}$. All K -groups in this article are Quillen algebraic K -groups. A coherent sheaf on a regular noetherian scheme is viewed in K -theory through a finite locally free resolution, or equivalently through the comparison $K \simeq G$.

The injectivity part of the rational Gersten conjecture asserts that, for every regular local ring R with fraction field F and every $n \geq 0$,

$$K_n(R)_{\mathbf{Q}} \longrightarrow K_n(F)_{\mathbf{Q}}$$

is injective. Its failure is sufficient to disprove the augmented rational Gersten conjecture.

Theorem 12.1. *Let p be an odd prime and set*

$$A = \left(\mathbf{Z}_{(p)}[x, y, z]/(p + xy + z^2) \right)_{(x, y, z)}.$$

There exists an integral class $c \in K_3(A)$ of infinite order whose image in $K_3(\text{Frac } A)$ is zero. In particular,

$$\ker(K_3(A)_{\mathbf{Q}} \longrightarrow K_3(\text{Frac } A)_{\mathbf{Q}}) \neq 0.$$

Only $p = 3$ is needed for a counterexample. The proof is given for odd p because its algebraic geometry is identical for all such primes.

In the case $p = 3$, Section 13 gives the following fully specified class:

$$(12.1) \quad c_{\text{exp}} = 2 \left\{ \frac{1+z}{2}, 1+x, 1 + \frac{(1+x)y}{4} \right\} \in K_3(A).$$

Braces mean the product of the three indicated unit classes in Quillen K_1 . All entries are units in A . The factor two makes generic vanishing an integral assertion.

Put

$$V = \mathbf{Z}_{(p)}, \quad W = V[u]/(u^2 + p), \quad E = \mathbf{Q}(u), \quad \sigma(u) = -u.$$

Thus W is a discrete valuation ring (DVR) with uniformizer u and residue field $\mathbf{F}_p$, and E is an imaginary quadratic number field. Write

$$A_0 = V[x, y, z]/(p + xy + z^2), \quad \mathfrak{m} = (x, y, z), \quad S = A_0 \setminus \mathfrak{m}.$$

Then $A = S^{-1}A_0$. Let

$$\begin{aligned} B_0 &= A_0 \otimes_V W = W[x, y, z]/(xy + z^2 - u^2), \\ B &= A \otimes_V W = S^{-1}B_0, \\ Q &= \text{Spec } E[x, y, z]/(xy + z^2 - u^2), \\ T &= \text{Spec } B[1/u]. \end{aligned}$$

For $s \in S$, let Q_s denote the corresponding principal open of Q .

We use the following standard inputs.

- (1) Quillen localization and dévissage, the projection formula for a regular closed immersion, and flat base change for the associated pushforward. These may be formulated in G -theory and transported to K -theory on the regular schemes involved; see [40, Sections 5–7].
- (2) $K_2(\mathbf{F}_p) = 0$ [41][50, IV, Corollary 1.13]. Consequently $K_3(W) \rightarrow K_3(E)$ is surjective, integrally.
- (3) Borel’s theorem gives a class in $K_3(E)$ with nonzero regulator at a complex embedding. The comparison of the Borel and Beilinson regulators identifies this, up to a nonzero scalar, with a class in

$$H_{\mathcal{D}, \text{an}}^1(\text{Spec } \mathbf{C}, \mathbf{R}(2)) = \mathbf{C}/\mathbf{R}(2), \quad \mathbf{R}(j) = (2\pi i)^j \mathbf{R}.$$

For an imaginary quadratic field, conjugating the embedding negates this regulator. See [12, Proposition 12.2] and [14, Section 6].

- (4) The analytic version of the Beilinson regulator is natural and multiplicative and has the usual first Chern-class normalization. A precise reference is [14, Theorem 2.31]. We use its weight-three component

$$r_3 : K_3(U) \longrightarrow H_{\mathcal{D}, \text{an}}^3(U, \mathbf{R}(3))$$

on smooth complex fibers, and its weight-two component on number fields.

- (5) Algebraic K -theory of rings commutes with filtered colimits [50, IV, Section 6.4]. In particular,

$$(12.2) \quad K_3(T) = \text{colim}_{s \in S} K_3(Q_s).$$

Remark 12.2. The site in [14] consists of regular separated schemes of finite type over $\mathbf{Z}$. Although E and Q_s are not finite type over $\mathbf{Z}$, each finite collection of coefficients, K -classes, line bundles, and morphisms used below spreads to a smooth model over $\mathcal{O}_E[1/N]$ for a suitable positive integer N . These models are regular and of finite type over $\mathbf{Z}$, and all further integer localizations have the same complex fiber. Regulator naturality and filtered-colimit compatibility therefore give precisely the maps used here. Projection from the conjugation-invariant collection of complex components to a chosen embedding preserves products. No regulator on the infinite scheme T is required.

12.2. Regularity and the supported class.

Lemma 12.3. *The ring A is a three-dimensional regular local domain with residue field $\mathbf{F}_p$. Its maximal ideal is generated by x, y, z , its fraction field is $\mathbf{Q}(x, z)$, and $p \in \mathfrak{m}^2$.*

Proof. The ambient ring $V[x, y, z]_{(p, x, y, z)}$ is regular local of dimension four. The element $p + xy + z^2$ belongs to its maximal ideal but not to its square: its initial linear term is p . Quotienting by this element gives a regular local ring of dimension three. The relation $p = -xy - z^2$ gives both the asserted generators of the maximal ideal and $p \in \mathfrak{m}^2$. Solving for y after inverting x gives the fraction field $\mathbf{Q}(x, z)$. Alternatively, the defining polynomial is primitive and irreducible as a polynomial of degree one in y , so the quotient is a domain. $\square$

The regular principal divisor $x = 0$ has coordinate ring

$$R = A/(x) = \left(V[y, z]/(z^2 + p) \right)_{(y, z)} \cong W[y]_{(u, y)},$$

where the map $W \rightarrow R$ sends u to z . Let

$$i : \operatorname{Spec} R \hookrightarrow \operatorname{Spec} A$$

be the closed immersion.

Lemma 12.4. *There is an integral class $\beta \in K_3(W)$ such that*

$$\sigma\beta = -\beta \quad \text{integrally,} \quad r_2(\beta|_E) \neq 0$$

at a fixed complex embedding $E \hookrightarrow \mathbf{C}$.

Proof. Choose a class of $K_3(E)$ with nonzero Borel–Beilinson regulator. Localization gives

$$K_3(W) \longrightarrow K_3(E) \longrightarrow K_2(\mathbf{F}_p) = 0,$$

so choose an integral lift $b_0 \in K_3(W)$. Set $\beta = b_0 - \sigma b_0$. Since $\sigma^2 = 1$, we have $\sigma\beta = -\beta$ without rationalization. The nontrivial embedding is complex conjugation, and on $\mathbf{C}/\mathbf{R}(2)$ this negates the nonzero regulator. Thus $r_2(\beta) = 2r_2(b_0) \neq 0$. $\square$

Define

$$(12.3) \quad c = i_*(\beta|_R) \in K_3(A).$$

The pushforward exists because A and R are regular, and the closed immersion is a regular Cartier immersion.

Lemma 12.5. *The image of c in $K_3(\operatorname{Frac} A)$ is zero integrally.*

Proof. By construction, c comes from a class with support on $V(x)$. The localization sequence therefore sends it to zero in $K_3(A[1/x])$, and hence in the fraction field. $\square$

The remaining question is whether c has infinite order. Merely knowing that $V(x)$ is principal does not answer this: a trivial normal bundle implies the vanishing of the self-intersection operator i^*i_* , not the vanishing of i_* itself.

12.3. The arithmetic node and its divisor class.

Lemma 12.6. *The ring B is a normal local domain of dimension three, with maximal ideal*

$$\mathfrak{n} = (u, x, y, z).$$

It is regular away from its closed point, and

$$\mathrm{gr}_{\mathfrak{n}} B \cong \mathbf{F}_p[U, X, Y, Z]/(XY + Z^2 - U^2).$$

Proof. The map $A \rightarrow B$ is finite free of rank two. Its closed fiber is $\mathbf{F}_p[u]/(u^2)$, with a unique maximal ideal, so B is local. In particular, $B = (B_0)_{\mathfrak{n}}$, not merely an unspecified localization of B_0 . The polynomial $u^2 + p$ is irreducible over the purely transcendental extension $\mathbf{Q}(x, z)$, hence B is a domain.

At a point where u is invertible, the ring is on a smooth affine quadric over E . At a point of the special fiber other than the origin, one of x, y, z is invertible, and the relative gradient $(y, x, 2z)$ has a unit component. Thus the punctured spectrum is regular. Being a hypersurface, B satisfies S_2 ; regularity in codimension one gives normality. The displayed associated graded ring follows from the nonzero quadratic initial form of the equation in the regular ambient local ring $W[x, y, z]_{(u, x, y, z)}$. $\square$

Consider the height-one primes

$$P_+ = (x, z - u), \quad P_- = (x, z + u).$$

Both have multiplicity one in $\mathrm{div}_B(x)$, so

$$(12.4) \quad \mathrm{div}_B(x) = P_+ + P_-.$$

Lemma 12.7. *Let W' be a DVR with uniformizer ϖ and residue field k , and let $0 \neq d \in \varpi W'$. In the normal local domain*

$$C = (W'[x, y, w]/(xy + w(w + d)))_{(\varpi, x, y, w)},$$

the height-one prime $P = (x, w)$ has infinite order in $\mathrm{Cl}(C)$. There is no restriction on the residue characteristic or on the positive valuation of d .

Proof. The defining polynomial is irreducible, and C is a hypersurface. Its generic fiber is smooth since $d \neq 0$. On the special fiber, a prime containing x and y also contains w ; hence every nonclosed point of that fiber has x or y invertible. Thus C is regular off its closed point, and S_2 and R_1 give normality.

Blow up the ideal $I = (x, w)$:

$$\rho : Y = \mathrm{Bl}_I(\mathrm{Spec} C) \longrightarrow \mathrm{Spec} C.$$

Its two standard charts are localizations of the following polynomial rings:

$$\begin{aligned} W'[x, t], \quad w = xt, \quad y = -t(xt + d), \\ W'[q, y], \quad w = -qy - d, \quad x = -q(qy + d). \end{aligned}$$

These are the actual Rees charts $C[w/x]$ and $C[x/w]$ inside the fraction field. For example, introducing $t = w/x$ gives $x(y + t(xt + d)) = 0$, and x -saturation gives exactly the first polynomial ring. The second is obtained by w -saturation. There are no extra equations because each indicated pair is a transcendence basis over $\mathrm{Frac}(W')$.

Consequently Y is regular. The closed fiber has equations (ϖ, x) in the first chart and (ϖ, y) in the second. Its two affine charts are $k[t]$ and $k[q]$, with $q = t^{-1}$ on their overlap. Hence it is a reduced curve

$$F \cong \mathbf{P}_k^1.$$

The ideal I is invertible off the closed point. Outside $V(I)$ this is immediate. At a nonclosed prime containing I , either y is a unit, giving $I = (w)$, or ϖ is a unit; then d and $w + d$ are units and $I = (x)$. The blowup is therefore an isomorphism off the closed point. Its only exceptional locus is the codimension-two curve F , so there are no exceptional prime divisors.

The standard tautological identity for a blowup is $I\mathcal{O}_Y = \mathcal{O}_Y(1)$, whose restriction to F is $\mathcal{O}_{\mathbf{P}_k^1}(1)$; see [44, Lemma 31.33.4, Tag 02OS]. Since I has multiplicity one at the generic point of P and there are no exceptional divisors,

$$I\mathcal{O}_Y = \mathcal{O}_Y(-\tilde{P}), \quad \mathcal{O}_Y(\tilde{P})|_F = \mathcal{O}_{\mathbf{P}_k^1}(-1).$$

If $\operatorname{div}_C(f) = NP$ for a nonzero integer N , then the same rational function on Y has divisor exactly $N\tilde{P}$. Restricting its trivial divisor line bundle to F gives

$$\mathcal{O}_{\mathbf{P}_k^1}(-N) \cong \mathcal{O}_{\mathbf{P}_k^1},$$

which forces $N = 0$, a contradiction. $\square$

Corollary 12.8. *The class $[P_+]$ is not torsion in $\operatorname{Cl}(B)$.*

Proof. Set $w = z - u$ and $d = 2u$. Then B is the ring in Lemma 12.7, with $W' = W$ and $\varpi = u$, and $P_+ = (x, w)$. $\square$

Remark 12.9. Blowing up the maximal ideal instead gives the regular resolution with exceptional quadric $\mathbf{P}_{\mathbf{F}_p}^1 \times \mathbf{P}_{\mathbf{F}_p}^1$. Its normal bundle is $\mathcal{O}(-1, -1)$, while the strict transform of P_+ restricts to one ruling, $\mathcal{O}(1, 0)$. If NP_+ were principal, its divisor upstairs would have the form $N\tilde{P}_+ + aF$, whose restriction would be $\mathcal{O}(N - a, -a)$. Triviality again forces $N = 0$. The small resolution above avoids this larger exceptional divisor and also works in residue characteristic two.

Lemma 12.10. *Restriction induces an isomorphism*

$$\operatorname{Cl}(B) \xrightarrow{\sim} \operatorname{Cl}(B[1/u]).$$

In particular, the image of $[P_+]$ is not torsion in $\operatorname{Pic}(T)$.

Proof. The quotient

$$B/(u) = \mathbf{F}_p[x, y, z]_{(x, y, z)} / (xy + z^2)$$

is a domain. The only height-one prime removed by inverting u is therefore the principal prime (u) . The localization sequence for divisor class groups is surjective with kernel generated by the removed prime classes, so its kernel is zero. Since $B[1/u]$ is regular, its class group equals its Picard group. $\square$

Remark 12.11. One can also compute $\operatorname{Cl}(B) = \mathbf{Z}$. Indeed, $B[1/x]$ is a localization of the unique factorization domain $W[x, x^{-1}, z]$, so the class group is generated by P_+ and P_- . Equation (12.4) gives their sum as zero, and Lemma 12.8 excludes any further relation. The argument needs only non-torsion, not this complete computation.

12.4. Persistence of Picard and Betti classes. Let D_+ and D_- denote the divisors of Q with equations

$$D_+ = (x, z - u), \quad D_- = (x, z + u).$$

They are disjoint affine lines over E .

Lemma 12.12. *We have $\operatorname{Pic}(Q) = \mathbf{Z}[D_+]$, and the same statement holds after extension to $\overline{E}$ or $\mathbf{C}$. The absolute Galois group of E acts trivially on the geometric Picard group. At a complex embedding,*

$$H^2(Q(\mathbf{C}), \mathbf{Z}) = \mathbf{Z},$$

with $c_1(\mathcal{O}_Q(D_+))$ a primitive generator after the usual Tate normalization.

Proof. The projective closure

$$\overline{Q} = \{xy + z^2 = u^2t^2\} \subset \mathbf{P}_E^3$$

is split: it has the rank-one matrix presentation

$$\det \begin{pmatrix} x & ut - z \\ ut + z & y \end{pmatrix} = 0.$$

Thus $\overline{Q} \cong \mathbf{P}^1 \times \mathbf{P}^1$. The divisor at infinity is a smooth divisor of type $(1, 1)$; after an automorphism of one factor it is the diagonal. Hence

$$Q \cong (\mathbf{P}^1 \times \mathbf{P}^1) \setminus \Delta, \quad \text{Pic}(Q) = \mathbf{Z}^2 / \mathbf{Z}(1, 1) = \mathbf{Z}.$$

The closure of D_+ is a ruling, and gives a generator. This calculation is unchanged over algebraically closed extensions. Because the generator is defined over E , the absolute Galois action is trivial.

Projection to one factor makes $Q(\mathbf{C}) \rightarrow \mathbf{P}^1(\mathbf{C})$ a locally trivial affine-line bundle, so it is a homotopy equivalence. Equivalently, the topological localization sequence for the diagonal gives the same quotient $\mathbf{Z}^2 / \mathbf{Z}(1, 1)$ in degree two. The first Chern class of a ruling maps to its primitive generator. $\square$

Lemma 12.13. *The restriction map $\text{Pic}(Q) \rightarrow \text{Pic}(T)$ is injective.*

Proof. The generator $[D_+]$ maps to the image of $[P_+]$ in $\text{Cl}(B[1/u])$. The latter has infinite order by Lemmas 12.8 and 12.10. $\square$

Lemma 12.14. *For each $s \in S$, every geometrically irreducible curve in $Q_{\mathbf{C}} \setminus (Q_s)_{\mathbf{C}}$ has zero class in $\text{Pic}(Q_{\mathbf{C}})$.*

Proof. Let D be an irreducible divisor over E in $Q \setminus Q_s$. It is absent from T , so $\mathcal{O}_Q(D)|_T$ is trivial. Lemma 12.13 implies $[D] = 0$ in $\text{Pic}(Q)$.

In characteristic zero the reduced divisor D becomes a reduced union of finitely many geometric irreducible components. These components are permuted transitively by the absolute Galois group of E . Its action on $\text{Pic}(Q_{\overline{E}}) = \mathbf{Z}$ is trivial, so all components have the same integer class, say a . If there are $r > 0$ components, their sum has class ra . This sum is the pullback of $[D] = 0$, so $ra = 0$ in the torsion-free group $\mathbf{Z}$, and therefore $a = 0$. Extending further to $\mathbf{C}$ preserves the calculation. $\square$

Remark 12.15. A geometric boundary component Γ is defined over some finite extension E'/E . Put $L = \mathcal{O}_Q(D_+)$ and write $[\mathcal{O}(\Gamma)] = a[L_{E'}]$ in $\text{Pic}(Q_{E'}) = \mathbf{Z}[L_{E'}]$. This line bundle is trivial on $T_{E'}$. Its finite-flat norm (equivalently, determinants after restriction of scalars) therefore makes $L_T^{a[E':E]}$ trivial. The infinite order of L_T forces $a = 0$, giving a second proof that each geometric boundary component has zero Picard class.

Remark 12.16. It is not enough that $\text{div}(s)$, as a whole, is principal. For instance $\text{div}(x) = D_+ + D_-$ is principal, while its two components have opposite nonzero classes. Removing $D_+ \cup D_-$ kills $H^2(Q, \mathbf{R})$. The element x is not in the permitted multiplicative set S .

Lemma 12.17. *For every $s \in S$, restriction is injective:*

$$H^2(Q_{\mathbf{C}}, \mathbf{R}) \longrightarrow H^2((Q_s)_{\mathbf{C}}, \mathbf{R}).$$

Proof. Let $Z = Q_{\mathbf{C}} \setminus (Q_s)_{\mathbf{C}}$ with its reduced support. The kernel is the image of the support map

$$H_Z^2(Q_{\mathbf{C}}, \mathbf{R}) \longrightarrow H^2(Q_{\mathbf{C}}, \mathbf{R}).$$

Remove from the ambient surface the finite set of singular points and intersections of the reduced curve components of Z . Support cohomology in finitely many points of a real four-manifold vanishes below degree four, so this changes neither the degree-two support group nor the ambient H^2 . The remaining components are disjoint smooth connected complex curves. The tubular-neighborhood calculation in the proof of Proposition 5.1 identifies degree-two support cohomology with one copy of $\mathbf{R}$ for each component. There the canonical transverse section of the divisor line bundle identifies the image of its support generator with the global first Chern class of the component. The isomorphism on ambient H^2 transports this identification across the deleted finite set. Every such class is zero by Lemma 12.14. Thus the support map is zero. $\square$

12.5. Analytic Deligne detection. Throughout this section, $H_{\mathcal{D},\text{an}}$ denotes analytic Deligne cohomology.

Lemma 12.18. *For a smooth complex surface U , there is a natural isomorphism*

$$(12.5) \quad H_{\mathcal{D},\text{an}}^3(U, \mathbf{R}(3)) \cong H^2(U, \mathbf{C})/H^2(U, \mathbf{R}(3)).$$

Proof. The analytic Deligne complex is

$$[\mathbf{R}(3) \longrightarrow \mathcal{O}_U \longrightarrow \Omega_U^1 \longrightarrow \Omega_U^2].$$

The holomorphic de Rham complex resolves the constant sheaf $\mathbf{C}$ by the holomorphic Poincaré lemma, and there are no holomorphic forms in degree three. The Deligne complex is consequently quasi-isomorphic to the constant sheaf $\mathbf{C}/\mathbf{R}(3)$ placed in degree one. Equivalently, it is the shifted cone of

$$R\Gamma(U, \mathbf{R}(3)) \longrightarrow R\Gamma(U, \mathbf{C}).$$

Since real cohomology injects into its complexification in every degree, the long exact sequence of this cone gives (12.5). All constructions are natural for restriction to open subsets. Properness is not used. $\square$

Remark 12.19. On a smooth normal-crossings compactification, the logarithmic de Rham complex also has $F^3 = 0$. Thus Deligne–Beilinson cohomology agrees with the analytic group in (12.5); compare [23, Remark 2.13]. The direct analytic formulation above already suffices for the proof.

Lemma 12.20. *For every $s \in S$, restriction induces an injection*

$$H_{\mathcal{D},\text{an}}^3(Q_{\mathbf{C}}, \mathbf{R}(3)) \longrightarrow H_{\mathcal{D},\text{an}}^3((Q_s)_{\mathbf{C}}, \mathbf{R}(3)).$$

Proof. Combine Lemmas 12.17 and 12.18. Explicitly, an injection of real vector spaces $V \hookrightarrow W$ induces an injection

$$\frac{V \otimes_{\mathbf{R}} \mathbf{C}}{(2\pi i)^3 V} \longrightarrow \frac{W \otimes_{\mathbf{R}} \mathbf{C}}{(2\pi i)^3 W},$$

because the intersection of the image of $V \otimes \mathbf{C}$ with $(2\pi i)^3 W$ is exactly $(2\pi i)^3 V$. $\square$

Set

$$\eta = [\mathcal{O}_{D_+}] \in K_0(Q), \quad \gamma = (\beta|_E)\eta \in K_3(Q).$$

The structure sheaf class is a perfect-complex class, with

$$\eta = 1 - [\mathcal{O}_Q(-D_+)], \quad \text{rank}(\eta) = 0, \quad \text{ch}_1(\eta) = c_1(\mathcal{O}_Q(D_+)).$$

Lemma 12.21. *The class γ has nonzero weight-three analytic Deligne regulator.*

Proof. Naturality and multiplicativity give

$$(12.6) \quad r_3(\gamma) = r_2(\beta|_E) \cup c_1(\mathcal{O}_Q(D_+)).$$

There are no additional terms of total weight three: a degree-three K -class pulled back from a point has the relevant regulator only in $H_{\mathcal{D}}^1(\text{pt}, \mathbf{R}(2))$, and the rank of η is zero.

Let h be a real primitive generator of $H^2(Q_{\mathbf{C}}, \mathbf{R})$, normalized so that $c_1(\mathcal{O}_Q(D_+))$ has Betti image $2\pi i h$. A nonzero element of $\mathbf{C}/\mathbf{R}(2)$ is represented by it for a nonzero real t . Under (12.5), the product in (12.6) is represented, up to the harmless convention for the sign of h , by

$$(it)(2\pi i)h = -2\pi t h.$$

This is a nonzero real multiple of h , whereas $H^2(Q_{\mathbf{C}}, \mathbf{R}(3))$ is the imaginary real line generated by h . The quotient class is therefore nonzero.

One may equivalently compute this cup product on $\mathbf{P}^1$: the projection of the affine-line torsor $Q \rightarrow \mathbf{P}^1$ induces the required degree-two Betti isomorphism, and cup product with the hyperplane class is the usual projective-bundle map. $\square$

Proposition 12.22. *The restriction $\gamma|_T \in K_3(T)$ has infinite order.*

Proof. By regulator naturality and Lemma 12.20,

$$r_3(\gamma|_{Q_s}) \neq 0 \quad \text{for every } s \in S.$$

If a positive integer multiple $N\gamma$ vanished in $K_3(T)$, continuity (12.2) would make it vanish already in $K_3(Q_s)$ for some s . Applying the regulator there would yield $Nr_3(\gamma|_{Q_s}) = 0$ in a real vector space, which is impossible. $\square$

12.6. Flat base change and nonvanishing.

Lemma 12.23. *The image of the supported class (12.3) in $K_3(T)$ equals $2\gamma|_T$.*

Proof. The map $A \rightarrow B[1/u]$ is flat, being a finite free base change followed by localization. Its pullback of the divisor $x = 0$ is the disjoint union

$$(D_+ \cap T) \amalg (D_- \cap T).$$

The original coefficient map $W \rightarrow R$ sends u to z . Therefore its two restrictions after the splitting are the coefficient maps $u \mapsto u$ and $u \mapsto -u$. Flat base change and the projection formula give

$$\begin{aligned} c|_T &= (\beta|_E)[\mathcal{O}_{D_+ \cap T}] + (\sigma\beta|_E)[\mathcal{O}_{D_- \cap T}] \\ &= (\beta|_E)([\mathcal{O}_{D_+ \cap T}] - [\mathcal{O}_{D_- \cap T}]). \end{aligned}$$

The two divisors are disjoint since $2u$ is a unit on T . Moreover their union is the principal divisor $V(x)$. The resolution by multiplication by x gives

$$[\mathcal{O}_{D_+ \cap T}] + [\mathcal{O}_{D_- \cap T}] = [\mathcal{O}_{T/(x)}] = 0 \quad \text{in } K_0(T).$$

Consequently $c|_T = 2(\beta|_E)[\mathcal{O}_{D_+ \cap T}] = 2\gamma|_T$. $\square$

Proof of Theorem 12.1. Lemma 12.3 gives the required regular local ring. By Proposition 12.22 and Lemma 12.23, the pullback of c to $K_3(T)$ has infinite order; hence c itself has infinite order. Lemma 12.5 shows that its generic image is zero integrally. Tensoring with $\mathbf{Q}$ therefore gives the asserted nonzero rational kernel. $\square$

The nonzero-product assertion in this proof is not deduced from a nonzero Picard class alone. The essential extra step is that its regulator remains nonzero *after every permitted finite localization*. This is what makes the filtered-colimit argument valid.

12.7. The unsplit generic fiber. The following calculation identifies the coefficient of the supported class on the unsplit generic fiber. Put

$$X = \operatorname{Spec} \mathbf{Q}[x, y, z]/(xy + z^2 + p).$$

Its divisor $Z = V(x)$ is $\mathbf{A}_E^1$, while

$$X \setminus Z \cong \mathbb{G}_{m, \mathbf{Q}} \times \mathbf{A}_{\mathbf{Q}}^1.$$

For every $n \geq 0$, localization, homotopy invariance, and the fundamental theorem give

$$K_n(E) \longrightarrow K_n(X) \longrightarrow K_n(\mathbf{Q}) \oplus K_{n-1}(\mathbf{Q}) \xrightarrow{\partial} K_{n-1}(E).$$

The boundary is zero on $K_n(\mathbf{Q})$ and is restriction $\operatorname{res}_{E/\mathbf{Q}}$ on the second summand: that summand is represented by $\{x\} \cdot b$, and x cuts Z with multiplicity one. The preceding boundary has image $\operatorname{res} K_n(\mathbf{Q})$ for the same reason.

After rationalization, restriction is injective because transfer composed with restriction is multiplication by two. Constants split the resulting short exact sequence, giving

$$(12.7) \quad K_n(X)_{\mathbf{Q}} \cong K_n(\mathbf{Q})_{\mathbf{Q}} \oplus K_n(E)_{\overline{\mathbf{Q}}}.$$

Here the second summand is identified using $\operatorname{res} \operatorname{Tr} = 1 + \sigma$ and enters through Z -Gysin. In degree three, Borel's calculation [12, Proposition 12.2] yields

$$K_3(\mathbf{Q})_{\mathbf{Q}} = 0, \quad K_3(E)_{\overline{\mathbf{Q}}} \cong \mathbf{Q}.$$

Thus the generic-fiber Gysin map identifies $K_3(E)_{\mathbf{Q}}$ with $K_3(X)_{\mathbf{Q}}$. The supported class is already a nonzero one-dimensional summand before localization at $\mathfrak{m}$. The main proof shows that this particular summand cannot disappear at that localization.

13. THE EXPLICIT THREE-ADIC CLASS AND ITS COMPLEX PERIOD

This section gives a geometric proof of the explicit class at $p = 3$. It uses the same geometric persistence lemmas, but obtains nonvanishing by an explicit period rather than by selecting a Borel class. Put

$$A = \left(\mathbf{Z}_{(3)}[x, y, z]/(3 + xy + z^2) \right)_{(x, y, z)},$$

and define

$$a = \frac{1+z}{2}, \quad s_0 = a(1-a) = 1 + \frac{xy}{4}, \quad \lambda = 1+x, \quad \mu = 1 + \frac{(1+x)y}{4}.$$

Every one of $a, 1-a, s_0, \lambda, \mu$ is a unit of A , and

$$(13.1) \quad 1 + x\mu = \lambda s_0.$$

13.1. Dennis–Stein notation and integral generic vanishing. We use the modern Dennis–Stein convention: $\langle r, t \rangle$ is defined if $1 - rt$ is a unit, and, if r is a unit,

$$\langle r, t \rangle = \{r, 1 - rt\}.$$

If t is a unit, antisymmetry gives $\langle r, t \rangle = \{1 - rt, t\}$. The relation used below is

$$(13.2) \quad \langle r, t \rangle + \langle r, v \rangle = \langle r, t + v - rtv \rangle, \quad \langle r, 1 \rangle = 0.$$

The modern convention and relations are those of [50, III, Definition 5.11 and (D1)–(D3)]; the historical source is [22]. We do not need a presentation theorem for K_2 .

Let

$$\delta = \langle -x, y/4 \rangle \in K_2(A), \quad \alpha = [a]\delta \in K_3(A).$$

On $A[1/x]$, the symbol is $\{-x, s_0\}$. Since a and $1-a$ are units, the Steinberg relation and graded commutativity give

$$2\{a, s_0\} = 2\{a, a(1-a)\} = 2[a]^2 + 2\{a, 1-a\} = 0.$$

Reordering products therefore gives

$$(13.3) \quad 2\alpha|_{A[1/x]} = 2\{a, -x, s_0\} = 0 \quad \text{integrally.}$$

The Dennis–Stein class can also be rewritten using units alone. By (13.2),

$$\delta = \langle -x, \mu \rangle = \{1 + x\mu, \mu\} = \{\lambda s_0, \mu\}.$$

Multiplying by $2[a]$ kills the $\{a, s_0, \mu\}$ term and yields

$$(13.4) \quad 2\alpha = 2\{a, \lambda, \mu\} = c_{\text{exp}}.$$

Thus the explicitly displayed class (12.1) has zero generic image integrally.

13.2. A finite open on which to calculate the period. Keep $E = \mathbf{Q}(u)$ with $u^2 = -3$, and fix $u = i\sqrt{3}$. Let

$$Q = \text{Spec } E[x, y, z]/(3 + xy + z^2), \quad U = Q[(1 - z^2)^{-1}].$$

The classes δ and α are defined on U , since $a, 1-a, s_0$ are units there. Both lines D_+ and D_- lie entirely inside U , because on them $1 - z^2 = 4$. Equation (13.3) holds on $U[1/x]$ as well.

Localization therefore provides a class supported on $D = V(x)$ that maps to $2\alpha \in K_3(U)$. Since the entire divisor D lies in U , excision identifies the supported groups for Q and U . Equivalently, apply the localization sequence with $K_3(D)$ and the naturality of its pushforward. Thus there is a class

$$(13.5) \quad \tilde{c} \in K_3(Q), \quad \tilde{c}|_U = 2\alpha.$$

No coefficient of this supported lift has to be chosen or computed explicitly.

13.3. The curvature and the Deligne product. The weight-two analytic regulator of δ has the holomorphic curvature

$$(13.6) \quad \omega = \frac{dx \wedge d(y/4)}{1 + xy/4}.$$

Indeed, where $x \neq 0$, multiplicativity for the two unit classes gives

$$d \log(-x) \wedge d \log s_0 = \frac{dx \wedge d(y/4)}{s_0}.$$

The Hodge endpoint is a canonical holomorphic two-form: in this degree $F^2 A^1 = 0$, so there is no preceding filtered form that could change it by a boundary. Both sides therefore extend as actual holomorphic two-forms on U , and agreement on the dense open proves (13.6). The logarithmic normalization follows from the ordinary unit regulator and [14, Theorem 2.31 and Lemma 4.5]. An independent algebraic differential calculation for Dennis–Stein characters over smooth commutative $\mathbf{Q}$ -algebras, with its own symbol conventions, appears in [26, Section 3.2, Theorem 3.4 and Corollary 3.6]; the preceding dense-open calculation fixes the regulator sign and gives the needed form.

Suppose that a has a single-valued holomorphic logarithm L on an analytic open $N \subset U(\mathbf{C})$. Then the regulator of $\alpha = [a]\delta$ on N is represented in

$$H_{\mathcal{D}, \text{an}}^3(N, \mathbf{R}(3)) = H^2(N, \mathbf{C}/\mathbf{R}(3))$$

by the closed holomorphic two-form

$$(13.7) \quad -L\omega.$$

We use the cone-coordinate convention of [14, Proposition 2.8]: an interval form Θ maps to

$$(\Theta|_0 \oplus \Theta|_1, -\int_{[0,1]} \Theta).$$

Let v denote the auxiliary interval coordinate and put $g = i \operatorname{Im} L + v \operatorname{Re} L$. By [14, Lemma 4.5], the unit class $[a]$ is represented by dg . We may replace this by $d(vL)$ in the same absolute interval complex:

$$dg - d(vL) = d((1 - v)i \operatorname{Im} L).$$

The primitive is allowed in $IDR(1)$: its endpoint at zero is $\mathbf{R}(1)$ -valued, and its endpoint at one is zero, as required by $F^1 A^0 = 0$. Thus this replacement does not substitute a different relative class for the original absolute regulator. Choose a closed interval representative Θ_δ whose Hodge endpoint is ω . The product $d(vL) \wedge \Theta_\delta$ has zero endpoint at zero and endpoint $dL \wedge \omega = 0$ at one, since N is a complex surface. Multiplicativity and Stokes' formula therefore give

$$-\int_{[0,1]} d(vL) \wedge \Theta_\delta = -L\omega + d \int_{[0,1]} vL \Theta_\delta.$$

Thus the cone coordinate is (13.7), for every chosen closed representative of the original absolute class. The form $L\omega$ is closed because N has complex dimension two.

13.4. A strictly positive real period. Inside $U(\mathbf{C})$ consider the smooth sphere

$$\Sigma : \quad z = i\sqrt{3}\tau, \quad x = \sqrt{3}\sqrt{1 - \tau^2} e^{i\vartheta}, \quad y = -\bar{x}, \quad -1 \leq \tau \leq 1.$$

The polar coordinates degenerate only at the poles; intrinsically this is the sphere $|x|^2 + (\operatorname{Im} z)^2 = 3$. Along it,

$$a = \frac{1 + i\sqrt{3}\tau}{2}, \quad s_0 = \frac{1 + 3\tau^2}{4} > 0.$$

There is an analytic neighborhood N of Σ on which $\operatorname{Re} a > 0$, so the principal $L = \log a$ is single-valued. On Σ ,

$$L = \frac{1}{2} \log \frac{1 + 3\tau^2}{4} + i \arctan(\sqrt{3}\tau), \quad \omega = -\frac{6i\tau}{1 + 3\tau^2} d\tau \wedge d\vartheta.$$

Consequently,

$$(13.8) \quad \int_{\Sigma} \operatorname{Re}(L\omega) = 2\pi \int_{-1}^1 \frac{6\tau \arctan(\sqrt{3}\tau)}{1+3\tau^2} d\tau > 0.$$

The integrand is positive except at $\tau = 0$. With the specified Bunke–Tamme cone convention, the real period of $r_3(2\alpha)$ is the negative double of this positive integral. Since $\mathbf{R}(3)$ is the imaginary axis, this detects a nonzero class in $H^2(N, \mathbf{C}/\mathbf{R}(3))$. Thus $r_3(2\alpha) \neq 0$ on U .

For comparison, integration by parts expresses (13.8) as

$$4\pi D(e^{i\pi/3}),$$

with the displayed orientation and logarithmic convention. Strict positivity alone is sufficient; neither the special value nor Borel’s rank theorem is needed in this argument.

13.5. A normalized supported coefficient. The exact coefficient can also be specified rationally. Put $\zeta = (1+u)/2$, and define $\beta_{\text{BT}} = \frac{1}{2} \operatorname{bl}(2[\zeta]) \in K_3(E)_{\mathbf{Q}}$ using the Bloch symbol map of [14, Theorem 4.9 and (4.8)], first on the full integer ring $\mathcal{O}_E = \mathbf{Z}[\zeta]$ and then by restriction to E . Both ζ and $1-\zeta = \zeta^{-1}$ are units. The doubled wedge boundary is zero even with the antisymmetric-tensor convention, so this definition avoids any integral two-torsion convention for the Bloch group. At the chosen embedding its cone-coordinate regulator is $-iD(\zeta)$. Borel injectivity [14, Section 6] makes this rational class independent of the integral lift.

A supported lift as in (13.5) has the form $\tilde{c} = \gamma\eta_+$, with $\gamma \in K_3(E)$: use homotopy invariance on the two affine lines and $\eta_- = -\eta_+$. The two generators of $I = (x, z-u)$ give a map $f: Q \rightarrow \mathbf{P}_E^1$ with $f^*\mathcal{O}(1) = I$. On Σ its coordinate is

$$\frac{z-u}{x} = -i\sqrt{\frac{1-\tau}{1+\tau}} e^{-i\vartheta}.$$

Both the radial derivative and the angular derivative are negative, so this is a map of degree $+1$ for the displayed orientation. Hence

$$\int_{\Sigma} c_1(I) = 2\pi i, \quad \int_{\Sigma} \operatorname{ch}_1(\eta_+) = -2\pi i.$$

Multiplicativity gives period $(-iD(\zeta))(-2\pi i) = -2\pi D(\zeta)$ for $\beta_{\text{BT}}\eta_+$, whereas (13.8) gives period $-8\pi D(\zeta)$ for $\tilde{c}$. Borel injectivity on $K_3(E)_{\mathbf{Q}}$ therefore gives

$$(13.9) \quad \tilde{c} = 4\beta_{\text{BT}}\eta_+ \quad \text{in } K_3(Q)_{\mathbf{Q}}.$$

This is a regulator-normalized identity for every supported lift. It does not identify β_{BT} with the Besser–de Jeu class used in Section 16.5; that latter comparison will only use a nonzero rational scalar.

13.6. Persistence and the explicit conclusion. By (13.5), the regulator of $\tilde{c}$ is nonzero on Q . Lemma 12.20 keeps it nonzero on every Q_s for $s \in S$. The element $1-z^2$ belongs to S , so T lies in U and the restrictions of $\tilde{c}$ and 2α agree on T . By (13.4), this is also the image of $c_{\text{exp}} \in K_3(A)$.

If a positive multiple of c_{exp} vanished in $K_3(A)$, the same multiple of $\tilde{c}$ would vanish in $K_3(T)$, and hence on some finite Q_s by continuity. Its nonzero real regulator rules this out. We have therefore obtained the explicit version of the counterexample:

$$(13.10) \quad \boxed{0 \neq 2 \left\{ \frac{1+z}{2}, 1+x, 1 + \frac{(1+x)y}{4} \right\} \in \ker(K_3(A) \rightarrow K_3(\operatorname{Frac} A)),}$$

where nonzero is strengthened to infinite order by the period argument. This conclusion depends on the geometric persistence proof in Sections 12.3–12.5, in addition to the explicit calculation in this section.

13.7. An independent algebraic detection modulo three. There is a short independent test that proves c_{exp} is nonzero integrally, and even that it is not divisible by three. This test does not prove infinite order, so it does not replace the regulator argument for the rational conjecture.

Let

$$R = A[T]_{\mathfrak{m}_A[T]}, \quad V' = \mathbf{Z}_{(3)}[T]_{(3)}.$$

Then R is a local V' -algebra with residue field $\mathbf{F}_3(T)$. Its module of relative differentials is presented by

$$\Omega_{R/V'}^1 = \frac{R dx \oplus R dy \oplus R dz}{R(y dx + x dy + 2z dz)},$$

so

$$\Omega_{R/V'}^3 = (R/(x, y, 2z)) dx \wedge dy \wedge dz \cong \mathbf{F}_3(T) dx \wedge dy \wedge dz.$$

For a local ring with infinite residue field, the usual Milnor symbol product

$$\iota : K_3^M(R) \longrightarrow K_3(R)$$

has a comparison homomorphism $\phi : K_3(R) \rightarrow K_3^M(R)$ satisfying $\phi\iota = 2$. This is the degree-three case of the Nesterenko–Suslin comparison recorded for general local rings in [31, Proposition 4.2.8(5),(6)], after identifying ordinary and improved Milnor K -theory for the infinite residue field of R .

Write $c_M = 2\{a, \lambda, \mu\} \in K_3^M(R)$. The logarithmic differential is well defined on Milnor symbols, since it kills every Steinberg relation. A direct calculation gives

$$d \log(c_M) = \frac{1}{2(1+z)\mu} dz \wedge dx \wedge dy.$$

Consequently the composite additive homomorphism

$$K_3(A) \longrightarrow K_3(R) \xrightarrow{\phi} K_3^M(R) \xrightarrow{d \log} \Omega_{R/V'}^3$$

sends c_{exp} to

$$\frac{1}{(1+z)\mu} dx \wedge dy \wedge dz = dx \wedge dy \wedge dz \neq 0.$$

The equality is in $\Omega_{R/V'}^3$, where x, y, z vanish in the coefficient ring and $\mu = 1$. Since this target is killed by three, $c_{\text{exp}} \notin 3K_3(A)$.

In particular, the coefficient exact sequence injects the nonzero image of this class in $K_3(A)/3$ into $K_3(A; \mathbf{Z}/3)$. Its generic image is zero by the integral calculation in (13.3). Thus this explicit example also fails Gersten injectivity with $\mathbf{Z}/3$ -coefficients, independently of the analytic regulator.

Remark 13.1. The same conclusion is already visible in

$$C = A/\mathfrak{m}^2 \cong \mathbf{F}_3[\epsilon_x, \epsilon_y, \epsilon_z]/(\epsilon_x, \epsilon_y, \epsilon_z)^2.$$

To check this without assuming that the finite residue field is large enough for an ordinary Milnor comparison, pass to

$$D = C[T]_{\mathfrak{m}_C C[T]} \cong k[\epsilon_x, \epsilon_y, \epsilon_z]/(\epsilon_x, \epsilon_y, \epsilon_z)^2, \quad k = \mathbf{F}_3(T).$$

Let $M = ke_x \oplus ke_y \oplus ke_z$ be a D -module through the residue map. The k -derivation $d : D \rightarrow M$ given by $d\epsilon_i = e_i$ induces

$$K_3^M(D) \longrightarrow \bigwedge_k^3 M, \quad \{g_1, g_2, g_3\} \longmapsto \frac{dg_1}{g_1} \wedge \frac{dg_2}{g_2} \wedge \frac{dg_3}{g_3}.$$

This kills the Steinberg relations because dg/g and $d(1-g)/(1-g)$ are proportional. The three entries of c_{exp} map to $2(1+\epsilon_z)$, $1+\epsilon_x$, and $1+\epsilon_y$. After the comparison $K_3(D) \rightarrow K_3^M(D)$, the image of c_{exp} is therefore

$$4e_z \wedge e_x \wedge e_y = e_x \wedge e_y \wedge e_z \neq 0.$$

In fact its image in $K_3(C)$ has exact order three: it is nonzero by this test, and $(1 + \epsilon_x)^3 = 1$ kills its triple product by three. Both the henselization and the completion of A have the same quotient C . Hence the explicit class also remains nonzero and not divisible by three in their K_3 -groups, with zero generic image. This observation alone does not prove infinite order in either ring.

14. QUADRATIC AND EISENSTEIN FAMILIES

14.1. Residue characteristic two. The small resolution in Lemma 12.7 allows the same argument at the prime two. This variant is not needed for the explicit counterexample in (12.1).

Proposition 14.1. *For the regular local ring*

$$A_2 = \left(\mathbf{Z}_{(2)}[x, y, z] / (1 + xy + z^2) \right)_{(2, x, y, z-1)},$$

the map $K_3(A_2)_{\mathbf{Q}} \rightarrow K_3(\text{Frac } A_2)_{\mathbf{Q}}$ has nonzero kernel.

Proof. Put $v = z - 1$. The defining equation is $2 + 2v + v^2 + xy$, whose linear term in the regular ambient local ring is 2. Thus A_2 is regular of dimension three, with maximal ideal generated by x, y, v and residue field $\mathbf{F}_2$.

Take $W_2 = \mathbf{Z}_{(2)}[i]$, $E_2 = \mathbf{Q}(i)$, and $\varpi = 1 - i$. The equation $\varpi^2 - 2\varpi + 2 = 0$ is Eisenstein, so W_2 is a DVR. The divisor $x = 0$ in $\text{Spec } A_2$ is regular, with coordinate ring $W_2[y]_{(\varpi, y)}$. Choose an integral anti-invariant Borel class $\beta \in K_3(W_2)$ exactly as in Lemma 12.4, and push it forward from this divisor.

After the finite free quadratic base change, put $w = z - i$. The local ring becomes

$$(W_2[x, y, w] / (xy + w(w + 2i)))_{(\varpi, x, y, w)}.$$

Here $d = 2i = -\varpi^2$ is nonzero and has positive valuation. Lemma 12.7 proves that (x, w) has infinite order in the class group. The quotient by ϖ is the domain

$$\mathbf{F}_2[x, y, w]_{(x, y, w)} / (xy + w^2),$$

so inverting ϖ preserves this class group.

The generic fiber over E_2 is the split smooth quadric $xy + z^2 + 1 = 0$. Its Picard group, geometric Galois action, and Betti cohomology are exactly those used in Section 12.4. The argument of Lemma 12.14 therefore proves persistence under every denominator from the displayed localization defining A_2 . Flat base change along $A_2 \rightarrow A_2 \otimes_{\mathbf{Z}_{(2)}} W_2$, followed by inversion of ϖ , gives twice the product of β with the ruling class, and the analytic regulator in twist three proves that this product has infinite order. Its generic vanishing follows from its original support on $x = 0$, as before. $\square$

14.2. Every higher odd K -degree. The regulator argument also has the following extension. It is stated separately to keep the degree-three counterexample independent of extra indexing conventions.

Proposition 14.2. *Let p be an odd prime and let $n = 2m - 1 \geq 3$. Set $\varepsilon = (-1)^{m+1}$ and*

$$A_{p,m} = \left(\mathbf{Z}_{(p)}[x, y, z] / (xy + z^2 - \varepsilon p) \right)_{(x, y, z)}.$$

There is an integral element of infinite order in

$$\ker(K_n(A_{p,m}) \longrightarrow K_n(\text{Frac } A_{p,m})).$$

Thus the family with $xy + z^2 + p$ works in degrees $3, 7, 11, \dots$, and the family with $xy + z^2 - p$ works in degrees $5, 9, 13, \dots$.

Proof. Take $W = \mathbf{Z}_{(p)}[u] / (u^2 - \varepsilon p)$, $E = \mathbf{Q}(u)$, and $\sigma(u) = -u$. The regularity and small-resolution arguments above are unchanged: after base change the equation is $xy + z^2 - u^2$, with $d = 2u$ in Lemma 12.7.

If m is even, E is imaginary quadratic. Borel's regulator identifies $K_{2m-1}(E) \otimes \mathbf{R}$ with a one-dimensional real space, and the involution acts by -1 : its target $\mathbf{C}/\mathbf{R}(m)$ is the imaginary

quotient. If m is odd, E is real quadratic. The regulator target is a pair of real lines indexed by the two real embeddings, and σ exchanges the two coordinates. Its anti-invariant part is again one-dimensional. These regulator ranks and conjugation conventions are those of [12, Proposition 12.2] and [14, Section 6].

In either case choose $b \in K_n(E)$ with nonzero anti-invariant regulator. Since $n - 1 > 0$ is even, $K_{n-1}(\mathbf{F}_p) = 0$ and localization lifts b to $K_n(W)$. Antisymmetrizing gives an integral β with $\sigma\beta = -\beta$ and

$$r_m(\beta) \neq 0 \quad \text{in} \quad H_{\mathcal{D},\text{an}}^1(\text{Spec } \mathbf{C}, \mathbf{R}(m)).$$

Push β forward from the regular divisor $x = 0$ of $\text{Spec } A_{p,m}$. The resulting class is zero in the fraction field. On the split quadric localization it becomes $2\beta\eta$.

The relevant regulator component now has target

$$H_{\mathcal{D},\text{an}}^{2(m+1)-n}(Q_s, \mathbf{R}(m+1)) = H_{\mathcal{D},\text{an}}^3(Q_s, \mathbf{R}(m+1)).$$

Since $m+1 \geq 3 > \dim_{\mathbf{C}} Q_s$, the same holomorphic Poincaré-lemma calculation gives

$$H_{\mathcal{D},\text{an}}^3(Q_s, \mathbf{R}(m+1)) = H^2(Q_s, \mathbf{C})/H^2(Q_s, \mathbf{R}(m+1)).$$

The proof of Lemma 12.17, applied to each geometric boundary curve, therefore gives the same injection. Multiplication by $c_1(D_+) = 2\pi i h$ identifies the nonzero coefficient in $\mathbf{C}/\mathbf{R}(m)$ with a nonzero coefficient in $\mathbf{C}/\mathbf{R}(m+1)$. No other source regulator component contributes: for a complex point and a positive twist, analytic Deligne cohomology is concentrated in degree one. The target regulator is nonzero at every finite stage, and K -theoretic continuity proves infinite order as before. $\square$

Remark 14.3. The same argument gives a *single* arithmetic local ring with a nonzero rational generic kernel in every odd degree $2n - 1 \geq 3$. Fix an odd prime p , a prime $\mathfrak{v}$ of $\mathbf{Z}_{(p)}[i]$ above p , and put

$$F = \mathbf{Q}(i), \quad V = (\mathbf{Z}_{(p)}[i])_{\mathfrak{v}}, \quad R = \left(V[x, y, z]/(p + xy + z^2) \right)_{(x,y,z)}.$$

Here V is an unramified DVR with uniformizer p , so R is regular of dimension three. Set $E = F(u)$, $u^2 = -p$, $W = V[u]$, and let σ fix F and send u to $-u$. Eisenstein's criterion makes W a DVR and gives $[E : F] = 2$. Both F and E are totally imaginary, with respectively one and two complex places. Borel's theorem [12, Proposition 12.2] gives $\dim_{\mathbf{Q}} K_{2n-1}(F)_{\mathbf{Q}} = 1$ and $\dim_{\mathbf{Q}} K_{2n-1}(E)_{\mathbf{Q}} = 2$, for every $n \geq 2$. Restriction and transfer satisfy $\text{res tr} = 1 + \sigma$ and $\text{tr res} = 2$. Thus the σ -fixed subspace is exactly the restriction of $K_{2n-1}(F)_{\mathbf{Q}}$, and the anti-invariant subspace has dimension one.

Choose a nonzero class there, clear denominators, lift it to $K_{2n-1}(W)$, and antisymmetrize. The lift exists because $K_{2n-2}(k_V) = 0$ [50, Chapter IV, Corollary 1.13]. This produces an actual integral β_n with $\sigma\beta_n = -\beta_n$ and a nonzero regulator at some complex embedding. Its pushforward from $R/(x) = W[y]_{(u,y)}$ is generically zero and becomes $2\beta_n(1 - [\mathcal{O}(-D_+)])$ on the split generic quadric. The small resolution and the persistence proved in Lemma 12.17 are unchanged. In twist $n+1 > 2$, the detecting target is

$$H_{\mathcal{D},\text{an}}^3(Q_s, \mathbf{R}(n+1)) = H^2(Q_s, \mathbf{C})/H^2(Q_s, \mathbf{R}(n+1)).$$

Cup with the ruling multiplies the point coefficient $\mathbf{C}/\mathbf{R}(n)$ by $2\pi i$, giving an isomorphism onto the ruling line modulo $\mathbf{R}(n+1)$. The product is nonzero at every finite stage, hence has infinite order after localization. Here σ is *not* complex conjugation: it fixes i . The invariant-rank argument, and multiplication by $2\pi i$, work for both parities of n [12, 14, 40].

14.3. Eisenstein polynomials and arbitrary kernel rank. The quadratic construction is one instance of a root-lattice calculation. This section records the stronger statement and its geometric proof.

Theorem 14.4. *Let p be any prime, let $f \in \mathbf{Z}_{(p)}[z]$ be a monic Eisenstein polynomial of degree $N \geq 2$, and put*

$$A_f = \left(\mathbf{Z}_{(p)}[x, y, z]/(xy + f(z)) \right)_{(x,y,z)}, \quad W = \mathbf{Z}_{(p)}[\theta]/(f(\theta)), \quad K = \mathbf{Q}(\theta).$$

The ring A_f is regular local of dimension three and has fraction field $\mathbf{Q}(x, z)$. Let (r_1, r_2) be the signature of K . For $n \geq 3$ odd, consider

$$g_n : K_n(W)_{\mathbf{Q}} \longrightarrow K_n(A_f)_{\mathbf{Q}}, \quad \beta \longmapsto i_*(\beta|_{A_f/(x)}),$$

where i is the regular principal divisor $x = 0$. Its image is contained in the generic kernel. Moreover,

n	$\ker g_n$	$\dim_{\mathbf{Q}} \operatorname{im} g_n$
3 (mod 4)	0	r_2
1 (mod 4)	$\operatorname{res}_{K/\mathbf{Q}} K_n(\mathbf{Q})_{\mathbf{Q}}$	$r_1 + r_2 - 1$.

In the second row $n \geq 5$, and $K_n(W)_{\mathbf{Q}}$ is identified with $K_n(K)_{\mathbf{Q}}$ by localization.

14.3.1. *The divisor lattice after splitting.* Let O be a DVR with uniformizer π , residue field k , and fraction field L . Suppose that $a_1, \dots, a_N$ are distinct elements of πO . Define

$$B = \left(O[x, y, z] / \left(xy + \prod_{i=1}^N (z - a_i) \right) \right)_{(\pi, x, y, z)}, \quad P_i = (x, z - a_i).$$

The same normality argument as in Lemma 12.7 shows that B is normal and regular off its closed point. Since $B[1/x]$ is a localization of $O[x, x^{-1}, z]$, the primes P_i generate $\operatorname{Cl}(B)$, with the relation

$$\operatorname{div}(x) = P_1 + \dots + P_N.$$

To prove that there are no further relations, successively blow up the strict transforms of $P_1, \dots, P_{N-1}$. The resulting regular scheme Y has N charts, each a localization of $O[u_i, v_i]$, on which

$$\begin{aligned} z &= a_i + u_i v_i, \\ x &= u_i \prod_{k < i} (z - a_k), & y &= -v_i \prod_{k > i} (z - a_k). \end{aligned}$$

For completeness, before the i th blowup the unresolved chart is

$$O[s_{i-1}, y, z] / \left(s_{i-1}y + \prod_{k \geq i} (z - a_k) \right), \quad s_0 = x.$$

Blowing up $(s_{i-1}, z - a_i)$ gives the displayed i th polynomial chart and the next unresolved chart, where $s_i = s_{i-1}/(z - a_i)$. These are the actual Rees charts inside the common function field. A future strict transform P_j , $j > i$, is absent from the resolved chart, since its generic point has $z = a_j$ and cannot satisfy $u_i = 0$, $z = a_i$. Thus the successive centers glue globally and do not modify previous resolved charts.

The adjacent transition is

$$u_{i+1} = v_i^{-1}, \quad v_{i+1} = v_i(a_i - a_{i+1} + u_i v_i).$$

The exceptional fiber is a reduced chain $C_1, \dots, C_{N-1}$ of projective lines. Its ideal is (π, u_1) on the first chart, (π, v_N) on the last, and $(\pi, u_i v_i)$ on an interior chart. In particular it has dimension one, and the resolution has no exceptional divisors. The two charts on C_j have coordinates v_j, u_{j+1} with $u_{j+1} = v_j^{-1}$.

Write $\tilde{P}_\ell$ for the strict transform of P_ℓ . On the i th chart its defining equation is

$$d_{\ell i} = \begin{cases} z - a_\ell, & \ell < i, \\ u_i, & \ell = i, \\ 1, & \ell > i. \end{cases}$$

These equations also show that no vertical divisor has been added: when $\ell < i$, the quotient by $u_i v_i + a_i - a_\ell$ is a domain. Taking transition functions of $\mathcal{O}_Y(\tilde{P}_\ell)$ gives

$$\deg(\mathcal{O}_Y(\tilde{P}_\ell)|_{C_j}) = \delta_{\ell, j+1} - \delta_{\ell, j}.$$

Therefore a principal relation $\sum n_i P_i$ forces $n_{j+1} = n_j$ for every j . Conversely equal coefficients give a power of x . We have proved the integral presentation

$$(14.1) \quad \text{Cl}(B) = \mathbf{Z}^N / \mathbf{Z}(1, \dots, 1).$$

Since $B/\pi B$ is the domain $k[x, y, z]_{(x, y, z)} / (xy + z^N)$, inverting π removes just the principal prime (π) . Thus (14.1) is also $\text{Pic}(B[1/\pi])$. On the full generic surface

$$Q_L = \text{Spec } L[x, y, z] / \left(xy + \prod_i (z - a_i) \right),$$

divisor localization at x gives the same presentation: the units of $L[x, x^{-1}, z]$ are $L^\times x^{\mathbf{Z}}$. Consequently

$$(14.2) \quad \text{Pic}(Q_L) \xrightarrow{\sim} \text{Pic}(B[1/\pi]).$$

14.3.2. Betti classes and finite localizations. Assume that L embeds in $\mathbf{C}$. The complex surface $Q_L(\mathbf{C})$ is the blowup of $\mathbf{A}_{\mathbf{C}}^2$ at the N points $(0, a_i)$ with the strict transform of the line $x = 0$ removed. Indeed, blow up the ideal $(x, \prod(z - a_i))$; its x -chart is precisely Q_L , and the complement is that strict transform.

The blowup has $H^2 = \mathbf{Z}^N$ on the exceptional classes E_i , and no other positive-degree cohomology. The removed line is isomorphic to $\mathbf{A}^1$, with class $-\sum E_i$, since its sum with the exceptional divisors is the principal divisor of x . The Gysin sequence therefore gives

$$H^2(Q_L(\mathbf{C}), \mathbf{Z}) = \mathbf{Z}^N / \mathbf{Z}(1, \dots, 1),$$

and first Chern class identifies this with $\text{Pic}(Q_L)$. All the root divisors are defined over L , so the absolute Galois action on the geometric Picard lattice is trivial.

For any denominator allowed in the localization $B[1/\pi]$, every removed L -prime divisor has zero Picard class by (14.2). Its geometric components are a single Galois orbit; they have equal classes in a torsion-free lattice, and their sum is zero. Each component therefore has zero class. The degree-two support argument of Section 12.4 proves

$$H^2(Q_L(\mathbf{C}), \mathbf{R}) \hookrightarrow H^2((Q_L)_s(\mathbf{C}), \mathbf{R})$$

for every such finite denominator s . The analytic Deligne injection in every twist greater than two follows exactly as in Section 12.5.

14.3.3. The arithmetic rank calculation. We now prove Theorem 14.4. Eisenstein's condition makes W a DVR with uniformizer θ and residue field $\mathbf{F}_p$. It also makes the defining equation of A_f have nonzero linear term p in its regular four-dimensional ambient local ring. Thus A_f is regular local of dimension three, and

$$A_f/(x) = W[y]_{(\theta, y)}.$$

Solving for y after inverting x gives $\text{Frac } A_f = \mathbf{Q}(x, z)$. Generic vanishing of g_n follows from localization.

Choose a splitting field L of f and a prime above p , and let O be the corresponding coefficient DVR. Every root a_i has positive valuation. The map $A_f \rightarrow B$ used above is flat: O is torsion-free over $\mathbf{Z}_{(p)}$, followed by localization. There is no need for O itself to be a finite $\mathbf{Z}_{(p)}$ -module. After inverting π , flat base change expresses the Gysin image as

$$(14.3) \quad g_n(\beta)|_{B[1/\pi]} = \sum_{i=1}^N \sigma_i(\beta) \eta_i, \quad \eta_i = [\mathcal{O}_{(x, z-a_i)}],$$

where $\sigma_i : K \hookrightarrow L$ are the embeddings indexed by the roots.

Write $n = 2m - 1$. At a fixed complex embedding of L , the regulator in weight $m + 1$ sends (14.3) to the vector of Borel coefficients

$$(r_m(\sigma_1 \beta), \dots, r_m(\sigma_N \beta)) \quad \text{modulo the diagonal line,}$$

with the common nonzero Tate normalization coming from first Chern class. Its target is

$$H_{\mathcal{D}, \text{an}}^3((Q_L)_s, \mathbf{R}(m+1)) = H^2((Q_L)_s, \mathbf{C}) / H^2((Q_L)_s, \mathbf{R}(m+1)),$$

because $m + 1 \geq 3$. The preceding finite-localization injection ensures that a nonzero vector modulo the diagonal remains nonzero at every finite stage, hence in $K_n(B[1/\pi])_{\mathbf{Q}}$.

If m is even, the two coefficients at conjugate embeddings have opposite signs, and a real embedding contributes zero. Borel's theorem [12, Proposition 12.2] identifies the real scalar extension of $K_n(K)_{\mathbf{Q}}$ with this r_2 -dimensional coefficient space. A vector in it is diagonal only when it is zero. Thus g_n is injective for $n \equiv 3 \pmod{4}$.

If m is odd, conjugate coefficients agree, and the coefficient space has dimension $r_1 + r_2$. Its diagonal line is exactly the real span of restriction from $K_n(\mathbf{Q})_{\mathbf{Q}}$, which is one-dimensional. Since this line is generated by a rational class and Borel's real regulator is injective [14, Section 6], its intersection with $K_n(K)_{\mathbf{Q}}$ is exactly that rational restriction line. Finally, that line really is killed by g_n : its classes lift from $K_n(\mathbf{Z}_{(p)})_{\mathbf{Q}}$, and the projection formula multiplies them by

$$[\mathcal{O}_{V(x)}] = [\mathcal{O}] - [\mathcal{O}] = 0 \quad \text{in } K_0(A_f).$$

This proves both the exact kernel and the stated image dimensions.

Corollary 14.5. *For every prime p and integer $r \geq 1$, the three-dimensional regular local ring*

$$\left(\mathbf{Z}_{(p)}[x, y, z]/(xy + z^{2r} + p) \right)_{(x, y, z)}$$

has a rational degree-three generic kernel of dimension at least r . Thus these kernel dimensions are unbounded in fixed local dimension three. Moreover the single ring with $xy + z^3 + p$ has a nonzero rational generic kernel in every odd degree $n \geq 3$: its number field has signature $(1, 1)$.

15. EXPLICIT TRIPLES OF UNITS AT EVERY RESIDUE PRIME

The preceding geometry also detects explicit symbols. The following formulas use the Dennis–Stein convention of Section 13; unlike (12.1), they require no prefactor.

Theorem 15.1. *For an odd prime p , put*

$$A_p = \left(\mathbf{Z}_{(p)}[x, y, z]/(xy + \Phi_{2p}(z - 1)) \right)_{(x, y, z)}.$$

Then

$$(15.1) \quad c_p = \{2 - z, 1 - x, 1 + (1 - x)yz((z - 1)^p - 1)\} \in K_3(A_p)$$

has infinite order and maps to zero in $K_3(A_p[1/x])$ integrally. For $p = 2$, put

$$A_2 = \left(\mathbf{Z}_{(2)}[x, y, t]/(xy + t^2 + 1) \right)_{(2, x, y, t-1)}, \quad H(t) = t^{10} - t^8 + t^6 - t^4 + t^2 - 1.$$

The same assertions hold for

$$(15.2) \quad c_2 = \{1 + t + t^2, 1 - x, 1 + (1 - x)yH(t)\} \in K_3(A_2).$$

All entries in these formulas are units. Each A_p is regular local of dimension three, with residue field $\mathbf{F}_p$, so the displayed classes give nonzero rational generic kernels.

15.1. The integral identities. We first record the common calculation. Suppose $1 - xb = u^n$, and put

$$\delta = \langle x, b \rangle, \quad \lambda = 1 - x, \quad \mu = 1 + \lambda b.$$

Assume u, g, λ, μ are units and $\{g, u^n\} = 0$ in K_2 . Relation (13.2), together with $\langle x, 1 \rangle = 0$, gives

$$\delta = \langle x, b + 1 - xb \rangle = \langle x, \mu \rangle = \{\lambda u^n, \mu\}, \quad 1 - x\mu = \lambda u^n.$$

Consequently

$$(15.3) \quad [g]\delta = \{g, \lambda, \mu\}, \quad [g]\delta|_{D(x)} = \{g, x, u^n\} = -[x]\{g, u^n\} = 0.$$

These are integral identities in Quillen K -theory.

For odd p , write $a = z - 1$ and set

$$H(a) = (a + 1)(a^p - 1), \quad b = yH(a), \quad g = 1 - a, \quad n = 2p.$$

The polynomial identity

$$a^{2p} - 1 = H(a)\Phi_{2p}(a) = -xyH(a)$$

gives $1 - xb = a^{2p}$. At the closed point, $a = -1$, $g = 2$, and $\lambda = \mu = 1$ in $\mathbf{F}_p$; these elements are units. Steinberg gives $\{1 - a, a^{2p}\} = 2p\{1 - a, a\} = 0$. Thus (15.3) proves (15.1) and its integral vanishing after inverting x . Moreover,

$$\Phi_{2p}(z - 1) = \sum_{k=0}^{p-1} (1 - z)^k = \sum_{j=1}^p (-1)^{j+1} \binom{p}{j} z^{j-1}$$

is monic Eisenstein of degree $p - 1$, with constant term p . The regularity assertion follows as in Theorem 14.4.

At two, take $u = t$, $n = 12$, $b = yH(t)$ and $g = 1 + t + t^2$. Here

$$(t^2 + 1)H(t) = t^{12} - 1, \quad 1 - xb = t^{12}.$$

All required units have nonzero residue at $t = 1$. To prove $\{g, t^{12}\} = 0$ integrally, use the finite étale local extension $A'_2 = A_2[\zeta]$, where $\zeta^2 + \zeta + 1 = 0$. The factor $1 - \zeta t$ is a unit, since

$$(1 - \zeta t)(1 - \zeta^2 t) = g, \quad (\zeta t)^{12} = t^{12}.$$

Steinberg and the transfer projection formula therefore give

$$0 = \text{Tr}_{A'_2/A_2}\{1 - \zeta t, t^{12}\} = \{\text{Norm}_{A'_2/A_2}(1 - \zeta t), t^{12}\} = \{g, t^{12}\}.$$

There is no factor of two in this formula. Indeed, multiplication by $1 - \zeta t$ in the basis $1, \zeta$ has matrix $\begin{pmatrix} 1 & t \\ -t & 1+t \end{pmatrix}$, elementary-equivalent to $\text{diag}(1, g)$; its K_1 transfer is exactly $[g]$. See [50, Chapter III, Definition 5.11 and relations (D1)–(D3)]. Equation (15.3) now proves (15.2). Finally, with $v = t - 1$, the equation becomes $xy + v^2 + 2v + 2 = 0$, which proves regularity at the specified maximal ideal.

15.2. A supported lift on a finite surface. For odd p let $E = \mathbf{Q}(\zeta_{2p})$, and at two let $E = \mathbf{Q}(i)$. Use the following smooth split surfaces and finite opens:

	Q	U
p odd	$\text{Spec } E[x, y, a]/(xy + \Phi_{2p}(a))$	$D(a(1 - a))$
$p = 2$	$\text{Spec } E[x, y, t]/(xy + t^2 + 1)$	$D(t(1 + t + t^2))$.

The Dennis–Stein product $\alpha_p = [g]\delta$ is defined on U and vanishes on $U[1/x]$ by the preceding calculation. At two the same norm argument uses $E(\zeta)/E$. The entire divisor $D_x = V(x) \subset Q$ lies in U : its coordinates are primitive $2p$ -th roots in the odd case, and $t = \pm i$, $g(\pm i) = \pm i$, at two. Localization and dévissage supply $\xi_p \in K_3(D_x)$ such that its pushforward to U is α_p . Pushing the same class into Q gives

$$(15.4) \quad \tilde{c}_p = i_* \xi_p \in K_3(Q), \quad \tilde{c}_p|_U = \alpha_p.$$

Only this supported lift will be needed; its coefficients need not be specified.

We calculate the regulator of α_p using its Dennis–Stein expression. The entries λ, μ need only be units in the arithmetic local ring; they are not needed on the analytic cycle below. On $D(x) \subset U$ one has $\delta = \{x, u^n\}$, and its holomorphic weight-two curvature extends to U as

$$\omega = n \frac{dx}{x} \wedge \frac{du}{u} = -\frac{dx \wedge db}{1 - xb}.$$

Whenever $g = e^L$ on an analytic neighborhood, the interval product calculation (13.7) gives $r_3(\alpha_p) = [-L\omega]$ there. This uses the multiplicativity and unit normalization of [14, Proposition 2.8, Theorem 2.31 and Lemma 4.5].

15.3. Two nonzero matching-sphere periods. We use a single endpoint formula. Let $\gamma : [-1, 1] \rightarrow \mathbf{C}$ be a smooth embedded arc joining two distinct simple roots of the relevant polynomial $P(u)$, avoiding the other roots. On $xy + P(u) = 0$, take

$$u = \gamma(\tau), \quad x = \sqrt{|P(\gamma(\tau))|} e^{i\vartheta}, \quad y = -P(\gamma(\tau))/x.$$

Adding $x = y = 0$ at both ends gives a smooth sphere Σ_γ , oriented by $d\tau \wedge d\vartheta$. Smoothness at an end follows by using $|x|^2$ as a radial coordinate and writing y as a smooth nonzero phase times $\bar{x}$.

Suppose L and F are holomorphic near the arc, with $e^L = g$ and $dF = -L du/u$. On the sphere away from its ends,

$$L\omega = n d \left(F \frac{dx}{x} \right).$$

Stokes' theorem on a truncated cylinder, followed by shrinking its two missing caps, gives

$$(15.5) \quad \int_{\Sigma_\gamma} r_3(\alpha_p) = -2\pi i n (F(\gamma(1)) - F(\gamma(-1))) \pmod{\mathbf{R}(3)}.$$

The caps contribute zero because $L\omega$ extends smoothly. This calculation uses dx/x on the cylinder and never requires a logarithm of x . Since $\mathbf{R}(3) = i\mathbf{R}$, a nonzero real period detects the regulator.

For odd p , put $\theta = \pi/p$, $r = e^{i\theta}$, and choose

$$\gamma(\tau) = \cos \theta + i \sin \theta \tau.$$

This chord joins $\bar{r}$ to r , lies in $0 < \Re a < 1$, and its interior lies strictly inside the unit circle. It therefore avoids $0, 1$ and all other roots. On one neighborhood of the whole arc, take the principal branches

$$L = \text{Log}(1 - a), \quad F = \text{Li}_2(a).$$

Here Li_2 is the classical dilogarithm, initially given by $\sum_{r \geq 1} a^r/r^2$ on the unit disc. The chosen branches satisfy the required differential equation. Writing D for the Bloch–Wigner dilogarithm, $F(r) - F(\bar{r}) = 2iD(r)$, so (15.5) yields

$$(15.6) \quad \int_{\Sigma_\gamma} \Re r_3(\alpha_p) = 8\pi p D(e^{i\pi/p}) > 0.$$

For example, positivity follows from $D(e^{i\theta}) = -\int_0^\theta \log(2 \sin(v/2)) dv$ for $0 < \theta \leq \pi/3$.

At two, the straight segment from $-i$ to i would meet $t = 0$. Instead use

$$\gamma(\tau) = \frac{1}{2}(1 - \tau^2) + i\tau.$$

Its interior has positive real part and modulus less than one. Thus the arc avoids 0 , the roots of g , and the other twelfth roots of unity. With $\zeta = e^{2\pi i/3}$, both $1 - \zeta\gamma(\tau)$ and $1 - \zeta^2\gamma(\tau)$ have positive real part, including at the endpoints. Hence one neighborhood supports the principal branches

$$L = \text{Log}(1 - \zeta t) + \text{Log}(1 - \zeta^2 t), \quad F = \text{Li}_2(\zeta t) + \text{Li}_2(\zeta^2 t).$$

Here $e^L = g$, $dF = -L dt/t$, and $F(-i) = \overline{F(i)}$. The absolutely convergent dilogarithm series on the unit circle gives

$$\text{Li}_2(w) + \text{Li}_2(\zeta w) + \text{Li}_2(\zeta^2 w) = \frac{1}{3} \text{Li}_2(w^3).$$

Taking imaginary parts at $w = i$ therefore gives $D(\zeta i) + D(\zeta^2 i) = -\frac{4}{3}D(i)$, and

$$(15.7) \quad \int_{\Sigma_\gamma} \Re r_3(\alpha_2) = -64\pi D(i) \neq 0.$$

The number $D(i) = \sum_{k \geq 0} (-1)^k / (2k + 1)^2$ is positive.

15.4. Persistence in the arithmetic local ring. We finish the proof of Theorem 15.1. For odd p , take the coefficient DVR $\mathcal{O} = \mathbf{Z}_{(p)}[\zeta_{2p}]$ and use the original coordinate $z = a + 1$. All roots $1 + r_i$ of $\Phi_{2p}(z - 1)$ lie in its maximal ideal. At two take $\mathcal{O} = \mathbf{Z}_{(2)}[i]$ and the coordinate $v = t - 1$; its two roots $\pm i - 1$ again lie in the maximal ideal. The split local rings therefore satisfy (14.1) and (14.2). The Betti and analytic Deligne persistence argument of Section 14.3 applies to every permitted finite localization of Q .

The open U is one of those permitted localizations: $a(1 - a)$ has nonzero residue -2 for odd p , and tg has nonzero residue 1 at two. On the generic fiber T of the split arithmetic local ring, the unit triple c_p , the Dennis–Stein product α_p , and the lift $\tilde{c}_p$ in (15.4) have the same image. Equations (15.6) and (15.7) show that the regulator of $\tilde{c}_p$ is nonzero already on Q . The cited persistence makes it nonzero on every further permitted finite open. A torsion relation on c_p would, after passage to T , descend to one such finite open by (12.2), contradicting that real regulator. Thus c_p has infinite order, as asserted.

16. COMPLETION AND ARITHMETIC p -ADIC DETECTION

There is a separate route to completed local rings. It uses the arithmetic comparison theorem for dagger varieties of Colmez–Nizioł, rather than an interchange of a localization limit with a p -adic inverse limit. We spell out the map and the order of limits, and distinguish actual K_3 -classes from classes in a completed spectrum.

16.1. A ruling-cup injection on the completed node.

Lemma 16.1. *Let $E/\mathbf{Q}_p$ be finite, let $W = \mathcal{O}_E$, and let $0 \neq d \in \mathfrak{m}_W$. Put*

$$\hat{B}_d = W[[x, y, w]]/(xy + w(w + d)), \quad T_d = \text{Spec } \hat{B}_d[1/d].$$

The ideal $I = (x, w)$ is invertible on T_d , and cup product induces an injection

$$H^1(E, \mathbf{Q}_p(2)) \longrightarrow H_{\text{cont}, \text{ét}}^3(T_d, \mathbf{Q}_p(3)), \quad b \longmapsto b \smile c_1(I).$$

Here continuous cohomology is formed by taking the derived inverse limit of finite-coefficient cohomology and then inverting p .

Proof. At a prime containing x, w , the element $w + d$ is a unit, and the equation gives $I = (x)$. Outside $V(I)$ the ideal is the unit ideal. It is therefore an invertible ideal, generated globally by two elements, and defines a morphism

$$f_T : T_d \longrightarrow \mathbf{P}_E^1, \quad f_T^* \mathcal{O}(1) = I.$$

Consider the smooth dagger affinoid

$$U^\dagger = \text{Sp}^\dagger E\langle X, Y, Z \rangle^\dagger / (XY + Z(Z + 1)).$$

The two roots $0, -1$ are distinct in every residue characteristic. Fix

$$\rho = |d|^{-1/2} \in \sqrt{|E^\times|}, \quad 1 < \rho < |d|^{-1},$$

and let U_ρ be the closed radius- ρ domain of the same algebraic quadric. The substitutions

$$(16.1) \quad x = dX, \quad y = dY, \quad w = dZ$$

define a homomorphism $\hat{B}_d[1/d] \rightarrow \mathcal{O}(U_\rho)$. Indeed every formal series with coefficients in W converges on this *same fixed domain*: terms of total degree n have weighted norm at most $(|d|\rho)^n$, which tends to zero. A fixed power of d^{-1} does not change the convergence radius.

For completeness, a noetherian integral model of this fixed affinoid is

$$A_\rho = W\langle a, b, c, r, s, t \rangle / (a^2 - dr, b^2 - ds, c^2 - dt, ab + c(c + d)).$$

Its generic fiber is U_ρ under $X = a/d, Y = b/d, Z = c/d$: the last three auxiliary variables impose $|dX^2|, |dY^2|, |dZ^2| \leq 1$. The elements a, b, c are topologically nilpotent in A_ρ , since their squares lie in dA_ρ . Substitution therefore defines $\hat{B}_d \rightarrow A_\rho$ before inverting d . The pair $(A_\rho, (p))$ is a noetherian henselian pair; removing its W -torsion, if desired, preserves the generic fiber.

Pull scheme cohomology to $\mathrm{Spec} \mathcal{O}(U_\rho)$, then to the affinoid étale topos. The latter map exists and is compatible with all finite coefficients, including p -power coefficients; see the Gabber–Fujiwara comparison in [1, Theorem 6.2.1], applied to a complete noetherian integral model of this one affinoid. At this fixed ρ , take the derived inverse limit over p^ν and invert p . Continuous étale and pro-étale cohomology agree for the affinoid, as recalled in [18, Section 3.2.3]. Finally map into the homotopy colimit over shrinking strict neighborhoods, which is the dagger pro-étale complex in that reference. This constructs

$$(16.2) \quad H_{\mathrm{cont}, \mathrm{ét}}^3(T_d, \mathbf{Q}_p(3)) \longrightarrow H_{\mathrm{proét}}^3(U^\dagger, \mathbf{Q}_p(3)).$$

It respects pullback from $\mathbf{P}_E^1$ and first Chern classes. We have not exchanged the inverse limit over coefficients with a direct limit over opens. The reverse comparison from dagger étale to dagger pro-étale cohomology is not assumed.

The ruling morphism $f : U^\dagger \rightarrow \mathbf{P}_E^1$ obtained from (X, Z) has an explicit two-chart description. Above the two standard closed discs of $\mathbf{P}^1$ its charts are dagger bidiscs

$$\begin{aligned} U_0 : (X, t), & \quad Z = Xt, & Y = -t(1 + Xt), \\ U_\infty : (s, Y), & \quad Z = -1 - sY, & X = -s(1 + sY). \end{aligned}$$

Their intersection is a dagger unit annulus times a disc, with

$$s = t^{-1}, \quad Y = -t - Xt^2.$$

The bidiscs have no positive-degree de Rham cohomology, and the intersection has $H^1 = E d \log t$. These facts follow by coefficientwise integration of overconvergent power and Laurent series: division by integers costs only a slight reduction of the overconvergence radius, and the $t^{-1} dt$ term remains. Coherent acyclicity for dagger affinoids [27, Proposition 3.1] and Mayer–Vietoris give

$$H_{\mathrm{dR}}^2(U^\dagger) = E[d \log t], \quad H_{\mathrm{dR}}^3(U^\dagger) = 0, \quad f^* : H_{\mathrm{dR}}^2(\mathbf{P}_E^1) \xrightarrow{\sim} H_{\mathrm{dR}}^2(U^\dagger).$$

The arithmetic Theorem 1.3(2), or Theorem 7.9(2) together with Proposition 5.9(2), of [18] gives a natural boundary

$$\partial_3 : \tilde{H}^2(R\Gamma_{\mathrm{dR}}(U^\dagger)/F^3) \longrightarrow \tilde{H}_{\mathrm{proét}}^3(U^\dagger, \mathbf{Q}_p(3)).$$

We verify that it is an isomorphism, so no passage from a monomorphism in a topological heart to ordinary cohomology is needed. The space $U^\dagger$ has the smooth weak formal model $\mathcal{O}_E\langle X, Y, Z \rangle^\dagger / (XY + Z(Z + 1))$; the two factors $Z, Z + 1$ generate the unit ideal. The ideal (X, Z) is invertible on this model and extends the ruling morphism to the weak formal projective line over $\mathcal{O}_E$. Let $F_0 = \mathrm{Frac} W(k_E)$. The functorial Hyodo–Kato to de Rham comparison of [18, equation (5.4) and Propositions 5.5–5.6] becomes a strict quasi-isomorphism after tensoring from F_0 to E . Our de Rham calculation, and its vanishing in degree three, therefore give

$$f^* : H_{\mathrm{HK}}^2(\mathbf{P}_E^1) \xrightarrow{\sim} H_{\mathrm{HK}}^2(U^\dagger), \quad H_{\mathrm{HK}}^3(U^\dagger) = 0.$$

Indeed scalar extension to the field E is faithful. Both degree-two groups are the rank-one slope-one isocrystal $F_0 h$ with $\varphi = p\sigma$ and $N = 0$. Its $\varphi = p^2$ eigenspace is zero by valuation. The second exact sequence of Theorem 1.3(2) now makes the term after $\tilde{H}_{\mathrm{proét}}^3$ vanish, on both $U^\dagger$ and $\mathbf{P}_E^1$. Since $F^3 = 0$ on this surface, the first exact sequence gives isomorphisms

$$\begin{aligned} H_{\mathrm{dR}}^2(U^\dagger) &\xrightarrow{\sim} \tilde{H}_{\mathrm{proét}}^3(U^\dagger, \mathbf{Q}_p(3)), \\ H_{\mathrm{dR}}^2(\mathbf{P}_E^1) &\xrightarrow{\sim} \tilde{H}_{\mathrm{proét}}^3(\mathbf{P}_E^1, \mathbf{Q}_p(3)). \end{aligned}$$

Here the sources are classical. Applying the classical-part functor of [18, Section 3.1.1] to these isomorphisms gives the corresponding isomorphisms on ordinary cohomology. Naturality and the de Rham ruling isomorphism show that

$$f^* : H_{\mathrm{ét}}^3(\mathbf{P}_E^1, \mathbf{Q}_p(3)) \longrightarrow H_{\mathrm{proét}}^3(U^\dagger, \mathbf{Q}_p(3))$$

is an isomorphism. Proper pro-étale and analytic étale cohomology are identified by [18, Section 7.2.4]. For the scheme $\mathbf{P}_E^1$, the torsion algebraic–analytic comparison of [3, Theorem 6.1.1]

gives the same identification, first for every p^ν and then by derived inverse limit. These comparisons preserve pullbacks and Kummer Chern classes. For finite coefficients, put $\Lambda = \mathbf{Z}/m$, where $m > 0$ is invertible in E , and write $\text{pr} : \mathbf{P}_E^1 \rightarrow \text{Spec } E$. The Kummer boundary defines $\xi_m = c_1(\mathcal{O}(1)) \in H_{\text{ét}}^2(\mathbf{P}_E^1, \Lambda(1))$ [36, Example 7.9(a)]. For every integer r , pullback and cup product give a morphism of étale complexes

$$(16.3) \quad \Lambda(r) \oplus \Lambda(r-1)[-2] \xrightarrow{(\text{pr}^*, \text{pr}^*(-) \smile \xi_m)} R\text{pr}_* \Lambda(r).$$

To check that it is a quasi-isomorphism, take a geometric stalk. Proper base change identifies the target with the cohomology complex of a geometric projective line, whose only nonzero cohomology is in degrees zero and two [36, Proposition 14.2 and Theorems 17.7 and 17.10]. The degree-zero map is the identity. In degree two, the Kummer calculation sends the generator $\mathcal{O}(1)$ of $\text{Pic}(\mathbf{P}^1) = \mathbf{Z}$ to a generator modulo m ; this is also the Chern normalization in [36, Theorems 23.2 and 23.3(b)]. Thus the map is a quasi-isomorphism on geometric stalks and hence on $\text{Spec } E$. This argument works over any field in which m is invertible.

The Kummer classes make these maps compatible for $m = p^\nu$. Assume $p \neq \text{char}(E)$. Applying $R\Gamma(E, -)$, then $R\varprojlim_\nu$, and then inverting p preserves this finite direct sum; all $\mathbf{Q}_p$ -cohomology groups here, including $H^1(E, \mathbf{Q}_p(2))$, are understood as continuous cohomology obtained by this construction. Taking $r = 3$ gives a split injection

$$H^1(E, \mathbf{Q}_p(2)) \xrightarrow{\smile c_1(\mathcal{O}(1))} H_{\text{cont,ét}}^3(\mathbf{P}_E^1, \mathbf{Q}_p(3)).$$

Composing the ruling-cup map on T_d with (16.2) is this split injection followed by the already proved isomorphism f^* . Hence the ruling-cup map on T_d is injective. $\square$

16.2. Simultaneous ruling classes. A map from a completed quadratic node isolates any chosen pair of roots. The resulting detector records their coefficients modulo the diagonal.

Lemma 16.2. *Let $E/\mathbf{Q}_p$ be finite and let $a_1, \dots, a_N$ be distinct elements of $\mathfrak{m}_{\mathcal{O}_E}$, with $N \geq 2$. Put*

$$\widehat{B} = \mathcal{O}_E[[x, y, z]] / \left(xy + \prod_{i=1}^N (z - a_i) \right), \quad T = \text{Spec } \widehat{B}[1/p], \quad J_i = (x, z - a_i).$$

The J_i are invertible ideals on T , their product is (x) , and the map

$$(16.4) \quad \frac{H^1(E, \mathbf{Q}_p(2))^N}{\text{diag } H^1(E, \mathbf{Q}_p(2))} \longrightarrow H_{\text{cont,ét}}^3(T, \mathbf{Q}_p(3)), \quad [(\alpha_i)] \longmapsto \sum_i \alpha_i \smile c_1(J_i)$$

is injective.

Proof. At a point with $x = z - a_i = 0$, every other root factor is a unit, since each nonzero $a_i - a_k$ is invertible after inverting p . The equation then gives $J_i = (x)$ locally. Away from its vanishing locus J_i is the unit ideal. This proves invertibility and, by the same local calculation, $\prod_i J_i = (x)$.

Fix distinct indices i, j , and set

$$d = a_j - a_i, \quad H_{ij}(z) = \prod_{k \neq i, j} (z - a_k),$$

where the empty product is 1. Consider the completed quadratic node

$$C_{ij} = \mathcal{O}_E[[u, v, w]] / (uv + w(w - d)), \quad T_{ij} = \text{Spec } C_{ij}[1/p].$$

There is a continuous homomorphism of complete local $\mathcal{O}_E$ -algebras

$$(16.5) \quad \widehat{B} \longrightarrow C_{ij}, \quad x \longmapsto u, \quad y \longmapsto H_{ij}(a_i + w)v, \quad z \longmapsto a_i + w.$$

All three images lie in the maximal ideal, so substitution defines every formal series. The defining relation maps to

$$H_{ij}(a_i + w)(uv + w(w - d)) = 0.$$

In particular no inverse of the cofactor H_{ij} is used. Inverting p gives a morphism $q_{ij} : T_{ij} \rightarrow T$.

The generated ideals obtained from J_i, J_j are

$$I_i = (u, w), \quad I_j = (u, w - d).$$

For $k \neq i, j$, put $b = a_k - a_i$, so $b \neq 0, d$. In $C_{ij}[1/p]$ the identity

$$b(b - d) = -uv - (w - b)(w + b - d)$$

shows that $(u, w - b)$ is the unit ideal, since $b(b - d) \in E^\times$.

These computations identify the pullbacks of the line bundles even though q_{ij} need not be flat. The pullback of each invertible J_k surjects onto its generated ideal. For $k = i, j$ that image is an invertible ideal by the quadratic-node calculation, and for every other k it is the unit ideal. A surjection between invertible modules is an isomorphism. Hence

$$q_{ij}^* J_i \cong I_i, \quad q_{ij}^* J_j \cong I_j, \quad q_{ij}^* J_k \cong \mathcal{O} \quad (k \neq i, j).$$

Also $I_i I_j = (u)$, so $c_1(I_j) = -c_1(I_i)$.

Pulling the cup sum to T_{ij} therefore gives

$$q_{ij}^* \left(\sum_k \alpha_k \smile c_1(J_k) \right) = (\alpha_i - \alpha_j) \smile c_1(I_i).$$

Lemma 16.1, with parameter $-d$, makes the right-hand cup map injective. Its prescribed inversion is the same as inversion of p , because d is a nonzero element of the maximal ideal of the DVR $\mathcal{O}_E$. Thus vanishing of the original sum forces $\alpha_i = \alpha_j$. Taking the pairs $(1, j)$ proves that its kernel is the diagonal. The diagonal already vanishes because $\prod_i J_i = (x)$, proving (16.4). $\square$

16.3. Actual local coefficients and the line-bundle formula.

Lemma 16.3. *Let $E/\mathbf{Q}_p$ be finite and let R be an E -algebra. The higher Chern maps used below have natural values in continuous étale cohomology, and their finite-coefficient line-bundle product identity holds in that continuous target. No vanishing of $\varprojlim^1$ for $\mathrm{Spec} R$ is required.*

Proof. Write $R = \mathrm{colim} R_\lambda$ over its finitely generated E -subalgebras. For $X_\lambda = \mathrm{Spec} R_\lambda$, geometric étale finiteness and finiteness of local Galois cohomology, through Hochschild–Serre, show that every $H_{\mathrm{ét}}^q(X_\lambda, \mu_{p^\nu}^{\otimes i})$ is finite. See [36, Theorem 19.1(b)] and [35, Chapter I, Corollary 2.3]; the relevant field has characteristic zero, so this includes p -power coefficients and $p = 2$. Jannsen’s exact sequence [30, equation (0.2)] consequently identifies continuous cohomology with the ordinary inverse limit at each such finite-type stage. The compatible finite-coefficient Chern classes therefore have unique continuous lifts there, and their product identities hold there as well.

Now use $K_3(R) = \mathrm{colim} K_3(R_\lambda)$ and pull those continuous classes to $\mathrm{Spec} R$. An element, an equality of elements, and any finite collection of projective modules descend to one common stage. This proves independence of the stage, naturality, and the product identity. It is the finite-type descent construction of [33, Section 5.1], applied here to affine E -algebras. In particular it applies to the completed generic rings and the affinoid algebras in this section. We do not identify $H_{\mathrm{cont}}^3(\mathrm{Spec} R, \mathbf{Z}_p(3))$ with $\varprojlim H_{\mathrm{ét}}^3(\mathrm{Spec} R, \mu_{p^\nu}^{\otimes 3})$, and do not interchange these cohomological limits with the filtered limit of rings. $\square$

Lemma 16.4. *For a finite extension $E/\mathbf{Q}_p$, the images of actual elements of $K_3(\mathcal{O}_E)$ under the rational étale Chern map*

$$c_{2,1} : K_3(\mathcal{O}_E) \longrightarrow H^1(E, \mathbf{Q}_p(2))$$

span the target over $\mathbf{Q}_p$. If E/F is quadratic, actual integral anti-invariant elements span its anti-invariant subspace, whose $\mathbf{Q}_p$ -dimension is $[F : \mathbf{Q}_p]$.

Proof. Levine’s finite-coefficient theorem [33, Theorem 4.12], also stated as the exact sequence in [49, Section 5], gives

$$K_3^M(E) \longrightarrow K_3(E; \mathbf{Z}/p^\nu) \xrightarrow{c_{2,1}} H^1(E, \mu_{p^\nu}^{\otimes 2}) \longrightarrow 0,$$

with the induced map on the indecomposable quotient an isomorphism, including $p = 2$. Milnor K_3 of a local field is divisible [50, Chapter III, Exercise 7.4]; hence $K_3^M(E)/p^\nu = 0$. The first image is therefore zero, and the displayed Chern map is a natural isomorphism.

The coefficient sequence for $G = K_3(E)$ is [50, IV, Section 2]

$$0 \longrightarrow G/p^\nu G \longrightarrow K_3(E; \mathbf{Z}/p^\nu) \longrightarrow K_2(E)[p^\nu] \longrightarrow 0.$$

The transition on the right is multiplication by p . By [50, III, Theorem 6.2.4], with the characteristic-zero torsion assertion attributed there to [34], $K_2(E)$ is the direct sum of a uniquely divisible group and a finite cyclic group. Hence that right-hand inverse limit is zero. Each $G/p^\nu G$ is finite, since it injects into the finite middle group, and this inverse system has vanishing $\varprojlim^1$. The finite groups $H^0(E, \mu_{p^\nu}^{\otimes 2})$ also have vanishing $\varprojlim^1$, so the inverse limit of the H^1 groups is the continuous group $H^1(E, \mathbf{Z}_p(2))$. Taking inverse limits therefore gives the natural identification

$$\varprojlim_\nu G/p^\nu G \cong H^1(E, \mathbf{Z}_p(2)).$$

The actual group G is dense in its completion, so its rational Chern image spans $H^1(E, \mathbf{Q}_p(2))$. There is no Tate-module contribution from K_2 hidden in this argument.

Localization lifts every element of $K_3(E)$ to $K_3(\mathcal{O}_E)$, since $K_2(k_E) = 0$. The natural Bloch–Kato isomorphism

$$\exp_E : E \xrightarrow{\sim} H^1(E, \mathbf{Q}_p(2))$$

identifies the anti-invariant subspace for a quadratic E/F with the trace-zero subspace of E , of dimension $[F : \mathbf{Q}_p]$. Apply $1 - \sigma$ to actual lifts whose images span that space. This gives integral classes β with $\sigma\beta = -\beta$, without dividing by two. Naturality of the Chern maps and the exponential ensures all these assertions are equivariant. For the exponential and its normalization, see [29, Section 1.3]. $\square$

We also need the precise line-bundle product. If L is a line bundle and $\beta \in K_3$, then

$$(16.6) \quad c_{3,3}([L] - 1)\beta = -2c_1(L) \smile c_{2,1}(\beta).$$

This is [49, Theorem 3.11(i)]. The K -degree is three, so the exceptional degree-two correction in that paper does not apply, even at $p = 2$. Lemma 16.3 puts this formula in continuous cohomology before passing to $\mathbf{Q}_p$. The nonzero factor -2 is harmless for rational detection.

16.4. Complete quadratic examples over every p -adic base.

Theorem 16.5. *Let $V = \mathcal{O}_F$ for a finite extension $F/\mathbf{Q}_p$, and let $f(z) \in V[z]$ be an Eisenstein quadratic polynomial. Then*

$$\hat{A}_f = V[[x, y, z]]/(xy + f(z))$$

is a regular local ring of dimension three, and

$$\dim_{\mathbf{Q}} \ker(K_3(\hat{A}_f)_{\mathbf{Q}} \longrightarrow K_3(\text{Frac } \hat{A}_f)_{\mathbf{Q}}) \geq [F : \mathbf{Q}_p].$$

Taking $V = \mathbf{Z}_p$, this applies to $f = z^2 + p$ for every prime p , including $p = 2$.

Proof. Write $E = F(\theta)$ for a root of f , $W = V[\theta] = \mathcal{O}_E$, and σ for the quadratic involution. Eisenstein’s condition proves regularity exactly as before, and the divisor $x = 0$ has regular coordinate ring $W[[y]]$. Choose actual anti-invariant classes $\beta_j \in K_3(W)$ whose étale Chern classes are $\mathbf{Q}_p$ -independent; Lemma 16.4 supplies $[F : \mathbf{Q}_p]$ such classes. Push them forward from $x = 0$ to obtain $c_j \in K_3(\hat{A}_f)$. Each generic image is zero by localization.

After the finite free base change from V to W , put $w = z - \theta$ and $d = \theta - \sigma\theta$. Since both roots are topologically nilpotent and distinct, this is a valid formal change of variables and $0 \neq d \in \mathfrak{m}_W$. The equation becomes

$$xy + w(w + d) = 0.$$

After inverting d , the two components of $x = 0$ are disjoint. With $I = (x, w)$ and $\eta_+ = [\mathcal{O}_{(x,w)}] = 1 - [I]$, flat base change gives

$$c_j|_{T_d} = \beta_j \eta_+ + \sigma(\beta_j) \eta_- = 2\beta_j \eta_+, \quad \eta_+ + \eta_- = 0.$$

Applying (16.6) gives

$$c_{3,3}(c_j|_{T_d}) = 4c_1(I) \smile c_{2,1}(\beta_j).$$

These classes are $\mathbf{Q}_p$ -independent by Lemma 16.1. Thus the integral classes c_j are $\mathbf{Q}$ -independent after rationalization, proving the assertion. $\square$

Remark 16.6. The simplest residue-two complete example in this theorem is $\mathbf{Z}_2[[x, y, z]]/(2 + xy + z^2)$. It is different from the noncomplete residue-two example of Section 14.1, which uses the imaginary quadratic number field $\mathbf{Q}(i)$ to obtain an archimedean source. The completed theorem uses local étale regulator classes and requires no imaginary number field.

16.5. The explicit class also survives completion at three.

Proposition 16.7. *For*

$$\hat{A} = \mathbf{Z}_3[[x, y, z]]/(3 + xy + z^2),$$

the explicit class c_{exp} of (12.1) has infinite order and zero generic image. Consequently the same explicit class has infinite order in the henselization of the original ring A as well.

Proof. Let $E_0 = \mathbf{Q}(u)$, $u^2 = -3$, and retain the supported lift $\tilde{c} \in K_3(Q)$ constructed in Section 13. Since D_+ and D_- are affine lines over E_0 , homotopy invariance and their pushforwards give

$$\tilde{c} = \gamma_+ \eta_+ + \gamma_- \eta_- = (\gamma_+ - \gamma_-) \eta_+ = \gamma \eta_+, \quad \gamma \in K_3(E_0).$$

The positive complex period in (13.8) implies $\gamma \neq 0$ in $K_3(E_0)_{\mathbf{Q}}$. Borel's theorem [12, Proposition 12.2] gives $\dim_{\mathbf{Q}} K_3(E_0)_{\mathbf{Q}} = 1$.

Put $\zeta = (1 + u)/2$, so $1 - \zeta = \zeta^{-1}$. The special-unit Bloch cycle $[\zeta]_2$ gives a rational class $\beta_{\text{cyc}} \in K_3(E_0)_{\mathbf{Q}}$, compatible with its integral model at three and with the embedding $E_0 \hookrightarrow E = \mathbf{Q}_3(u)$. Besser–de Jeu's Theorem 1.10(2), in its unconditional weight-two case, computes its local syntomic regulator as

$$\pm L_{\text{mod},2}(\zeta) = \pm \text{Li}_2(\zeta),$$

because $\log \zeta = \log(1 - \zeta) = 0$ [8]. We verify directly that this value is nonzero.

Normalize $v_3(3) = 1$ and put

$$\epsilon = 1 + \zeta = \frac{3 + u}{2}.$$

Then $v_3(\epsilon) = 1/2$ and $\epsilon^3 = 3u$. Coleman's reflection and inversion identities, and the duplication formula, give

$$\text{Li}_2(\epsilon) = -\text{Li}_2(-\zeta), \quad \text{Li}_2(\zeta) + \text{Li}_2(-\zeta) = \frac{1}{2} \text{Li}_2(\zeta^2) = -\frac{1}{2} \text{Li}_2(-\zeta).$$

Here $1 - \epsilon = -\zeta$, $(-\zeta)^{-1} = \zeta^2$, and all logarithms of roots of unity vanish. See [8, equation (1.2) and (2.4)] and [7, p. 377]. Consequently

$$\frac{\text{Li}_2(\zeta)}{u} = \frac{3}{2u} \text{Li}_2(\epsilon) = \frac{3}{2u} \sum_{n \geq 1} \frac{\epsilon^n}{n^2}.$$

The term $n = 3$ is exactly $1/2$. Every other term has valuation

$$\frac{n+1}{2} - 2v_3(n) \geq 1.$$

For $v_3(n) = 0$ this is immediate; for $v_3(n) = 1$ and $n \neq 3$ use $n \geq 6$; for $r = v_3(n) \geq 2$ use $(3^r + 1)/2 - 2r \geq 1$. The valuations tend to infinity. Galois equivariance and inversion show that the quotient $\text{Li}_2(\zeta)/u$ lies in $\mathbf{Q}_3$. Intersecting the resulting congruence modulo $3\mathcal{O}_E$ with $\mathbf{Q}_3$ therefore gives

$$(16.7) \quad \boxed{\frac{\text{Li}_2(\zeta)}{u} \in \frac{1}{2} + 3\mathbf{Z}_3, \quad v_3(\text{Li}_2(\zeta)) = \frac{1}{2}.}$$

The syntomic-to-étale map for $\text{Spec } \mathcal{O}_E$ is compatible with Chern classes, and under the identification

$$H_{\text{syn}}^1(\mathcal{O}_E, 2) \cong E$$

it is the Bloch–Kato exponential. Precise references are [29, Example 2.2.4 and Propositions 2.2.7, 2.2.9, 2.3.4]; see also the proof of [47, Corollary 5.19]. These statements allow ramified $E/\mathbf{Q}_3$. Since the exponential for $\mathbf{Q}_3(2)$ is an isomorphism, (16.7) gives

$$c_{2,1}(\beta_{\text{cyc}}|_E) \neq 0.$$

In particular β_{cyc} is nonzero globally. The one-dimensionality of $K_3(E_0)_{\mathbf{Q}}$ now gives $\gamma = q\beta_{\text{cyc}}$ in $K_3(E_0)_{\mathbf{Q}}$ for some $q \in \mathbf{Q}^\times$. No exact comparison scalar between a Dennis–Stein presentation and a Bloch lift has been assumed or is needed.

On the completed split generic ring T the image of c_{exp} equals the image of $\tilde{c}$: indeed $1 - z^2$ is a unit in $\hat{A}$, and the equality $\tilde{c}|_U = 2\alpha$ from Section 13 pulls back there. Thus

$$c_{\text{exp}}|_T = q\beta_{\text{cyc}}\eta_+ \quad \text{in } K_3(T)_{\mathbf{Q}}.$$

Its étale Chern class is nonzero by (16.6) and Lemma 16.1, with $w = z - u$, $d = 2u$. Therefore c_{exp} has infinite order in $K_3(\hat{A})$. Its generic image is zero by the integral identity (13.3), which holds after completion. The map from the original local ring to its completion factors through its henselization, so infinite order after completion also proves infinite order there. $\square$

16.6. The explicit class also survives completion at two.

Lemma 16.8. *For the Coleman dilogarithm on $E = \mathbf{Q}_2(i)$, $i^2 = -1$, normalized by its power series at zero and with $\log 2 = 0$, one has*

$$(16.8) \quad \frac{\text{Li}_2(i)}{i} \in \frac{1}{2} + \mathbf{Z}_2, \quad v_2(\text{Li}_2(i)) = -1, \quad v_2(2) = 1.$$

Proof. The modified dilogarithm $D(z) = \text{Li}_2(z) + \frac{1}{2} \log z \log(1 - z)$ satisfies $D(1 - z) = -D(z)$ and $D(z^{-1}) = -D(z)$; see [7, equation (1.8), p. 307, and p. 377]. Put $q = 1 - i$, so $v_2(q) = 1/2$. Since $\log i = 0$, reflection gives the convergent expansion

$$\text{Li}_2(i) = -\text{Li}_2(q) = -\sum_{n \geq 1} \frac{q^n}{n^2}.$$

Now $q^2 = -2i$, $q^4 = -4$, and $q^8 = 16$. Thus the fourth and eighth terms cancel exactly, while the second term is $-i/2$. For $n \notin \{2, 4, 8\}$,

$$v_2(q^n/n^2) = n/2 - 2v_2(n) \geq 0.$$

Indeed, writing $n = 2^r k$ with k odd proves this immediately for $r \leq 3$; for $r \geq 4$ use $2^{r-1}k - 2r \geq 2^{r-1} - 2r \geq 0$. These valuations tend to infinity. Hence $\text{Li}_2(i) - i/2 \in \mathcal{O}_E$, proving the asserted valuation. Galois equivariance and inversion give $\sigma(\text{Li}_2(i)) = -\text{Li}_2(i)$ for $\sigma(i) = -i$; see [8, Remark 2.3 and equation (1.2)]. Thus $\text{Li}_2(i)/i \in \mathbf{Q}_2$, and intersecting the integral congruence with $\mathbf{Q}_2$ proves (16.8). $\square$

Proposition 16.9. *The explicit triple c_2 of (15.2) has infinite order and zero generic image in*

$$\hat{A}_2 = \mathbf{Z}_2[[x, y, w]]/(xy + w^2 + 2w + 2), \quad t = 1 + w.$$

The same holds in the henselization of its original local ring.

Proof. Let $E_0 = \mathbf{Q}(i)$ and let $\tilde{c}_2$ be the supported lift of (15.4). Homotopy invariance on the two affine-line divisors and $\eta_- = -\eta_+$ give

$$\tilde{c}_2 = \gamma\eta_+, \quad \gamma \in K_3(E_0)_{\mathbf{Q}}, \quad \eta_+ = 1 - [(x, t - i)].$$

Its nonzero period (15.7) implies $\gamma \neq 0$.

The rational Bloch cycle $[i]_2$ defines $\beta_i \in K_3(E_0)_{\mathbf{Q}}$. Interpret it over $\mathbf{Z}[i]_{(1-i)}$ by the rational localization isomorphism. Theorem 1.12 of [8] applies to every root of unity, including p -power order, and gives its local syntomic regulator as $\pm \text{Li}_2(i)$ in $E = \mathbf{Q}_2(i)$. The theorem has no odd-prime or $p > \text{weight}$ restriction. The Chern-compatible syntomic-to-étale map at the point $\text{Spec } \mathbf{Z}_2[i]$ is the Bloch–Kato exponential, also at $p = 2$; see [29, Example 2.2.4 and Propositions 2.2.7, 2.3.4] and [47, Corollary 5.19 and its proof]. This exponential is an isomorphism

in twist two, so Lemma 16.8 proves $c_{2,1}(\beta_i|_E) \neq 0$. Borel's equality $\dim_{\mathbf{Q}} K_3(\mathbf{Q}(i))_{\mathbf{Q}} = 1$ [12, Proposition 12.2] now gives $\gamma = a\beta_i$ for some $a \in \mathbf{Q}^\times$.

After finite free coefficient extension to $W = \mathbf{Z}_2[i]$, put

$$s = t - i = w - (i - 1), \quad d = 2i.$$

Because $i - 1$ is topologically nilpotent, this is a valid formal coordinate change, and the split completed ring is

$$\widehat{B} = W[[x, y, s]]/(xy + s(s + d)).$$

Let $T = \text{Spec } \widehat{B}[1/2]$ and $I = (x, s)$ on T . The open used for the supported lift is $D(tg)$, where $g = 1 + t + t^2$. Both $t = 1 + w$ and $g = 3 + 3w + w^2$ are units after completion. Thus (15.4) pulls back to $c_2|_T = a\beta_i(1 - [I])$. Equation (16.6) gives

$$c_{3,3}(c_2|_T) = 2a c_1(I) \smile c_{2,1}(\beta_i|_E) \neq 0$$

by Lemma 16.1, with parameter $d = 2i$. Its scaled model is $XY + Z(Z + 1) = 0$, obtained from $x = dX$, $y = dY$, $s = dZ$; hence that lemma applies in residue characteristic two. This proves infinite order in $K_3(\widehat{A}_2)$. The integral generic-zero identity for c_2 remains valid after completion. Finally the completion map factors through the henselization, giving the same conclusion there. $\square$

16.7. Cyclotomic local values and a common normalization.

Lemma 16.10. *Let p be odd, $r \geq 1$, and let ρ be a primitive p^r -th root of unity in $E = \mathbf{Q}_p(\rho)$. For the Coleman dilogarithm normalized by its series at zero and $\log p = 0$, one has*

$$v_p(\text{Li}_2(\rho)) = \frac{p}{p-1} - 2r, \quad \text{Li}_2(-\rho) \neq 0.$$

Proof. Put $e = p^{r-1}(p-1)$ and $q = 1 - \rho$, so $v_p(q) = 1/e$. Coleman reflection for $D(z) = \text{Li}_2(z) + \frac{1}{2} \log z \log(1-z)$ and $\log \rho = 0$ give

$$\text{Li}_2(\rho) = - \sum_{n \geq 1} \frac{q^n}{n^2};$$

see [7, p. 377]. Writing $n = p^s k$ with $p \nmid k$, its term valuation is $p^s k/e - 2s$. For $k = 1$, the consecutive differences are $p^{s-r+1} - 2$, negative for $s < r$ and positive for $s \geq r$. Thus the unique least term is $n = p^r$, with the asserted valuation; every other term has valuation at least one more, and the valuations tend to infinity.

Let $\sigma_2(\rho) = \rho^2$. Galois equivariance and duplication [8, Remark 2.3 and equation (2.4)] give

$$\text{Li}_2(-\rho) = \frac{1}{2} \sigma_2(\text{Li}_2(\rho)) - \text{Li}_2(\rho).$$

If this vanished, iteration through the finite order d of σ_2 would give $(2^d - 1) \text{Li}_2(\rho) = 0$, a contradiction. $\square$

Lemma 16.11. *Fix the Besser-de Jeu weight-two symbol map with one natural sign convention. For a root of unity a of order N in a number field F , with $a, 1-a \in \mathcal{O}_F^\times$, let $\beta_{\text{BDJ}}(a)$ denote its rational symbol class and set*

$$\beta_{\text{BT}}(a) = \frac{1}{N} \text{bl}(N[a]) \in K_3(F)_{\mathbf{Q}}$$

using [14, Theorem 4.9]. *There is one $s \in \mathbf{Q}^\times$, independent of F and a , such that*

$$\beta_{\text{BDJ}}(a) = s \beta_{\text{BT}}(a).$$

Proof. The multiple $N[a]$ has zero integral wedge boundary because $a^N = 1$, including with the antisymmetric-tensor convention. The BT class is unique rationally by Borel injectivity [14, Section 6] and has regulator $(-iD_{\text{BW}}(\sigma a))_\sigma$.

The field map in [8, equation (1.1), Theorem 1.10(1)] is de Jeu's map. Its complex regulator is computed in [21, Proposition 4.1 and Remark 5.2], also restated in [20, Theorem 2.3],

simultaneously at all embeddings, by $P_2 = D_{\text{BW}}$. Transporting its degree-dependent target identifications to the BT target therefore gives one nonzero real constant C with

$$r_{\text{BT}}(\beta_{\text{BDJ}}(a))_\sigma = C i D_{\text{BW}}(\sigma a).$$

The sign choice is made once in the fixed weight and degree, as in [8, Remark 1.7].

Calibrate at $\zeta = (1 + \sqrt{-3})/2$ in $\mathbf{Q}(\sqrt{-3})$. Both classes have nonzero regulator and lie in its rank-one rational K_3 [12, Proposition 12.2], so $\beta_{\text{BDJ}}(\zeta) = s\beta_{\text{BT}}(\zeta)$ for some $s \in \mathbf{Q}^\times$. Their regulator values imply $C = -s$. For every F, a , the two classes in the statement now have the same regulator at all embeddings. Borel injectivity proves their equality. $\square$

16.8. The explicit odd-prime triples survive completion. We identify all supported coefficients before applying the local detector. This avoids using a single complex embedding to identify a class in a higher-rank number-field K -group.

Lemma 16.12. *Let p be odd, let $E_0 = \mathbf{Q}(\mu_{2p})$, and enumerate its primitive $2p$ -th roots as $a_1, \dots, a_{p-1}$. On*

$$Q = \text{Spec } E_0[x, y, a]/(xy + \Phi_{2p}(a))$$

put $D_i = (x, a - a_i)$, $J_i = \mathcal{O}_Q(-D_i)$ and $\eta_i = [\mathcal{O}_{D_i}] = 1 - [J_i]$. Normalize the rational Bloch classes $\beta_{\text{BT}}(a_i)$ by

$$r_2(\beta_{\text{BT}}(a_i))_\sigma = -i D_{\text{BW}}(\sigma(a_i))$$

in the Bunke–Tamme cone coordinate, at every complex embedding σ of E_0 . Then every supported lift $\tilde{c}_p$ in (15.4) satisfies

$$(16.9) \quad \tilde{c}_p = -2p \sum_{i=1}^{p-1} \beta_{\text{BT}}(a_i) \eta_i \quad \text{in } K_3(Q)_{\mathbf{Q}}.$$

Proof. The stated normalization defines actual rational classes: $a_i, 1 - a_i$ are global units, because $\Phi_{2p}(1) = 1$, and $2p[a_i]$ lies in the kernel of $[a] \mapsto a \wedge (1 - a)$. One may therefore take

$$\beta_{\text{BT}}(a_i) = \frac{1}{2p} \text{bl}(2p[a_i])|_{E_0}$$

using [14, Theorem 4.9 and identification (4.8)]. Borel injectivity [14, Section 6] makes this rational class independent of the integral choice and equivariant under field embeddings.

Homotopy invariance on the affine-line components of D_x and the projection formula give $\tilde{c}_p = \sum_i \gamma_i \eta_i$, with $\gamma_i \in K_3(E_0)$. Fix distinct i, j and an embedding $\sigma : E_0 \hookrightarrow \mathbf{C}$. The chord from $\sigma(a_i)$ to $\sigma(a_j)$ avoids zero: primitive $2p$ -th roots cannot be antipodal when p is odd. Its interior lies inside the unit circle and meets no other root, and the whole chord has $\Re a < 1$. Thus the principal $\text{Log}(1 - a)$ and $\text{Li}_2(a)$ are defined on one neighborhood of it.

Orient the associated matching sphere from the i th root to the j th root as in (15.5). That formula gives the real period

$$2\pi(2p)(D_{\text{BW}}(\sigma(a_j)) - D_{\text{BW}}(\sigma(a_i))).$$

The sphere meets D_i with oriented intersection $+1$, D_j with intersection -1 , and misses the other divisors. Indeed, near its starting pole $\tau + 1$ is a positive smooth multiple of $|x|^2$, whereas near its ending pole $1 - \tau$ is. Consequently the periods of $\text{ch}_1(\eta_i)$ and $\text{ch}_1(\eta_j)$ are $+2\pi i$ and $-2\pi i$.

Write $r_2(\gamma_k)_\sigma = iq_k(\sigma)$ with $q_k(\sigma)$ real. Multiplicativity therefore gives

$$q_i(\sigma) - q_j(\sigma) = 2p(D_{\text{BW}}(\sigma(a_i)) - D_{\text{BW}}(\sigma(a_j))).$$

Since this holds at every complex embedding, Borel injectivity implies

$$\gamma_i - \gamma_j = -2p(\beta_{\text{BT}}(a_i) - \beta_{\text{BT}}(a_j)) \quad \text{in } K_3(E_0)_{\mathbf{Q}}.$$

Thus $\gamma_i + 2p\beta_{\text{BT}}(a_i)$ is one common coefficient. Its contribution is zero because $\sum_i \eta_i = [\mathcal{O}_{V(x)}] = 0$, proving (16.9). $\square$

Proposition 16.13. *For every odd prime p , the explicit triple c_p in (15.1) has infinite order and zero generic image in*

$$\hat{A}_p = \mathbf{Z}_p[[x, y, z]]/(xy + \Phi_{2p}(z - 1)).$$

The same assertions hold in the henselization of its original arithmetic local ring.

Proof. Put $E = \mathbf{Q}_p(\mu_{2p})$ and $W = \mathcal{O}_E$. After the finite free coefficient extension to W , the completed ring has the split form

$$\hat{B} = W[[x, y, z]] / \left(xy + \prod_i (z - (1 + a_i)) \right).$$

All $1 + a_i$ lie in the maximal ideal of W . Let $T = \text{Spec } \hat{B}[1/p]$ and still write $J_i = (x, z - (1 + a_i))$ on T . The finite open $U = D(a(1 - a))$, with $a = z - 1$, contains this generic scheme, since a and $1 - a$ are units in $\hat{B}$. Thus (16.9) pulls back to $c_p|_T$.

By Lemma 16.11, the classes $\beta_{\text{BT}}(a_i)$ are one common nonzero rational multiple of the de Jeu classes represented by $[a_i]_2$. Theorem 1.12 of [8], followed by the point syntomic-to-étale comparison and the Bloch–Kato isomorphism, therefore gives

$$c_{2,1}(\beta_{\text{BT}}(a_i)|_E) = \kappa \exp_E(\text{Li}_2(a_i))$$

with one $\kappa \in \mathbf{Q}_p^\times$, independent of i . Lemma 16.10 proves $\text{Li}_2(a_i) \neq 0$, and inversion gives $\text{Li}_2(a_i^{-1}) = -\text{Li}_2(a_i)$. The displayed coefficient vector is consequently non-diagonal.

The line-bundle formula (16.6) gives

$$c_{3,3}(c_p|_T) = -4p \sum_i c_1(J_i) \smile c_{2,1}(\beta_{\text{BT}}(a_i)|_E) \neq 0$$

by Lemma 16.2. This proves infinite order after completion. The integral vanishing identity (15.3) remains valid after every base change, so its generic image is zero. Finally, the completion map factors through the henselization, which proves both assertions there as well. $\square$

17. COMPLETED EISENSTEIN AND WILD CYCLOTOMIC FAMILIES

Pairwise ruling maps detect all root coefficients modulo the diagonal. We apply this detector first to arbitrary Eisenstein polynomials and then to explicit cyclotomic coefficients with independent wild conjugates.

17.1. Eisenstein polynomials and unbounded rank. Lemma 16.2 applies to every Eisenstein polynomial, including wild degrees. We combine it with actual local coefficients to obtain the following rank bound.

Theorem 17.1. *Let $F/\mathbf{Q}_p$ be finite, let $V = \mathcal{O}_F$, and let $f(z) \in V[z]$ be an Eisenstein polynomial of degree $N \geq 2$. Then*

$$A_f = V[[x, y, z]]/(xy + f(z))$$

is a complete regular local ring of dimension three with residue field k_F , and

$$\dim_{\mathbf{Q}} \ker (K_3(A_f)_{\mathbf{Q}} \longrightarrow K_3(\text{Frac } A_f)_{\mathbf{Q}}) \geq [F : \mathbf{Q}_p](N - 1).$$

The lower bound is witnessed by actual integral classes whose generic images vanish integrally.

Proof. In the regular local ring $V[[x, y, z]]$, the equation $xy + f(z)$ has a nonzero linear term in a uniformizer of V . Indeed the constant coefficient of f has valuation one, while all other terms lie in the square of the maximal ideal. Its quotient is therefore regular of dimension three, with maximal ideal (x, y, z) and residue field k_F .

Choose a root θ of f and put

$$K = F(\theta), \quad W = V[\theta] = \mathcal{O}_K.$$

Eisenstein's criterion gives $[K : F] = N$, and the divisor $x = 0$ has regular coordinate ring

$$A_f/(x) \cong W[[y]].$$

For an actual $\beta \in K_3(W)$ define

$$c(\beta) = i_*(\beta|_{W[[y]]}) \in K_3(A_f).$$

Quillen localization [40] makes its image in $K_3(A_f[1/x])$, hence in $K_3(\text{Frac } A_f)$, zero.

Let E/F be a splitting field of f , let $\sigma_1, \dots, \sigma_N : K \hookrightarrow E$ be its F -embeddings, and write $a_i = \sigma_i(\theta)$. Characteristic zero makes the roots distinct; Eisenstein's condition puts every a_i in $\mathfrak{m}_{\mathcal{O}_E}$. Finite free coefficient extension gives

$$A_f \otimes_V \mathcal{O}_E \cong \widehat{B} = \mathcal{O}_E[[x, y, z]] / \left(xy + \prod_i (z - a_i) \right).$$

Set $T = \text{Spec } \widehat{B}[1/p]$. This scheme is regular: it is the finite étale base change of $\text{Spec } A_f[1/p]$ along the separable field extension E/F .

After this flat base change and inversion of p , the divisor $W[[y]]$ splits into the disjoint regular Cartier divisors

$$D_i = V(x, z - a_i) \subset T, \quad D_i \cong \text{Spec } \mathcal{O}_E[[y]][1/p].$$

Let $J_i = (x, z - a_i)$ and $\eta_i = [\mathcal{O}_{D_i}] = 1 - [J_i]$. Flat base change and the projection formula give

$$(17.1) \quad c(\beta)|_T = \sum_{i=1}^N \sigma_i(\beta) \eta_i.$$

Here $\sigma_i(\beta)$ denotes restriction to $K_3(K)$ followed by extension along σ_i , and then the coefficient pullback from $K_3(E)$ to $K_3(T)$. The displayed formal divisor rings are localized power-series rings; they are not being replaced by $E[[y]]$.

Put

$$\alpha = c_{2,1}(\beta|_K) \in H^1(K, \mathbf{Q}_p(2)), \quad a = \exp_K^{-1}(\alpha) \in K.$$

Naturality of the Bloch–Kato exponential gives

$$(17.2) \quad (c_{2,1}(\sigma_i(\beta)))_i = (\exp_E(\sigma_i(a)))_i.$$

The continuous Chern maps of Lemma 16.3 and the product formula (16.6) turn (17.1) into

$$c_{3,3}(c(\beta)|_T) = 2 \sum_i c_{2,1}(\sigma_i(\beta)) \smile c_1(J_i).$$

The factor 2 remains nonzero over $\mathbf{Q}_p$ for every p .

By Lemma 16.2, the cup detector kills exactly the diagonal coefficient vectors. By (17.2), this means that all the $\sigma_i(a)$ are equal. This happens exactly when $a \in F$: every element of $\text{Gal}(E/F)$ restricts to one of the embeddings σ_i , so equality of all conjugates makes a fixed by $\text{Gal}(E/F)$. Consequently the map

$$g : K \longrightarrow E^N / E(1, \dots, 1), \quad a \longmapsto [(\sigma_i(a))_i]$$

has kernel F . Its image has $\mathbf{Q}_p$ -dimension $[K : \mathbf{Q}_p] - [F : \mathbf{Q}_p] = [F : \mathbf{Q}_p](N - 1)$.

By Lemma 16.4, the values $\exp_K^{-1} c_{2,1}(\beta)$ of actual $\beta \in K_3(W)$ span K over $\mathbf{Q}_p$. Choose $[F : \mathbf{Q}_p](N - 1)$ actual integral coefficient classes whose images under g are $\mathbf{Q}_p$ -independent. The displayed detector makes their pushforwards $\mathbf{Q}$ -independent after rationalization: a rational relation would give a $\mathbf{Q}_p$ -linear relation among those detected coefficient vectors. Each generic image already vanishes integrally by localization, proving the theorem. $\square$

In particular, for every fixed prime p and every integer $N \geq 2$, with no restriction on divisibility by p ,

$$A_N = \mathbf{Z}_p[[x, y, z]] / (p + xy + z^N)$$

has residue field $\mathbf{F}_p$, dimension three, and a rational generic kernel in K_3 of dimension at least $N - 1$. Thus these dimensions are unbounded with the prime, residue field, and local dimension all fixed.

17.2. Explicit wild-orbit families at every residue prime. We prove a local dilogarithm independence statement before applying the geometric detector. It gives the whole minus space at the primes two and three. For larger primes the assertion concerns one wild orbit; no nonvanishing assertion for the other odd tame characters is needed. Throughout, the Coleman dilogarithm has its usual series at zero and the logarithm is chosen with $\log p = 0$.

Lemma 17.2. *Let ρ_r be a primitive p^r -th root with $\rho_r^p = \rho_{r-1}$, put $R_r = \text{Li}_2(\rho_r)$, and normalize $v_p(p) = 1$. For odd p and $r \geq 2$, write $e = p^{r-1}(p-1)$. Then*

$$v_p(R_r - p^{-2}R_{r-1}) = \begin{cases} -3/e, & p = 3, \\ -(p-2)/e, & p \geq 5. \end{cases}$$

For $p = 2$ and $r \geq 3$ one has

$$v_2(R_r - \frac{1}{4}R_{r-1}) = -2^{-(r-2)}.$$

Proof. Put $F_p(z) = \text{Li}_2(z) - p^{-2}\text{Li}_2(z^p)$. Theorem 2.7(3) and equation (3.3) of [6] identify it with the analytic continuation of $\sum_{p \nmid n} z^n/n^2$ to $|1-z| > p^{-1/(p-1)}$. These statements apply to every prime, including two. For $j \geq 0$ put

$$S_j(t) = \sum_{m \geq 0} m^j t^m = \frac{P_j(t)}{(1-t)^{j+1}}, \quad P_j \in \mathbf{Z}[t], \quad P_0 = 1, \quad A_j(z) = \sum_{a=1}^{p-1} \frac{z^a}{a^{j+2}},$$

with $m^0 = 1$ also at $m = 0$. Expanding $(a+pm)^{-2}$ first on $|z| < 1$ gives

$$(17.3) \quad F_p(z) = \sum_{j \geq 0} (-1)^j (j+1) p^j A_j(z) S_j(z^p).$$

For any rational $0 < a < 1$, the right side converges uniformly on

$$|z| \leq 1, \quad v_p(1-z^p) \leq a,$$

since its j th summand has valuation at least $j - (j+1)a$. This domain is the connected annulus $p^{-a/p} \leq |1-z| \leq 1$, and lies in the quoted domain of F_p . The rigid analytic identity principle proves (17.3) there. For all the roots under consideration one may take $a = 2/3$.

At $p = 2$ one has $A_j(z) = z$. Write $d = v_2(1 - \rho_r^2) = 2^{-(r-2)} \leq 1/2$. The $j = 0$ term is $\rho_r/(1 - \rho_r^2)$, of valuation $-d$; every $j \geq 1$ term has valuation at least $j - (j+1)d = -d + j(1-d) > -d$. This proves the assertion at two.

At $p = 3$ the numerator of the $j = 0$ term is $\rho_r + \rho_r^2/4$, which reduces to 2. With $d = v_3(1 - \rho_r^3) = 3/e \leq 1/2$, the same argument gives $-3/e$.

Suppose now that $p \geq 5$. For $A = A_0$, reduction modulo p gives

$$A(1) = A'(1) = 0, \quad A''(1)/2 = -1/2 \quad \text{in } \mathbf{F}_p.$$

Indeed these are the usual sums of inverse squares, inverses, and $(a-1)/(2a)$ over $a \in \mathbf{F}_p^\times$. Since $q = 1 - \rho_r$ has valuation $1/e$ and $2/e < 1$, the unique least term of $A(1-q)$ is its quadratic term. Thus $v_p(A(\rho_r)) = 2/e$ and the $j = 0$ term has valuation $(2-p)/e$. For $j \geq 1$ the difference between its lower valuation bound $j - (j+1)p/e$ and $(2-p)/e$ is

$$j(1-p/e) - 2/e \geq 1 - (p+2)/e > 0,$$

because $e \geq p(p-1) > p+2$. Again the $j = 0$ term is uniquely minimal. $\square$

Proposition 17.3. *For odd p , let $E_r = \mathbf{Q}_p(\rho_r)$ and $\Gamma_r = \text{Gal}(E_r/E_1)$. The p^{r-1} elements*

$$\{\sigma \text{Li}_2(-\rho_r) : \sigma \in \Gamma_r\}$$

are linearly independent over $\mathbf{Q}_p$. For $p = 3$ they form a basis of E_r^- .

At $p = 2$, let $r \geq 2$, $E_r = \mathbf{Q}_2(\rho_r)$, and $\Gamma_r = \text{Gal}(E_r/\mathbf{Q}_2(i))$. Put

$$\Psi_2(\rho_r) = \frac{1}{3} \text{Li}_2(\rho_r^3) - \text{Li}_2(\rho_r).$$

Its 2^{r-2} conjugates under Γ_r form a $\mathbf{Q}_2$ -basis of E_r^- .

Proof. For odd p , write $\text{Gal}(E_r/\mathbf{Q}_p) = \Delta_r \times \Gamma_r$, with $|\Delta_r| = p - 1$, and let $\omega : \Delta_r \rightarrow \mathbf{Q}_p^\times$ be the Teichmüller character: $\omega(\sigma_a) \equiv a \pmod{p}$ when $\sigma_a(\rho_r) = \rho_r^a$. Use its idempotent

$$e_\omega = \frac{1}{p-1} \sum_{\sigma \in \Delta_r} \omega(\sigma)^{-1} \sigma.$$

If $y \neq 0$ has valuation m/e with $m \equiv 1 \pmod{p-1}$, then $e_\omega y$ has the same valuation as y . In fact, write $y = q^m u$ with u a unit. Since the extension is totally ramified, $\sigma_a(u)/u$ reduces to 1, while $\sigma_a(q)/q$ reduces to a . Each summand of $e_\omega y/y$ reduces to 1, as does their average.

The integer numerators -3 and $-(p-2)$ in Lemma 17.2 are congruent to 1 modulo $p-1$. Consequently the ω projection of every depleted value is nonzero. The base value satisfies $v_p(R_1) = -(p-2)/(p-1)$ by Lemma 16.10, and its ω projection is nonzero for the same reason.

Put $C_r = p^{2r-1} R_r$. Distribution [6, Proposition 2.10(1)] gives

$$\text{Tr}_{E_r/E_{r-1}} R_r = p^{-1} R_{r-1}, \quad p^{-1} \text{Tr}_{E_r/E_{r-1}} C_r = C_{r-1}.$$

Hence the successive rational isotypic projections of $e_\omega C_r$ under the cyclic group Γ_r are

$$e_\omega C_1, \quad e_\omega(C_2 - C_1), \quad \dots, \quad e_\omega(C_r - C_{r-1}).$$

They are all nonzero. The normal basis theorem identifies $e_\omega E_r$ with the regular $\mathbf{Q}_p[\Gamma_r]$ -module [35, Chapter I, p. 34]. Its irreducible constituents have minimal polynomials $T - 1, \Phi_p(T), \dots, \Phi_{p^{r-1}}(T)$, each once; these are irreducible over $\mathbf{Q}_p$. Thus $e_\omega R_r$ is cyclic and its orbit is a basis. Projecting any relation shows that the orbit of R_r itself is independent. Duplication gives

$$\text{Li}_2(-\rho_r) = (\tfrac{1}{2}\sigma_2 - 1)R_r.$$

The displayed operator is invertible, commutes with Γ_r , and preserves independence: σ_2 has finite order whereas 2 is not a root of unity. Inversion makes every value anti-invariant. At $p = 3$, $\Delta_r = \{1, \text{inversion}\}$, so this is the whole minus space.

At $p = 2$ use the base level $r = 2$ and $\text{Gal}(E_r/\mathbf{Q}_2) = \{1, \text{inversion}\} \times \Gamma_r$. The group Γ_r is cyclic of order 2^{r-2} , generated by σ_5 . Its minus space is its regular representation [35, Chapter I, p. 34]. Set again $C_r = 2^{2r-1} R_r$; distribution gives average trace $C_r \mapsto C_{r-1}$. The base C_2 is nonzero by Lemma 16.8, and every $C_j - C_{j-1}$ for $j \geq 3$ is nonzero by Lemma 17.2. The same multiplicity-one argument, now using $T - 1, \Phi_2(T), \dots, \Phi_{2^{r-2}}(T)$ over $\mathbf{Q}_2$, shows that R_r is a cyclic vector in E_r^- . Finally $\Psi_2(\rho_r) = (\sigma_3/3 - 1)R_r$, an invertible operator commuting with Γ_r , so it too is cyclic. $\square$

Theorem 17.4. *For odd p and $r \geq 1$ put*

$$A_{p,r} = \left(\mathbf{Z}_{(p)}[x, y, z] / (xy + \Phi_{2p^r}(z - 1)) \right)_{(p, x, y, z)}.$$

Write $a = z - 1$, $n = 2p^r$, and

$$H = \frac{a^n - 1}{\Phi_{2p^r}(a)}, \quad \lambda = 1 - x, \quad \mu = 1 + \lambda y H.$$

For $0 \leq j < p^{r-1}$, let k_j be the unique odd integer with $1 \leq k_j < 2p^r$ and $k_j \equiv (1+p)^j \pmod{p^r}$. Then the explicit classes

$$c_j = k_j \{1 - a^{k_j}, \lambda, \mu\} \in K_3(A_{p,r})$$

are linearly independent over $\mathbf{Q}$ after tensoring with $\mathbf{Q}$, and each has zero generic image integrally. Their images retain these properties in the henselization and completion of $A_{p,r}$.

At $p = 2$, for $r \geq 2$ put

$$A_{2,r} = \left(\mathbf{Z}_{(2)}[x, y, t] / (xy + \Phi_{2^r}(t)) \right)_{(2, x, y, t-1)}, \quad n = 3 \cdot 2^r,$$

and set $H = (t^n - 1)/\Phi_{2^r}(t)$, $\lambda = 1 - x$, $\mu = 1 + \lambda y H$. Let k_j be the least positive representative of 5^j modulo 2^r , for $0 \leq j < 2^{r-2}$. The same assertions hold for the 2^{r-2} classes

$$c_j = k_j \{1 + t^{k_j} + t^{2k_j}, \lambda, \mu\}.$$

All the displayed local rings are regular of dimension three.

Proof. The polynomials $\Phi_{2p^r}(z-1)$ for odd p and $\Phi_{2^r}(1+z)$ at two are Eisenstein. Their constant terms are respectively p and 2 ; modulo that prime they are powers of z . This proves regularity and dimension as in Theorem 14.4. Every entry of each displayed triple is a unit: at the closed point $a = -1$ for odd p , and $t = 1$ at two.

For either prime let u denote a or t , and put $b = yH$. Then $1 - xb = u^n$ and the Dennis–Stein identity (15.3) applies to $\delta = \langle x, b \rangle$. In the odd case, for any allowed k ,

$$k\{1 - a^k, a^n\} = \{1 - a^k, (a^k)^n\} = 0.$$

At two, use the finite étale quadratic extension adjoining ζ_3 . Steinberg and the transfer projection formula give

$$\begin{aligned} 0 &= \text{Tr}\{1 - \zeta_3 t^k, (\zeta_3 t^k)^n\} \\ &= \{1 + t^k + t^{2k}, t^{kn}\} = k\{1 + t^k + t^{2k}, t^n\}. \end{aligned}$$

Here $3 \mid n$, and the norm of $1 - \zeta_3 t^k$ is exactly $1 + t^k + t^{2k}$, with no extra factor of two. Only oddness of k is needed; k may be divisible by 3. If g_k denotes the first entry of the corresponding triple, then

$$k[g_k]\delta = k\{g_k, \lambda, \mu\}, \quad k[g_k]\delta|_{D(x)} = 0.$$

This proves integral generic vanishing before and after every base change.

We specify the coefficient calculation needed for independence. On the split finite surface $Q = \text{Spec } F[x, y, u]/(xy + P(u))$, with $F = \mathbf{Q}(\mu_{2p^r})$ for odd p and $F = \mathbf{Q}(\mu_{2^r})$ at two, write the roots as u_i , and put $\eta_i = 1 - [(x, u - u_i)]$. All roots lie in the finite open $D(ug_k)$. Localization and dévissage give a supported lift of $k[g_k]\delta$ to $K_3(Q)$. The matching-sphere calculation (15.5) identifies any such lift, rationally, as

$$(17.4) \quad \tilde{c}_k = -n \sum_i b_k(u_i) \eta_i,$$

where

$$b_k(u) = \begin{cases} \beta_{\text{BT}}(u^k), & p \text{ odd}, \\ \text{Tr}_{F(\zeta_3)/F}^K \beta_{\text{BT}}(\zeta_3 u^k), & p = 2. \end{cases}$$

For completeness, choose an arc between each pair of roots whose interior lies in the punctured open unit disc. There $\text{Log}(1 - u^k)$, or the sum $\text{Log}(1 - \zeta_3 u^k) + \text{Log}(1 - \zeta_3^2 u^k)$, has a holomorphic branch. Its primitive for $-L du/u$ is respectively

$$\frac{1}{k} \text{Li}_2(u^k), \quad \frac{1}{k} (\text{Li}_2(\zeta_3 u^k) + \text{Li}_2(\zeta_3^2 u^k)).$$

The factor k in c_k cancels this denominator. The endpoint periods therefore have the common factor n , independent of k . Checking the periods at every complex embedding, and then applying Borel injectivity as in Lemma 16.12, determines all coefficient differences. The remaining diagonal coefficient vanishes because $\sum_i \eta_i = 0$. This proves (17.4).

The common normalization of Lemma 16.11, the root-of-unity formula [8, Theorem 1.12], and the point comparison give, with one nonzero scalar for the whole vector, the local coefficient values

$$c_{2,1}(b_k(u_i)) = \begin{cases} \kappa \exp_E(\text{Li}_2(u_i^k)), & p \text{ odd}, \\ \kappa \exp_E(\Psi_2(u_i^k)), & p = 2. \end{cases}$$

At two this follows by restriction-transfer and distribution: $\text{Li}_2(\zeta_3 u^k) + \text{Li}_2(\zeta_3^2 u^k) = \Psi_2(u^k)$. All these roots and their complements used in β_{BT} are global units. No root-dependent normalization is introduced.

After completion and the finite free extension to $\mathcal{O}_E$, every shifted root lies in its maximal ideal, so Lemma 16.2 and the line-bundle Chern formula apply to (17.4). They kill exactly the diagonal coefficient vectors. Every vector displayed above changes sign when u_i is replaced by u_i^{-1} . A diagonal linear combination must therefore be zero. Looking at a single fixed primitive root now turns a relation among the c_j into a relation among the wild-orbit values in Proposition 17.3. Its independence forces every coefficient to vanish, even over $\mathbf{Q}_p$ at the regulator level. Thus the actual K_3 classes are $\mathbf{Q}$ -linearly independent after completion. The

arithmetic local ring and its henselization map to that completion, proving the corresponding assertions there. $\square$

Remark 17.5. The explicit lower bounds are p^{r-1} for odd p and 2^{r-2} at two. They equal the cyclotomic minus dimension at $p = 3$ and $p = 2$. For $p \geq 5$ this argument proves independence through the Teichmüller component; it does not assert full minus rank $(p-1)p^{r-1}/2$.

18. HENSELIZATION AND INFINITE RANK

Actual local coefficients descend to henselian arithmetic valuation rings. Their finite families give the same rank bounds before completion and produce infinite rank after strict henselization.

18.1. Actual coefficients over a henselian valuation ring. The coefficient classes needed by the completed argument can already be chosen over henselian number-field valuation rings. The finite-quotient assertion in Levine's theorem gives this directly.

Lemma 18.1. *Let W_0 be a discrete valuation ring whose fraction field is a number field and whose residue field has characteristic p . Put*

$$H = \text{Frac}(W_0^h), \quad K = \text{Frac}(\widehat{W_0}).$$

For every $\nu \geq 1$, the natural map

$$K_3(W_0^h) \longrightarrow K_3(K)/p^\nu$$

is surjective. In particular the rational étale Chern classes of actual elements of $K_3(W_0^h)$ span $H^1(K, \mathbf{Q}_p(2))$ over $\mathbf{Q}_p$.

Proof. The henselization W_0^h is a discrete valuation ring with the same residue field and completion as W_0 ; see [46, Section 15.46, Lemmas 15.46.1, 15.46.3 and 15.46.11]. Its fraction field embeds densely in K , and

$$H = \{a \in K : a \text{ is algebraic over } \mathbf{Q}\}.$$

Indeed H is algebraic over $\text{Frac}(W_0)$. Conversely a henselian rank-one valued field is separably algebraically closed in its completion: approximate a separable algebraic root by an element of the field closely enough to apply Hensel's lemma, and isolate that root from the finitely many other roots. The Henselian root then equals the original root. Characteristic zero makes every algebraic extension separable.

The second assertion of [33, Theorem 4.13, p. 336], applied to K and its field of constants H , gives

$$K_3(H)^{\text{ind}}/p^\nu \xrightarrow{\sim} K_3(K)^{\text{ind}}/p^\nu.$$

Milnor $K_3(K)$ is p -divisible, as used in Lemma 16.4. Its image in $K_3(K)$ is therefore contained in $p^\nu K_3(K)$, so $K_3(K)^{\text{ind}}/p^\nu = K_3(K)/p^\nu$. Consequently $K_3(H) \rightarrow K_3(K)/p^\nu$ is surjective. Quillen localization lifts every element of $K_3(H)$ to $K_3(W_0^h)$, because $K_2(k_{W_0}) = 0$.

The proof of Lemma 16.4 identifies $\varprojlim_\nu K_3(K)/p^\nu$ with $H^1(K, \mathbf{Z}_p(2))$. The surjectivity just proved makes the image of the actual group $K_3(W_0^h)$ dense in this inverse limit. Its rational Chern image therefore spans $H^1(K, \mathbf{Q}_p(2))$. $\square$

Theorem 18.2. *Let F_0 be a number field, let $V_0 = (\mathcal{O}_{F_0})_{\mathfrak{p}}$ for a prime above p , and let F be its completion. Let $f(z) \in V_0[z]$ be Eisenstein of degree $N \geq 2$, and put*

$$R_0 = (V_0[x, y, z]/(xy + f(z)))_{(\mathfrak{p}, x, y, z)}, \quad R = R_0^h.$$

Then R is a henselian regular local ring of dimension three with residue field k_F , and

$$\dim_{\mathbf{Q}} \ker (K_3(R)_{\mathbf{Q}} \longrightarrow K_3(\text{Frac } R)_{\mathbf{Q}}) \geq [F : \mathbf{Q}_p](N-1).$$

The lower bound is witnessed by actual integral classes whose generic images vanish integrally.

Proof. Choose a root θ of f and put

$$W_0 = V_0[\theta], \quad K = F(\theta), \quad W = \mathcal{O}_K.$$

Eisenstein's condition makes W_0 a discrete valuation ring with uniformizer θ , residue field k_F , and completion W . It also makes R_0 regular of dimension three, by the same linear-term calculation as in Theorem 17.1. Regularity, dimension and the residue field are preserved by henselization, and

$$\widehat{R} = \mathcal{O}_F[[x, y, z]]/(xy + f(z)).$$

These assertions, including the faithful flatness of $R \rightarrow \widehat{R}$, follow from [46, Section 15.46].

Henselization commutes with the quotient by x , so the regular divisor $D = \text{Spec}(R/(x))$ has coordinate ring

$$R/(x) \cong \left(W_0[y]_{(\theta, y)}\right)^h.$$

Here we use [46, Section 10.156, Lemma 10.156.2]. The local map $W_0 \rightarrow R/(x)$ extends to $W_0^h \rightarrow R/(x)$. On completion this map is compatible with $W \rightarrow W[[y]] = \widehat{R}/(x)$, by the universal property of henselization.

For $\beta \in K_3(W_0^h)$ define

$$c^h(\beta) = i_*(\beta|_D) \in K_3(R), \quad i : D \hookrightarrow \text{Spec } R.$$

Localization makes its image in $K_3(R[1/x])$, hence in $K_3(\text{Frac } R)$, zero integrally. Flat base change along $R \rightarrow \widehat{R}$ sends $c^h(\beta)$ to exactly the class $c(\widehat{\beta})$ constructed in Theorem 17.1, where $\widehat{\beta}$ is the image in $K_3(W)$.

By Lemma 18.1, the values

$$a(\beta) = \exp_K^{-1} c_{2,1}(\beta|_K)$$

of actual $\beta \in K_3(W_0^h)$ span K over $\mathbf{Q}_p$. The completed detector in Theorem 17.1 has kernel precisely F on this coefficient space. Choose $[F : \mathbf{Q}_p](N - 1)$ actual coefficient classes whose values are independent in K/F over $\mathbf{Q}_p$. Their completed pushforwards have independent detected classes. Any rational relation among their henselian pushforwards would remain a relation after completion and would contradict that independence. This proves the asserted lower bound. $\square$

In particular, for every prime p and every $N \geq 2$, the henselization

$$\left(\left(\mathbf{Z}_{(p)}[x, y, z]/(p + xy + z^N) \right)_{(x, y, z)} \right)^h$$

has residue field $\mathbf{F}_p$, dimension three, and rational generic K_3 -kernel of dimension at least $N - 1$.

18.2. One complete strictly henselian ring with infinite rational rank.

Lemma 18.3. *Lemma 16.1 remains valid when E is a complete discretely valued field of mixed characteristic $(0, p)$ with perfect residue field, containing a fixed finite extension of $\mathbf{Q}_p$ as a valued subfield.*

Proof. We check the points in that proof where the base could matter. The valuation ring $\mathcal{O}_E$ is still a noetherian complete DVR. The fixed rational-radius models used in (16.1) are therefore noetherian henselian models to which [1, Theorem 6.2.1] applies, including for p -power coefficients. The order of operations in (16.2) is unchanged: finite coefficients at one fixed radius, derived inverse limit, inversion of p , and finally the dagger homotopy colimit. The two-chart calculation by integration of overconvergent series gives the same isomorphism

$$f^* : H_{\text{dR}}^2(\mathbf{P}_E^1) \xrightarrow{\sim} H_{\text{dR}}^2(U^\dagger).$$

The standing hypotheses of [18], stated at the beginning of its introduction, allow any complete mixed-characteristic DVR with perfect residue field. Thus its Theorem 1.3(2) still gives the injective boundary ∂_3 on $U^\dagger$. For $\mathbf{P}_E^1$ the boundary is an isomorphism: $H_{\text{HK}}^2(\mathbf{P}^1) = F_0 h$ with $\varphi(ah) = p\sigma(a)h$, where $F_0 = \text{Frac } W(k_E)$, while $H_{\text{HK}}^3(\mathbf{P}^1) = 0$. The eigenspace $\varphi = p^2$ is zero,

since $p\sigma(a) = p^2a$ for $a \neq 0$ contradicts $v_p(\sigma(a)) = v_p(a)$. This argument does not require finite order of σ . Proper étale and pro-étale cohomology agree by [18, Section 7.2.4].

We also remove the distinction between topological and ordinary cohomology in the target. The unit quadric has the smooth weak formal model $\mathrm{Spwf} \mathcal{O}_E\langle X, Y, Z \rangle^\dagger / (XY + Z(Z + 1))$, and its ruling extends to the weak formal projective line: the ideal (X, Z) is already invertible on the integral model. The natural Hyodo–Kato comparison in [18, equation (5.4) and the following strict quasi-isomorphism, Propositions 5.5–5.6], together with faithful extension from F_0 to E , transports the de Rham degree-two isomorphism to

$$H_{\mathrm{HK}}^2(\mathbf{P}_E^1) \xrightarrow{\sim} H_{\mathrm{HK}}^2(U^\dagger); \quad H_{\mathrm{HK}}^3(U^\dagger) = 0.$$

Thus the same Hyodo–Kato term vanishes for $U^\dagger$, and $\partial_{3,U}$ is an isomorphism. Its de Rham source is classical by [18, Section 5.1], even if E has infinite dimension over $\mathbf{Q}_p$. Hence its target is classical too. The classical-part functor of [18, Section 3.1.1] sends $\tilde{H}^i$ to ordinary H^i ; applying it to this isomorphism gives an isomorphism on ordinary cohomology. Naturality now proves injectivity of pullback from $H_{\mathrm{cont},\mathrm{ét}}^3(\mathbf{P}_E^1, \mathbf{Q}_p(3))$ on the ordinary cohomology used by the Chern maps.

For the last step, the splitting (16.3) with $r = 3$ gives the split injection

$$H^1(E, \mathbf{Q}_p(2)) \xrightarrow{\sim c_1(\mathcal{O}(1))} H_{\mathrm{cont},\mathrm{ét}}^3(\mathbf{P}_E^1, \mathbf{Q}_p(3)).$$

Its finite-coefficient splitting survives derived inverse limits and inversion of p . No vanishing assertion about $H^3(E, \mathbf{Q}_p(3))$ is required. Its pullback is the ruling-cup map, which is therefore injective. $\square$

Theorem 18.4. *Set*

$$V_\infty = W(\overline{\mathbf{F}}_3), \quad A_\infty = V_\infty[[x, y, z]]/(3 + xy + z^2).$$

Then A_∞ is a complete strictly henselian regular local ring of dimension three, and

$$\dim_{\mathbf{Q}} \ker(K_3(A_\infty)_{\mathbf{Q}} \longrightarrow K_3(\mathrm{Frac} A_\infty)_{\mathbf{Q}}) = \infty.$$

For every $m \geq 1$ this kernel contains m independent classes represented by actual integral elements of $K_3(A_\infty)$.

Proof. The Witt ring V_∞ is a complete DVR with uniformizer 3. The equation has nonzero linear term 3 in the cotangent space of $V_\infty[[x, y, z]]$, so its quotient is regular local of dimension three, with maximal ideal (x, y, z) . Completeness and the algebraically closed residue field imply strict henselianity.

For each positive integer m , put

$$V_m = W(\mathbf{F}_{3^m}), \quad F_m = \mathrm{Frac} V_m, \quad A_m = V_m[[x, y, z]]/(3 + xy + z^2).$$

The canonical local map $A_m \rightarrow A_\infty$ is flat by miracle flatness: both rings are regular of dimension three and the closed fiber has dimension zero [45, Tag 00R4].

Choose $u^2 = -3$ and write

$$E_m = F_m(u), \quad W_m = V_m[u], \quad E_\infty = (\mathrm{Frac} V_\infty)(u), \quad W_\infty = V_\infty[u].$$

The field E_∞ is the completion of E_m^{ur} for every m . Indeed its Witt coefficients can be approximated to every finite precision in finite residue subfields. Completion of a henselian discretely valued field preserves its category of finite separable extensions [42, Lemma 4.2.1]; consequently

$$G_{E_\infty} \simeq I_{E_m}.$$

Restriction therefore induces an injection

$$(18.1) \quad H^1(E_m, \mathbf{Q}_3(2)) \hookrightarrow H^1(E_\infty, \mathbf{Q}_3(2)).$$

Here is a direct verification. For $G = G_{E_m}$ and $I = I_{E_m}$, $\mathbf{Q}_3(2)^I = 0$ because the cyclotomic character squared is nontrivial on inertia. A cocycle trivial in $H^1(I, \mathbf{Q}_3(2))$ can be changed by a coboundary to vanish on I . For such a cocycle b , $ib(g) = b(ig) = b(g(g^{-1}ig)) = b(g)$, so every $b(g)$ belongs to $\mathbf{Q}_3(2)^I$ and is zero.

By Lemma 16.4, choose actual integral anti-invariant classes $\beta_{m,1}, \dots, \beta_{m,m} \in K_3(W_m)$ whose Chern classes

$$\alpha_{m,j} = c_{2,1}(\beta_{m,j}) \in H^1(E_m, \mathbf{Q}_3(2))$$

are $\mathbf{Q}_3$ -independent. Pull these classes to the regular divisor $A_m/(x) \simeq W_m[[y]]$ and push forward to obtain $c_{m,j} \in K_3(A_m)$. Localization makes them zero after inverting x . Their images $c_{m,j}^\infty \in K_3(A_\infty)$ are therefore also generically zero.

Make the finite free base change $V_\infty \rightarrow W_\infty$ and put

$$T_\infty = \text{Spec}(W_\infty[[x, y, z]]/(xy + z^2 - u^2))[1/u], \quad I_+ = (x, z - u), \quad \eta_+ = 1 - [I_+].$$

Flat base change and the projection formula give

$$c_{m,j}^\infty|_{T_\infty} = 2 \beta_{m,j}|_{E_\infty} \eta_+.$$

All rings here are algebras over the fixed finite extension $\mathbf{Q}_3(u)$, so Lemma 16.3 applies without any new finiteness assumption over E_∞ . Using (16.6) gives

$$c_{3,3}(c_{m,j}^\infty|_{T_\infty}) = 4 c_1(I_+) \smile \text{res}_{E_\infty/E_m}(\alpha_{m,j}).$$

The coefficient classes remain $\mathbf{Q}_3$ -independent by (18.1). Lemma 18.3, with $w = z - u$ and $d = 2u$, proves independence of these Chern images. Hence the actual integral classes $c_{m,1}^\infty, \dots, c_{m,m}^\infty$ are $\mathbf{Q}$ -independent after rationalization. Since m is arbitrary and A_∞ is fixed, the asserted kernel is infinite dimensional. $\square$

Remark 18.5. This proof uses arbitrarily large finite families from finite unramified coefficient fields. It neither requires that actual regulators exhaust E_∞ nor identifies A_∞ with a filtered colimit of the rings A_m .

18.3. Infinite rank already in an arithmetic strict henselization.

Lemma 18.6. *For a prime p , set*

$$A_{0,p} = \left(\mathbf{Z}_{(p)}[x, y, z]/(p + xy + z^2) \right)_{(x,y,z)}, \quad S_p = A_{0,p}^{sh},$$

using the geometric residue field $\overline{\mathbf{F}}_p$. There is a canonical compatible local map

$$S_p \longrightarrow A_{\infty,p} := W(\overline{\mathbf{F}}_p)[[x, y, z]]/(p + xy + z^2),$$

and it identifies $A_{\infty,p}$ with the completion of S_p . In particular S_p is noetherian regular local of dimension three with residue field $\overline{\mathbf{F}}_p$.

Proof. The target is a complete local ring with algebraically closed residue field, hence strictly henselian. The local map from $A_{0,p}$ extends to S_p by the universal property with the prescribed geometric residue field. The assertions about S_p follow from [46, Section 15.46].

Write $\mathfrak{m} = (x, y, z)$. Fix $r \geq 1$ and choose s with $2s \geq r$. Since $p \in \mathfrak{m}^2$, the Artinian local ring $R_r = A_{0,p}/\mathfrak{m}^r$ is a $\mathbf{Z}/p^s$ -algebra, and

$$A_{\infty,p}/\mathfrak{m}^r \cong R_r \otimes_{\mathbf{Z}/p^s} W_s(\overline{\mathbf{F}}_p).$$

Here W_s denotes Witt vectors of length s . The equality

$$W_s(\overline{\mathbf{F}}_p) = \text{colim}_m W_s(\mathbf{F}_{p^m}),$$

with m ordered by divisibility, follows because a finite Witt vector has only finitely many residue-field coordinates. Each $R_r \otimes_{\mathbf{Z}/p^s} W_s(\mathbf{F}_{p^m})$ is the finite étale local extension of R_r with residue field $\mathbf{F}_{p^m}$. Finite étale extensions of a henselian local ring are determined by their residue extensions [46, Tag 04GK]. Thus $A_{\infty,p}/\mathfrak{m}^r$ is the strict henselization of R_r . By quotient compatibility [46, Section 10.156, Lemma 10.156.4],

$$A_{\infty,p}/\mathfrak{m}^r \cong S_p/\mathfrak{m}^r S_p.$$

These identifications respect the transition maps and the chosen residue field. Taking inverse limits proves $\widehat{S}_p \cong A_{\infty,p}$. $\square$

Corollary 18.7. *Let*

$$S = \left(\left(\mathbf{Z}_{(3)}[x, y, z]/(3 + xy + z^2) \right)_{(x, y, z)} \right)^{sh}.$$

Then

$$\dim_{\mathbf{Q}} \ker (K_3(S)_{\mathbf{Q}} \longrightarrow K_3(\text{Frac } S)_{\mathbf{Q}}) = \infty.$$

For every $m \geq 1$ there are m independent classes represented by actual integral elements of $K_3(S)$ whose generic images vanish integrally.

Proof. For each m , choose a monic polynomial $g_m \in \mathbf{Z}[T]$ whose reduction modulo 3 is irreducible of degree m , and choose a residue root in $\mathbf{F}_{3^m} \subset \overline{\mathbf{F}}_3$. The ring

$$V_{m,0} = \mathbf{Z}_{(3)}[T]/(g_m)$$

is a finite étale local discrete valuation ring with number-field fraction field, residue field $\mathbf{F}_{3^m}$, and completion $V_m = W(\mathbf{F}_{3^m})$. The chosen residue root gives compatible embeddings of $V_{m,0}$ into $\mathbf{Z}_{(3)}^{sh}$ and into V_m . Put

$$R_{m,0} = A_{0,3} \otimes_{\mathbf{Z}_{(3)}} V_{m,0}, \quad R_m = R_{m,0}^h.$$

The finite étale algebra $R_{m,0}$ is local, because its closed fiber is the field $\mathbf{F}_{3^m}$. There are compatible local maps

$$R_m \longrightarrow S \longrightarrow A_{\infty,3}, \quad \widehat{R}_m = A_m = V_m[[x, y, z]]/(3 + xy + z^2) \longrightarrow A_{\infty,3}.$$

The two maps from R_m to $A_{\infty,3}$ agree by the universal property of henselization. The second row is the usual inclusion of Witt coefficients after completion.

Set $u^2 = -3$ and $W_{m,0} = V_{m,0}[u]$. Lemma 18.1, followed by $1 - \sigma$ for $\sigma(u) = -u$, supplies actual anti-invariant classes

$$\beta_{m,1}^h, \dots, \beta_{m,m}^h \in K_3(W_{m,0}^h)$$

whose Chern classes in $H^1(\text{Frac}(V_m)(u), \mathbf{Q}_3(2))$ are $\mathbf{Q}_3$ -independent. Pull them to $R_m/(x)$ and push forward along $x = 0$ as in Theorem 18.2. Their resulting classes in $K_3(R_m)$ vanish after inverting x , and so do their images in $K_3(S)$.

After mapping to $A_{\infty,3}$ these are exactly the completed finite-level classes detected in Theorem 18.4. Its proof applies to arbitrary independent anti-invariant coefficient classes from each finite unramified level. It therefore proves that these m images are $\mathbf{Q}$ -independent. Any rational relation in $K_3(S)$ would remain a relation in $K_3(A_{\infty,3})$, so the classes in $K_3(S)$ are independent as well. Since m is arbitrary, the result follows. $\square$

19. CYCLOTOMIC COEFFICIENTS IN HIGHER ODD DEGREES

The following construction supplies actual higher K -classes without requiring a general surjectivity assertion from actual K -theory to its p -completed realization. Fix a weight $n \geq 2$. We write $c_{n,1} : K_{2n-1} \rightarrow H^1(-, \mathbf{Q}_p(n))$ and $c_{n+1,3} : K_{2n-1} \rightarrow H^3(-, \mathbf{Q}_p(n+1))$; the second subscript denotes the cohomological degree.

Lemma 19.1. *Let $E/\mathbf{Q}_p$ be finite and $n \geq 2$. For a root of unity $\rho \in E$, the number-field cyclotomic class $b_n(\rho) \in K_{2n-1}(\mathcal{O}_E)_{\mathbf{Q}}$ has*

$$(19.1) \quad c_{n,1}(b_n(\rho)) = \exp_{E,n}(\kappa_n \text{Li}_n(\rho)), \quad \kappa_n \in \mathbf{Q}^{\times},$$

where κ_n is common to all roots and embeddings under one fixed symbol convention. The exponential is an isomorphism for $n \geq 2$. For an invertible module L and an actual class β of degree $2n-1$, one has

$$(19.2) \quad c_{n+1,3}(\beta(1 - [L])) = n c_{n,1}(\beta) \cup c_1(L).$$

Finally, Lemmas 16.1 and 16.2 hold with coefficient group $H^1(E, \mathbf{Q}_p(n))$ and target twist $n+1$.

Proof. Theorem 1.12 of [8] constructs the actual rational cyclotomic class and includes roots of p -power order. Its normalized syntomic value is $\pm(n-1)! \text{Li}_n(\rho)$. To compare the normalizations, put $q = p^f = |k_E|$, let E_0 be the maximal unramified subfield of E , and write σ for its Frobenius. Set

$$A = 1 - p^{-n}\sigma, \quad N = \sum_{j=0}^{f-1} p^{-nj}\sigma^j, \quad P = 1 - q^{-n}.$$

Then $NA = AN = P$. The point syntomic complex of [29, Example 2.2.4] is $[E_0 \rightarrow E \oplus E_0]$, with $c \mapsto (c, Ac)$, and its canonical E -coordinate is $\eta([a, b]) = a - A^{-1}b$. The comparison of [5, Proposition 8.6(2)–(3), its proof and equation (8.1)] uses N to pass from the semilinear p -Frobenius to the linear q -Frobenius. On the point the latter acts as the identity on E , and the raw modified-cone coordinate is

$$\epsilon = Pa - Nb = P\eta([a, b]).$$

Definition 4.6 of [8] sends this coordinate to

$$(19.3) \quad (1 - \varphi_q^*/q^n)^{-1}\epsilon = P^{-1}\epsilon = \eta([a, b]).$$

Thus [5, Corollary 9.10 and Proposition 9.11], or equivalently [29, Proposition 2.3.4], identifies the normalized regulator with the Chern-compatible Bloch–Kato exponential. This proves (19.1); the only remaining constant is the fixed rational symbol normalization in Theorem 1.12 and Remark 1.7 of [8]. The calculation applies to arbitrary ramification and to $p = 2$.

Formula (3.10) and Theorem 3.11(i) of [49], applied to $[L] - 1$ in degree zero and β in degree $2n - 1$, have only the term $i_1 = 1, i_2 = n$ in weight $n + 1$. Its coefficient is $-n$; negating $[L] - 1$ gives (19.2). The exceptional K -degree two case does not occur. The finite-type descent proof of Lemma 16.3 applies unchanged in degree $2n - 1 \geq 3$, so the identity holds in continuous cohomology.

For the cup assertion use the same fixed-radius map (16.2) and the same ruling morphism $f : U^\dagger \rightarrow \mathbf{P}_E^1$. The boundary with twist $n + 1$ in [18, Theorem 1.3] is injective in degree three when $n = 2$ and is an isomorphism when $n \geq 3$, since $3 \leq (n + 1) - 1$. On $\mathbf{P}_E^1$ the boundary is an isomorphism also for $n = 2$, as in Lemma 16.1. Its naturality and the de Rham ruling isomorphism make f^* injective on $H_{\text{proét}}^3(-, \mathbf{Q}_p(n + 1))$. The splitting (16.3), with $r = n + 1$, identifies cup product with $c_1(\mathcal{O}(1))$ as a split injection from $H^1(E, \mathbf{Q}_p(n))$ into its source. Composing with the injective f^* proves the quadratic assertion. The pairwise formal maps in Lemma 16.2 then give the simultaneous assertion with the same diagonal kernel. $\square$

Lemma 19.2. *For every even integer $n \geq 2$ and $i^2 = -1$,*

$$\frac{\text{Li}_n(i)}{i} \in \frac{1}{2} + \mathbf{Z}_2.$$

In particular $v_2(\text{Li}_n(i)) = -1$.

Proof. Choose a primitive cube root ζ in an unramified quadratic extension and put $D_n = \text{Li}_n(\zeta i) + \text{Li}_n(\zeta^2 i)$. Write $F_{2,n}(z) = \text{Li}_n(z) - 2^{-n} \text{Li}_n(z^2)$. Distribution and inversion give

$$D_n = -(1 + 3^{1-n}) \text{Li}_n(i), \quad \text{Li}_n(-\zeta) + \text{Li}_n(-\zeta^2) = 0.$$

Thus $D_n = F_{2,n}(\zeta i) + F_{2,n}(\zeta^2 i)$. For $S_j(t) = \sum_{m \geq 0} m^j t^m = P_j(t)/(1 - t)^{j+1}$, expand the odd denominators by the binomial series first on $|z| < 1$. The terms converge uniformly when $|z| \leq 1$ and $v_2(1 - z^2) \leq 2/3$, with valuation at least $j - (j + 1)2/3$. The rigid identity principle and [6, Theorem 2.7(3)] therefore give

$$\frac{D_n}{i} = \sum_{j \geq 0} (-1)^j \binom{n + j - 1}{j} 2^j A_j, \quad A_j = \sum_{a=1}^2 \zeta^a S_j(-\zeta^{2a}).$$

It applies because $1 + \zeta^{2a}$ is a unit. Every A_j belongs to $\mathbf{Z}[\zeta]$ and is fixed by $\zeta \mapsto \zeta^2$, hence lies in $\mathbf{Z}$. Directly, $A_0 = -2$ and $A_1 = 1$. Consequently

$$D_n/i \in -2 - 2n + 4\mathbf{Z}_2 = -2 + 4\mathbf{Z}_2.$$

Since $n - 1$ is odd, $3^{n-1} \equiv 3 \pmod{8}$ and $v_2(1 + 3^{1-n}) = 2$. The first displayed identity now gives $\text{Li}_n(i)/i \in \mathbf{Q}_2$ of valuation -1 , which is exactly the asserted coset. $\square$

Proposition 19.3. *Let $r \geq 3$, let ρ_r be a primitive 2^r -th root with $\rho_r^2 = \rho_{r-1}$, and put $E_r = \mathbf{Q}_2(\rho_r)$ and $D = 2^{r-2}$. For every $n \geq 2$,*

$$(19.4) \quad v_2(\text{Li}_n(\rho_r) - 2^{-n} \text{Li}_n(\rho_{r-1})) = -2^{-(r-2)}.$$

The $D - 1$ elements

$$\text{Li}_n(\rho_r^{5^{j+1}}) - \text{Li}_n(\rho_r^{5^j}), \quad 0 \leq j \leq D - 2,$$

are $\mathbf{Q}_2$ -linearly independent, and their span intersects $\mathbf{Q}_2 \subset E_r$ only in zero. If n is even, the D elements

$$\text{Li}_n(\rho_r^{5^j}), \quad 0 \leq j \leq D - 1,$$

form a $\mathbf{Q}_2$ -basis of the inversion-odd subspace E_r^- .

Proof. Put $F_{2,n}(z) = \text{Li}_n(z) - 2^{-n} \text{Li}_n(z^2)$. With $S_j(t) = P_j(t)/(1-t)^{j+1}$ as above, expansion on $|z| < 1$ gives

$$F_{2,n}(z) = z \sum_{j \geq 0} (-1)^j \binom{n+j-1}{j} 2^j S_j(z^2).$$

The sum converges uniformly on $|z| \leq 1$, $v_2(1 - z^2) \leq 2/3$, since its j th term has valuation at least $j - (j+1)2/3 \rightarrow \infty$. This is a connected annulus inside $|1 - z| > 1/2$; [6, Theorem 2.7(3)] and the rigid identity principle extend the equality there. At ρ_r put $d = v_2(1 - \rho_r^2) = 2^{-(r-2)} \leq 1/2$. The $j = 0$ term has valuation $-d$, while every $j \geq 1$ term has valuation at least $j - (j+1)d > -d$. This proves (19.4).

Write $R_s = \text{Li}_n(\rho_s)$ and $\Gamma_r = \text{Gal}(E_r/\mathbf{Q}_2(i)) = \langle \gamma \rangle$, with $\gamma(\rho_r) = \rho_r^5$ and $|\Gamma_r| = D$. Inversion acts on R_r by $\epsilon = (-1)^{n+1}$. The normal-basis theorem makes E_r^ϵ the regular $\mathbf{Q}_2[\Gamma_r]$ -module [35, Chapter I, p. 34]. Distribution gives average trace $R_s \mapsto 2^{-n} R_{s-1}$. For every $3 \leq s \leq r$, the rational conductor 2^{s-2} projection of R_r is therefore

$$2^{-n(r-s)}(R_s - 2^{-n} R_{s-1}) \neq 0.$$

Each nontrivial rational irreducible of the cyclic group occurs once, and its polynomial Φ_{2^a} is irreducible over $\mathbf{Q}_2$. Hence the augmentation ideal acts injectively on R_r . Its basis $\gamma^{j+1} - \gamma^j$, $0 \leq j \leq D - 2$, gives the asserted independent values. Every such value has Γ_r -average zero, so their span meets $\mathbf{Q}_2$ only in zero. If n is even, the trivial Γ_r projection is $2^{-n(r-2)} \text{Li}_n(i) \neq 0$ by Lemma 19.2. All rational blocks then occur, so the whole orbit is a basis of E_r^- . $\square$

Theorem 19.4. *For $r \geq 3$ put $D = 2^{r-2}$ and*

$$A_r = \mathbf{Z}_2[[x, y, z]]/(xy + \Phi_{2^r}(1 + z)).$$

For every $n \geq 2$,

$$\dim_{\mathbf{Q}} \ker(K_{2n-1}(A_r)_{\mathbf{Q}} \longrightarrow K_{2n-1}(\text{Frac } A_r)_{\mathbf{Q}}) \geq \begin{cases} D, & n \text{ even}, \\ D - 1, & n \text{ odd}. \end{cases}$$

More precisely, let $W_r = \mathbf{Z}_2[\rho_r]$, identify $A_r/(x) = W_r[[y]]$, and let i be this closed immersion. For even n , use the D classes

$$i_* b_n(\rho_r^{5^j}), \quad 0 \leq j \leq D - 1.$$

For odd n , use the $D - 1$ classes

$$i_*(b_n(\rho_r^{5^{j+1}}) - b_n(\rho_r^{5^j})), \quad 0 \leq j \leq D - 2.$$

The coefficients are first pulled to $W_r[[y]]$. The indicated classes are independent in the displayed kernel.

Proof. The polynomial $\Phi_{2r}(1+z)$ is Eisenstein at two. Its linear uniformizer term makes A_r regular local of dimension three, with residue field $\mathbf{F}_2$. Localization makes each supported class generically zero. After the finite free coefficient extension from $\mathbf{Z}_2$ to W_r , the roots are $a_\sigma = \sigma(\rho_r) - 1$. On the regular generic open, flat base change and projection give

$$i_*(\beta)|_T = \sum_{\sigma \in \text{Gal}(E_r/\mathbf{Q}_2)} \sigma(\beta)(1 - [J_\sigma]).$$

Base change may be performed in G -theory before restricting to this regular open, so regularity of the split integral model is not required. By Lemma 19.1, its Chern class is

$$n \sum_{\sigma} \exp_{E_r, n}(\kappa_n \sigma(R)) \cup c_1(J_\sigma),$$

where R is the corresponding difference in Proposition 19.3, or the individual value there when n is even. The simultaneous cup detector kills a coefficient vector exactly when it is diagonal. Since the exponential is injective, a relation among these detectors would put the corresponding $\mathbf{Q}_2$ -linear combination of the R 's in $\mathbf{Q}_2$. Proposition 19.3 makes that combination zero (using $E_r^- \cap \mathbf{Q}_2 = 0$ in even weight), and then makes all its coefficients zero. The detectors are $\mathbf{Q}_2$ -independent, hence the actual rational K -classes are $\mathbf{Q}$ -independent. A common integer clears the denominators of this finite family and gives integral classes whose generic images are zero integrally. $\square$

For odd weight the trivial wild-character block lies in the constant field and is deliberately removed by taking differences. No assertion about surjectivity of the actual local higher K -regulator onto its completed target is needed.

Corollary 19.5. *For the single regular local ring*

$$R = \mathbf{Z}_2[[x, y, z]]/(xy + z^4 + 4z^3 + 6z^2 + 4z + 2),$$

the generic restriction has nonzero rational kernel in every odd degree at least three. In degree $4j - 1$ its kernel has $\mathbf{Q}$ -dimension at least two, and in degree $4j + 1$, $j \geq 1$, at least one.

Proof. Take $r = 3$ in Theorem 19.4 and use $\Phi_8(1+z) = (1+z)^4 + 1$. $\square$

19.1. Higher polylogarithms on odd-prime wild orbits. Fix an odd prime p , put $M = p-1$, and choose compatible primitive p^r -th roots ρ_r . Write

$$E_r = \mathbf{Q}_p(\rho_r), \quad e_r = Mp^{r-1}, \quad R_r = \text{Li}_n(\rho_r), \quad n \geq 2.$$

Let Δ be the tame subgroup and $\omega : \Delta \rightarrow \mu_M$ its Teichmüller character. Put $\chi_n = \omega^{n-1}$ and $\Gamma_r = \text{Gal}(E_r/E_1)$, generated by $\gamma(\rho_r) = \rho_r^{1+p}$.

Proposition 19.6. *For even n , let $\ell \in \{1, 3, \dots, p-2\}$ satisfy $\ell \equiv n-1 \pmod{M}$. Then*

$$(19.5) \quad v_p(R_1) = \frac{\ell}{p-1} - 1 - v_p(n-1).$$

For either parity of n , let $q \in \{0, \dots, p-2\}$ satisfy $q \equiv n \pmod{M}$. For $r \geq 2$,

$$(19.6) \quad v_p(R_r - p^{-n}R_{r-1}) = \frac{q-p}{e_r}.$$

The χ_n -projection preserves the leading term in both formulas. Writing $D = p^{r-1}$, the following families are $\mathbf{Q}_p$ -linearly independent:

$$\{\gamma^j R_r : 0 \leq j < D\} \quad (n \text{ even}),$$

and

$$\{\gamma^{j+1} R_r - \gamma^j R_r : 0 \leq j < D-1\} \quad (n \text{ odd}).$$

The span of either displayed family has zero intersection with $\mathbf{Q}_p \subset E_r$.

Proof. Set $F_{p,n}(z) = \text{Li}_n(z) - p^{-n} \text{Li}_n(z^p)$ and $S_j(t) = P_j(t)/(1-t)^{j+1}$, $P_j \in \mathbf{Z}[t]$, where $S_j(t) = \sum_{h \geq 0} h^j t^h$ and $S_0(t) = 1/(1-t)$. Grouping the prime-to- p denominators as $a + ph$ gives

$$(19.7) \quad F_{p,n}(z) = \sum_{j \geq 0} (-1)^j \binom{n+j-1}{j} p^j \sum_{a=1}^{p-1} \frac{z^a}{a^{n+j}} S_j(z^p).$$

On $|z| \leq 1$, $v_p(1 - z^p) \leq b < 1$, the j th term has valuation at least $j - (j+1)b$, so the series converges uniformly. This domain is the annulus $p^{-b/p} \leq |1 - z| \leq 1$ and lies in the continuation region of [6, Theorem 2.7(3)]. Both sides agree on $|z| < 1$, hence on the annulus. Taking $b = 2/3$ includes all points evaluated below.

Suppose first that n is even, and put $\rho = \rho_1$. Distribution and inversion [6, Proposition 2.10] give

$$\sum_{\xi \in \mu_M \setminus \{1\}} \text{Li}_n(\xi \rho) = -(1 + M^{1-n}) \text{Li}_n(\rho).$$

Since $(\xi \rho)^p = \xi$ and $\sum_{\xi \neq 1} \text{Li}_n(\xi) = 0$ by inversion, the sum on the left can be replaced by $\sum_{\xi \neq 1} F_{p,n}(\xi \rho)$. All terms of (19.7) with $j \geq 1$ then belong to $p\mathcal{O}_{E_1}$. Its $j = 0$ sum is

$$P_n(\rho) = \sum_{a=1}^{p-1} \frac{C_a \rho^a}{a^n}, \quad C_a = \sum_{\xi \neq 1} \frac{\xi^a}{1 - \xi} = a - \frac{p}{2}.$$

Indeed, pairing inverse roots gives $\sum_{\xi \neq 1} (1 - \xi)^{-1} = (M - 1)/2$; hence $C_1 = 1 - p/2$, and $C_{a+1} - C_a = 1$ for $1 \leq a < M$.

For any integer b , the coefficient of $(Z - 1)^k$ in $\sum_{a=1}^{p-1} a^b Z^a \in \mathbf{F}_p[Z]$ is $\sum_a a^b \binom{a}{k}$. If q is the least nonnegative residue of $-b$ modulo M , the first nonzero coefficient is at $k = q$ and equals $-1/q!$: for $k < q$ no power occurring in the degree- k polynomial $a^b \binom{a}{k}$ has exponent divisible by M , while at q only its leading coefficient contributes. Applying this with $b = 1 - n$ shows that $\overline{P}_n$ has a zero of exact order ℓ at 1. Thus

$$v_p P_n(\rho) = \ell/M < 1.$$

The lower Taylor coefficients are divisible by p , and the higher terms have strictly greater valuation. They and the $j \geq 1$ remainder cannot cancel this term. Since $n - 1$ is odd, the elementary lifting-the-exponent identity gives

$$v_p(1 + M^{1-n}) = 1 + v_p(n - 1),$$

proving (19.5).

For $r \geq 2$ and either parity, the $j = 0$ term in (19.7) is

$$\frac{\sum_{a=1}^{p-1} a^{-n} \rho_r^a}{1 - \rho_r^p}.$$

The same Taylor calculation gives numerator valuation q/e_r , including $q = 0$. The denominator has valuation p/e_r . For $j \geq 1$ the valuation is at least

$$j - (j+1)p/e_r \geq j - (j+1)/(p-1) \geq 0,$$

and this bound tends to infinity. The $j = 0$ term has negative valuation, so it is the unique leading contribution. This proves (19.6).

It remains to verify the character assertion, rather than merely nonvanishing of the total values. Put $\pi_r = \rho_r - 1$. If $\tau \in \Delta$ has $\overline{\omega(\tau)} = a$, then $\tau(\pi_r)/\pi_r \equiv a \pmod{\pi_r}$. For $0 \neq x \in E_r$ with $v_{\pi_r}(x) = m$, the leading term therefore transforms by a^m . The idempotent

$$e_{\chi_n} = \frac{1}{M} \sum_{\tau \in \Delta} \chi_n(\tau)^{-1} \tau$$

preserves it if $m \equiv n - 1 \pmod{M}$. For the base value this congruence follows from $m = \ell - M(1 + v_p(n - 1))$; for the wild difference it follows from $m = q - p$.

The normal basis theorem identifies $E_r^{\chi_n}$ with the regular $\mathbf{Q}_p[\Gamma_r]$ -module [35, Chapter I, p. 34]. Its rational character blocks occur once, with polynomials Φ_{p^j} , $0 \leq j \leq r-1$. If $H_r = \text{Gal}(E_r/E_{r-1})$, distribution gives

$$\frac{1}{p} \sum_{h \in H_r} hR_r = p^{-n} R_{r-1}.$$

Thus (19.6) supplies a nonzero projection to the new block $\Phi_{p^{r-1}}$, and the averaging identity preserves every previously nonzero block. For even n , (19.5) starts this induction with a nonzero trivial wild block, giving a cyclic vector of the full regular module. For odd n , all nontrivial blocks occur, so the augmentation ideal maps injectively. Applying e_{χ_n} to a relation now proves the two asserted independence statements for the unprojected values themselves. Finally, for even n inversion is odd, excluding $\mathbf{Q}_p$; for odd n every displayed difference has zero Γ_r -average, which likewise excludes $\mathbf{Q}_p$. $\square$

Remark 19.7. The tame idempotent is used only on the regulator target. Actual coefficients may remain the root classes $b_n(\rho_r^{(1+p)^j})$ or their adjacent differences; no $\mathbf{Q}_p$ -linear projector is applied inside actual rational K -theory. With the common normalization (19.1), their usual local Chern values are the corresponding displayed values followed by multiplication by $\kappa_n \neq 0$ and the Bloch–Kato exponential. Since these operations are $\mathbf{Q}_p$ -linear isomorphisms, the actual root family has $\mathbf{Q}$ -span of dimension D in even weight, and the actual difference family has $\mathbf{Q}$ -span of dimension $D-1$ in odd weight: any rational relation would give a forbidden $\mathbf{Q}_p$ -linear relation of regulator values. The proposition detects one tame character and does not assert nonvanishing of all tame character components.

Corollary 19.8. *For an odd prime p and $r \geq 2$ put $D = p^{r-1}$ and*

$$A_{p,r} = \mathbf{Z}_p[[x, y, z]]/(xy + \Phi_{p^r}(1+z)).$$

For every $n \geq 2$, the rational generic kernel in degree $2n-1$ has dimension at least D when n is even and at least $D-1$ when n is odd. The classes are the pushforwards from $x=0$ of $b_n(\rho_r^{(1+p)^j})$ in even weight, or their consecutive differences in odd weight, after one common denominator is cleared. They vanish after inverting x integrally. In particular, fixing $r=2$ gives one complete regular local ring with residue field $\mathbf{F}_p$ for which rational generic injectivity fails in every odd degree at least three.

Proof. The polynomial is Eisenstein and the equation has nonzero linear term p , so the displayed ring is regular local of dimension three. Apply the proof of Theorem 19.4, with Proposition 19.6 in place of Proposition 19.3. The simultaneous ruling-cup detector in Lemma 19.1 kills exactly the diagonal coefficient vectors. The regulator spans just computed meet the constant field only in zero, so the same proof gives precisely these ranks and the stated integral supported representatives. $\square$

20. CONTINUOUS CYCLIC CHARACTERS AND EXPLICIT FAMILIES

This section supplies an independent proof for the explicit completed three-adic class, and a direct endpoint criterion for all the explicit prime-power families. The first proof uses an elementary convergent series. The second uses Coleman functions only as analytic primitives. The coefficient-cup argument at the end independently supplies actual local coefficients for existence and rank constructions.

20.1. A direct detector in continuous cyclic homology. For a spectrum M , write $M_{\mathbf{Q}_p} = M_p^\wedge[1/p]$. Thus $K(R; \mathbf{Q}_p)$ denotes a rationalized completed spectrum; it does not denote the tensor product of each actual K -group with $\mathbf{Q}_p$. Products of units below are products of actual Quillen K_1 -classes.

Let $W = \mathcal{O}_E$, where $E/\mathbf{Q}_p$ is finite, and let π be a uniformizer. If $\mathfrak{X} = \text{Spf } B$ is smooth and topologically of finite presentation over $\text{Spf } W$, put

$$\Omega_B^j = \widehat{\Omega}_{B/W}^j[1/p].$$

The hat denotes continuous differentials, completed before inverting p . Cohomology is ordinary cohomology in $D(E)$, with differential images taken without closure.

Lemma 20.1. *Suppose that $\mathfrak{X} = \mathrm{Spf} B$ is smooth, affine, topologically of finite presentation, and of relative dimension at most two over $\mathrm{Spf} W$. There is a natural relative map*

$$\chi_B : K(B, \pi B) \longrightarrow \Sigma HC^{\mathrm{cts}}(B/W; \mathbf{Q}_p).$$

For $v \in 1 + \pi B$ and $b_1, b_2 \in B^\times$, the top HKR component of its degree-three value is

$$(20.1) \quad \chi_B([v][b_1][b_2])_{\mathrm{top}} = \log(v) d \log b_1 \wedge d \log b_2 \quad \text{in } \Omega_B^2/d\Omega_B^1,$$

up to one universal sign determined by the norm and suspension conventions. Here $[v]$ has its relative lift furnished by its reduction to the identity unit.

Proof. The fiber square of [2, Proposition 4.10] has rows

$$\begin{array}{ccc} K^{\mathrm{cts}}(\mathfrak{X}; \mathbf{Q}_p) & \longrightarrow & K(B/\pi; \mathbf{Q}_p) \\ \downarrow & & \downarrow \\ TC^{\mathrm{cts}}(\mathfrak{X}; \mathbf{Q}_p) & \longrightarrow & TC(B/\pi; \mathbf{Q}_p) \\ \downarrow & & \downarrow \\ HC^{-, \mathrm{cts}}(B/W; \mathbf{Q}_p) & \longrightarrow & HP^{\mathrm{cts}}(B/W; \mathbf{Q}_p). \end{array}$$

The last horizontal fiber is $\Sigma HC^{\mathrm{cts}}(B/W; \mathbf{Q}_p)$, not negative cyclic homology. Mapping the actual relative spectrum to these fibers gives χ_B . Consequently relative K_3 has target HC_2 , and its top component is $\Omega_B^2/d\Omega_B^1$.

We record why the fiber comparison respects multiplication by absolute K -classes. Set $H = THH(B)$, a commutative algebra in cyclotomic spectra. If TC_{cyc} denotes the functor on that category, then $TC_{\mathrm{cyc}}(H) = TC(B)$; no second THH is intended. All objects $H \otimes D$ and maps $\mathrm{id}_H \otimes f$ in [2, Proposition 2.8, Corollary 2.10, Theorem 2.12] are H -modules and module maps. The fixed-point and Tate functors, their canonical maps, and the norm triangle preserve the induced $TC(B)$ -actions [38, Corollary I.4.3]. Thus the inverses of the equivalences used there are module inverses. The comparisons from derived to ordinary reduction and then to B/π are induced by B -algebra maps. The projection $HH(B) \rightarrow HH(B/W)$ in the proof of Proposition 4.10 also preserves the action. These inverses and fiber equivalences are used after p -completion and rationalization.

The multiplicative cyclotomic trace [9, Theorem 7.4] now gives $K(B)$ -linearity, with target action through the canonical map $K(B) \rightarrow HC^{-, \mathrm{cts}}(B/W; \mathbf{Q}_p)$. At the cyclic-complex level, the compatibility of the norm boundary and its suspension with the HC^- -module action is the Connes-sequence compatibility of [10, Proposition 2.3]. The identification of the spectral norm with the Connes operator is [28, Theorem 2.1, Lemma 2.2, Theorem 2.3]; this includes the usual graded sign of a connecting map. In particular, no assumption that an unspecified additive fiber equivalence is multiplicative is being made.

To normalize a relative unit, first take $B = W\langle T \rangle$ and $v = 1 + \pi T$. Its relative character is a function $f \in HC_0^{\mathrm{cts}}(B/W; \mathbf{Q}_p) = E\langle T \rangle$. Choose the norm convention in which its image in HC_1^- is df . The left vertical arrow in [2, Theorem 2.12] is canonical; its projection to Hochschild homology takes a unit to the Dennis trace $v^{-1} \otimes v$. Hence $df = d \log v$. Evaluation at $T = 0$ gives $f(0) = 0$. Continuous differentiation on $E\langle T \rangle$ has only constants in its kernel, and therefore

$$f = \log(1 + \pi T) = \sum_{n \geq 1} (-1)^{n+1} \pi^n T^n / n.$$

The series converges beyond the closed unit disc. Substitution $T = (v - 1)/\pi$ proves the assertion for every relative unit under consideration.

Finally use the normalized cyclic complex $C^+ = E[t, t^{-1}]/tE[t] \otimes C_*$, with $|t| = -2$. The $HC^- \times HC$ action is shuffle plus t times cyclic shuffle [10, Definition 2.1 and Remark 2.2]. Starting with the column-zero representative $\log v$, all positive columns and cyclic-shuffle

corrections vanish in C^+ . After each of the two unit multiplications only the ordinary shuffle remains. Under

$$b_0 \otimes \cdots \otimes b_n \mapsto b_0 db_1 \wedge \cdots \wedge db_n / n!,$$

the two shuffles in degree two cancel the factor $2!$, giving (20.1), with the suspension signs just specified. This calculation involves only degrees through two and commutes with the continuous, relative-to- W projection. It does not assume a multiplicative splitting of an infinite HKR filtration. $\square$

Lemma 20.2. *Let $a_0, a_1 \in W$ have unit difference, and put*

$$B = W\langle X, Y, Z \rangle / (XY + (Z - a_0)(Z - a_1)).$$

For $F \in E\langle Z \rangle$, the form $\omega_F = (dX/X) \wedge dF$ is globally regular. If it is exact in $\Omega_B^2/d\Omega_B^1$, then $F(a_0) = F(a_1)$.

Proof. The two root-deletion affine opens have completed coordinates

$$Z = a_i + Xt_i, \quad Y = -t_i(a_i - a_{1-i} + Xt_i) \quad (i = 0, 1).$$

They are the completions of $D(X) \cup D(Z - a_{1-i})$, cover the formal scheme, and have intersection $W\langle X, X^{-1}, Z \rangle$. The forms

$$q_i = -(F(Z) - F(a_i)) dX/X$$

are regular on the respective bidiscs, satisfy $dq_i = \omega_F$, and have difference $(F(a_1) - F(a_0))dX/X$.

A closed bidisc form $A(X, t)dX + B(X, t)dt$ has zero Laurent residue after $t = c/X$, for every $|c| \leq 1$. Indeed its residue is the convergent series

$$\sum_{m \geq 0} (a_{m, m+1} - b_{m+1, m}) c^{m+1} = 0,$$

because closedness equates $(m+1)a_{m, m+1}$ and $(m+1)b_{m+1, m}$. Division by $m+1$ is used only in this individual coefficient equality, not to construct a bounded primitive. If $\omega_F = dq$ globally, apply this to $q_i - q$ after restricting the overlap to $Z = 0$, where $c = -a_i$. The difference then has residue zero, which is the required endpoint equality. This argument includes $c = 0$ and works over the residue field $\mathbf{F}_2$. $\square$

Proposition 20.3. *Let $A = \mathbf{Z}_3[[x, y, z]]/(3 + xy + z^2)$. The actual element*

$$c = 2 \left\{ \frac{1+z}{2}, 1+x, 1 + \frac{(1+x)y}{4} \right\} \in K_3(A)$$

has infinite order and has zero image in $K_3(A[1/x])$. The ring A is regular local of dimension three.

Proof. This is the completed case of Theorem 2.1. The proof there constructs the relative class on the smooth formal quadric, uses the actual vanishing $K_4(C) = 0$ to pass to absolute K -theory, and detects the resulting class by an endpoint difference in the ordinary continuous de Rham quotient. $\square$

The finite continuous HKR calculation. For the three-adic quadric, put

$$B_0 = W[X, Y, Z]/(XY + Z^2 - 1), \quad B = \widehat{B}_{03}, \quad \Omega_B^i = \widehat{\Omega}_{B/W}^i[1/3].$$

The algebra B_0 is smooth over W of relative dimension two. The normalized HKR map on its Hochschild complex is

$$b_0 \otimes \cdots \otimes b_n \mapsto \frac{1}{n!} b_0 db_1 \wedge \cdots \wedge db_n \quad (0 \leq n \leq 2),$$

and is zero in higher degrees. It is defined over W , since 2 is a unit, and intertwines the Hochschild differential with zero and the Connes operator with d . Smoothness makes it a quasi-isomorphism of mixed complexes: the Hochschild homology is $\Omega_{B_0/W}^n$ for $0 \leq n \leq 2$ and vanishes above degree two. The comparison with the continuous invariant is supplied by [2, Proposition 4.2]. In the cyclic, negative cyclic, and periodic total complexes of this finite mixed complex, each total degree has only finitely many differential-form terms. Their reduction

towers are surjective, so derived three-adic completion is computed termwise. After inverting 3, this gives in particular

$$HC_0^{\text{cts}}(B/W; \mathbf{Q}_3) = \Omega_B^0, \quad HC_2^{\text{cts}}(B/W; \mathbf{Q}_3) = \frac{\Omega_B^2}{d\Omega_B^1} \oplus \ker(d : \Omega_B^0 \rightarrow \Omega_B^1).$$

The norm map from the degree-zero cyclic term to the degree-one negative cyclic term is $f \mapsto df$, with the chosen global suspension sign. In particular, the displayed quotient is the ordinary quotient by the image of d ; its image is not replaced by its closure. The same calculation in relative dimension one applies to the universal unit in $W\langle T \rangle$.

The ordinary quotient is essential. The target in [2, Constructions 4.6 and 4.11] uses the ordinary derived category. For example, in the preceding quadric put $\omega_0 = (dX/X) \wedge dZ$. The globally regular primitives $-(Z - Z^{3^j})dX/X$ satisfy

$$d\left(-(Z - Z^{3^j})\frac{dX}{X}\right) = \omega_0 - 3^j Z^{3^j-1} \omega_0 \longrightarrow \omega_0.$$

The ruling form is therefore in the closure of exact forms although the residue lemma shows it is not exact. No assertion about Hausdorff-reduced cohomology or the order at each finite Artin stage is made here.

20.1.1. The same computation for the explicit families. For odd p and $r \geq 1$, put $M = p^r$, $n = 2M$, and

$$A_{p,r} = \mathbf{Z}_p[[x, y, z]]/(xy + \Phi_{2M}(z - 1)), \quad u = z - 1, \quad P(u) = \Phi_{2M}(u).$$

For $p = 2$, take $r \geq 2$, $M = 2^r$, $n = 3M$, and

$$A_{2,r} = \mathbf{Z}_2[[x, y, w]]/(xy + \Phi_M(1 + w)), \quad u = 1 + w, \quad P(u) = \Phi_M(u).$$

Let $H(u) = (u^n - 1)/P(u)$, $b = yH(u)$, $\lambda = 1 - x$, and $\mu = 1 + \lambda b$. For positive odd k prime to p , put

$$g_k = \begin{cases} 1 - u^k, & p \text{ odd}, \\ 1 + u^k + u^{2k}, & p = 2, \end{cases} \quad c_k = k\{g_k, \lambda, \mu\} \in K_3(A_{p,r}).$$

Every displayed entry is a unit. We use Coleman Li_2 with its series at zero and $\log p = 0$, and at roots set

$$D_k(a) = \begin{cases} \text{Li}_2(a^k), & p \text{ odd}, \\ \frac{1}{3} \text{Li}_2(a^{3k}) - \text{Li}_2(a^k), & p = 2. \end{cases}$$

Proposition 20.4. *The classes c_k vanish integrally after inverting x . If a rational relation $\sum_k q_k c_k = 0$ holds in $K_3(A_{p,r}) \otimes \mathbf{Q}$, then*

$$\sum_k q_k D_k(a) = 0$$

for every root a of P .

Proof. Write $s = 1 - xb = u^n$. The Dennis–Stein identity gives $\delta = \langle x, b \rangle = \{\lambda s, \mu\}$. At odd p , Steinberg applied to u^k gives $k\{g_k, s\} = 0$. At two, apply Steinberg to $\zeta_3 u^k$ after the finite étale extension adjoining ζ_3 and transfer: since $3 \mid n$, the second entry comes from u^{kn} and the norm of the first is exactly g_k . Thus the same relation holds, without an extra factor two. It follows that $c_k = k[g_k]\delta$, and the unit expression for δ on $D(x)$ makes its generic image zero.

Fix distinct roots a, a' in $E = \mathbf{Q}_p(\mu_M)$, set $W = \mathcal{O}_E$, $d = a' - a$, and $C(u) = P(u)/((u - a)(u - a'))$. There is an actual convergent map to the smooth good reduction model

$$B = W\langle X, Y, Z \rangle / (XY + Z(Z + 1)), \quad u \mapsto a - dZ, \quad x \mapsto dX, \quad y \mapsto dC(a - dZ)Y.$$

All roots reduce to -1 at odd p and to 1 at two, so d and the images of the original formal parameters are topologically nilpotent. No inverse of C is used. The special fiber is again an affine-line torsor over $\mathbf{P}^1$, so its K_3 and K_4 vanish after p -completion and rationalization.

Set $m = p - 1$ at odd p and $m = 1$ at two. Then $g_k^m \in 1 + \pi B$, and $k[g_k^m][\lambda][\mu]$ is an actual relative lift of mc_k . Let $L_k = \log g_k$ and choose F_k with $dF_k = L_k d \log s = nL_k du/u$. These functions converge beyond the closed Z -disc. The differential identity

$$d \log \lambda \wedge d \log \mu = d \log x \wedge d \log s - d \log s \wedge d \log \mu = -dx \wedge db/\mu$$

gives the following global representative of the relative character:

$$\tau_k = mk \frac{dX}{X} \wedge dF_k - d(mkF_k d \log \mu).$$

At odd p , u^k lies in the nonsingular residue disc of -1 ; the analytic differential equation gives $F_k = -(n/k) \text{Li}_2(u^k) + \text{constant}$. At two, use the rigid analytic sum

$$\text{Li}_2(\zeta_3 u^k) + \text{Li}_2(\zeta_3^2 u^k).$$

Its two arguments lie in nonsingular residue discs, its derivative is $-kL_k du/u$, and it descends from the unramified quadratic extension. Distribution gives its endpoint values $D_k(a)$. These are only identities of analytic functions; the necessary analyticity, distribution and inversion are [6, Theorem 2.7(1), Proposition 4.4, Proposition 2.10]. No assertion of analyticity for the individual Li_2 terms across the residue disc of 1 is used at two.

Lemma 20.2, with roots 0, -1 , now computes the endpoint obstruction of mc_k as

$$\pm mn(D_k(a') - D_k(a)).$$

The prefactor k has canceled the $1/k$ in the primitive, leaving one common scalar. A rational relation therefore makes $\sum_k q_k D_k(a)$ independent of a . Inversion sends roots to roots and changes this value's sign, since logarithms vanish at roots of unity. The common value must consequently be zero. $\square$

Corollary 20.5. *Using the purely analytic wild-orbit independence statement of Proposition 17.3, choose odd representatives $k_j \equiv (1 + p)^j \pmod{p^r}$ for $0 \leq j < p^{r-1}$ at odd p , and $k_j \equiv 5^j \pmod{2^r}$ for $0 \leq j < 2^{r-2}$ at two. The corresponding actual classes c_{k_j} are rationally independent in $K_3(A_{p,r})$ and have integral generic image zero. The same holds over the polynomial quotients with coefficient ring $\mathbf{Z}_{(p)}$, localized at the ideal generated by p and the displayed formal parameters, and over their henselizations.*

Proof. Apply Proposition 20.4 at one fixed primitive root and then the stated analytic independence. A rational relation in a preceding local ring or its henselization would map to one in the completion. The generic-zero Dennis–Stein relation holds already in those local rings. $\square$

21. FINITE MIXED HKR AND CLASSES IN EVERY HIGHER DEGREE

Lemma 21.1. *Let W be a complete mixed-characteristic DVR with perfect residue field, and let B be the p -adic completion of a smooth affine W -algebra B_0 of relative dimension $D < \infty$. The continuous cyclic, negative-cyclic, and periodic complexes of B/W , completed before inverting p , are computed by the finite mixed complex of continuous forms $(\Omega_B^{\leq D}, 0, d)$. In particular the highest form component of HC_D^{cts} is $\Omega_B^D/d\Omega_B^{D-1}$. For $v \in 1 + \pi B$ and $b_1, \dots, b_D \in B^\times$, the character of Lemma 20.1 satisfies*

$$\chi_B([v]_{\text{rel}}[b_1] \cdots [b_D])_{\text{top}} = \pm \log(v) d \log b_1 \wedge \cdots \wedge d \log b_D.$$

The sign is the same norm and suspension convention as before.

Proof. An integral comparison suffices to control all completion operations. In Hochschild degree $j \leq D$ use the map

$$a_0 \otimes \cdots \otimes a_j \mapsto \frac{D!}{j!} a_0 da_1 \wedge \cdots \wedge da_j,$$

and use zero in higher degrees. It intertwines b with zero and the Connes operator with d . Smooth HKR identifies its map on every nonzero Hochschild homology group with multiplication by $D!$. The cone thus has homology in degrees $0, \dots, D$, all killed by p^a , where $a = v_p(D!)$; flatness of the form modules is used here.

This error can be killed before taking any circle construction. Its Postnikov tower lies in the category $\text{Fun}(BS^1, D(W))$ of complexes with circle action. Each graded piece has trivial circle action: the automorphism space of a module in the heart is discrete, while BS^1 is simply connected. Hence p^a kills that piece in the equivariant category. In a triangle $A \rightarrow M \rightarrow C$, if p^r kills A and p^s kills C , then $p^s \text{id}_M$ factors through A , so p^{r+s} kills M . Induction on the finite Postnikov tower shows that $p^{a(D+1)}$ kills the whole equivariant cone. The exact orbit, fixed-point, Tate, and derived completion functors preserve this null map. Their comparison errors therefore vanish after inverting p .

In every total degree the cyclic complexes of the finite form model contain only finitely many form modules. These modules are W -flat and their reduction towers are surjective. Derived completion is consequently termwise completion, also in the unbounded total complexes: the cone-of-1 – shift model for the inverse limit is degreewise exact for these surjections. Proposition 4.2 of [2], including its relative version, identifies this with the asserted continuous invariant. No exchange of rationalization with an uncontrolled infinite product is used.

The module comparison and the one-variable normalization of a relative unit in Lemma 20.1 do not depend on the relative dimension. Its ordinary cyclic representative $\log v$ starts in column zero. At every unit multiplication the positive columns and the extra t in the cyclic-shuffle correction vanish in C^+ . Only the ordinary shuffles of the lowest Dennis unit traces remain. Their $D!$ permutations cancel the $1/D!$ normalized HKR factor, proving the formula. $\square$

21.1. The comparison for completed essentially smooth models.

Lemma 21.2. *Let $W_0 = W(\overline{\mathbf{F}}_p)$ and let W be a finite totally ramified extension, with uniformizer π and fraction field E . Let B_0 be a localization of a smooth W -algebra, with $\Omega_{B_0/W}^1$ finite projective of constant rank D , and put $B = \widehat{B}_{0,p}$. There is a natural actual relative character*

$$\chi_B : K(B, \pi B) \longrightarrow \Sigma HC^{\text{cts}}(B/W; \mathbf{Q}_p),$$

which factors through an equivalence from $K(B, \pi B)_p^\wedge[1/p]$. It is $K(B)$ -linear through the negative-cyclic character, and the target has the finite continuous mixed-HKR model, with ordinary de Rham quotients.

Proof. We justify the extension beyond topological finite presentation. For each r , the ring W_0/p^r is the filtered union of the finite étale $\mathbf{Z}/p^r$ -algebras $W_r(\mathbf{F}_{p^m})$. Flat base change for cotangent complexes therefore gives $\widehat{L}_{W_0/\mathbf{Z},p} = 0$. Write $W = W_0[\pi]$ and let f be the Eisenstein polynomial of π . Then

$$\widehat{L}_{W/\mathbf{Z},p} \simeq [W \xrightarrow{f'(\pi)} W]$$

in cohomological degrees $-1, 0$. Choose a with $p^a \in f'(\pi)W$; multiplication by p^a on this complex is null-homotopic.

Cotangent transitivity, after derived p -completion, gives

$$M \longrightarrow \widehat{L}_{B_0/\mathbf{Z},p} \longrightarrow P, \quad M = [B \xrightarrow{f'(\pi)} B], \quad P = \widehat{\Omega}_{B_0/W}^1.$$

The target P is finite projective in degree zero, so this triangle splits as a triangle of underlying B -module complexes. The error in the i th derived exterior power of its projection is a sum of terms $(\mathbf{L}\Lambda^r M) \otimes \Lambda^{i-r} P$, $1 \leq r \leq i$. The r th such term is killed by p^{ar} as an object: derived exterior power sends the null-homotopy of $p^a \text{id}_M$ to one for p^{ar} .

The connective HKR filtration on Hochschild homology has graded terms $(\mathbf{L}\Lambda^i L)[i]$. Any fixed homotopy range involves only finitely many of these terms, even after completion. Indeed derived p -completion preserves connectivity: for a c -connective spectrum X , its only possible new bottom group is $\lim_r^1(\pi_c X/p^r) = 0$, because the quotient tower is surjective. Thus the increasingly connective HKR tails remain increasingly connective. The preceding bounds and a finite extension argument therefore make

$$HH(B_0; \mathbf{Z}_p) \longrightarrow HH(B_0/W; \mathbf{Z}_p)$$

a quasi-isogeny in each finite range. Its rationalization is an equivalence. The bounds use the base different and the range; they are unaffected by the localization set. No interchange of an infinite localization colimit with completion is used.

To pass to cyclic homology, we control the comparison cone in the equivariant derived category. Let C be the cone of the map of completed Hochschild complexes. For every integer N , the preceding argument gives an integer a_N killing each of the finitely many nonzero homotopy modules of $\tau_{\leq N}C$. The pointwise t -structure on $\text{Fun}(BS^1, D(\mathbf{Z}_p))$ gives its equivariant Postnikov tower. Each heart object has trivial circle action: the automorphism space of a module in the heart is discrete and BS^1 is simply connected. A finite extension argument therefore gives an integer b_N such that p^{b_N} annihilates $\tau_{\leq N}C$ as an equivariant object, not merely on its underlying homotopy groups.

Taking homotopy orbits preserves this scalar null-homotopy. The omitted Postnikov tail remains $(N+1)$ -connective after homotopy orbits and derived p -completion. Indeed, homotopy orbits of a connective spectrum are connective, and in derived p -completion the potentially new bottom $\lim^1$ term is zero because the bottom coefficient tower is the surjective tower $\pi_{N+1}(-)/p^r$. Thus the cone of the corresponding map of completed cyclic complexes has bounded p -torsion in every fixed degree, and vanishes after inverting p .

Equivalently, one can take a sufficiently high finite bar skeleton for the circle-orbit construction in each degree. Its building blocks are finite colimits of $(S^1)_+^j \wedge X$, so the comparison commutes with derived completion. This argument requires no uniform bound b_N as N varies and no interchange of rationalization with an infinite product. The norm fiber sequence then makes the square between absolute and relative $HC^- \rightarrow HP$ cartesian. It does not require that the two vertical maps of that square be equivalences individually.

By [2, Proposition 4.2], including the relative variant stated just before Proposition 4.10, both B_0 and B compute the same continuous p -completed HH and cyclic theories. The ordinary ring B is henselian along (p) , so [2, Theorem A] applies. The map $B/p \rightarrow B/\pi$ has nilpotent kernel, and [2, Corollary 3.8] identifies its K-theory and TC after rationalized p -completion. Composing the actual-to-completed map on relative K-theory with the resulting fiber equivalence gives χ_B .

The module argument is unchanged: for $H = THH(B)$, the objects $H \otimes D$ and maps $\text{id}_H \otimes f$ in [2, Proposition 2.8, Corollary 2.10, and Theorem 2.12] form a diagram of H -modules in cyclotomic spectra. The lax monoidal TC and circle functors give module diagrams, their equivalences can be inverted there, and the relative HH projection preserves the action. Together with the multiplicative cyclotomic trace this gives the asserted actual $K(B)$ -linearity. This construction has no finite presentation requirement.

Finally, the integral mixed HKR map in degree $j \leq D$ has coefficient $D!/j!$ and induces multiplication by $D!$ on $HH_j(B_0/W) = \Omega_{B_0/W}^j$. Its equivariant cone has a Postnikov tower of length $D+1$ whose graded objects are killed by $p^{v_p(D!)}$. These heart objects have trivial circle action; the extension argument kills the entire equivariant cone by $p^{(D+1)v_p(D!)}$. Thus circle orbits, fixed points, Tate constructions, and derived completion retain only bounded torsion error. The rational mixed-HKR comparison follows. Each total degree has finitely many flat form modules, with surjective reduction towers, so completion is termwise and cohomology uses the ordinary image of d . $\square$

21.2. Continuous residues and higher unit products.

Lemma 21.3. *Let W be a complete DVR of mixed characteristic with uniformizer π and fraction field E . Let B be a smooth formal W -algebra of topologically finite type. Take a nonzero $g \in W[T_1, \dots, T_s]$ whose nonzero coefficients are units, and set*

$$B_g = (B[T_1^{\pm 1}, \dots, T_s^{\pm 1}, 1/g])_{\pi}^{\wedge}.$$

There are E -linear maps on continuous forms, with base forms placed before the logarithmic differentials,

$$\text{Res}_s : \Omega_{B_g[1/p]/E, \text{cts}}^{q+s} \longrightarrow \Omega_{B[1/p]/E, \text{cts}}^q,$$

such that

$$\mathrm{Res}_s(d\eta) = d\mathrm{Res}_s(\eta), \quad \mathrm{Res}_s(\omega \wedge d\log T_1 \wedge \cdots \wedge d\log T_s) = \omega.$$

Consequently wedging with these logarithmic differentials gives a split injection on ordinary continuous de Rham cohomology. There is also a degree-zero projection P_0 which takes the constant coefficient of the summand containing no $d\log T_i$. It commutes with d and is the identity on continuous base forms.

Proof. Put $R_0 = W[T_1^{\pm 1}, \dots, T_s^{\pm 1}, 1/g]$. There is an integral formal expansion

$$R_0 \longrightarrow W((T_1)) \cdots ((T_s)).$$

To construct it, write $g = T_s^a(g_a + T_s g_{a+1} + \cdots)$ with a minimal. By induction g_a is a unit in the preceding iterated Laurent ring: the recursion ends at one of the unit coefficients of g . Expand the remaining inverse as a formal geometric series in T_s . This construction divides by no power of π . Taking the coefficient of T_s^0 , then T_{s-1}^0 , and so on, defines a W -linear map

$$\mathrm{CT} : R_0 \longrightarrow W, \quad \mathrm{CT}(1) = 1, \quad \mathrm{CT}(T_i \partial_{T_i} h) = 0.$$

It is uniformly π -adically continuous, since $\mathrm{CT}(\pi^a R_0) \subseteq \pi^a W$ for every a . The B -linear extension therefore extends uniquely to $B_g \rightarrow B$. The same construction works for every finite projective module of continuous base forms, by taking a direct summand of a finite free module.

Decompose forms into base forms and the ordered $d\log T_i$. Retain only the summand containing all s logarithmic factors, and apply this extended constant-term map to its coefficient. This defines Res_s integrally before inverting p . The base differential commutes with constant-term extraction; every term from a Laurent-variable derivative has zero constant term. The two displayed identities follow, first on the dense uncompleted modules and then by continuity. They show directly that boundaries map to boundaries and that the stated composite is the identity on forms and cohomology. For P_0 , a Laurent-variable derivative always introduces a logarithmic factor and is discarded, proving its chain identity by the same calculation.

No embedding of the completed ring into the ordinary Laurent ring is asserted. For example $\sum_{a \geq 0} \pi^a T_s^{-a}$ has no lower bound on its Laurent exponents. If a completed series target is desired, it is $\varprojlim_a (B/\pi^a)((T_1)) \cdots ((T_s))$. Nor need the formal expansion of $1/g$ converge analytically when every T_i has norm one. $\square$

Lemma 21.4. *Let X be a smooth variety of dimension d over a perfect field of characteristic p . For every $j > d$,*

$$\pi_j K(X)_p^\wedge = \pi_j K(X; \mathbf{Q}_p) = 0.$$

In particular, if $s \geq 1$, $0 \neq \bar{g} \in \mathbf{F}_3[T_1^{\pm 1}, \dots, T_s^{\pm 1}]$, and

$$C_s = \mathrm{Spec}(\mathbf{F}_3[X, Y, Z, T_1^{\pm 1}, \dots, T_s^{\pm 1}, 1/\bar{g}]/(XY + Z^2 - 1)),$$

then

$$\pi_{s+3} K(C_s; \mathbf{Q}_3) = \pi_{s+4} K(C_s; \mathbf{Q}_3) = 0.$$

Proof. Geisser–Levine prove

$$K_j(X; \mathbf{Z}/p^a) = 0 \quad (a \geq 1, j > d)$$

in [25, Theorem 8.4, p. 491]. These are the finite-coefficient K -groups, with the usual coefficient long exact sequence; see their Section 2, p. 464. In particular the assertion includes the torsion contribution from the preceding actual K -group. The homotopy-limit exact sequence is

$$0 \longrightarrow \varprojlim_a^1 K_{j+1}(X; \mathbf{Z}/p^a) \longrightarrow \pi_j K(X)_p^\wedge \longrightarrow \varprojlim_a K_j(X; \mathbf{Z}/p^a) \longrightarrow 0.$$

Both outside terms vanish when $j > d$. Inverting p proves the first assertion. The displayed C_s is smooth of dimension $s + 2$, so its two asserted groups lie in this range. $\square$

Theorem 21.5. *Put*

$$A_0 = \left(\mathbf{Z}_{(3)}[x, y, z] / (3 + xy + z^2) \right)_{(x, y, z)}, \quad \mathfrak{m}_0 = (x, y, z)A_0.$$

For $s \geq 0$, let

$$R_s = A_0[T_1, \dots, T_s]_{\mathfrak{m}_0 A_0[T_1, \dots, T_s]}.$$

This is a regular local ring of dimension three and residue field $\mathbf{F}_3(T_1, \dots, T_s)$. The actual class

$$c_s = 2 \left\{ \frac{1+z}{2}, 1+x, 1 + \frac{(1+x)y}{4}, T_1, \dots, T_s \right\} \in K_{s+3}(R_s)$$

has infinite order and restricts to zero in $K_{s+3}(R_s[1/x])$ integrally. Consequently rational Gersten injectivity fails in each degree $j \geq 3$ on an explicitly specified ring of dimension three.

Proof. Polynomial extension and localization preserve regularity; the chosen prime has height three and the displayed residue field. Every T_i is a unit. The Dennis–Stein calculation of Proposition 20.3, followed by multiplication by these units, proves integral vanishing after inverting x . The case $s = 0$ has already been proved.

Suppose that $Nc_s = 0$ for some positive integer N and $s \geq 1$. Continuity of actual K -theory makes this relation hold over $A_0[T_i^{\pm 1}, 1/\bar{f}]$ for one finite denominator with $\bar{f} \neq 0$ in $\mathbf{F}_3[T_1, \dots, T_s]$. Use $W = \mathbf{Z}_3[u]/(u^2 + 3)$ and the original good quadric $B = W\langle X, Y, Z \rangle / (XY + Z^2 - 1)$. Choose $g \in W[T_1, \dots, T_s]$ by lifting exactly the nonzero coefficients of $\bar{f}$ to units, and form B_g as in Lemma 21.3. Under $(x, y, z) = (uX, uY, uZ)$, the image of f is congruent to g modulo u . Thus $f/g \in 1 + uB_g$ is a unit, and the finite relation has an actual image in $K_{s+3}(B_g)$.

The specified image has the actual relative lift

$$\xi_s = [a^2]_{\text{rel}}[\lambda][\mu][T_1] \cdots [T_s].$$

The formal algebra B_g has relative dimension $D = s + 2$ over W . Lemma 21.1 gives its top character as

$$\chi_{B_g}(\xi_s)_{\text{top}} = \pm \tau_c \wedge d \log T_1 \wedge \cdots \wedge d \log T_s,$$

where $[\tau_c] \neq 0$ was calculated in Proposition 20.3. The residue of Lemma 21.3 sends this class back to $\pm[\tau_c]$, so it is nonzero in the ordinary quotient by exact forms.

By Lemma 21.4, $\pi_{s+4}K(B_g/u; \mathbf{Q}_3) = 0$. The long exact sequence of the completed relative spectrum therefore makes

$$\pi_{s+3}K(B_g, uB_g; \mathbf{Q}_3) \longrightarrow \pi_{s+3}K(B_g; \mathbf{Q}_3)$$

injective. The nonzero character of ξ_s shows that its absolute image has infinite order, contrary to the image of $Nc_s = 0$. This proves the assertion for the actual class in R_s . Only the auxiliary special fiber is subject to the finite-coefficient vanishing theorem; its perfect ground field is $\mathbf{F}_3$ even though the source ring has the stated imperfect residue field. $\square$

22. COEFFICIENT PRODUCTS AND INFINITE RANK

The continuous cyclic character also detects products of actual point classes with a ruling class. This gives an independent construction of the coefficient classes used in the rank arguments.

22.1. Coefficient products in cyclic homology. Write $K(R; \mathbf{Q}_p) = K(R)_p^\wedge[1/p]$; this notation denotes a rationalized completed spectrum. Let $E/\mathbf{Q}_p$ be finite, let $W = \mathcal{O}_E$, and let k be its residue field. By [2, Proposition 4.10 and Example 4.3], the point has a natural relative character

$$r_W : K_3(W) \longrightarrow \pi_3 K(W; \mathbf{Q}_p) \xrightarrow{\sim} HC_2^{\text{cts}}(W/W; \mathbf{Q}_p) = E.$$

The isomorphism uses $\pi_3 K(k; \mathbf{Q}_p) = \pi_4 K(k; \mathbf{Q}_p) = 0$.

Lemma 22.1. *Let $E/\mathbf{Q}_p$ be finite, let $W = \mathcal{O}_E$, and let $n \geq 2$ and $s = 2n - 1$. Write*

$$K(R)_p^\wedge = \text{holim}_\nu K(R)/p^\nu, \quad K(R; \mathbf{Q}_p) = K(R)_p^\wedge[1/p].$$

There is a natural $\mathbf{Q}_p$ -linear map

$$\tilde{c}_{n,E} : \pi_s K(W; \mathbf{Q}_p) \longrightarrow H^1(E, \mathbf{Q}_p(n))$$

such that the usual étale Chern class on actual K -theory is the composite

$$K_s(W)_{\mathbf{Q}} \longrightarrow \pi_s K(W; \mathbf{Q}_p) \xrightarrow{\tilde{c}_{n,E}} H^1(E, \mathbf{Q}_p(n)).$$

These maps are natural under finite extensions of p -adic fields. Consequently, actual classes with $\mathbf{Q}_p$ -linearly independent étale Chern values have $\mathbf{Q}_p$ -linearly independent images in $\pi_s K(W; \mathbf{Q}_p)$.

Proof. First construct the factor over the field E , where p is invertible. For every $\nu \geq 1$ the finite-coefficient Chern class is an additive map

$$c_{n,1}^{(\nu)} : K_s(E; \mathbf{Z}/p^\nu) \longrightarrow H^1(E, \mu_{p^\nu}^{\otimes n}).$$

This includes $p = 2$: the exceptional K -degree in [49, Proposition 2.4, p. 10] is two, whereas $s \geq 3$. The maps are compatible with $p^{\nu+1} \rightarrow p^\nu$ reduction. Indeed, take $q = p$ and $r = p^\nu$ in the middle square of [49, Proposition 2.8, p. 12]:

$$\begin{array}{ccc} K_s(E; \mathbf{Z}/p^{\nu+1}) & \longrightarrow & K_s(E; \mathbf{Z}/p^\nu) \\ c_{n,1}^{(\nu+1)} \downarrow & & \downarrow c_{n,1}^{(\nu)} \\ H^1(E, \mu_{p^{\nu+1}}^{\otimes n}) & \longrightarrow & H^1(E, \mu_{p^\nu}^{\otimes n}). \end{array}$$

The homotopy-limit projections therefore define a homomorphism

$$(22.1) \quad \pi_s K(E)_p^\wedge \longrightarrow \varprojlim_\nu K_s(E; \mathbf{Z}/p^\nu) \xrightarrow{\varprojlim_\nu c_{n,1}^{(\nu)}} \varprojlim_\nu H^1(E, \mu_{p^\nu}^{\otimes n}).$$

No assertion about a source $\varprojlim^1$ term is required: (22.1) uses only the canonical projections. On the target, the groups $H^0(E, \mu_{p^\nu}^{\otimes n})$ are finite. Their inverse system satisfies the Mittag-Leffler condition, so [30, equation (0.2), p. 208] gives

$$\varprojlim_\nu H^1(E, \mu_{p^\nu}^{\otimes n}) \cong H_{\text{cont}}^1(E, \mathbf{Z}_p(n)).$$

The resulting map is $\mathbf{Z}_p$ -linear. For clarity, the source homotopy group has its natural $\mathbf{Z}_p = \pi_0 \mathbb{S}_p^\wedge$ action. Each composite to the ν th cohomology group is additive and has target annihilated by p^ν . If $\lambda \in \mathbf{Z}_p$ and $\lambda = a + p^\nu b$ with $a \in \mathbf{Z}$ and $b \in \mathbf{Z}_p$, that composite sends λx to a times the image of x , which is also λ times that image. Passing to the inverse limit proves linearity. Inverting p and using the definition of rational continuous cohomology [30, equation (0.4)] gives

$$\bar{c}_{n,E} : \pi_s K(E; \mathbf{Q}_p) \longrightarrow H^1(E, \mathbf{Q}_p(n)).$$

Define $\tilde{c}_{n,E}$ by preceding this map with the natural map $\pi_s K(W; \mathbf{Q}_p) \rightarrow \pi_s K(E; \mathbf{Q}_p)$.

By [49, Lemma 2.3, p. 10], at each finite level the usual Chern class on $K_s(E)$ is the finite-coefficient class after reduction. The target inverse-limit identification then gives the asserted equality on actual classes. Naturality follows from [49, Proposition 2.1.1] and the naturality of every map above. Finally, applying the $\mathbf{Q}_p$ -linear map $\tilde{c}_{n,E}$ to a relation proves the independence assertion. $\square$

Remark 22.2. This lemma does not assert that actual $K_s(W)$ surjects onto the homotopy group of its completed spectrum. It also does not identify any cyclic or syntomic coordinate with a polylogarithm. If an independently constructed $\mathbf{Q}_p$ -linear isomorphism $\pi_s K(W; \mathbf{Q}_p) \cong E$ is available, the lemma only transports independence to that coordinate.

Lemma 22.3. *The values of r_W on actual classes span E over $\mathbf{Q}_p$. For a quadratic extension E/F , actual integral anti-invariant classes span E^- under this map.*

Proof. Let $d = [E : \mathbf{Q}_p]$ and put $V = \pi_3 K(\mathcal{O}_E; \mathbf{Q}_p)$. The point comparison of [2, Proposition 4.10] identifies V with E , so $\dim_{\mathbf{Q}_p} V = d$. Independently, the Bloch–Kato exponential identifies $H^1(E, \mathbf{Q}_p(2))$ with E , also of dimension d . The finite-coefficient Chern construction gives a natural linear map

$$\tilde{c}_{2,E} : V \longrightarrow H^1(E, \mathbf{Q}_p(2))$$

through which the Chern map on actual $K_3(\mathcal{O}_E)$ factors. This is the case $n = 2$ of Lemma 22.1; its proof uses only finite-coefficient Chern additivity, coefficient compatibility, and the canonical map from the homotopy limit to the limit of homotopy groups.

By Lemma 16.4, whose proof uses Levine's finite-coefficient theorem, Milnor K_3 -divisibility, the Moore–Merkurjev description of $K_2(E)$, and localization, the actual Chern values span $H^1(E, \mathbf{Q}_p(2))$. Let V_{act} be the span in V of the images of actual classes. Then $\tilde{c}_{2,E}(V_{\text{act}}) = H^1(E, \mathbf{Q}_p(2))$. Both ambient vector spaces have dimension d , whence $V_{\text{act}} = V$ and $\tilde{c}_{2,E}$ is an isomorphism. Thus the actual AMMN point values span E .

All maps are natural for finite extensions of local fields. For a quadratic E/F , apply $1 - \sigma$ to actual integral classes. Their images span E^- since $(1 - \sigma)E = E^-$ as $\mathbf{Q}_p$ -vector spaces. This argument divides by no integer inside actual integral K -theory, and applies equally at $p = 2$. $\square$

Proposition 22.4. *Let $0 \neq d \in \mathfrak{m}_W$, and set*

$$T = (W[[x, y, w]]/(xy + w(w + d)))[1/p], \quad J = (x, w)\mathcal{O}_T.$$

If $\beta \in K_3(W)$ satisfies $r_W(\beta) \neq 0$, then $\beta(1 - [J])$ has infinite order in $K_3(T)$. Actual coefficients with $\mathbf{Q}_p$ -independent r_W -values give rationally independent products.

Proof. Put

$$B = W\langle X, Y, Z \rangle / (XY + Z(Z + 1)), \quad I = (X, Z), \quad C = B/\pi.$$

The substitution $(x, y, w) = (dX, dY, dZ)$ converges for every formal series and defines a ring map to B . On the generic fiber it takes J to a line bundle isomorphic to I . The special fiber C is an affine-line torsor over $\mathbf{P}_k^1$, so $K(C) \simeq K(k) \oplus K(k)$ and $\pi_3 K(C; \mathbf{Q}_p) = \pi_4 K(C; \mathbf{Q}_p) = 0$.

Let $b = r_W(\beta)$ and $\eta = 1 - [I]$. The module compatibility of the AMMN fiber character, verified in Lemma 20.1, identifies the relative image of $\beta\eta$ with the action of $\text{ch}^-(\eta)$ on $t^{-1}b \in HC_2$, where $|t| = -2$. Write a normalized representative of the former as $c_0 + tc_2 + t^2c_4 + \dots$. Its rank term c_0 is zero. In the shuffle-plus-cyclic-shuffle product [10, Section 2, Remark 2.2], every cyclic-shuffle correction puts $b \in E$ in a normalized bar slot and hence is zero. The remaining product in ordinary cyclic homology is exactly bc_2 : terms of positive t -degree vanish. Its two-form component is

$$-\lambda_E b c_1^{\text{dR}}(I), \quad \lambda_E \in E^\times,$$

where the nonzero normalization follows from [2, Proposition 4.12]. In weight one, the map $HC_0^- \rightarrow HP_0$ induces $H^2(\Omega_B^{\geq 1}) \rightarrow H^2(\Omega_B^\bullet)$; both groups are $\Omega_B^2/d\Omega_B^1$, so this is an isomorphism. Consequently the periodic Chern component determines precisely the two-form component used above. Thus no comparison with an étale or syntomic regulator, and no additional exact normalization of $K_0 \rightarrow HC^-$, is needed.

The class $c_1^{\text{dR}}(I)$ is represented, up to sign, by the globally regular form

$$\omega_0 = \frac{dX}{X} \wedge dZ = \text{Tr}(P dP \wedge dP), \quad P = \begin{pmatrix} -Z & X \\ Y & Z + 1 \end{pmatrix}.$$

To verify its algebraic nonvanishing, use the two root-deletion bidiscs with $Z = Xt_0$ and $Z = -1 + Xt_{-1}$. Their intersection is $E\langle X, X^{-1}, Z \rangle$. Local primitives are

$$q_0 = -Z dX/X, \quad q_{-1} = -(Z + 1)dX/X, \quad q_{-1} - q_0 = -dX/X.$$

If a global primitive existed, subtract it from these primitives, restrict to $Z = 0$, and take the Laurent residue in X . A closed bidisc form $A(X, t)dX + B(X, t)dt$ has zero residue on $t = c/X$ for every $|c| \leq 1$: its residue is

$$\sum_{m \geq 0} (a_{m, m+1} - b_{m+1, m})c^{m+1} = 0,$$

by closedness and convergence of the Tate coefficients. Here the two values of c are 0 and 1, so the argument also applies at $p = 2$. The displayed difference has residue -1 , a contradiction. This also proves that $b \mapsto b[\omega_0]$ is injective.

The fiber square therefore detects $\beta\eta$ nontrivially in $\pi_3 K(B; \mathbf{Q}_p)$. Finally, localization preserves its rational nonvanishing in $K_3(B[1/p])$: its possible kernel comes from $K_3(C)$, a finite group. This is the image of $\beta(1 - [J])$. The argument is linear in the coefficient and proves the independence assertion as well. $\square$

Remark 22.5. The target in this proof is the ordinary de Rham quotient $\Omega^2/d\Omega^1$ of the termwise completed complex, as in [2, Constructions 4.6 and 4.11]. It is not the quotient by the closure of $d\Omega^1$. Indeed, with $h_j(Z) = -(-Z)^{p^j}$, the regular primitives $-(Z - h_j(Z))dX/X$ have derivatives converging to ω_0 , although ω_0 is not exact. This proof gives nonzero images of actual K -classes in a rationalized completed spectrum; it does not assert a prescribed order in every finite Artin approximation.

22.2. Infinite rank over a strict henselian base at every prime. We first extend the coefficient-cup calculation to the infinite unramified base. Throughout this subsection, $K(-; \mathbf{Q}_p)$ means the rationalization of the p -completed spectrum.

Lemma 22.6. *Let E be a complete discretely valued field of mixed characteristic $(0, p)$ whose residue field k is algebraic over $\mathbf{F}_p$, and let $W = \mathcal{O}_E$. The AMMN point character*

$$r_W : K_3(W) \longrightarrow \pi_3 K(W; \mathbf{Q}_p) \xrightarrow{\sim} HC_2^{\text{cts}}(W/W; \mathbf{Q}_p) = E$$

is defined as before and is natural for continuous local maps of these coefficient DVRs. Proposition 22.4 holds for this W : any finite set of actual coefficients with $\mathbf{Q}_p$ -independent r_W -values gives rationally independent products with the ruling class on the completed split node.

Proof. The standing hypotheses of [2, Section 4.1] require a perfect residue field, and Proposition 4.10 requires a smooth formal scheme; neither requires a finite residue field. Example 4.3 applies to the affine p -complete rings in question. Naturality when the coefficient ring changes follows by keeping the canonical maps used in the proof of Proposition 4.10: a map of pairs $(W, R) \rightarrow (W', R')$ gives a commutative square

$$\begin{array}{ccc} HH(R; \mathbf{Q}_p) & \longrightarrow & HH(R/W; \mathbf{Q}_p) \\ \downarrow & & \downarrow \\ HH(R'; \mathbf{Q}_p) & \longrightarrow & HH(R'/W'; \mathbf{Q}_p). \end{array}$$

The absolute fiber square, cyclic constructions, and horizontal fibers are natural as well. At the point, the cyclic scalar is sent by $E \rightarrow E'$. Thus $r_{W'}(\beta|_{W'}) = r_W(\beta)|_{E'}$.

Quillen's finite-field calculation and filtered-colimit continuity show that $K_i(k)$ is uniquely p -divisible for $i > 0$, and that $K_3(k)$ is torsion. The coefficient exact sequence gives $K(k)/p^a \simeq H\mathbf{Z}/p^a$ directly, hence $K(k; \mathbf{Z}_p) \simeq H\mathbf{Z}_p$. In particular the two adjacent positive homotopy groups needed to define the point fiber lift vanish.

For $B = W\langle X, Y, Z \rangle / (XY + Z(Z + 1))$, its reduced special fiber C is an affine-line torsor over $\mathbf{P}_k^1$. Therefore $K(C) \simeq K(k) \oplus K(k)$. The relative-to-absolute step in Proposition 22.4 still has vanishing π_3 and π_4 on this special fiber. Its ordinary de Rham calculation and two-chart residue proof work over the complete field E verbatim. In particular $b \mapsto b[\omega_0]$ is injective, where $\omega_0 = (dX/X) \wedge dZ$ is globally regular. Finally $K_3(C)$ is torsion, which is enough for localization to give $K_3(B)_{\mathbf{Q}} \hookrightarrow K_3(B[1/p])_{\mathbf{Q}}$. The module calculation and the nonzero scalar of [2, Proposition 4.12] therefore detect exactly the same coefficient products. No Hausdorff quotient of de Rham cohomology is taken. $\square$

Theorem 22.7. *For every prime p , the complete strictly henselian regular local ring*

$$A_{\infty, p} = W(\overline{\mathbf{F}}_p)[[x, y, z]]/(p + xy + z^2)$$

has dimension three and infinite-dimensional rational generic K_3 -kernel. More precisely, for every $m \geq 1$ there are m actual integral classes with zero generic image whose rational classes are independent.

Proof. Put $V_m = W(\mathbf{F}_{p^m})$, $V_{\infty} = W(\overline{\mathbf{F}}_p)$, $F_m = \text{Frac } V_m$, and choose $u^2 = -p$. Write $W_m = V_m[u]$, $E_m = F_m(u)$, $W_{\infty} = V_{\infty}[u]$, and $E_{\infty} = \text{Frac } W_{\infty}$. Eisenstein's criterion makes

each W_m and W_∞ a complete DVR with uniformizer u . The linear term p of the equation in the regular ambient power-series ring proves regularity and dimension three for $A_{\infty,p}$. Its residue field is algebraically closed.

By Lemma 22.3, actual r_{W_m} -values span E_m . Apply $1 - \sigma$, where $\sigma(u) = -u$, and choose $\beta_{m,1}, \dots, \beta_{m,m} \in K_3(W_m)$ with $\sigma\beta_{m,j} = -\beta_{m,j}$ whose values are independent in $E_m^- = uF_m$. This construction involves no division by two in actual K -theory. Lemma 22.6 carries these values by the field inclusion into E_∞ , preserving their $\mathbf{Q}_p$ -independence.

Set $A_m = V_m[[x, y, z]]/(p + xy + z^2)$. Pull each coefficient to $A_m/(x) = W_m[[y]]$ and push forward along $x = 0$, obtaining $c_{m,j} \in K_3(A_m)$. These vanish on $A_m[1/x]$ integrally. The local map $A_m \rightarrow A_{\infty,p}$ is flat by miracle flatness: both rings are regular of dimension three, and its closed fiber is a field [45, Tag 00R4]. Thus their images are the corresponding supported pushforwards on $A_{\infty,p}$.

After the finite free quadratic coefficient extension and inversion of p , put

$$T = (W_\infty[[x, y, z]]/(xy + (z - u)(z + u)))[1/p], \quad J_+ = (x, z - u), \quad \eta_+ = 1 - [J_+].$$

The two components of $x = 0$ are disjoint Cartier divisors. Their structure-sheaf classes sum to zero because their union is the principal divisor x . Flat base change and the projection formula therefore give exactly

$$c_{m,j}|_T = \beta_{m,j}\eta_+ + \sigma\beta_{m,j}\eta_- = 2\beta_{m,j}\eta_+.$$

Use $w = z - u$ and $d = 2u$ in Lemma 22.6. The substitution into the good quadric is $x = dX$, $y = dY$, $z = u + dZ$; all original formal variables have norm at most $|u| < 1$, including at $p = 2$. The coefficient detector gives m independent images, so the original rational classes are independent. Their generic images are zero by localization at x . Since m is arbitrary, the asserted kernel is infinite dimensional. $\square$

Corollary 22.8. *For every prime p , the arithmetic strict henselization*

$$S_p = \left(\left(\mathbf{Z}_{(p)}[x, y, z]/(p + xy + z^2) \right)_{(x,y,z)} \right)^{sh}$$

also has infinite-dimensional rational generic K_3 -kernel, represented by actual integral classes with integral generic vanishing.

Proof. Choose a finite étale local DVR $V_{m,0}/\mathbf{Z}_{(p)}$ with number-field fraction field, residue $\mathbf{F}_{p^m}$, completion V_m , and a compatible residue embedding. Set $W_{m,0} = V_{m,0}[u]$, where $u^2 = -p$. The fraction field of $W_{m,0}^h$ is the field of constants in E_m , by Krasner's lemma.

The second assertion of [33, Theorem 4.13] and $K_3^M(E_m)/p^a = 0$ imply that $K_3(\text{Frac } W_{m,0}^h)$ surjects onto $K_3(E_m)/p^a$. The Milnor vanishing follows from norm-residue [50, VI, Theorem 4.1] and local p -cohomological dimension two, also at $p = 2$ [35, Chapter I, discussion preceding Theorem 2.8]. Localization, using $K_2(\mathbf{F}_{p^m}) = 0$, then gives

$$K_3(W_{m,0}^h) \twoheadrightarrow K_3(W_m)/p^a \quad (a \geq 1).$$

Here the passage from E_m back to W_m does not change the finite quotient because its kernel comes from prime-to- p torsion $K_3(\mathbf{F}_{p^m})$. The actual-completion argument of Lemma 22.3 now shows directly that the henselian values of the AMMN point map span E_m . Applying $1 - \sigma$ supplies m actual anti-invariant coefficients with independent AMMN values. No comparison with an étale-selected coefficient class is used.

Let $R_m = (A_{0,p} \otimes_{\mathbf{Z}_{(p)}} V_{m,0})^h$. Its divisor quotient by x receives $W_{m,0}^h$, its completion is A_m , and it maps compatibly to S_p . Push the chosen coefficients forward along $x = 0$ and then map to S_p . Their images under $S_p \rightarrow A_{\infty,p}$ are precisely the completed classes detected in Theorem 22.7; the map and completion identification are supplied uniformly in p by Lemma 18.6. Hence these m rational classes on S_p are independent. They vanish after inverting x integrally. Letting m grow proves the corollary. $\square$

23. HIGHER POINT COORDINATES AND UNRAMIFIED COEFFICIENTS

Lemma 23.1. *Let W be a complete mixed-characteristic discrete valuation ring with uniformizer π , fraction field E , and residue field k algebraic over $\mathbf{F}_p$. Let B be a smooth affine formal W -algebra, topologically of finite presentation and of relative dimension two, and put $C = B/\pi B$. Fix $n \geq 2$ and suppose that*

$$K_{2n}(C) = 0.$$

Let $\eta \in K_0(B)$ have rank zero. Write

$$\mathrm{ch}^-(\eta) = t\gamma, \quad \gamma \in \widehat{\Omega}_{B/W}^2[1/p]/d\widehat{\Omega}_{B/W}^1[1/p], \quad |t| = -2,$$

for its negative-cyclic character in the finite mixed-HKR model, and assume $\gamma \neq 0$.

The AMMN point comparison gives a $\mathbf{Q}_p$ -linear isomorphism

$$\pi_{2n-1}K(W; \mathbf{Q}_p) \xrightarrow{\sim} HC_{2n-2}^{\mathrm{cts}}(W/W; \mathbf{Q}_p) = t^{1-n}E.$$

For an actual class $\beta \in K_{2n-1}(W)$, denote its value by $t^{1-n}r_n(\beta)$. If $r_n(\beta) \neq 0$, then the actual class

$$\eta\beta_B \in K_{2n-1}(B)$$

has infinite order. More generally, if a finite family of actual classes β_j has $\mathbf{Q}_p$ -linearly independent values $r_n(\beta_j)$, then the classes $\eta(\beta_j)_B$ are linearly independent in $K_{2n-1}(B) \otimes_{\mathbf{Z}} \mathbf{Q}_p$.

Proof. All occurrences of $K(-; \mathbf{Q}_p)$ denote rationalized spectral p -completion, and write $\Omega_B^j = \widehat{\Omega}_{B/W}^j[1/p]$. Since k is algebraic over $\mathbf{F}_p$, Quillen's finite-field computation and filtered colimits give

$$K_{2m}(k) = 0 \quad (m > 0), \quad K_{2m-1}(k) \text{ is prime-to-} p \text{ torsion.}$$

The coefficient exact sequences also show that $K(k)_p^\wedge$ has no positive homotopy groups. The point isomorphism in the statement therefore follows from the fiber square of [2, Proposition 4.10]; [2, Example 4.3] identifies the continuous K-theory of $\mathrm{Spf} W$ with the spectral completion used here.

First handle one actual class β . Its reduction in $K_{2n-1}(k)$ need not be zero. Choose an integer N , prime to p , which kills that reduction. The actual relative exact sequence, together with $K_{2n}(k) = 0$, gives a unique actual lift

$$\kappa \in K_{2n-1}(W, \pi W) \quad \text{of} \quad N\beta.$$

By compatibility with completion, its relative AMMN character is

$$\chi_W(\kappa) = Nt^{1-n}r_n(\beta).$$

This constructs an actual relative class before using its completed character; no completed relative lift is being treated as an actual lift.

The AMMN fiber comparison is $K(B)$ -linear, by the module construction through [2, Proposition 2.8, Corollary 2.10, and Theorem 2.12], the relative-to- W projection in Proposition 4.10, and the multiplicative cyclotomic trace [9, Theorem 7.4]. More explicitly, with $H = THH(B)$, the objects $H \otimes D$ and arrows $\mathrm{id}_H \otimes f$ in that construction are H -modules and module maps in cyclotomic spectra. The lax monoidal circle and TC functors preserve the induced $TC(B)$ actions, and the equivalences used there are inverted in the module category. The relative Hochschild projection is also H -linear. Thus this assertion uses the actual module maps in the construction, rather than the naturality of an unspecified additive equivalence. Its target action is the usual action of negative cyclic homology on the suspended ordinary cyclic fiber. The norm compatibility is [10, Proposition 2.3], with the spectral norm identified with the Connes operator by [28, Theorem 2.1, Lemma 2.2, and Theorem 2.3].

The finite mixed-HKR model in relative dimension two gives

$$HC_0^-(B/W; \mathbf{Q}_p) = H_{\mathrm{dR}}^0(B/W) \oplus tH_{\mathrm{dR}}^2(B/W),$$

where $H_{\mathrm{dR}}^2 = \Omega_B^2/d\Omega_B^1$ is the ordinary quotient of continuous forms. Rank zero removes the first component of $\mathrm{ch}^-(\eta)$, and higher form degrees vanish. Thus this character is exactly $t\gamma$.

The pullback of $\chi_W(\kappa)$ is the scalar cyclic class $Nt^{1-n}r_n(\beta)$ on B . Scalar multiplication has no cyclic shuffle correction: in the normalized relative bar complex any such correction contains a bar occupied by this scalar in E , and hence is zero. Consequently the actual relative product

$$\xi = \eta \kappa_B \in K_{2n-1}(B, \pi B)$$

has character

$$(23.1) \quad \chi_B(\xi) = Nt^{2-n}r_n(\beta)\gamma \quad \text{in } HC_{2n-2}^{\text{cts}}(B/W; \mathbf{Q}_p).$$

There is no constant component. For $n \geq 2$, the displayed two-form component is an ordinary de Rham cohomology summand; since $r_n(\beta) \neq 0$ and E is a field, it is nonzero. Thus ξ has infinite order.

Now use the actual relative exact sequence

$$K_{2n}(C) = 0 \longrightarrow K_{2n-1}(B, \pi B) \longrightarrow K_{2n-1}(B).$$

It is injective at the middle term, and the absolute image of ξ is exactly $N\eta\beta_B$. Hence $\eta\beta_B$ has infinite order. No inverse to multiplication by N on actual K-groups is used.

For a finite family, choose one common prime-to- p integer N killing all reductions, and form the unique actual lifts κ_j . Tensoring the last injection with the flat $\mathbf{Z}$ -module $\mathbf{Q}_p$ preserves injectivity. Applying the $\mathbf{Q}_p$ -linear extension of χ_B to a relation between the relative products gives

$$N \left(\sum_j q_j r_n(\beta_j) \right) \gamma = 0.$$

The assumed independence forces every $q_j = 0$. □

Remark 23.2. For the good all-prime quadric

$$B = W\langle X, Y, Z \rangle / (XY + Z(Z+1)), \quad I = (X, Z), \quad \eta = 1 - [I],$$

the special fiber is an affine-line torsor over $\mathbf{P}_k^1$. Homotopy invariance, descent, and the projective-bundle formula give $K(C) \simeq K(k) \oplus K(k)$, so the required actual even K-group vanishes. The idempotent

$$e = \begin{pmatrix} -Z & X \\ Y & 1 + Z \end{pmatrix}$$

has image I and satisfies

$$\text{Tr}(e \, de \wedge de) = \frac{dX}{X} \wedge dZ.$$

Thus γ is, up to the consistent first-Chern sign, the nonzero ruling form detected by the root-chart residue lemma. The same formal coefficient calculation works in every fixed degree $2n-1$: changing n only changes the cyclic column. For $n=1$, by contrast, the prospective term $t\gamma$ is killed in ordinary cyclic homology, so this argument gives no degree-one nonvanishing statement.

Remark 23.3. Let B_g be the completed Laurent test algebra obtained from a smooth algebraic model of B by adjoining $T_1^{\pm 1}, \dots, T_s^{\pm 1}$ and inverting a polynomial g in those variables with coefficients in W . Its relative dimension is $D = s + 2$. For the actual relative lift κ above, the same module calculation gives

$$\chi_{B_g}(\eta \kappa_{B_g}[T_1] \cdots [T_s])_{\text{top}} = Nt^{2-n}r_n(\beta) \gamma \wedge d \log T_1 \wedge \cdots \wedge d \log T_s$$

in the $t^{2-n}H_{\text{dR}}^D$ component of $HC_{2n+s-2}^{\text{cts}}(B_g/W; \mathbf{Q}_p)$.

When $n \geq 3$, the starting cyclic column is negative, so the positive columns cannot simply be discarded on the basis of the ordinary cyclic quotient. Instead use the following chain-degree calculation. Write a negative-cyclic cycle for η as $\sum_{r \geq 1} t^r e_{2r}$ and one for each unit as $\sum_{a_i \geq 0} t^{a_i} u_{2a_i+1}$. The first sum starts at $r=1$: its degree-zero term represents the rank in $HH_0(B_g/W) = B_g$, and $b: C_1 \rightarrow C_0$ is zero for a commutative algebra. Hence rank zero forces that term itself to vanish.

The scalar point cup has zero cyclic-shuffle correction. After the remaining s unit multiplications, a term using indices $r, a_1, \dots, a_s$ and c cyclic-shuffle corrections has Hochschild degree

$$2r + s + 2 \sum_i a_i + 2c.$$

The finite normalized HKR map in dimension $D = s + 2$ kills this term unless $r = 1$, all $a_i = 0$, and $c = 0$. Thus only the first Chern component, the lowest unit Dennis traces, and ordinary shuffles survive. Their normalized image is exactly the displayed wedge, with no additional degree-dependent factor. This argument allows the unwanted terms to survive in the ordinary cyclic complex; it kills them by their final form degree.

For a test polynomial g chosen as the same-support unit lift of its residual polynomial, the integral continuous Laurent residue maps this wedge to γ . It is therefore nonzero whenever $\gamma \neq 0$. If the special fiber of B_g is smooth of dimension D over k , Geisser–Levine’s finite-coefficient vanishing [25, Theorem 8.4] implies

$$\pi_{2n+s} K(B_g/\pi; \mathbf{Q}_p) = 0,$$

since $2n + s > D$. The completed relative-to-absolute map is thus injective in degree $2n + s - 1$. In this Laurent extension one uses this completed injection; no actual even K-group vanishing of the higher-dimensional special fiber is asserted.

Remark 23.4. The lemma does not assert surjectivity from actual K-theory to spectral completion, and it does not identify r_n with a Bloch–Kato, syntomic, or polylogarithmic coordinate. Independently known actual classes with independent étale Chern values may be used once the separate factorization through spectral completion has been established. The hypothesis on k in this lemma is algebraicity over $\mathbf{F}_p$; perfectness alone does not imply the actual even K-group vanishing used in its proof.

23.1. Unramified actual coefficients at every prime.

Proposition 23.5. *Let p be any prime, $m \geq 2$, and put*

$$k_m = \mathbf{F}_{p^m}, \quad V_m = W(k_m), \quad F_m = \text{Frac } V_m.$$

Take $u = i$ if $p = 2$, and take u to be a primitive p -th root if $p > 2$. Put $E_m = F_m(u)$, $O_m = V_m[u]$, and let σ fix F_m and invert u . For each $n \geq 2$ there exist Teichmüller roots $\zeta_1, \dots, \zeta_{m-1} \in \mu_{p^m-1}(V_m) \setminus \{1\}$ such that the actual rational classes

$$\beta_{n,j} = b_n(\zeta_j u) - b_n(\zeta_j u^{-1}) \in K_{2n-1}(O_m)_{\mathbf{Q}}, \quad 1 \leq j \leq m-1,$$

are σ -anti-invariant and have $\mathbf{Q}_p$ -linearly independent images in $H^1(E_m, \mathbf{Q}_p(n))$ under $c_{n,1}$. For each fixed p, m, n , a common integer multiple is represented by strictly anti-invariant integral classes.

Proof. Put $\delta = u - u^{-1} \neq 0$, $\epsilon = u^p = u^{-p}$, and $F_{p,n}(z) = \text{Li}_n(z) - p^{-n} \text{Li}_n(z^p)$. The equal p th powers make the depleted terms cancel exactly:

$$\text{Li}_n(\zeta u) - \text{Li}_n(\zeta u^{-1}) = F_{p,n}(\zeta u) - F_{p,n}(\zeta u^{-1}).$$

For $1 \leq a \leq p-1$ the Laurent polynomial

$$Q_a(u) = \frac{u^a - u^{-a}}{u - u^{-1}} = \sum_{b=0}^{a-1} u^{a-1-2b}$$

is integral and reduces to a . Grouping denominators as $a + p\ell$ and using the binomial series gives, with $S_j(t) = P_j(t)/(1-t)^{j+1}$ and $P_j \in \mathbf{Z}[t]$,

$$\begin{aligned} a_{n,p}(\zeta) &:= \frac{\text{Li}_n(\zeta u) - \text{Li}_n(\zeta u^{-1})}{\delta} \\ (23.2) \quad &= \sum_{j \geq 0} (-1)^j \binom{n+j-1}{j} p^j \sum_{a=1}^{p-1} \frac{\zeta^a Q_a(u)}{a^{n+j}} S_j(\epsilon \zeta^p) \in O_m. \end{aligned}$$

Here $S_0(t) = 1/(1-t)$. The series identity first holds for $|z| < 1$ and extends uniformly to $|z| \leq 1$, $v_p(1-z^p) \leq 2/3$. The rigid identity principle and [6, Theorem 2.7(3)] identify it with Coleman continuation. At the displayed points $1 - \epsilon\zeta^p$ is a unit, so every j th summand has valuation at least j .

If $t = \bar{\zeta}$, reduction of (23.2) is

$$h_n(t) = \frac{\sum_{a=1}^{p-1} a^{1-n} t^a}{1 - t^p}, \quad t \in k_m^\times \setminus \{1\}.$$

The numerator has t -coefficient one and degree $p-1$. Thus h_n is a nonconstant rational map of degree at most p . If $p > 2$, each finite fiber has at most p elements, so

$$\#h_n(k_m^\times \setminus \{1\}) \geq \left\lceil \frac{p^m - 2}{p} \right\rceil = p^{m-1}.$$

Its $\mathbf{F}_p$ -span has dimension at least $m-1$. If $p=2$, then $h_n(t) = t/(1+t)^2 = w^2 + w$, where $w = (1+t)^{-1}$ ranges through $k_m \setminus \{0, 1\}$. Its image is exactly the nonzero part of $\ker \text{Tr}_{k_m/\mathbf{F}_2}$, whose span has dimension $m-1$. Choose $m-1$ independent image values and Teichmüller lifts ζ_j of corresponding inputs. A nonzero $\mathbf{Q}_p$ -linear relation among $a_{n,p}(\zeta_j)$ can be normalized to coefficients in $\mathbf{Z}_p$, with one a unit. Reduction would contradict the chosen $\mathbf{F}_p$ -independence. The coordinates are therefore $\mathbf{Q}_p$ -independent; they need not belong to F_m .

These are actual number-field coefficients. For $N = p^m - 1$ put $L_m = \mathbf{Q}(\mu_N, u)$. The order of p modulo N is m , so the chosen completion of $\mathbf{Q}(\mu_N)$ is F_m , and the completion of L_m is E_m . The shifted cyclotomic polynomial is Eisenstein in the ramified factor, so its integer ring is O_m . Theorem 1.12 of [8] gives $b_n(\zeta_j u)$ in $K_{2n-1}(L_m)_{\mathbf{Q}}$. Localization at the selected prime gives its rational lift to the local integer ring, since the positive residue-field K -groups are torsion. Pullback gives a class over O_m . The global automorphism fixing μ_N and inverting u preserves the selected prime and induces σ . Naturality proves exact anti-invariance of $\beta_{n,j}$. Clearing denominators of the individual $b_n(\zeta_j u)$, then subtracting their integral σ -images, gives the asserted integral representatives.

Finally the common normalization (19.1) gives

$$c_{n,1}(\beta_{n,j}) = \exp_{E_m,n}(\kappa_n \delta a_{n,p}(\zeta_j)).$$

For $n \geq 2$ this exponential is a $\mathbf{Q}_p$ -linear isomorphism. The independence of the coordinates and $\kappa_n \delta \neq 0$ prove the assertion. No actual-to-completed K -theory surjectivity is used. $\square$

This proposition is a finite-level coefficient statement. Its use over one fixed ring with infinite perfect residue field requires the compatible ruling detector after that base change.

23.2. One strict henselian ring in every odd degree. Put $k = \bar{\mathbf{F}}_p$, $V_\infty = W(k)$, and define

$$f_p(z) = \begin{cases} \Phi_p(1+z), & p > 2, \\ \Phi_4(1+z) = z^2 + 2z + 2, & p = 2. \end{cases}$$

The polynomial has Eisenstein degree $p-1$ for odd p and degree two for $p=2$.

Theorem 23.6. *For every prime p , the rings*

$$R_{\infty,p} = V_\infty[[x, y, z]]/(xy + f_p(z)), \quad S_p = \left((\mathbf{Z}_{(p)}[x, y, z]/(xy + f_p(z)))_{(p,x,y,z)} \right)^{\text{sh}}$$

are strictly henselian regular local rings of dimension three, and $\hat{S}_p \cong R_{\infty,p}$ after fixing their residue identifications with k . For every $n \geq 2$, each of these two rings R satisfies

$$\dim_{\mathbf{Q}} \ker(K_{2n-1}(R)_{\mathbf{Q}} \longrightarrow K_{2n-1}(\text{Frac } R)_{\mathbf{Q}}) = \infty.$$

More precisely, for every $m \geq 2$ the kernel contains $m-1$ rationally independent classes represented by actual integral K -classes whose restrictions to $R[1/x]$ are zero integrally. The ring for a fixed p is independent of both m and n .

Proof. Fix $n \geq 2$ and $m \geq 2$, and write $q = 2n - 1$, $V_m = W(\mathbf{F}_{p^m})$, and $F_m = \text{Frac } V_m$. Take $\rho = i$ for $p = 2$, and a primitive p th root for $p > 2$. Put $W_m = V_m[\rho]$, $E_m = F_m(\rho)$ and $W_\infty = V_\infty[\rho]$, $E_\infty = \text{Frac } W_\infty$. These are complete coefficient DVRs and their fraction fields; $\rho - 1$ is a uniformizer in each ramified DVR. Let σ fix the unramified coefficient ring and invert ρ .

Proposition 23.5 supplies $m - 1$ actual coefficients $\beta_j \in K_q(W_m)$, after one common integer multiplication, with $\sigma\beta_j = -\beta_j$ and independent $\mathbf{Q}_p$ -valued étale Chern classes. Lemma 22.1 implies that their images in $\pi_q K(W_m; \mathbf{Q}_p)$ are $\mathbf{Q}_p$ -independent. Under the AMMN point isomorphism used in Lemma 23.1, denote their coordinates by

$$a_j = r_{n, W_m}(\beta_j) \in E_m.$$

These coordinates are independent and $\sigma a_j = -a_j$. Naturality of that point calculation under $W_m \rightarrow W_\infty$ sends them by the field inclusion $E_m \hookrightarrow E_\infty$, preserving independence. This argument only transports independence; it does not identify the AMMN coordinates with a prescribed polylogarithm normalization.

Set $R_m = V_m[[x, y, z]]/(xy + f_p(z))$. Weierstrass division identifies its regular divisor $x = 0$ with $W_m[[y]]$, by $1 + z \mapsto \rho$. Pull back β_j and push forward along this divisor to obtain $g_j \in K_q(R_m)$. Localization makes $g_j|_{R_m[1/x]} = 0$ integrally. Both R_m and $R_{\infty, p}$ are regular local of dimension three: the equation has linear term p in the regular ambient ring, so x, y, z are regular parameters. The closed fiber of $R_m \rightarrow R_{\infty, p}$ is k ; hence the map is flat by [45, Tag 00R4]. Its pullback therefore preserves the supported pushforward.

After the finite free extension $V_\infty \rightarrow W_\infty$ and inversion of p , the polynomial splits with distinct roots $a_g = g(\rho) - 1$, indexed by $G = \text{Gal}(E_m/F_m)$. Write $I_g = (x, z - a_g)$ and $\eta_g = 1 - [I_g]$ on this generic ring. The root components of $x = 0$ are disjoint Cartier divisors; their structure-sheaf classes sum to zero because their union is principal. Flat base change and the projection formula give

$$g_j = \sum_{g \in G} g(\beta_j) \eta_g.$$

It suffices to retain the fixed inverse-root pair. Put $d = \rho - \rho^{-1} \neq 0$, $H(z) = \prod_{g \neq 1, \sigma} (z - a_g)$, and substitute

$$z = (\rho - 1) + dZ, \quad x = dX, \quad y = dH((\rho - 1) + dZ)Y$$

into the smooth formal quadric $B = W_\infty \langle X, Y, Z \rangle / (XY + Z(Z + 1))$. All original formal variables are topologically nilpotent, so the substitution defines a ring map. It uses $z = t - 1$, including at two. The two selected root ideals pull to $J = (X, Z)$ and $J' = (X, Z + 1)$, up to the invertible generic scalar d . Every omitted root ideal becomes trivial: if $b = (a_g - a_1)/d \notin \{0, -1\}$, then

$$b(b + 1) = -XY - (Z - b)(Z + b + 1) \in (X, Z - b).$$

No inversion of H or flatness of this last substitution is used. The pulled invertible module surjects onto the displayed invertible ideal, hence is isomorphic to it. The two remaining structure-sheaf classes sum to zero, so on $B[1/p]$ the class becomes exactly $2\beta_j(1 - [J])$. Lemma 23.1 detects these products by their independent coordinates $2a_j$ in $K_q(B)$. Here $K_q(B/\pi)$ is torsion, since the special fiber has the K -theory of $\mathbf{P}_k^1$ and k is algebraic over $\mathbf{F}_p$. Localization therefore makes $K_q(B)\mathbf{Q} \rightarrow K_q(B[1/p])\mathbf{Q}$ injective, so these products remain independent on the generic test ring. Thus the g_j are rationally independent on $R_{\infty, p}$.

For the arithmetic assertion choose $N = p^m - 1$ and a prime v above p in $\mathbf{Q}(\mu_N)$ compatible with k . The essentially étale local DVR $V_{m,0} = (\mathbf{Z}_{(p)}[\zeta_N])_v$ has completion V_m , since p has multiplicative order m modulo N ; put $W_{m,0} = V_{m,0}[\rho]$. The coefficients in Proposition 23.5 come from the actual rational K_q of $\text{Frac } W_{m,0} = \mathbf{Q}(\mu_N, \rho)$. The localization map from $K_q(W_{m,0})$ is surjective [40, Sections 5–7], since $K_{q-1}(\mathbf{F}_{p^m}) = 0$ by [41][50, IV, Corollary 1.13]. Clear denominators before lifting and applying $1 - \sigma$. This produces the same independent finite family from actual classes on $W_{m,0}$. The chosen pointed étale neighborhood gives $V_{m,0} \rightarrow S_p$, and $\rho \mapsto 1 + z$ gives $W_{m,0} \rightarrow S_p/(x)$. Pull the coefficients to this regular divisor and push to S_p . They vanish after inverting x integrally.

The universal property gives a compatible local map $S_p \rightarrow R_{\infty,p}$. Both rings have residue k and regular parameters x, y, z ; the induced associated-graded map is the identity on $k[x, y, z]$. It follows that $\widehat{S}_p \cong R_{\infty,p}$. Completion is flat, so these arithmetic supported classes map to the already detected completed classes and are independent. Letting m grow proves both assertions. $\square$

24. ONE FIXED RING IN EVERY DEGREE AT LEAST THREE

We use the ordinary continuous de Rham quotient throughout. First we record why introducing one Laurent variable preserves both the coefficient–ruling products and their products with that unit.

Lemma 24.1. *Let W be the integers of a finite extension $E/\mathbf{Q}_p$, with residue field k and uniformizer π . Put*

$$B = W\langle X, Y, Z \rangle / (XY + Z(Z + 1)), \quad J = (X, Z), \quad \eta = 1 - [J].$$

Let $0 \neq g \in W[T]$ have unit nonzero coefficients, and put $B_g = (B[T^{\pm 1}, 1/g])_{\pi}^{\wedge}$. For $n \geq 2$, suppose actual classes $\beta_1, \dots, \beta_r \in K_{2n-1}(W)$ have $\mathbf{Q}_p$ -linearly independent AMMN point coordinates $a_1, \dots, a_r \in E$. Then each of the two families

$$\eta\beta_j \in K_{2n-1}(B_g[1/p])_{\mathbf{Q}}, \quad \eta\beta_j[T] \in K_{2n}(B_g[1/p])_{\mathbf{Q}}$$

is rationally independent.

Proof. Choose a common integer N , prime to p , which kills the reductions of the β_j in $K_{2n-1}(k)$. Their unique actual relative lifts have point characters $Nt^{1-n}a_j$, where $|t| = -2$. Write $\gamma = \text{ch}_1^-(\eta)$ for the nonzero ruling two-form class of Lemma 23.1. The module construction of that lemma, in the finite mixed-HKR model of relative dimension three, gives the relative characters

$$(24.1) \quad Nt^{2-n}a_j\gamma, \quad Nt^{2-n}a_j\gamma \wedge d \log T.$$

Here is the degree check for the second formula, including $n > 2$. The rank-zero negative Chern cycle has terms $t^r e_{2r}$ with $r \geq 1$; a unit cycle has terms $t^a u_{2a+1}$ with $a \geq 0$. A cyclic-shuffle correction increases Hochschild degree by two. Dimension three therefore retains only $r = 1$, $a = 0$, and the ordinary shuffle. Normalized HKR gives exactly the second term in (24.1). This argument does not discard positive cyclic columns merely because the initial column might be negative.

The degree-zero projection P_0 of Lemma 21.3 sends the first form coefficient in (24.1) back to $Na_j\gamma$; its residue sends the second coefficient there. The independence of the a_j thus detects every nonzero rational linear combination of either relative family. The special fiber $C_g = B_g/\pi$ is smooth of dimension three. Lemma 21.4 gives $\pi_{q+1}K(C_g; \mathbf{Q}_p) = 0$ for every $q \geq 3$. Consequently the relative-to-absolute map is injective in all the degrees used here after spectral completion and rationalization. This proves independence in the actual groups of B_g .

The final generic localization requires a separate actual K -theory observation. Let $U_g = \text{Spec } k[T^{\pm 1}, 1/\bar{g}]$. The ruling and projective-bundle formula give $K(C_g) \simeq K(U_g) \oplus K(U_g)$. For $q \geq 2$, localization on $\mathbf{A}_k^1$ [40, Sections 5–7] puts $K_q(U_g)$ between a quotient of $K_q(k)$ and a subgroup of the finite direct sum $\bigoplus_v K_{q-1}(k(v))$, where the omitted points v have finite residue fields. These groups are finite by [41][50, IV, Corollary 1.13]. Thus $K_q(C_g)$ is torsion, and localization makes

$$K_q(B_g)_{\mathbf{Q}} \longrightarrow K_q(B_g[1/p])_{\mathbf{Q}}$$

injective. This proves the stated generic assertions. $\square$

Theorem 24.2. *Fix a prime p , let $S = \mathbf{Z}_{(p)}^{\text{sh}}$ with residue $k = \overline{\mathbf{F}}_p$, and set*

$$V_0 = (S[T])_{(p)}, \quad f_p(z) = \begin{cases} \Phi_p(1+z), & p > 2, \\ \Phi_4(1+z) = z^2 + 2z + 2, & p = 2, \end{cases}$$

$$R_p = (V_0[x, y, z]/(xy + f_p(z)))_{(x, y, z)}.$$

This is one Noetherian regular local ring of dimension three, with residue field $k(T)$, and for every $j \geq 3$,

$$\dim_{\mathbf{Q}} \ker (K_j(R_p)_{\mathbf{Q}} \longrightarrow K_j(\text{Frac } R_p)_{\mathbf{Q}}) = \infty.$$

Each of the finite independent families used to establish this assertion consists of actual integral classes which vanish after inverting x integrally.

Proof. The strict henselization S is a Noetherian DVR with uniformizer p ; see [46, Tags 06LJ and 0AP3]. The localization V_0 is a DVR with the same uniformizer and residue field $k(T)$. The polynomial f_p is Eisenstein of degree at least two. In the regular ambient local ring of dimension four, the equation has nonzero linear term p , proving the assertions about R_p . Equivalently,

$$(24.2) \quad R_p = (S[T, x, y, z]/(xy + f_p(z)))_{(p, x, y, z)}.$$

Fix $n \geq 2$ and $m \geq 2$, and take $\rho = i$ at two and a primitive p th root otherwise. Choose a compatible prime v of $\mathbf{Q}(\mu_{p^m-1})$ and put

$$V_{m,0} = (\mathbf{Z}_{(p)}[\zeta_{p^m-1}])_v \subset S, \quad W_{m,0} = V_{m,0}[\rho].$$

The actual number-field classes of Proposition 23.5 lift to $K_{2n-1}(W_{m,0})$ after a common integer multiplication: the obstruction is $K_{2n-2}(\mathbf{F}_{p^m}) = 0$. Choose the integral lifts before applying $1 - \sigma$, where σ fixes $V_{m,0}$ and inverts ρ . We obtain $m - 1$ strictly anti-invariant classes β_j . Their finite-local-field étale Chern classes are independent; Lemma 22.1 and the AMMN point isomorphism therefore give independent point coordinates in $E_m = \text{Frac } W(\mathbf{F}_{p^m})(\rho)$.

The regular divisor $x = 0$ in R_p receives $W_{m,0}$ by $\rho \mapsto 1 + z$. Pull back and push forward to obtain $a_j \in K_{2n-1}(R_p)$, and put $b_j = a_j[T] \in K_{2n}(R_p)$. Here T is a unit. Both families vanish on $R_p[1/x]$ integrally, by localization and the projection formula.

We check independence after any putative finite relation. Using (24.2) and filtered-ring continuity of K -theory [50, IV, Section 6.4], the coefficients, the classes, and an actual integral relation all descend to

$$(V_{\lambda,0}[T, x, y, z]/(xy + f_p(z)))[1/F], \quad \bar{F}(T) \neq 0,$$

where $V_{\lambda,0} \subset S$ is an essentially étale local DVR containing $V_{m,0}$ and all constants in the finite data. We include T in the inverted product F when necessary. Its completion is an unramified coefficient DVR V_ℓ with finite residue field; put $W_\ell = V_\ell[\rho]$ and $E_\ell = \text{Frac } W_\ell$. Point-character naturality sends the independent coordinates by the field inclusion $E_m \hookrightarrow E_\ell$.

Over W_ℓ , retain the two roots $\rho - 1, \rho^{-1} - 1$ of f_p . With $d = \rho - \rho^{-1}$ and $H(z) = \prod_{a \neq \rho-1, \rho^{-1}-1} (z - a)$, use

$$z = (\rho - 1) + dZ, \quad x = dX, \quad y = dH((\rho - 1) + dZ)Y.$$

Choose $g(T)$ by lifting precisely the nonzero coefficients of $\bar{F}(T)$ to units in W_ℓ and form B_g as in Lemma 24.1. All three displayed images are topologically nilpotent. The image of F is $g + \pi h$, hence a unit of B_g . Thus this finite relation has an actual image in the test ring; no map from all of R_p to a fixed test algebra is asserted.

Flat coefficient base change and the projection formula, followed by inversion of p , split the supported class as $\sum_\tau \tau(\beta_j)(1 - [I_\tau])$. The pair substitution makes every omitted root ideal trivial. The two remaining disjoint components have structure-sheaf classes summing to zero, and anti-invariance therefore gives

$$a_j \longmapsto 2\eta\beta_j, \quad b_j \longmapsto 2\eta\beta_j[T] \quad \text{in } K_*(B_g[1/p]).$$

Lemma 24.1 excludes the asserted relation in either degree. Hence both families have rational rank $m - 1$ in R_p . As m is arbitrary and the degrees $2n - 1, 2n$ with $n \geq 2$ cover every integer at least three, the theorem follows. $\square$

24.1. A single complete ring in every degree at least three. The finite-witness proof of Theorem 24.2 does not itself pass to arbitrary completed coefficients. We now give a complete-ring construction, using Lemma 21.2 and a constant-term map defined on all primitive denominators.

Lemma 24.3. *Let W be a complete DVR with uniformizer π , and set*

$$V_W = \widehat{W[T]_{(\pi)}}^\pi, \quad B' = W\langle X, Y, Z \rangle / (XY + Z(Z+1)), \\ \mathcal{B} = V_W\langle X, Y, Z \rangle / (XY + Z(Z+1)).$$

There are continuous W -linear maps on the relative form complexes, a degree-zero chain projection P_0 and a degree-minus-one residue, which fix base forms and satisfy

$$P_0(\omega) = \omega, \quad \text{Res}_T(\omega \wedge d \log T) = \omega \quad (\omega \in \hat{\Omega}_{B'/W}^\bullet).$$

They preserve actual de Rham boundaries. The complexes here retain the differential dT : their base ring is W , not V_W .

Proof. For $a \geq 1$, put $L_a = (W/\pi^a)((T))$. Every $g \in W[T] \setminus (\pi)$ is a unit in L_a . Indeed its nonzero reduction is invertible in the field $k((T))$, and inverses lift across the nilpotent ideal (π) . Uniqueness makes these inverses compatible in a . Hence

$$W[T]_{(\pi)} \longrightarrow \varprojlim_a L_a$$

is an integral, π -adically continuous ring homomorphism. Constant-term extraction is compatible with these reduction maps, so extends to

$$\text{CT} : V_W \longrightarrow W, \quad \text{CT}(1) = 1, \quad \text{CT}(T \partial_T f) = 0.$$

The identity with the logarithmic derivative is coefficientwise at every level and then passes to the inverse limit.

Tensor with the quadric algebra before completion and use $\varprojlim_a (B'/\pi^a)((T))$ as target. The completed relative forms decompose into a component with no $d \log T$ and one with $d \log T$ at the right. Apply constant term to the first to define P_0 , and to the second to define Res_T . The derivative identity gives compatibility with the de Rham differentials (with the conventional shift sign for a residue). In particular an actual primitive maps to an actual primitive. All operations are integral and bounded before inverting p ; there is no completion of the image of the differential. $\square$

Theorem 24.4. *For a prime p , put $k = \overline{\mathbf{F}}_p$, $W_0 = W(k)$, and*

$$V = \widehat{W_0[T]_{(p)}}^p, \quad \mathcal{R}_p = V[[x, y, z]] / (xy + f_p(z)),$$

where $f_p(z) = \Phi_p(1+z)$ for $p > 2$, and $f_2(z) = \Phi_4(1+z) = z^2 + 2z + 2$. Then $\mathcal{R}_p$ is a complete Noetherian regular local domain of dimension three, with residue field $k(T)$. For every $j \geq 3$,

$$\dim_{\mathbf{Q}} \ker(K_j(\mathcal{R}_p)_{\mathbf{Q}} \longrightarrow K_j(\text{Frac } \mathcal{R}_p)_{\mathbf{Q}}) = \infty.$$

The finite independent families proving this assertion consist of actual integral classes with zero image in $K_j(\mathcal{R}_p[1/x])$.

Proof. The ring $W_0[T]_{(p)}$ is a Noetherian DVR with uniformizer p and residue $k(T)$. Its completion V has the same properties. Since f_p is Eisenstein of degree at least two, the quotient in the statement has nonzero linear term p in a regular ambient local ring of dimension four. It is therefore regular of dimension three, with maximal ideal (x, y, z) and the asserted residue field.

Let $\rho = i$ for $p = 2$, and otherwise let ρ be a primitive p th root of unity. Put

$$W = W_0[\rho], \quad \pi = \rho - 1, \quad E = \text{Frac } W.$$

There are compatible identifications

$$(24.3) \quad W_0[T]_{(p)}[\rho] = W[T]_{(\pi)}, \quad V[\rho] = \widehat{W[T]_{(\pi)}}^\pi.$$

To verify the first, the norm of a polynomial $g \in W[T] \setminus (\pi)$ reduces to $\bar{g}^{[W:W_0]}$ and is primitive over W_0 . The norm and adjugate identity therefore invert g in the left side. The reverse inclusion is immediate. Finite freeness over W_0 gives the second identity after completion; the p -adic and π -adic topologies are cofinal.

Fix $n \geq 2$ and an arbitrary $m \geq 2$. Proposition 23.5 and Lemma 22.1 supply $m - 1$ actual classes $\beta_1, \dots, \beta_{m-1}$ over $W_m = W(\mathbf{F}_{p^m})[\rho]$, exactly anti-invariant under $\sigma(\rho) = \rho^{-1}$, with independent AMMN point coordinates. Their images in $K_{2n-1}(W)$ have coordinates $b_1, \dots, b_{m-1} \in E$ still independent over $\mathbf{Q}_p$: the point character is natural under the coefficient inclusion $\text{Frac } W_m \hookrightarrow E$. No surjectivity from actual K -groups to all point coordinates is asserted.

The regular divisor $x = 0$ of $\mathcal{R}_p$ has coordinate ring

$$\mathcal{R}_p/(x) \simeq V[\rho][[y]], \quad \rho \mapsto 1 + z.$$

Pull back the β_ν and push forward to obtain $a_\nu \in K_{2n-1}(\mathcal{R}_p)$. Also put $a_\nu[T] \in K_{2n}(\mathcal{R}_p)$. These actual classes vanish after inverting x .

Consider the single test algebra

$$\mathcal{B} = V[\rho]\langle X, Y, Z \rangle / (XY + Z(Z + 1)), \quad J = (X, Z), \quad \eta = 1 - [J].$$

By (24.3) it is the completion of an essentially smooth W -algebra of relative dimension three. Lemma 21.2 applies with this perfect-constant coefficient ring W . Its continuous de Rham model includes the variable T .

Retain the inverse pair of roots $a = \rho - 1$, $a' = \rho^{-1} - 1$ of f_p . Set $d = \rho - \rho^{-1}$ and $H(z) = \prod_{\alpha \neq a, a'} (z - \alpha)$. The substitution

$$(24.4) \quad z = a + dZ, \quad x = dX, \quad y = dH(a + dZ)Y$$

defines an actual map $\mathcal{R}_p \rightarrow \mathcal{B}$. The equation becomes $d^2 H(a + dZ)(XY + Z(Z + 1)) = 0$. All three images lie in $\pi\mathcal{B}$, so arbitrary formal power series converge. The coefficient map is the actual inclusion $V \rightarrow V[\rho]$; no finite-stage descent of completed coefficients is required.

Flat coefficient base change first splits the divisor after inverting p . Pulling its root-ideal line bundles along (24.4) leaves only J and $J' = (X, Z + 1)$ on $\mathcal{B}[1/p]$. For an omitted root with normalized value $b = (\alpha - a)/d \notin \{0, -1\}$, the ideal $(X, Z - b)$ is the unit ideal, since

$$b(b + 1) = -XY - (Z - b)(Z + b + 1).$$

The two surviving disjoint components have classes summing to zero, because their union is the principal divisor $X = 0$. The projection formula and exact anti-invariance therefore give

$$(24.5) \quad a_\nu \mapsto 2\beta_\nu\eta, \quad a_\nu[T] \mapsto 2\beta_\nu\eta[T] \quad \text{on } \mathcal{B}[1/p].$$

It remains to check these classes in actual groups. The reduction of each β_ν lies in $K_{2n-1}(k)$, a torsion group of order prime to p element by element. Choose one common prime-to- p integer N killing the finite list of reductions. Since $K_{2n}(k) = 0$, each $N\beta_\nu$ has a unique actual relative lift at the coefficient point. Its character is $Nt^{1-n}b_\nu$. The module comparison and the finite dimension-three HKR calculation of Lemma 23.1 give

$$Nt^{2-n}b_\nu\gamma, \quad Nt^{2-n}b_\nu\gamma \wedge d \log T,$$

where γ is the nonzero ordinary ruling two-form class on $B' = W\langle X, Y, Z \rangle / (XY + Z(Z + 1))$. Lemma 24.3 sends these respectively to $Nb_\nu\gamma$. The independence of the b_ν detects every nonzero rational combination of either actual relative family.

Finally,

$$C = \mathcal{B}/\pi = k(T)[X, Y, Z]/(XY + Z(Z + 1)), \quad K(C) \simeq K(k(T)) \oplus K(k(T)).$$

For $q \geq 2$, the actual group $K_q(k(T))$ is torsion. Every element descends first to a finite field of constants and then to a finite open of its affine line. Quillen localization expresses the K_q -group of that open as an extension of a quotient of the finite group $K_q(k_m)$ by a subgroup of a finite sum of $K_{q-1}(k(v))$. These groups are finite, including at $q = 2$. Continuity proves the claim for $k(T)$.

The algebra $\mathcal{B}$ is Noetherian and flat over the complete DVR $V[\rho]$, and its special fiber is smooth. Consequently π is a nonzerodivisor and $\mathcal{B}$ is regular: at every maximal ideal the regular quotient by π gives the regularity criterion, and π belongs to the Jacobson radical. Thus $K_{q+1}(C)_{\mathbf{Q}} = K_q(C)_{\mathbf{Q}} = 0$ for every $q \geq 3$. The actual relative sequence and the regular-divisor localization sequence respectively give injections

$$K_q(\mathcal{B}, \pi)_{\mathbf{Q}} \hookrightarrow K_q(\mathcal{B})_{\mathbf{Q}}, \quad K_q(\mathcal{B})_{\mathbf{Q}} \hookrightarrow K_q(\mathcal{B}[1/p])_{\mathbf{Q}}.$$

The detected families therefore remain independent in the generic test ring. Equation (24.5) proves their independence on $\mathcal{R}_p$. Their ranks are at least $m - 1$ for arbitrary m , and the degrees $2n - 1$ and $2n$ cover every integer at least three. $\square$

25. MILNOR, MOTIVIC, AND HIGHER-CHOW CONSEQUENCES

Put

$$A = (\mathbf{Z}_{(3)}[x, y, z]/(3 + xy + z^2))_{(x, y, z)}, \quad R \in \{A, A^h, \widehat{A}, A^{\text{sh}}\}, \quad F_R = \text{Frac}(R).$$

In each ring use the units

$$a = \frac{1+z}{2}, \quad \lambda = 1+x, \quad \mu = 1 + \frac{(1+x)y}{4}, \quad s_0 = a(1-a) = 1 + \frac{xy}{4}.$$

Thus $1 + x\mu = \lambda s_0$ and $c_R = 2[a][\lambda][\mu] \in K_3(R)$ is the explicit class already constructed. The superscript M , naive below means the tensor algebra on $R^\times$ modulo the two-sided Steinberg ideal; graded commutativity is not added to its definition.

Corollary 25.1. *For every such R , each of the four generic restriction maps*

$$\begin{aligned} K_3^{M, \text{naive}}(R)_{\mathbf{Q}} &\longrightarrow K_3^M(F_R)_{\mathbf{Q}}, & \widehat{K}_3^M(R)_{\mathbf{Q}} &\longrightarrow K_3^M(F_R)_{\mathbf{Q}}, \\ H_B^3(R, \mathbf{Q}(3)) &\longrightarrow H_B^3(F_R, \mathbf{Q}(3)), & CH^3(R, 3)_{\mathbf{Q}} &\longrightarrow CH^3(F_R, 3)_{\mathbf{Q}} \end{aligned}$$

has a nonzero kernel. Explicit witnesses are, respectively, $2\{a, \lambda, \mu\}$, its image in improved Milnor theory, $2[a]_B \cup [\lambda]_B \cup [\mu]_B$, and twice the cubical graph cycle specified below. The Milnor symbols and the graph cycle vanish over F_R integrally.

Proof. We first record the stronger Quillen input

$$(25.1) \quad c_R|_{R[1/3]} \neq 0 \quad \text{in } K_3(R[1/3])_{\mathbf{Q}}.$$

Proposition 20.3 detects the actual image of c_R on $B = W\langle X, Y, Z \rangle / (XY + Z^2 - 1)$, where $W = \mathbf{Z}_3[u]/(u^2 + 3)$, by $(x, y, z) = (uX, uY, uZ)$. For A^{sh} , first map to its completion and use the same calculation over $W_\infty = W(\overline{\mathbf{F}}_3)[u]/(u^2 + 3)$: the nonzero endpoint difference remains nonzero in the larger coefficient field, and Lemma 20.2 applies there. The perfect-residue-field scope of [2, Proposition 4.10] permits this base change. In either case $C = B/u$ is an affine-line torsor over $\mathbf{P}_k^1$, with $k = \mathbf{F}_3$ or $\overline{\mathbf{F}}_3$. Hence $K_3(C)$ is torsion by the projective bundle formula, Quillen's finite-field computation, and filtered continuity. Localization and dévissage therefore make $K_3(B)_{\mathbf{Q}} \rightarrow K_3(B[1/3])_{\mathbf{Q}}$ injective. The nonzero detector image on $B[1/3]$ factors through $R[1/3]$, proving (25.1).

Milnor theory. There are natural symbol maps

$$K_3^{M, \text{naive}}(R) \xrightarrow{\eta_R} \widehat{K}_3^M(R) \xrightarrow{\iota_R} K_3(R),$$

whose composite sends a symbol to the product of its unit classes. Kerz's universal construction, Proposition 4.2.8(4)–(6), and Theorem 4.3.2 give these maps, surjectivity of η_R , and a map $r_R : K_3(R) \rightarrow \widehat{K}_3^M(R)$ with $r_R \iota_R = 2$; these assertions hold for arbitrary local rings [31, Theorem 4.2.5, Proposition 4.2.8, Theorem 4.3.2]. Thus ι_R is split injective rationally. Both displayed Milnor classes have infinite order because their Quillen image is c_R . For the first three rings the residue field is $\mathbf{F}_3$ and no equality of naive and improved theory is asserted. For A^{sh} the residue field is infinite, so η_R is an isomorphism integrally.

In the field F_R , Steinberg relations give

$$\{\lambda s_0, \mu\} = \{-x, \lambda s_0\} = \{-x, s_0\}, \quad 2\{a, s_0\} = 0.$$

The first equality uses $\{-x\mu, 1+x\mu\} = 0$, and the second cancellation uses $\{-x, 1+x\} = 0$. Consequently

$$2\{a, \lambda, \mu\} = 2\{a, \lambda s_0, \mu\} = -2\{a, s_0, -x\} = 0.$$

These calculations use $-x$ only in the field, where it is a unit.

Beilinson cohomology. All four rings are regular, noetherian, and finite dimensional. Cisinski–Déglise identify

$$H_B^q(X, \mathbf{Q}(p)) \simeq \mathrm{Gr}_\gamma^p K_{2p-q}(X)_{\mathbf{Q}}$$

for every regular X , compatibly with the multiplicative Adams decomposition [17, Corollaries 14.2.14 and 14.2.19]. A unit has Adams weight one, so c_R has pure weight three and corresponds to the asserted nonzero cup product. Its generic restriction is zero by the preceding symbol calculation.

Ordinary cubical higher Chow groups. Let $\square = \mathbf{P}^1 \setminus \{1\}$, with faces $0, \infty$. Intersect the graph of $(a, \lambda, \mu) : \mathrm{Spec} R \rightarrow (\mathbf{P}^1)^3$ with $\mathrm{Spec} R \times \square^3$, and call the resulting closed cycle Γ_R . It has codimension three and

$$\Gamma_R \simeq \mathrm{Spec} R[1/(xy)], \quad \gamma_R = 2[\Gamma_R] \in CH^3(R, 3).$$

Indeed $a-1$ is a unit, $\lambda-1=x$, and $\mu-1=\lambda y/4$; the three graph equations form a regular sequence. This is not a morphism from all of $\mathrm{Spec} R$ to $\square^3$. Every face intersection is empty because the three coordinates are units, so Γ_R is admissible and has zero boundary. Over F_R , Totaro’s field isomorphism sends the Milnor symbol to this graph; the integral vanishing already proved therefore gives $\gamma_R|_{F_R} = 0$ [48, Theorem 1 and Sections 1–2].

Set $U = \mathrm{Spec} R[1/3]$. This is a regular equal-characteristic zero scheme, and the compatible comparisons give

$$CH^3(U, 3)_{\mathbf{Q}} \simeq H_B^3(U, \mathbf{Q}(3)) \simeq K_3(U)_{\mathbf{Q}}^{(3)}.$$

Here U is affine, excellent, regular, noetherian, finite dimensional, and of equal characteristic zero. The comparison with the actual Bloch cycle complex in precisely this affine scope is recorded in [11, Remark 1.3.2]; see also [17, Introduction B, p. xxii; Theorem 11.1.24 and Examples 11.1.25 and 11.2.3]. The cubical and simplicial cycle complexes give the same higher Chow groups [32, Theorem 4.7]. The rational monoidal comparison with Beilinson motives is [17, Theorem 16.1.4 and Corollary 16.1.7]. The graph-symbol normalization and its compatibility with the product of unit classes can be checked on a smooth finite-type characteristic-zero model and then pulled back: each map $A \rightarrow R$ is flat, and flat pullback takes the graph cycle to the displayed graph cycle. Thus the restricted class $\gamma_R|_U$ corresponds to $c_R|_U$. Thus (25.1) implies $\gamma_R|_U \neq 0$, and hence $\gamma_R \neq 0$. Only open restriction on R is used in this last step; no comparison between the full mixed-characteristic cycle complex and Beilinson cohomology on R is assumed. $\square$

26. HIGHER ODD DEGREES FOR THE ARITHMETIC QUARTIC

The arithmetic root lattice also detects coefficients in every higher odd degree. In this section write $A_2 = A$ for the quartic ring of Theorem 3.1, let $W_{\mathrm{ar}} = \mathbf{Z}_{(5)}[\rho]$ and $E_{\mathrm{ar}} = \mathbf{Q}(\rho)$, and let $i : V(x) \hookrightarrow \mathrm{Spec} A_2$ be its regular divisor. We retain the root classes η_g and their real Chern classes e_g on the split affine surface $\mathcal{Q}$. Throughout this section $g \in \mathbf{F}_5^\times$, and σ_g sends ρ to ρ^g .

Theorem 26.1. *For the quartic arithmetic local ring A_2 and every $n \geq 2$, the kernel of $K_{2n-1}(A_2) \rightarrow K_{2n-1}(\mathrm{Frac} A_2)$ contains an actual infinite-order class. The rational kernel has dimension at least two for even n and at least one for odd n .*

Proof. Borel’s theorem gives $\dim_{\mathbf{Q}} K_{2n-1}(E_{\mathrm{ar}})_{\mathbf{Q}} = 2$ and $\dim_{\mathbf{Q}} K_{2n-1}(\mathbf{Q})_{\mathbf{Q}} = 0$ for even n , respectively 1 for odd n [12, Proposition 12.2]. Denote restriction and transfer by res and Tr , respectively; they satisfy $\mathrm{res} \mathrm{Tr} = \sum_g \sigma_g$ and $\mathrm{Tr} \mathrm{res} = 4$. Thus the invariant subspace is exactly the restriction from $\mathbf{Q}$, and its trace-zero complement V_n has the asserted dimension. Quillen localization [40, 50] gives $K_{2n-1}(W_{\mathrm{ar}})_{\mathbf{Q}} \simeq K_{2n-1}(E_{\mathrm{ar}})_{\mathbf{Q}}$. Choose an actual $\gamma \in K_{2n-1}(W_{\mathrm{ar}})$ with noninvariant Borel image and put $\beta_n = 4\gamma - \sum_g \sigma_g \gamma$. Its regulator is nonzero and its Galois sum is zero exactly.

Set $c_n = i_*\beta_n$. Flat base change again identifies its image on $\mathcal{T}$ with $\sum_g(\sigma_g\beta_n|_{E_{\text{ar}}})\eta_g$. At a complex embedding, represent the point regulators $b_g \in \mathbf{C}/\mathbf{R}(n)$ in the complementary real line $i\mathbf{R}(n)$ [14, Section 6]. They form a nonzero vector with sum zero, hence a nonconstant vector. For each finite open $U = \mathcal{Q}_s$, the product regulator is represented by $2\pi i \sum_g b_g e_g$ in

$$H_{\mathcal{D},\text{an}}^3(U, \mathbf{R}(n+1)) = H^2(U, \mathbf{C})/H^2(U, \mathbf{R}(n+1)), \quad n+1 \geq 3.$$

Multiplication by $2\pi i$ induces an isomorphism $\mathbf{C}/\mathbf{R}(n) \simeq \mathbf{C}/\mathbf{R}(n+1)$. The root lattice and Proposition 5.1 therefore make this class nonzero. If an integer killed c_n , continuity for the filtered arithmetic localization would make that equality hold on some $\mathcal{Q}_s$, contradicting the regulator. Localization at the source divisor gives integral vanishing on $A_2[1/x]$. Applying the argument to every nonzero vector of V_n proves injectivity on V_n and gives the asserted rank bounds. $\square$

The restriction $n \geq 2$ is essential here: at $n = 1$, the target twist is two and F^2 on a complex affine surface is not zero. The displayed quotient formula cannot be used in that degree.

27. AN EXPLICIT PRODUCT OF THREE UNITS

Retain $A_2 = A$ from Setup 4.2 and its completion $\hat{A}_2 = \hat{A}$ from Theorem 3.2. We identify an explicit product of three units with the weighted root class after pullback to the split surface and the completed test algebra.

Let N be a complex manifold of dimension two. Write $A^\bullet(N)$ for complex smooth forms and

$$I(p)^k = \{\xi \in A^k([0, 1] \times N) : \xi|_0 \in (2\pi i)^p A_{\mathbf{R}}^k(N), \quad \xi|_1 \in F^p A^k(N)\}.$$

Thus $H^k(I(p)) = H_{\mathcal{D},\text{an}}^k(N, \mathbf{R}(p))$, with multiplication induced by wedge product.

Lemma 27.1. *Let g be a nowhere-vanishing holomorphic function with a holomorphic logarithm L , and let $z \in H^2(I(2))$. If ξ is a closed representative of z , put $\omega = \xi|_1$. Then ω is a closed holomorphic two-form, is independent of the representative, and*

$$r_1(g)z = [-L\omega] \quad \text{in} \quad H_{\mathcal{D},\text{an}}^3(N, \mathbf{R}(3)) = H^2(N, \mathbf{C})/H^2(N, \mathbf{R}(3)).$$

Here $r_1(g)$ is the analytic Deligne first regulator of g , with the interval and cone conventions of [14, Definition 2.7 and Proposition 2.8].

Proof. Write $L = a + i\phi$, with a, ϕ real, and let τ denote the interval coordinate. The usual representative of $r_1(g)$ is

$$i d\phi + d(\tau a).$$

Its difference from $d(\tau L)$ is $d(i(1 - \tau)\phi)$. The primitive $i(1 - \tau)\phi$ belongs to $I(1)^0$: its value at 0 is purely imaginary, hence in $\mathbf{R}(1)$, and its value at 1 is zero. Consequently $d(\tau L)$ represents $r_1(g)$ in $I(1)$.

The endpoint $\omega = \xi|_1$ is closed and lies in $F^2 A^2(N)$, so it is a closed holomorphic two-form. Changing ξ by an $I(2)$ -boundary cannot change this endpoint, because $F^2 A^1(N) = 0$.

The product has representative

$$v = d(\tau L) \wedge \xi = d(\tau L \xi).$$

Its endpoint at 0 vanishes. Its endpoint at 1 is $dL \wedge \omega$, which vanishes by complex dimension two. Thus the cone map sends its class to $[-\int_0^1 v]$. Fiberwise Stokes gives

$$-\int_0^1 v = -L\omega + d_N \left(\int_0^1 \tau L \xi \right),$$

and hence gives the asserted representative. It is closed, since $d(L\omega) = dL \wedge \omega = 0$. The primitive $\tau L \xi$ has endpoint $L\omega$ at 1; as $F^3 A^2(N) = 0$, it generally does not belong to $I(3)^2$.

Finally $F^3 A^\bullet(N) = 0$, so the cone long exact sequence gives the stated quotient: the next real-to-complex cohomology map is injective. This proves the formula. $\square$

Proposition 27.2. *The triple*

$$(27.1) \quad c_{\exp} = \left\{ 2 + y, 1 - x, 1 + (1 - x)(x^3 + Ty^3)y((1 + y)^5 + 1) \right\} \in K_3(A_2)$$

has infinite order in A_2 and in $\widehat{A}_2$, and its restriction to the localization at x is zero integrally in both cases.

Proof. Integral vanishing. Put $t = -1 - y$, $v = x^3 + Ty^3$, and

$$H(t) = \frac{t^{10} - 1}{\Phi_{10}(t)} = (t^5 - 1)(t + 1) = y((1 + y)^5 + 1), \quad b = vH(t).$$

Then $xv = -\Phi_{10}(t)$, so $1 - xb = t^{10}$. Set $g = 1 - t$, $\lambda = 1 - x$, and $\mu = 1 + \lambda b$. These are units: their residues are 2, 1, 1 respectively, and t is also a unit. In the convention $\langle r, s \rangle = \{r, 1 - rs\}$ when r is a unit, the Dennis–Stein relations give

$$\delta = \langle x, b \rangle = \langle x, b + 1 - xb \rangle = \langle x, \mu \rangle = \{\lambda t^{10}, \mu\}.$$

Here $\langle x, 1 \rangle = 0$ and $1 - x\mu = \lambda t^{10}$; see [50, Chapter III, Definition 5.11 and relations (D1)–(D3)]. Since $\{g, t^{10}\} = 10\{1 - t, t\} = 0$, multiplication gives

$$(27.2) \quad [g]\delta = \{g, \lambda, \mu\} = c_{\exp}, \quad [g]\delta|_{D(x)} = \{g, x, t^{10}\} = 0.$$

All identities are integral and persist under every ring map in use.

The universal supported coefficients. Write $E_{\text{ar}} = \mathbf{Q}(\rho)$, where ρ is primitive of order five, and $t_g = -\rho^g$ for $g \in \mathbf{F}_5^\times$. On the smooth surface

$$Q^0 = \text{Spec } E_{\text{ar}}[X, V, t]/(XV + \Phi_{10}(t)), \quad U = Q^0[1/(t(1 - t))],$$

the class $\alpha = [1 - t]\langle X, VH(t) \rangle$ is defined and vanishes on $U[1/X]$. The entire divisor $D = V(X)$ is contained in U . Localization and dévissage therefore give an actual supported lift $\kappa \in K_3(Q^0)$ of α . Put $\eta_g = 1 - [(X, t - t_g)]$. Since the four components of D are affine lines, $\kappa = \sum_g \gamma_g \eta_g$ with $\gamma_g \in K_3(E_{\text{ar}})$, and $\sum_g \eta_g = 0$.

Independence of the supported lift. This lift is unique rationally as a class on Q^0 . Here is a direct K -theoretic check, independent of its periods, using localization and the Laurent fundamental theorem [50]. Put $Z = \mathbf{A}_{E_{\text{ar}}}^1 \setminus \{0, 1\}$. Then $U[1/X] \simeq \mathbf{G}_m \times Z$, and localization on the affine line, split by the units t and $1 - t$, gives

$$K_r(Z) \simeq K_r(E_{\text{ar}}) \oplus K_{r-1}(E_{\text{ar}})[t] \oplus K_{r-1}(E_{\text{ar}})[1 - t].$$

Together with the Laurent fundamental theorem this gives

$$\begin{aligned} K_4(U[1/X]) &\simeq K_4(E_{\text{ar}}) \oplus K_3(E_{\text{ar}})[t] \oplus K_3(E_{\text{ar}})[1 - t] \\ &\quad \oplus K_3(E_{\text{ar}})[X] \oplus K_2(E_{\text{ar}})[X][t] \oplus K_2(E_{\text{ar}})[X][1 - t]. \end{aligned}$$

The first three summands extend to U , so their boundary in $K_3(D) = K_3(E_{\text{ar}})^4$ is zero. For a number field, $K_2(E_{\text{ar}})_{\mathbf{Q}} = K_4(E_{\text{ar}})_{\mathbf{Q}} = 0$: localization reduces this to the rational even K -group vanishing for its number ring [12, Proposition 12.2] and the torsion odd K -groups of finite fields [41][50, IV, Corollary 1.13]. The only remaining boundary is that of $\beta[X]$, namely the diagonal tuple $\pm(\beta, \beta, \beta, \beta)$ with a common convention-dependent sign, since X has order one on each root component. Localization therefore identifies the kernel as

$$\ker(K_3(D)_{\mathbf{Q}} \rightarrow K_3(U)_{\mathbf{Q}}) = \Delta K_3(E_{\text{ar}})_{\mathbf{Q}}.$$

The diagonal pushes forward to zero on Q^0 , by the length-one free resolution of the principal divisor $X = 0$. Consequently any two actual supported lifts of α give the same class in $K_3(Q^0)_{\mathbf{Q}}$. Moreover, since $Q^0[1/X] \simeq \mathbf{G}_m \times \mathbf{A}_{E_{\text{ar}}}^1$, the same Laurent calculation gives $\ker(K_3(D) \rightarrow K_3(Q^0)) = \Delta K_3(E_{\text{ar}})$ already integrally. Thus its supported coefficients are unique modulo the diagonal.

Write bl for the Bloch-symbol map to algebraic K_3 in [14, Theorem 4.9]. For a root a of order ten define $\beta_{\text{BT}}(a) = \frac{1}{10} \text{bl}(10[a])$ using [14, Theorem 4.9 and (4.8)]. Both a and $1 - a$ are global

units, and $10[a]$ has zero wedge boundary. The normalization at every complex embedding σ is $r_2(\beta_{\text{BT}}(a))_\sigma = -iD_{\text{BW}}(\sigma a)$. We claim the exact rational identity

$$(27.3) \quad \kappa = -10 \sum_g \beta_{\text{BT}}(t_g) \eta_g \quad \text{in } K_3(Q^0)_{\mathbf{Q}}.$$

Determination of the supported coefficients. Fix a complex embedding $\sigma : E_{\text{ar}} \hookrightarrow \mathbf{C}$ and write $a_g = \sigma t_g$. The four roots are $e^{\pm i\pi/5}$ and $e^{\pm 3i\pi/5}$. For any distinct pair a_i, a_j , their chord avoids 0, 1, and the other roots. Choose one neighborhood of the chord on which the logarithm $L = \text{Log}(1 - t)$ and the classical dilogarithm $F = \text{Li}_2(t)$ are single-valued. We use the branches continued from their power series on the open unit disc; they satisfy $dF = -L dt/t$.

Over this chord the matching sphere is parametrized by

$$X = \sqrt{|\Phi_{10}(t)|} e^{i\vartheta}, \quad V = -\frac{\Phi_{10}(t)}{|\Phi_{10}(t)|} \overline{X},$$

with $X = V = 0$ at the two endpoints. This parametrization satisfies $XV = -\Phi_{10}(t)$ and is smooth at each pole: $|X|^2$ is a smooth positive multiple of the distance to that endpoint along the chord. Orient the sphere by $ds \wedge d\vartheta$, where s increases from a_i to a_j .

On $X \neq 0$, the weight-two holomorphic endpoint of the regulator of $\delta = \langle X, VH(t) \rangle$ is

$$\omega = 10 \frac{dX}{X} \wedge \frac{dt}{t} = -\frac{dX \wedge d(VH(t))}{t^{10}}.$$

The last expression extends holomorphically across both poles. It therefore identifies the endpoint on the entire neighborhood. Lemma 27.1 now gives $r_3(\alpha) = [-L\omega]$. In particular this calculation retains the full interval regulator, including its real endpoint. On the punctured sphere, $L\omega = 10d(F dX/X)$. The boundary circles at the two poles contribute $2\pi i F(a_j)$ and $-2\pi i F(a_i)$, and the integrals over shrinking smooth caps tend to zero. The resulting period is

$$(27.4) \quad \int_{\Sigma_{ij}} r_3(\alpha) = -20\pi i (F(a_j) - F(a_i)) \pmod{\mathbf{R}(3)}.$$

Since $|a_i| = |a_j| = 1$, the real part is $20\pi(D_{\text{BW}}(a_j) - D_{\text{BW}}(a_i))$. The root divisor has local normal coordinate X . Its intersection numbers with the oriented sphere are $+1$ at the initial pole and -1 at the terminal pole. Thus the Chern periods of η_i and η_j are $2\pi i$ and $-2\pi i$, respectively. Multiplicativity gives, for every σ ,

$$r_2(\gamma_i - \gamma_j)_\sigma = -10r_2(\beta_{\text{BT}}(t_i) - \beta_{\text{BT}}(t_j))_\sigma.$$

Borel injectivity [14, Section 6], applied to all embeddings, proves equality in $K_3(E_{\text{ar}})_{\mathbf{Q}}$. The remaining coefficient is common to all four components and contributes zero because $\sum_g \eta_g = 0$. This proves (27.3).

Comparison with the actual cyclotomic coefficient. Write $\beta_{\text{BDJ}}(a)$ for the field class supplied by the symbol map of [8, Theorem 1.10(1)–(2) and Remark 1.7]. The complex formula of [21, Proposition 4.1 and Remark 5.2] compares its regulator with that of $\beta_{\text{BT}}(a)$ by one nonzero real scalar, independent of the field, root, and embedding. This scalar is rational. Indeed, take $a = \zeta_6$ in $L = \mathbf{Q}(\sqrt{-3})$ and put $\beta_{\text{BT}}(a) = \frac{1}{6} \text{bl}(6[a])$. Both classes belong to the one-dimensional rational vector space $K_3(L)_{\mathbf{Q}}$ [12, Proposition 12.2] and are nonzero, since $D_{\text{BW}}(\zeta_6) > 0$. Consequently $\beta_{\text{BDJ}}(\zeta_6) = s\beta_{\text{BT}}(\zeta_6)$ for a unique $s \in \mathbf{Q}^\times$. Taking the complex regulator identifies s with the universal real scalar. Borel injectivity at all embeddings [14, Section 6] then yields

$$\beta_{\text{BDJ}}(a) = s\beta_{\text{BT}}(a)$$

for every root in use. The same field symbol map occurs in the complex and p -adic formulas of the cited theorem. Finite-residue localization identifies $K_3(W_{\text{ar}})_{\mathbf{Q}}$ with $K_3(E_{\text{ar}})_{\mathbf{Q}}$, so this comparison also holds before local completion. Consequently the coefficient and character argument of Proposition 10.1 applies to

$$\Gamma = \text{bl}(10[-\rho]) \in K_3(\mathcal{O}_{E_{\text{ar}}}), \quad \beta = \Gamma - \sigma_{-1}\Gamma \in K_3(W_{\text{ar}}).$$

This is an actual integral antisymmetrization. Its residue class vanishes because σ_{-1} acts trivially on the residue field. The identities $D_{\text{BW}}(a^{-1}) = -D_{\text{BW}}(a)$ at every embedding and Borel injectivity [14, Section 6] give

$$\beta_{\mathbf{Q}} = 20\beta_{\text{BT}}(-\rho) = \frac{20}{s}\beta_{\text{BDJ}}(-\rho).$$

The syntomic symbol formula therefore makes the point regulator of β a common nonzero rational multiple of $\text{Li}_2(-\rho)$. The two congruences in the proof of Proposition 10.1 imply that its e_1 and e_3 components are nonzero. The finite Chern factorization and point comparison in Proposition 10.1 transfer this statement to its AMMN point coordinates. Equation (27.3) becomes

$$(27.5) \quad \kappa = -\frac{1}{2} \sum_g \sigma_g(\beta) \eta_g.$$

Detection on the arithmetic ring. On the chord $e^{-i\pi/5} \rightarrow e^{i\pi/5}$, the real period in (27.4) is

$$40\pi D_{\text{BW}}(e^{i\pi/5}) > 0.$$

The inequality follows from $D_{\text{BW}}(e^{i\theta}) = -\int_0^\theta \log(2 \sin(v/2)) dv$ for $0 < \theta < \pi/3$. This sphere lies in $Q^0[1/(t+1)]$, identified with $\mathcal{Q}$ by $X = x$, $V = x^3 + Ty^3$, and $t = -1 - y$. The supported expression for κ makes its regulator a real root class. Proposition 5.1 keeps this class nonzero on every permitted finite localization. If a nonzero integer killed c_{exp} , continuity would realize that vanishing at one such finite localization, contradicting its regulator. This proves infinite order in A_2 directly.

Detection after completion. The same substitutions define a ring map $\mathcal{O}(Q^0) \rightarrow B[1/\pi]$, with $X \mapsto \pi X$, $t \mapsto -1 - \pi Y$. Both t and $1 - t$ are units already in B , and the root line $(X, t - t_g)$ pulls back to $J_g[1/\pi]$. The denominator and completion maps of Section 11 therefore give

$$c_{\text{exp}}|_{B[1/\pi]} = \widehat{c}_{\text{exp}}|_{B[1/\pi]} = -\frac{1}{2} \sum_g \sigma_g(\beta)(1 - [J_g])|_{B[1/\pi]}.$$

Proposition 9.1 and the actual assembly detect the right side as nonzero in rationalized completed K -theory. Thus both actual source classes have infinite order. The equality used here is on the split surface and its test pullback. $\square$

Corollary 27.3. *For either $A = A_2$ or $A = \widehat{A}_2$, the ordinary Milnor symbol*

$$m = \left\{ 2 + y, 1 - x, 1 + (1 - x)(x^3 + Ty^3)y((1 + y)^5 + 1) \right\} \in K_3^M(A)$$

has infinite order and has zero image in $K_3^M(\text{Frac } A)$, integrally. In particular the ordinary Milnor Gersten augmentation also fails after rationalization.

Proof. Write

$$\begin{aligned} t &= -1 - y, & g &= 1 - t, & \lambda &= 1 - x, & q &= t^{10}, \\ b &= (x^3 + Ty^3)y((1 + y)^5 + 1), & \mu &= 1 + \lambda b. \end{aligned}$$

The defining equation gives $1 - xb = q$ and hence $1 - x\mu = \lambda q$. The elements g, λ, μ, q are units in A : their residues are respectively 2, 1, 1, 1. In its fraction field, x and $x\mu$ are also nonzero. The two Steinberg relations and skew-symmetry in field Milnor K -theory give

$$\begin{aligned} 0 &= \{x\mu, \lambda q\}, & \{x, \lambda\} &= 0, \\ \{x, q\} &= -\{\mu, \lambda\} - \{\mu, q\} = \{\lambda q, \mu\}. \end{aligned}$$

Furthermore $\{g, q\} = 10\{1 - t, t\} = 0$. Therefore, without rationalizing,

$$m = \{g, \lambda, \mu\} = \{g, \lambda q, \mu\} = \{g, x, q\} = -\{g, q, x\} = 0 \quad \text{in } K_3^M(\text{Frac } A).$$

The natural homomorphism $K_3^M(A) \rightarrow K_3(A)$ induced by products of units sends m to the already detected infinite-order class c_{exp} . Thus no nonzero integer can kill m . $\square$

28. A TAME FAMILY AT EVERY RESIDUE PRIME

Theorem 28.1. *Let p be any prime and let $d \geq 3$ satisfy $p \nmid d$. Put*

$$A_{p,d} = \left(\mathbf{Z}_{(p)}[T, x, y] / (p + x^d + y^d + Txy^{d-1}) \right)_{(p,x,y)}.$$

This is a regular local ring of dimension two, with residue field $\mathbf{F}_p(T)$, and

$$\dim_{\mathbf{Q}} \ker(K_3(A_{p,d})_{\mathbf{Q}} \longrightarrow K_3(\text{Frac } A_{p,d})_{\mathbf{Q}}) \geq \lfloor d/2 \rfloor.$$

In particular these dimensions are unbounded with the residue prime and local dimension fixed. The theorem concerns the uncompleted rings displayed above.

Proof. The actual source and the flat splitting ring. Write $A = A_{p,d}$. Its regular ambient local ring has parameters p, x, y , and its defining equation has linear term p . Thus A is regular of dimension two with maximal ideal (x, y) . Solving the equation for T gives $\text{Frac } A = \mathbf{Q}(x, y)$.

Choose $\theta^d = -p$ and set

$$K = \mathbf{Q}(\theta), \quad W_0 = \mathbf{Z}_{(p)}[\theta], \quad A/(x) = W_0[T]_{(\theta)}.$$

Eisenstein's criterion gives $[K : \mathbf{Q}] = d$; W_0 is a DVR with uniformizer θ and residue field $\mathbf{F}_p$. Quillen localization gives $K_3(W_0)_{\mathbf{Q}} \simeq K_3(K)_{\mathbf{Q}}$, and every actual $K_3(K)$ class lifts to $K_3(W_0)$, since $K_2(\mathbf{F}_p) = 0$. Pull back such a coefficient β to $A/(x)$ and push it forward to obtain $c_{\beta} \in K_3(A)$. Its restriction to $A[1/x]$ is zero. We will prove that the resulting rational map $K_3(K)_{\mathbf{Q}} \rightarrow K_3(A)_{\mathbf{Q}}$ is injective.

Let $E = K(\mu_d)$ and choose a prime $\mathfrak{P}$ above p . Put $W = (\mathcal{O}_E)_{\mathfrak{P}}$ and let k be its finite residue field. The completion of E at this prime is $\mathbf{Q}_p(\mu_d, \theta)$. Since $p \nmid d$, adjoining μ_d is unramified, and $Z^d + p$ remains Eisenstein over that unramified extension. Consequently θ is a uniformizer of W , and the d roots of unity $(\zeta_i)_{i=1}^d$ in W have distinct reductions in k . The ring W is flat over $\mathbf{Z}_{(p)}$, but need not be finite. Use the explicit local ring

$$B = (A \otimes_{\mathbf{Z}_{(p)}} W)_{(\theta,x,y)} = \left(V[x, y] / (x^d + y^d + Txy^{d-1} - \theta^d) \right)_{(\theta,x,y)}, \quad V = W[T]_{(\theta)}.$$

The ideal indicated is maximal with residue field $k(T)$, and $A \rightarrow B$ is flat by tensor product followed by localization.

The exceptional curve and its root points. The projectivized tangent cone of B is

$$C : X^d + Y^d + TXY^{d-1} = U^d \subset \mathbf{P}_{k(T)}^2.$$

It is geometrically smooth. A singular point must have $U = 0$; then $Y \neq 0$ and $q(s) = s^d + Ts + 1$ would have a multiple root. The identity $dq - sq' = (d-1)Ts + d$ shows this can occur only at finitely many algebraic values of T , or never if $p \mid d-1$. Thus it cannot occur over $k(T)$.

The smooth plane curve C is geometrically integral, so the associated graded ring of B is a domain. Krull intersection then shows that B is a domain. The blow-up $\mathcal{Y} = \text{Bl}_{(\theta,x,y)} \text{Spec } B$ is regular and has the single reduced exceptional curve C . Here are the local details needed for this assertion. Its θ -chart has equation $X^d + Y^d + TXY^{d-1} = 1$, with $x = \theta X$, $y = \theta Y$; the other two strict-transform charts reduce to the corresponding charts of the smooth C . A nonzero differential tangent to the ambient exceptional plane therefore proves regularity at every point of C . The nonregular locus is closed by excellence. Its proper image in the local scheme $\text{Spec } B$, if nonempty, would contain the closed point, a contradiction. This proves regularity everywhere. Off the closed point the blow-up is an isomorphism; the hypersurface S_2 property then also shows B is normal.

The tautological ideal gives $\mathcal{O}_{\mathcal{Y}}(C)|_C = \mathcal{O}_C(-1)$. The root primes $(x, y - \theta\zeta_i)$ have strict transforms meeting C at $P_i = [1 : 0 : \bar{\zeta}_i]$, each with multiplicity one. Indeed, on the θ -chart the Y -derivative is a unit at $X = 0, Y = \zeta_i$, and the two regular parameters there are θ, X .

The exceptional Picard lattice. On $C^* = C \cap \{UY \neq 0\}$ the root-point classes have only the diagonal relation. To prove this, extend k to $\bar{k}$ and consider

$$M = \text{Spec } \bar{k}[X, Z, Y^{\pm 1}] / (XZ - (1 - Y^d)), \quad T = (Z - X^{d-1}) / Y^{d-1}.$$

On inverting X its ring is the Laurent UFD $\bar{k}[X^{\pm 1}, Y^{\pm 1}]$. Divisor localization and its units $cX^a Y^b$ give $\text{Pic}(M) = \mathbf{Z}^d / \mathbf{Z}(1, \dots, 1)$, generated by $(X, Y - \bar{\zeta}_i)$.

Every finite fiber $T = t$ is geometrically integral. Under $s = X/Y$, $w = 1/Y$ it is $w^d = s^d + ts + 1$, $w \neq 0$. The polynomial $q_t = s^d + ts + 1$ has $\deg \gcd(q_t, q'_t) \leq 1$ by $dq_t - sq'_t = (d-1)ts + d$ and $d \neq 0$ in $\bar{k}$. If it were an ℓ th power for a prime $\ell \mid d$, that degree would be at least $d - d/\ell \geq 2$, a contradiction. A rational-function power would already be a polynomial power. Since $\bar{k}$ contains μ_d and $p \nmid d$, this also proves irreducibility of $w^d - q_t$: the root extension is cyclic, and a proper degree would express q_t as such a power. Each removed T -fiber is therefore one principal prime divisor, of multiplicity one. Divisor localization gives $\text{Pic}(M) \simeq \text{Pic}(M_{\bar{k}(T)})$. The generic fiber of M is $C_{\bar{k}(T)}^*$, proving the root-point assertion.

Arithmetic Picard and Betti persistence. The generic affine surface and test localization are

$$\begin{aligned} Q &= \text{Spec } E[T, x, y, y^{-1}] / (p + x^d + y^d + Txy^{d-1}) \\ &\simeq \text{Spec } E[x, v, y^{\pm 1}] / (xv + y^d + p), \quad \mathcal{T} = \text{Spec } B[1/\theta, 1/y], \end{aligned}$$

where $v = x^{d-1} + Ty^{d-1}$. Inverting x again gives a Laurent UFD. Thus $\text{Pic}(Q) = \mathbf{Z}^d / \mathbf{Z}(1, \dots, 1)$, generated by $D_i = (x, y - \theta\zeta_i)$; this remains true after every coefficient field extension. The map $\text{Pic}(Q) \rightarrow \text{Pic}(\mathcal{T})$ is injective. Indeed, a root combination made principal on $\mathcal{T}$ can acquire on $\mathcal{Y}$ only strict divisors over $\theta = 0$ or $y = 0$, and a multiple of C . The former miss C^* and $\mathcal{O}_C(-1)$ is trivial on $U \neq 0$. Restriction to C^* gives the same combination of the P_i , so all coefficients are equal, which was already the zero relation on Q .

Write $\mathcal{O}(\mathcal{T}) = \varinjlim_s \mathcal{O}(Q_s)$, where s runs over the permitted arithmetic denominators. Each E -prime divisor removed by such an s has zero Picard class by the injection just proved. Its geometric components are one Galois orbit. The absolute Galois group over E acts trivially on the torsion-free root Picard lattice, so each geometric component also has zero class. The degree-two support sequence consequently gives

$$H^2(Q(\mathbf{C}), \mathbf{R}) \hookrightarrow H^2(Q_s(\mathbf{C}), \mathbf{R}) \quad \text{for every permitted } s.$$

The root Chern classes e_i span an injected copy of $\mathbf{R}^d / \mathbf{R}(1, \dots, 1)$: in the Gysin sequence of the D_i , the angular class of x has residue the diagonal and that of y has residue zero. This proves precisely the needed real root-class persistence.

Detection of every nonzero coefficient. Let $\iota_i : K \hookrightarrow E$ send θ to $\theta\zeta_i$, and put $\eta_i = 1 - [\mathcal{O}_Q(-D_i)]$. Flat base change for the actual Gysin class and the projection formula, followed by restriction to the regular $\mathcal{T}$, give

$$c_\beta|_{\mathcal{T}} = \sum_i \iota_i(\beta) \eta_i \Big|_{\mathcal{T}}.$$

At one complex embedding of E , the point regulators $b_i \in \mathbf{C}/\mathbf{R}(2)$ run through all embeddings of K . Choose their representatives in $i\mathbf{R}$. If $\beta \neq 0$ in $K_3(K)_{\mathbf{Q}}$, Borel injectivity [14, Section 6] makes this vector nonzero. Conjugate embeddings have opposite coefficients and real embeddings have coefficient zero, so it is not diagonal.

The absolute multiplicative regulator of Bunke–Tamme [14, Theorem 2.31] of the supported sum is the real root class $2\pi i \sum_i b_i e_i$, modulo $H^2(Q_s(\mathbf{C}), \mathbf{R}(3))$. It is nonzero on every permitted finite open by the preceding injection. Here the analytic Deligne target is $H^2(Q_s, \mathbf{C}) / H^2(Q_s, \mathbf{R}(3))$ because its complex dimension is two and $F^3 = 0$; a nonzero real class survives this quotient. Each individual regulator calculation spreads to a regular finite type arithmetic model after inverting one integer. The standard continuity $K_3(\mathcal{T}) = \varinjlim_s K_3(Q_s)$ now rules out any nonzero rational coefficient in the kernel of the Gysin map.

Finally $Z^d + p$ has one real root when d is odd and none when d is even. Borel’s rank calculation [12, Proposition 12.2] therefore gives $\dim_{\mathbf{Q}} K_3(K)_{\mathbf{Q}} = r_2(K) = \lfloor d/2 \rfloor$. Choosing actual lifts of a rational basis yields the asserted number of independent classes of infinite order in the generic kernel. All uses of localization, base change and regulators are the standard inputs cited above [40, 41, 12, 14, 50]. $\square$

For example, $(p, d) = (2, 3)$ gives $(\mathbf{Z}_{(2)}[T, x, y]/(2 + x^3 + y^3 + Txy^2))_{(2, x, y)}$. For any fixed p , arbitrarily large odd d prime to p give the unbounded rank assertion without enlarging the residue field $\mathbf{F}_p(T)$.

Remark 28.2. The condition $d \geq 3$ is used in the vertical-fiber calculation. When $d = 2$ and $p \neq 2$, the fiber at $t = 2$ has $s^2 + 2s + 1 = (s + 1)^2$, so the Kummer irreducibility argument is unavailable and the vertical fiber splits. Localization can then kill a root divisor difference. The proof therefore supplies no degree-two analogue of the displayed tame family.

Corollary 28.3. *For $A_{p,d}$ as in Theorem 28.1 and every $n \geq 2$, the rational kernel of $K_{2n-1}(A_{p,d}) \rightarrow K_{2n-1}(\text{Frac } A_{p,d})$ has dimension at least*

$$\begin{cases} \lfloor d/2 \rfloor, & n \text{ even,} \\ \lfloor d/2 \rfloor - 1, & n \text{ odd.} \end{cases}$$

These bounds are realized by actual classes of infinite order. For odd d the bound is $(d - 1)/2$ in every odd K -degree at least three.

Proof. Retain $K = \mathbf{Q}(\theta)$, W_0 , the splitting field E , and the d embeddings $\iota_i : K \hookrightarrow E$ from the theorem. Use the specified rational coefficient space

$$V_n = \ker \left(\text{Tr}_{K/\mathbf{Q}} : K_{2n-1}(K)_{\mathbf{Q}} \longrightarrow K_{2n-1}(\mathbf{Q})_{\mathbf{Q}} \right).$$

For an actual $\gamma \in K_{2n-1}(K)$ set $\beta = d\gamma - \text{res}_{K/\mathbf{Q}} \text{Tr}_{K/\mathbf{Q}} \gamma$. The identity $\text{Tr res} = d$ shows that $\text{Tr } \beta = 0$ integrally. These actual classes span V_n rationally, because the displayed operator acts as multiplication by d on V_n . Each β lifts to $K_{2n-1}(W_0)$, since $K_{2n-2}(\mathbf{F}_p) = 0$. Localization, flat base change and the projection formula give an actual class vanishing generically whose test image is $\sum_i \iota_i(\beta) \eta_i$.

At a fixed complex embedding of E let b_i be the point regulators, represented in the complementary real line $i\mathbf{R}(n)$ to $\mathbf{R}(n)$ in $\mathbf{C}$. Base change for field transfer gives $\sum_i \iota_i(\beta) = \text{res}_{E/\mathbf{Q}} \text{Tr}_{K/\mathbf{Q}} \beta = 0$; therefore $\sum_i b_i = 0$. If β is nonzero rationally, Borel injectivity [14, Section 6] makes the vector (b_i) nonzero and hence nondiagonal. Its line-bundle product is $2\pi i \sum_i b_i e_i$. The preceding Betti persistence keeps this root class nonzero on every permitted finite open. The relevant analytic Deligne group is

$$H_{\mathcal{D}, \text{an}}^3(Q_s, \mathbf{R}(n+1)) = H^2(Q_s, \mathbf{C})/H^2(Q_s, \mathbf{R}(n+1)),$$

because $n+1 \geq 3$ and $Q_s(\mathbf{C})$ is a surface. Multiplication by $2\pi i$ identifies $\mathbf{C}/\mathbf{R}(n)$ with $\mathbf{C}/\mathbf{R}(n+1)$, so the product regulator is nonzero. The same continuity argument proves injectivity of the supported map on V_n ; applying it to every rational relation gives independent actual representatives.

Rational transfer is surjective, since its composition with restriction is multiplication by d . Borel's ranks [12, Proposition 12.2] give

$$\dim V_n = \begin{cases} r_2(K), & n \text{ even,} \\ r_1(K) + r_2(K) - 1, & n \text{ odd.} \end{cases}$$

Here $r_2(K) = \lfloor d/2 \rfloor$, while $r_1(K)$ is one for odd d and zero for even d ; the bounds follow. $\square$

29. A FIXED ARITHMETIC RING WITH INFINITE RANK

Theorem 29.1. *Fix a prime p and an integer $d \geq 3$ prime to p . Let $V = \mathbf{Z}_{(p)}^{\text{sh}}$, with residue field $\overline{\mathbf{F}}_p$, and put*

$$R = \left(V[T, x, y]/(p + x^d + y^d + Txy^{d-1}) \right)_{(p, x, y)}.$$

Then R is a Noetherian regular local ring of dimension two, with residue field $\overline{\mathbf{F}}_p(T)$. For every $n \geq 2$,

$$\dim_{\mathbf{Q}} \ker(K_{2n-1}(R)_{\mathbf{Q}} \longrightarrow K_{2n-1}(\text{Frac } R)_{\mathbf{Q}}) = \infty.$$

The coefficient DVR V is strictly henselian; no henselianity of the displayed R is asserted. This theorem concerns this single uncompleted ring.

Proof. The coefficient and splitting rings. The strict henselization V is a DVR with uniformizer p , and $F = \text{Frac } V$ is algebraic over $\mathbf{Q}$; see the Stacks Project, [Tag 0AP3](#) and the proof of [Tag 09ET](#). It is excellent because it is a DVR of fraction characteristic zero ([Tag 07QW\(4\)](#)). The simple roots of $Z^N - 1$ in $\overline{\mathbf{F}}_p$ lift by Hensel's lemma whenever $p \nmid N$, so $\mu_N \subset V$. The ambient local ring defining R has parameters p, x, y , and the defining equation has initial term p . Thus R has the stated regularity, dimension and residue field.

Choose $\theta^d = -p$ and set

$$W = V[\theta], \quad E_\infty = F(\theta), \quad G = \text{Gal}(E_\infty/F) = \{\sigma_\zeta : \zeta \in \mu_d\}, \quad \sigma_\zeta(\theta) = \zeta\theta.$$

Eisenstein shows that W is a finite free DVR extension of V , with uniformizer θ , residue field $\overline{\mathbf{F}}_p$, and degree d . Its displayed Galois action preserves W . Moreover

$$D = \text{Spec}(R/(x)) = \text{Spec } W[T]_{(\theta)}$$

is regular. Pullback to D followed by its Gysin map $i_* : K_{2n-1}(D) \rightarrow K_{2n-1}(R)$ therefore constructs actual classes vanishing on $R[1/x]$.

The geometric detector over E_∞ . Use the flat splitting map to

$$B = (R \otimes_V W)_{(\theta, x, y)}, \quad \mathcal{T} = \text{Spec } B[1/\theta, 1/y],$$

and the smooth surface

$$Q = \text{Spec } E_\infty[x, v, y^{\pm 1}]/(xv + y^d + p), \quad v = x^{d-1} + Ty^{d-1}.$$

The geometric proof of [Theorem 28.1](#) applies over these coefficient rings. The required geometric calculations are as follows. The blow-up still has the single smooth exceptional curve

$$C : X^d + Y^d + TXY^{d-1} = U^d \subset \mathbf{P}_{\overline{\mathbf{F}}_p(T)}^2.$$

The same three charts prove its regularity along C , and excellence and properness prove regularity everywhere. The root strict transforms meet it transversely at $[1 : 0 : \zeta]$. On $C \cap \{UY \neq 0\}$ their Picard classes have the lattice $\mathbf{Z}^d/\mathbf{Z}(1, \dots, 1)$: the auxiliary surface $XZ = 1 - Y^d$, $Y \neq 0$, has that lattice, and every finite $T = (Z - X^{d-1})/Y^{d-1}$ fiber is integral by the $dq - sq'$ argument. These calculations already take place over $\overline{\mathbf{F}}_p$ and do not require finite residue constants.

Consequently, restriction of divisors to this exceptional open gives

$$\text{Pic}(Q) = \mathbf{Z}^d/\mathbf{Z}(1, \dots, 1) \hookrightarrow \text{Pic}(\mathcal{T}),$$

with generators $D_\zeta = (x, y - \zeta\theta)$. The Picard calculation is unchanged by coefficient field extension. Each E_∞ -prime divisor removed by a permitted finite localization Q_s has zero Picard class. Its geometric components form one Galois orbit and all have the same class, since the absolute Galois group over E_∞ fixes its root generators and the lattice is torsion-free. Each component class is therefore zero. The support sequence gives, for every embedding $j : E_\infty \hookrightarrow \mathbf{C}$,

$$(29.1) \quad H^2(Q_j(\mathbf{C}), \mathbf{R}) \hookrightarrow H^2((Q_s)_j(\mathbf{C}), \mathbf{R}).$$

The Gysin sequence with complement $(\mathbf{G}_m)^2$ identifies the root subspace with $\mathbf{R}^d/\mathbf{R}(1, \dots, 1)$, since the angular classes of x and y have residues respectively the diagonal vector and zero. This argument uses no regulator on an infinite arithmetic base.

Finite actual coefficient spaces of unbounded rank. For $N > 2$ divisible by d and prime to p , choose compatible roots of unity in V and set

$$L_N = \mathbf{Q}(\mu_N) \subset F, \quad K_N = L_N(\theta) \subset E_\infty.$$

The chosen p -adic place of L_N is unramified over $\mathbf{Q}_p$. Eisenstein there gives $[K_N : L_N] = d$, and $\text{Gal}(K_N/L_N)$ is the restriction of G . Both fields are totally imaginary. For fixed $n \geq 2$, Borel's ranks [[12](#), Proposition 12.2] and restriction–transfer identities give

$$(29.2) \quad V_{N,n} = \ker \left(\sum_{\sigma \in G} \sigma : K_{2n-1}(K_N)_{\mathbf{Q}} \rightarrow K_{2n-1}(K_N)_{\mathbf{Q}} \right), \quad \dim_{\mathbf{Q}} V_{N,n} = \frac{(d-1)\varphi(N)}{2}.$$

Indeed, the invariant subspace is exactly the restriction of $K_{2n-1}(L_N)_{\mathbf{Q}}$, because $\text{res Tr} = \sum_{\sigma} \sigma$ and $\text{Tr res} = d$. The two Borel ranks are $d\varphi(N)/2$ and $\varphi(N)/2$ in every degree under consideration.

Let $W_N = (\mathcal{O}_{K_N})_{\mathfrak{p}_N}$ be the DVR at the prime induced by W . It maps to W , and G preserves it: above the chosen L_N place the extension is totally ramified and has a unique prime. Finite-field localization gives

$$K_{2n-1}(W_N)_{\mathbf{Q}} \simeq K_{2n-1}(K_N)_{\mathbf{Q}},$$

as well as actual surjectivity to the field, since K_{2n-2} of the finite residue field is zero. Thus $V_{N,n}$ defines a rational linear map into $K_{2n-1}(R)_{\mathbf{Q}}$ by coefficient pullback and i_* . Every element of $V_{N,n}$ has an actual representative after clearing denominators. If exact integral trace zero is wanted, replace an actual lift γ by $d\gamma - \sum_{\sigma \in G} \sigma\gamma$; this operation acts as multiplication by d on $V_{N,n}$.

Detection, with both continuity passages explicit. Suppose $0 \neq \beta \in V_{N,n}$ mapped to zero in $K_{2n-1}(R)_{\mathbf{Q}}$. Borel injectivity [14, Section 6] supplies an embedding $j_0 : K_N \hookrightarrow \mathbf{C}$ at which its point regulator is nonzero. Extend j_0 to an embedding $j : E_{\infty} \hookrightarrow \mathbf{C}$, using that E_{∞}/K_N is algebraic. For $\sigma \in G$, let b_{σ} denote the point regulator at $j_0\sigma$, represented in the complementary real line $i\mathbf{R}(n)$ to $\mathbf{R}(n)$ in $\mathbf{C}$. These coefficients satisfy

$$b_1 \neq 0, \quad \sum_{\sigma \in G} b_{\sigma} = 0,$$

so this vector is not diagonal. The detecting embedding is chosen for this coefficient; no single embedding is required to detect every element of $V_{N,n}$.

Put $\eta_{\zeta} = 1 - [\mathcal{O}_Q(-D_{\zeta})]$. Flat Gysin base change and the projection formula identify the image of the supported class on $\mathcal{T}$ with the restriction of

$$\alpha = \sum_{\zeta \in \mu_d} (\sigma_{\zeta} \beta) \eta_{\zeta} \in K_{2n-1}(Q)_{\mathbf{Q}}.$$

The assumed vanishing therefore makes α zero on $\mathcal{T}$. First, K -theory continuity for the filtered localization $\mathcal{O}(\mathcal{T}) = \varinjlim_s \mathcal{O}(Q_s)$ gives a finite permitted denominator set s for which $\alpha|_{Q_s} = 0$ rationally. Second, E_{∞} is a filtered union of number fields. All coefficients of these finitely many denominators, the root ideals, and the chosen K -classes descend to a number field $E_{\text{ar}} \subset E_{\infty}$ containing K_N . Continuity for this filtered coefficient extension implies that the asserted vanishing already holds on the corresponding finite open over some larger number field $E_1 \subset E_{\infty}$.

Apply the published absolute multiplicative regulator to that finite stage, after spreading it to a regular finite-type arithmetic model by inverting one integer. Evaluate at $j|_{E_1}$. The complex open of this finite model is exactly $(Q_s)_j$, and naturality gives the product class

$$2\pi i \sum_{\zeta \in \mu_d} b_{\sigma_{\zeta}} e_{\zeta} \quad \text{in} \quad H_{\mathcal{D}, \text{an}}^3((Q_s)_j, \mathbf{R}(n+1)) = \frac{H^2((Q_s)_j, \mathbf{C})}{H^2((Q_s)_j, \mathbf{R}(n+1))}.$$

The equality holds because $n+1 \geq 3$ and the complex dimension is two. Multiplication by $2\pi i$ identifies $\mathbf{C}/\mathbf{R}(n)$ with $\mathbf{C}/\mathbf{R}(n+1)$. The non-diagonal root vector and (29.1) make this regulator nonzero, contradicting the finite-stage vanishing. This proves injectivity on every $V_{N,n}$. No commutation of analytic cohomology with an infinite coefficient colimit has been used.

Finally, take for example $N = d^a$ as $a \rightarrow \infty$. The dimensions in (29.2) are unbounded, while R remains fixed. This proves infinite rational rank in every degree $2n-1 \geq 3$. All coefficient lifts, transfers and products used above are actual Quillen K -theory operations [40, 41, 50, 12, 14]. $\square$

For each p one may fix $d = 3$ if $p \neq 3$, and $d = 4$ if $p = 3$. This gives one ring at each residue prime with the stated infinite-rank conclusion in all higher odd degrees. The construction here uses the displayed arithmetic local ring; the complete rings considered below use a separate continuous detector.

30. VISIBLE COEFFICIENT RANK AND ALL HIGHER ODD DEGREES

The elliptic contraction detects the e_3 -component of the point coefficient. We construct arbitrarily large families whose projections to this component are $\mathbf{Q}_5$ -linearly independent, and then use them on a single complete local ring. We write $\mathcal{K}(-) = K(-)_5^\wedge[1/5]$ for spectral 5-completion followed by inversion of 5.

For the continuous detector, write $W' = W(\overline{\mathbf{F}}_5)[\rho]$ for the coefficient ring denoted by R in Proposition 9.1, and put $E_\infty = W'[1/5]$. The automorphisms σ_g fix $W(\overline{\mathbf{F}}_5)$. We use the normalization of (9.6): the bounded functional has $\ell(\omega) = 1$, and the polarization fixes the common sign. Trace, contraction, and formal replacement are E_∞ -linear on the differential quotients. The same four-term calculation therefore gives, for every $b \in E_\infty$ with $\sigma_{-1}b = -b$,

$$L \operatorname{Tr}_h \left(\sum_{g \in \mathbf{F}_5^\times} \sigma_g(b) \gamma_g \right) = 4e_3(b)[\tilde{Q}^*\eta].$$

Here the operators on the scaled model are transported by formal replacement, as in the proof of Proposition 9.1. The vector $[\tilde{Q}^*\eta]$ is nonzero in the ordinary Cohen quotient by Proposition 8.4. This formula applies to coefficients in the finite local fields below and requires only their e_3 -projections. It uses the linear operators over the fixed scalar field, with no passage to a colimit of completed K -theory.

For each complete coefficient discrete valuation ring O below and $n \geq 2$, let $r_{n,O}(\beta)$ denote the scalar whose AMMN point character is $t^{1-n}r_{n,O}(\beta)$, with $|t| = -2$, using the point comparison in Lemma 23.1.

Lemma 30.1. *Fix $n \geq 2$ and $m \geq 2$. Put*

$$V_m = W(\mathbf{F}_{5^m}), \quad F_m = V_m[1/5], \quad O_m = V_m[\rho], \quad E_m = F_m(\rho), \quad \pi = \rho - 1,$$

where ρ is a primitive fifth root of unity. Let σ_g fix F_m and send ρ to ρ^g . Let $\omega_T : \mathbf{F}_5^\times \rightarrow \mathbf{Z}_5^\times$ be the Teichmüller character, and put

$$e_3 = \frac{1}{4} \sum_{g=1}^4 \omega_T(g)^{-3} \sigma_g, \quad \tau_3 = \sum_{g=1}^4 \omega_T(g)^{-3} \rho^g.$$

There are actual classes $\beta_j \in K_{2n-1}(O_m)$, indexed by $1 \leq j \leq m-1$, each of the form $(1-\sigma_{-1})\beta_{0,j}$, with the following properties: their reductions in $K_{2n-1}(\mathbf{F}_{5^m})$ vanish integrally, and their AMMN point coordinates $b_j = r_{n,O_m}(\beta_j) \in E_m$ have $\mathbf{Q}_5$ -linearly independent projections $e_3 b_j$.

Proof. Define $S_j(z) = (z d/dz)^j (1-z)^{-1} \in \mathbf{Z}[z, (1-z)^{-1}]$. The expansion and continuation argument in the proof of Proposition 10.1, now with $\binom{n+j-1}{j}$ in place of $j+1$, gives

$$F_n(z) := \operatorname{Li}_n(z) - 5^{-n} \operatorname{Li}_n(z^5) = \sum_{j \geq 0} (-1)^j \binom{n+j-1}{j} 5^j S_j(z^5) \sum_{a=1}^4 \frac{z^a}{a^{n+j}}.$$

The series is uniformly convergent on $|z| \leq 1$, $|1-z^5| = 1$, and agrees there with Coleman continuation by [6, Theorem 2.7(3)]. The fifth power acts on the polylogarithm variable.

For a Teichmüller root $\zeta \in \mu_{5^m-1}(V_m) \setminus \{1\}$, the value $(\zeta\rho)^5 = \zeta^5$ is fixed by every σ_g . The exact Gauss identity $e_3(\rho^a) = \frac{1}{4}\omega_T(a)^3\tau_3$ therefore gives

$$\begin{aligned} (30.1) \quad U_n(\zeta) &:= \frac{4e_3 \operatorname{Li}_n(\zeta\rho)}{\tau_3} \\ &= \sum_{j \geq 0} (-1)^j \binom{n+j-1}{j} 5^j S_j(\zeta^5) \sum_{a=1}^4 \frac{\zeta^a \omega_T(a)^3}{a^{n+j}} \in V_m. \end{aligned}$$

The terms with $j \geq 1$ belong to $5V_m$. Thus, writing $t = \bar{\zeta}$, its reduction is

$$h_{3,n}(t) = \frac{P_n(t)}{1 - t^5}, \quad P_n(t) = \sum_{a=1}^4 a^{3-n} t^a \in \mathbf{F}_5[t], \quad t \in \mathbf{F}_{5^m}^\times \setminus \{1\}.$$

The polynomial P_n is nonzero, has coefficient one at t , and has degree four. Hence $h_{3,n}$ is nonconstant and has rational-map degree at most five. Every finite fiber has at most five elements, so

$$\#h_{3,n}(\mathbf{F}_{5^m}^\times \setminus \{1\}) \geq \left\lceil \frac{5^m - 2}{5} \right\rceil = 5^{m-1}.$$

Its $\mathbf{F}_5$ -span has dimension at least $m - 1$. Choose $m - 1$ independent image values and corresponding Teichmüller roots ζ_j . Reduction of a normalized $\mathbf{Q}_5$ -linear relation shows that the $U_n(\zeta_j)$ are $\mathbf{Q}_5$ -independent. Since $\tau_3 \neq 0$, the values $e_3 \text{Li}_n(\zeta_j \rho)$ are independent as well.

These are regulators of actual rational number-field classes. Indeed, all $\zeta_j \rho$ lie in $L = \mathbf{Q}(\mu_{5^m-1}, \rho)$ and are roots of unity. At the chosen prime above 5 they and their complements are units, since $\bar{\zeta}_j \neq 1$. The rational symbols $[\zeta_j \rho]_n$ are cycles, and [8, Theorem 1.10(2)] maps them to a common nonzero rational multiple of $L_{\text{mod},n}(\zeta_j \rho) = \text{Li}_n(\zeta_j \rho)$. The point normalization in (19.3) identifies the normalized Besser–de Jeu coordinate with the ordinary syntomic coordinate, so the scalar is common to the chosen roots. The number-field case of the cited theorem applies unconditionally. The local completion at the chosen prime is E_m .

Localization at that coefficient prime lifts these rational classes to its integer ring, since $K_{2m-2}(\mathbf{F}_{5^m}) = 0$. Clear one common integer denominator and then form the actual differences $\beta_j = (1 - \sigma_{-1})\beta_{0,j}$, mapping them to $K_{2n-1}(O_m)$. Their residue reductions are zero because σ_{-1} fixes the residue field. Applying e_3 to their syntomic regulators multiplies each of the preceding values by the same nonzero rational scalar, including the factor two from the difference.

The ramified point comparison [29, Example 2.2.4 and Propositions 2.2.7, 2.2.9, 2.3.4] is Galois-equivariant and identifies syntomic and étale point cohomology in every twist $n \geq 2$. Thus the e_3 -projections of the étale Chern values are $\mathbf{Q}_5$ -independent. Lemma 22.1 gives a natural Galois-equivariant $\mathbf{Q}_5$ -linear map

$$\pi_{2n-1}(K(O_m)_5^\wedge[1/5]) \longrightarrow H^1(E_m, \mathbf{Q}_5(n))$$

through which the actual Chern class factors. The finite Chern maps are applied over E_m , where 5 is invertible, after the canonical projections to $K_{2n-1}(O_m; \mathbf{Z}/5^\nu)$ and restriction to the field. Compatibility with reduction gives the map to the inverse limit of H^1 . The finite H^0 groups are Mittag–Leffler, identifying this target with continuous cohomology. This construction uses the canonical source projections and allows a nonzero source $\lim^1$.

The point case of the AMMN comparison identifies the source with $t^{1-n}E_m$, naturally under σ_g . A relation among the $e_3 b_j$ would therefore induce the same relation among the independent projected Chern values. This proves the claim without identifying the AMMN and polylogarithm coordinates. $\square$

Remark 30.2. The rational-map degree bound cannot uniformly be replaced by four. When $n \equiv 3 \pmod{4}$ one has

$$h_{3,n}(t) = (t - 1)^{-5} - (t - 1)^{-1}.$$

Its image lies in the trace-zero hyperplane of $\mathbf{F}_{5^m}/\mathbf{F}_5$. The lower bound $m - 1$ accommodates this actual loss of one dimension.

Put

$$k = \overline{\mathbf{F}}_5, \quad W = W(k), \quad V = \widehat{W[T]_{(5)}}^{(5)}, \quad R = V[[x, y]]/(\Phi_5(1 + y) + x^4 + Txy^3).$$

Theorem 30.3. *The ring R is a complete regular local domain of dimension two, with maximal ideal (x, y) and residue field $k(T)$. For every $n \geq 2$,*

$$\dim_{\mathbf{Q}} \ker(K_{2n-1}(R)_{\mathbf{Q}} \longrightarrow K_{2n-1}(\text{Frac } R)_{\mathbf{Q}}) = \infty.$$

For each fixed $n, m \geq 2$, the kernel contains $m - 1$ independent classes represented by actual integral classes that vanish on $R[1/x]$ integrally. The ring R is independent of n, m .

Proof. The regularity argument is the same linear-term calculation as in Section 4. Set $W' = W[\rho]$ and $E_\infty = W'[1/5]$. For fixed n, m , take the actual coefficients of Lemma 30.1. Naturality of the AMMN point calculation sends their coordinates under $E_m \hookrightarrow E_\infty$; hence their visible projections remain $\mathbf{Q}_5$ -independent.

Use the same finite parameter cover and scaled formal test ring B as in (11.1), now over V . It is the completion of an essentially smooth W' -algebra of relative dimension two. The spectrum comparison underlying Lemma 21.2 gives, for each fixed n , the relative character to $HC_{2n-2}^{\text{cts}}(B/W'; \mathbf{Q}_5)$. In its finite mixed-HKR model the top-form summand is

$$(30.2) \quad t^{2-n} \left(\Omega_{B/W'}^2[1/5] / d\Omega_{B/W'}^1[1/5] \right).$$

Indeed its cycles in this summand are all two-forms, and its boundaries are precisely $t^{2-n}d\Omega^1$; form degrees above two vanish.

The zero residue reductions of the coefficients provide actual relative lifts at the point. Pull them to B and multiply by $\eta_g = 1 - [J_g]$. With the common scalar $\lambda_{W'} \in E_\infty^\times$ of Proposition 7.1, the negative-cyclic module product is

$$(t^{1-n}\sigma_g(b_j))(\lambda_{W'}t\gamma_g) = \lambda_{W'}t^{2-n}\sigma_g(b_j)\gamma_g.$$

Rank zero removes the scalar term, higher Chern forms vanish in dimension two, and the cyclic-shuffle terms vanish because the point coefficient is a scalar relative to W' . The scalar-bar calculation applies in every cyclic column, including the negative columns that occur for $n > 2$. The scalar occurs to the first power because it belongs to the first Chern component.

Trace and contraction in Proposition 9.1 are E_∞ -linear. They send the coefficient of (30.2) to

$$4\lambda_{W'}e_3(b_j)[\tilde{Q}^*\eta].$$

The fixed vector $[\tilde{Q}^*\eta]$ is nonzero over E_∞ after the generic Cohen completion, by Lemma 8.1 and Proposition 8.4. Multiplication by the common nonzero $\lambda_{W'}$ is an injective $\mathbf{Q}_5$ -linear map on this coefficient field. Thus these character images are $\mathbf{Q}_5$ -independent, without any rationality or Galois-invariance assumption on $\lambda_{W'}$.

Write $C = B/\pi$. As before, C is a filtered colimit of smooth affine surfaces over k . By [25, Theorem 8.4], $\pi_q\mathcal{K}(C) = 0$ for every $q \geq 3$. The relative and regular-divisor localization sequences therefore give, in the present degree,

$$\pi_{2n-1}\mathcal{K}(B, (\pi)) \hookrightarrow \pi_{2n-1}\mathcal{K}(B) \hookrightarrow \pi_{2n-1}\mathcal{K}(B[1/\pi]).$$

These use the vanishings in degrees $2n$ and $2n - 1$, respectively. These are vanishings of the rationalized completed spectrum.

The source divisor $x = 0$ has ring $V[\rho]$. Pull each β_j to that divisor and push forward to define $c_j \in K_{2n-1}(R)$. Localization gives $c_j|_{R[1/x]} = 0$ integrally. Finite flat coefficient extension $V \rightarrow V[\rho]$, followed by inversion of 5, splits the divisor into the four root components with coefficients $\sigma_g(\beta_j)$. Pullback of their line-bundle classes under the scaled map gives

$$c_j|_{B[1/\pi]} = \sum_g \sigma_g(\beta_j)(1 - [J_g])|_{B[1/\pi]}.$$

This is exactly the actual product detected above; the scaled map itself is not required to be flat. Consequently the c_j are rationally independent. Since m is arbitrary while R is fixed, the kernel has infinite dimension for this n . Applying the argument separately to each $n \geq 2$ proves the theorem. $\square$

31. ONE COMPLETE TWO-DIMENSIONAL RING IN EVERY DEGREE AT LEAST THREE

The additional parameter below is a unit of the source ring. Its logarithmic differential raises the degree of the detected class by one. Detecting the even products also proves independence of their odd factors, including degree three.

Lemma 31.1. *Let W' be the coefficient discrete valuation ring, with uniformizer π , used in the quartic construction. Let D_0 be a smooth affine W' -curve whose special fiber is integral, and put*

$$A_T = \widehat{(D_0)}_{(\pi)}^{(\pi)}, \quad A_{TS} = \widehat{(D_0[S])}_{(\pi)}^{(\pi)}.$$

Here localization at (π) inverts every element with nonzero reduction. Let B_T be the π -adic completion of a smooth affine relative curve over A_T , defined by a model over D_0 , and let B_{TS} be the corresponding completed model over A_{TS} . Suppose their continuous differentials over W' are finite projective of ranks two and three, respectively. There is a W' -linear operator

$$\text{Res}_S : \Omega_{B_{TS}/W'}^{r,\text{cts}} \longrightarrow \Omega_{B_T/W'}^{r-1,\text{cts}}$$

which is integral and π -adically continuous and satisfies

$$\text{Res}_S(d\alpha) = -d\text{Res}_S(\alpha), \quad \text{Res}_S(\theta \wedge d\log S) = \theta \quad \text{for every } \theta \in \Omega_{B_T/W'}^2.$$

In particular it sends actual exact three-forms to actual exact two-forms. The induced map

$$\frac{\Omega_{B_T/W'}^2[1/5]}{d\Omega_{B_T/W'}^1[1/5]} \longrightarrow \frac{\Omega_{B_{TS}/W'}^3[1/5]}{d\Omega_{B_{TS}/W'}^2[1/5]}, \quad [\theta] \longmapsto [\theta \wedge d\log S],$$

is injective; these are ordinary continuous quotients.

Proof. For every $a \geq 1$ there is a compatible ring map

$$B_{TS}/\pi^a \longrightarrow (B_T/\pi^a)((S)).$$

Indeed, if $g \in D_0[S]$ has nonzero reduction, its image modulo π is a nonzero polynomial over the field $\text{Frac}(D_0/\pi)$. It is therefore a unit in that field's Laurent series ring. The ideal generated by π in $(B_T/\pi^a)((S))$ is nilpotent, so this inverse lifts. This argument applies to every primitive denominator, including denominators whose literal lowest coefficient is not a unit. The lifts of the inverses are unique and commute with reduction in a . The maps extend to the completed algebra because the inverse-limit target is complete. Thus the relevant target is

$$\mathcal{L} = \varprojlim_a (B_T/\pi^a)((S)).$$

The negative Laurent bound may depend on a , which is why this inverse-limit target is used.

Use Laurent series of continuous base forms, with differential defined coefficientwise and with the usual S -derivative. These operations are compatible modulo π^a and preserve the π -adic filtration. The universal continuous derivation of B_{TS} therefore gives a map into this Laurent differential complex. Finite projectivity of the source form modules justifies extension to their completions and exterior powers.

Write a form of degree r in the target as $\alpha(S) + \beta(S) \wedge dS$, with base-form degrees r and $r - 1$. Define

$$\text{Res}_{S,r}(\alpha(S) + \beta(S) \wedge dS) = (-1)^{r-1} [S^{-1}] \beta(S).$$

Coefficient extraction is compatible with reduction and is integral. The coefficient of S^{-1} in the derivative of a Laurent series is zero. It follows directly that $\text{Res}_{S,r+1} d = -d\text{Res}_{S,r}$. For $r = 3$ its sign is positive, which gives the second displayed identity. Inverting 5 preserves these identities. All quotients in the conclusion use the ordinary image of the differential. $\square$

Put

$$k = \overline{\mathbf{F}}_5, \quad W_0 = W(k), \quad V_{TS} = W_0 \widehat{[T, S]}_{(5)}^{(5)},$$

and

$$R_{TS} = V_{TS}[[x, y]] / (\Phi_5(1 + y) + x^4 + Txy^3).$$

Theorem 31.2. *The ring R_{TS} is a complete regular local domain of dimension two, with maximal ideal (x, y) and residue field $k(T, S)$. For every integer $j \geq 3$,*

$$\dim_{\mathbf{Q}} \ker(K_j(R_{TS})_{\mathbf{Q}} \longrightarrow K_j(\text{Frac } R_{TS})_{\mathbf{Q}}) = \infty.$$

For every $n, m \geq 2$, the odd degree $2n - 1$ and the even degree $2n$ kernels contain $m - 1$ independent classes represented by actual integral classes which vanish on $R_{TS}[1/x]$ integrally. The ring R_{TS} is independent of n and m .

Proof. The ring $W_0[T, S]_{(5)}$ is a discrete valuation ring with uniformizer 5 and residue field $k(T, S)$. Its completion V_{TS} has the same properties. The defining equation of R_{TS} is congruent to 5 modulo $(5, x, y)^2$ in the regular local ambient ring of dimension three. Its quotient is therefore regular of dimension two, and is complete with maximal ideal (x, y) . In particular it is a domain. The element S is a unit of R_{TS} .

Set $W' = W_0[\rho]$, $\pi = \rho - 1$, and $E_\infty = W'[1/5]$. Fix $n, m \geq 2$, and take the $m - 1$ actual coefficients β_j from Lemma 30.1. Map them to $K_{2n-1}(W')$ and write $b_j \in E_\infty$ for their natural AMMN point coordinates. Their visible projections $e_3 b_j$ remain $\mathbf{Q}_5$ -independent. Their reductions vanish integrally, since each coefficient is an actual difference $(1 - \sigma_{-1})\beta_{0,j}$.

Choose the same finite parameter cover, independent of S , as in the one-parameter quartic detector. In the notation of Lemma 31.1, it gives the old coefficient ring A_T and the new one A_{TS} . The latter receives $V_{TS}[\rho]$: every primitive polynomial in T, S has nonzero image in $k(U)(S)$, and hence is a unit. Completion extends this map. Put

$$B_{TS} = A_{TS}\langle X, Y \rangle / \left(X^4 + \frac{\Phi_5(1 + \pi Y)}{\pi^4} + TXY^3 \right), \quad C_{TS} = B_{TS}/\pi.$$

The old algebra B_T is given by the same formula over A_T . They are the completions of essentially smooth W' -algebras of relative dimensions three and two, respectively. Smoothness holds near the entire special fiber; localizing a model at an element congruent to one modulo π leaves the completion unchanged. In particular B_{TS} and C_{TS} are regular and π is a non-zero-divisor.

For $g \in \mathbf{F}_5^\times$, let

$$a_g = \frac{\rho^g - 1}{\pi}, \quad J_g = (X, Y - a_g), \quad \eta_g = 1 - [J_g], \quad \gamma_g = -c_1(J_g).$$

These invertible ideals are pulled from B_T . We can therefore represent their Chern classes by two-forms pulled from the old continuous complex, independently of S .

The actual residue sequence at the point supplies actual relative lifts of all β_j ; here $K_{2n}(k) = 0$ by Quillen's finite-field computation and filtered colimits. Pull these lifts to B_{TS} and form their relative products

$$\xi_j = \sum_g \sigma_g(\beta_j) \eta_g[S] \quad \text{in } K_{2n}(B_{TS}, (\pi)),$$

where the coefficient symbols in this formula denote the specified relative lifts.

Apply the essentially smooth relative fiber comparison of Lemma 21.2 in each fixed degree. Its $K(B_{TS})$ -module compatibility gives the negative-cyclic action on the ordinary cyclic target. In relative dimension three the normalized finite mixed-HKR map uses only the denominators 2 and 6, both units at 5. Let $\lambda_{W'} \in E_\infty^\times$ be the common scalar of Proposition 7.1. The line bundles are pulled from the rank-two algebra B_T , so naturality pulls back their negative-cyclic characters $\lambda_{W'} t \gamma_g$. The point coefficient is $t^{1-n} \sigma_g(b_j)$, and the unit character starts with $d \log S$. The top character component is

$$\chi_{B_{TS}}(\xi_j)_{\text{top}} = \lambda_{W'} t^{2-n} \sum_g \sigma_g(b_j) \gamma_g \wedge d \log S \quad \text{in } HC_{2n-1}^{\text{cts}}(B_{TS}/W'; \mathbf{Q}_5).$$

Here $|t| = -2$. This formula also holds when the initial cyclic column is negative. To check it at chain level, the rank-zero line has underlying Hochschild degrees at least two; the lowest unit term has degree one. Every higher line or unit term, or cyclic-shuffle correction in their product, has final Hochschild degree at least five, and is killed by the dimension-three HKR map. The scalar point cup has no cyclic-shuffle correction because its scalar lies in the relative base. Thus precisely the displayed ordinary shuffle survives.

The top-form summand is

$$t^{2-n} (\Omega_{B_{TS}/W'}^3[1/5] / d\Omega_{B_{TS}/W'}^2[1/5]).$$

All its three-forms are cycles, and its boundaries are precisely the indicated ordinary exact forms. Apply Lemma 31.1 to their coefficients. It recovers $\lambda_{W'} \sum_g \sigma_g(b_j) \gamma_g$ on B_T . The weighted trace and elliptic contraction of Proposition 9.1 are E_∞ -linear and send these classes to

$$4\lambda_{W'} e_3(b_j) [\tilde{Q}^* \eta].$$

The fixed base vector $[\tilde{Q}^* \eta]$ survives the old generic Cohen completion by Lemma 8.1, and remains nonzero after the finite constant extension adjoining ρ by Proposition 8.4. Multiplication by the common nonzero $\lambda_{W'}$ preserves $\mathbf{Q}_5$ -linear independence. Consequently the characters of the ξ_j are $\mathbf{Q}_5$ -independent.

The special fiber C_{TS} is a filtered colimit of smooth affine threefolds over the perfect field k . Geisser–Levine [25, Theorem 8.4] gives

$$K_q(C_{TS}; \mathbf{Z}/5^\nu) = 0 \quad (q > 3, \nu \geq 1).$$

The Milnor sequence for spectral completion, using degrees q and $q + 1$, therefore gives $\pi_q \mathcal{K}(C_{TS}) = 0$ for $q \geq 4$. For the even degree under consideration the relative and regular-divisor localization sequences give

$$\pi_{2n} \mathcal{K}(B_{TS}, (\pi)) \hookrightarrow \pi_{2n} \mathcal{K}(B_{TS}) \hookrightarrow \pi_{2n} \mathcal{K}(B_{TS}[1/\pi]).$$

The first injection uses the vanishing in degree $2n + 1$, and the second that in degree $2n$; both are valid for $n \geq 2$. Here $\mathcal{K} = K_5^\wedge[1/5]$ denotes the rationalized spectrally completed K -theory.

It remains to match these detected even classes with actual classes of the fixed source. Its regular Cartier divisor $i : x = 0 \hookrightarrow \text{Spec } R_{TS}$ has ring $V_{TS}[\rho]$. Pull the coefficients to this divisor and set

$$c_{n,j} = i_*(\beta_j) \in K_{2n-1}(R_{TS}), \quad a_{n,j} = c_{n,j}[S] \in K_{2n}(R_{TS}).$$

Both classes vanish on $R_{TS}[1/x]$ integrally. Make the finite flat coefficient extension $V_{TS} \rightarrow V_{TS}[\rho]$, and then invert 5. The divisor $x = 0$ splits into its four root components with coefficients $\sigma_g(\beta_j)$. Their length-one Cartier resolutions and the projection formula express the resulting odd class as $\sum_g \sigma_g(\beta_j)(1 - [I_g])$, where $I_g = (x, y - (\rho^g - 1))$. Apply the continuous scaling

$$x \mapsto \pi X, \quad y \mapsto \pi Y, \quad R_{TS} \longrightarrow B_{TS}.$$

It exists because the defining equation maps to π^4 times the test equation and the maximal ideal maps into (π) . After inverting π , pullback of the invertible ideals I_g gives J_g . The resulting actual class is

$$a_{n,j}|_{B_{TS}[1/\pi]} = \sum_g \sigma_g(\beta_j) \eta_g[S]|_{B_{TS}[1/\pi]} \quad \text{in actual } K_{2n}.$$

Only the initial coefficient extension is asserted to be flat; the subsequent substitution pulls back line bundles. This equality and the two completed injections prove that the actual classes $a_{n,j}$ are rationally independent.

Finally, a rational relation among the actual odd classes $c_{n,j}$ would, after multiplication by the same actual unit class $[S]$ in the same ring, give a relation among the $a_{n,j}$. Thus the odd classes are independent as well. This argument includes $n = 2$ and requires no vanishing of $\pi_3 \mathcal{K}(C_{TS})$. Letting m grow proves infinite rank separately in degrees $2n - 1$ and $2n$. These pairs cover every $j \geq 3$. $\square$

Remark 31.3. No general injectivity of multiplication by $[S]$ is used: it is proved only on the displayed finite spans by detecting their products. In particular the odd degree-three conclusion does not use a degree-three generic-localization injection for the new special fiber. The residue field in this fixed-ring statement is $\overline{\mathbf{F}}_5(T, S)$.

APPENDIX A. LOW DEGREES AND COMPARISON WITH OTHER GERSTEN STATEMENTS

We use the arithmetic-quadric notation of Section 12 in this appendix.

A.1. A low-degree stress test. The analogous K_1 construction does not supply a counterexample. Its product regulator would have degree three and twist two. Here the filtration is the analytic truncation of the holomorphic de Rham complex, which need not vanish in filtration degree two on a complex surface.

For a smooth affine complex surface U , write $X = U^{\text{an}}$. An affine embedding realizes X as a closed complex submanifold of $\mathbf{C}^N$. Coordinate functions separate points and give local coordinates. They bound the holomorphic hull of each compact set inside a compact coordinate polydisc; the hull is closed there and hence compact. Together with second countability, these properties make X Stein. Cartan's theorem B [16, §5, proof of Theorem B, pp. 19–14–19–15] gives $H^q(X, \Omega_X^q) = 0$ for every $q > 0$. The holomorphic Poincaré lemma follows on coordinate balls by radial integration and identifies $\mathbf{C}_X$ with the holomorphic de Rham complex. Its hypercohomology is therefore computed by global holomorphic forms, in degrees zero through two. Thus

$$H^2(X, \mathbf{C}) = \Gamma(X, \Omega_X^2)/d\Gamma(X, \Omega_X^1), \quad H^3(X, \mathbf{C}) = 0.$$

For $F_{\text{an}}^2 \Omega_X^\bullet = \Omega_X^{\geq 2} = \Omega_X^2[-2]$, the map from degree-two hypercohomology to $H^2(X, \mathbf{C})$ is the displayed quotient map and is surjective. Moreover, $\mathbb{H}^3(X, F_{\text{an}}^2 \Omega_X^\bullet) = H^1(X, \Omega_X^2) = 0$. The constant real sheaf $\mathbf{R}(2)$ is a direct summand of the underlying real sheaf $\mathbf{C}$, so $H^3(X, \mathbf{R}(2)) = 0$ as well. The analytic Deligne cone sequence contains

$$\begin{aligned} H^2(X, \mathbf{R}(2)) \oplus \mathbb{H}^2(X, F_{\text{an}}^2 \Omega_X^\bullet) &\longrightarrow H^2(X, \mathbf{C}) \longrightarrow H_{\mathcal{D}, \text{an}}^3(X, \mathbf{R}(2)) \\ &\longrightarrow H^3(X, \mathbf{R}(2)) \oplus \mathbb{H}^3(X, F_{\text{an}}^2 \Omega_X^\bullet). \end{aligned}$$

Its first arrow is surjective and its final group is zero. Hence $H_{\mathcal{D}, \text{an}}^3(U, \mathbf{R}(2)) = 0$. This calculation uses the analytic truncation just defined; it makes no corresponding assertion about the logarithmic mixed-Hodge filtration. The elementary quotient in Lemma 12.18 is unavailable in twist two. This agrees with the fact that K_1 of a commutative local ring is its unit group and injects into the fraction field.

There is also an explicit algebraic version of this test. On the split quadric in coordinates

$$uv = t(1 - t),$$

the matrix

$$e = \begin{pmatrix} t & u \\ v & 1 - t \end{pmatrix}$$

is a rank-one idempotent. Up to choosing the ruling, its image is the line bundle $P = \mathcal{O}(D_+)$. For $\alpha \in E^\times$, $\alpha \neq 1$, the determinant-one matrix

$$M = \begin{pmatrix} \alpha^{-1} & 0 \\ 0 & 1 \end{pmatrix} (I + (\alpha - 1)e)$$

represents $\alpha([P] - 1)$ in K_1 . Its upper-left entry is s/α with $s = 1 + (\alpha - 1)t$. On $D(s)$ Gaussian elimination makes M elementary. The removed curve has

$$t = -\frac{1}{\alpha - 1}, \quad uv = -\frac{\alpha}{(\alpha - 1)^2} \neq 0,$$

so it is a geometrically irreducible principal curve isomorphic to $\mathbb{G}_m$. Hence ordinary Betti injectivity alone cannot protect a K_1 product. The condition $3 > \dim_{\mathbf{C}} U = 2$ in the main proof is essential.

A.2. Equal characteristic and other localizations. Replacing the arithmetic DVR by $k[t]_{(t)}$ and adjoining u with $u^2 = -t$ does not reproduce the same coefficient class. For every field k , restriction gives an isomorphism

$$K_3(k)_{\mathbf{Q}}^{(2)} \xrightarrow{\sim} K_3(k(u))_{\mathbf{Q}}^{(2)}.$$

To prove this, put $F = k(u)$ and compare the Milnor and Quillen rational-function residue sequences [50, III, Theorem 7.4; V, Corollary 6.7.1]. They have respective kernels $K_3^M(k)$ and $K_3(k)$ and surjective residue maps to $\bigoplus_P K_2^M(k[u]/P)$ and $\bigoplus_P K_2(k[u]/P)$, where P runs over

monic irreducible polynomials. The right-hand terms agree by Matsumoto's theorem [50, III, Theorem 6.1].

Their residue maps are compatible with the symbol maps. At each polynomial valuation, a product of unit classes extends over the valuation ring and has zero boundary, while for a uniformizer π and units a, b the boundary-product formula gives

$$\partial\{\pi, a, b\} = \{\bar{a}, \bar{b}\}.$$

Multilinearity and $\{\pi, \pi\} = \{-1, \pi\}$ reduce every Milnor symbol to unit symbols and symbols with exactly one uniformizer. These formulas therefore give its tame residue [50, III, Theorem 7.3; V, equation (6.1.1), Example 6.1.2, and Lemma 6.7.3].

Given $z \in K_3(F)$, surjectivity of the Milnor residue map supplies an image m of $K_3^M(F)$ with the same residues as z . Exactness of the Quillen sequence gives $z = \text{res}(\beta) + m$ for some $\beta \in K_3(k)$. The rational Milnor image has pure Adams weight three [50, IV, Example 5.9.1]. Naturality of the rational Adams decomposition [50, IV, Theorem 5.11] therefore makes projection to weight two surjective from $K_3(k)_{\mathbf{Q}}^{(2)}$. Injectivity follows from the injective constant-field map in the Quillen sequence, proving the displayed isomorphism. The involution $u \mapsto -u$ fixes k and hence acts trivially on this space; in characteristic two it is already the identity. Thus the needed rational weight-two anti-invariant regulator source is absent. The arithmetic number-field class is an essential ingredient.

The scheme T is regular and contains a field, but it is not semilocal: its nonzero Picard group already rules that out. Local Gersten theorems over fields therefore do not give an injection $K_3(T) \rightarrow K_3(\text{Frac } T)$ for this global scheme.

Sections 12.3–12.5 prove persistence for the precise noncomplete localization; they do not by themselves justify completion. Section 16 supplies a separate arithmetic p -adic proof for completed examples. Propositions 16.7, 16.9, and 16.13 verify that the particular explicit classes at every residue prime survive both completion and henselization. These arguments use arithmetic detectors after completion.

A.3. The distinction between the generic fiber and the fraction field. The rings considered here are regular and flat over $\mathbf{Z}$, and are consequently F -smooth in the sense relevant to [13]. Corollaries 5.12–5.13 of that reference apply to them: in degree i , the motivic comparison with $\mathbf{Z}/p^{\nu}(i)$ -coefficients injects into the cohomology of the entire generic fiber $A[1/p]$. This is compatible with the construction here, whose nonzero classes are detected on that generic fiber and become zero only after a further restriction to the fraction field. The generic fiber has positive dimension and cannot be replaced by its generic point in that statement.

The special fiber $A/(p)$ of the basic examples is the singular domain $k[x, y, z]/(xy + z^2)$, or its corresponding completion. Thus these maps are not smooth over their coefficient DVR. They cannot become smooth through a different local DVR coefficient presentation either: primality of (p) forces the base uniformizer to be associated to p , and the same singular special fiber remains. Absolute purity also does not assert that a principal-divisor Gysin map in higher K -theory is zero; triviality of the normal bundle gives the weaker self-intersection identity $i^*i_* = 0$.

A.4. Rational and finite coefficients. The regulator proves that the kernel contains an element of infinite order. It does not by itself prove that any fixed finite-coefficient Gersten map fails. In the explicit $p = 3$ case, Section 13.7 supplies a separate algebraic proof for $\mathbf{Z}/3$ -coefficients. That coefficient characteristic equals the residue characteristic; no prime-to-residue-characteristic conclusion is asserted here. A rationally nonzero class can otherwise become divisible in a sufficiently large group. In particular, one may not interchange a filtered colimit over shrinking opens with an inverse limit over coefficient exponents without a separate argument.

Feld's Theorem 1.1 [24] constructs a nonzero generic-kernel class in $K_2(A_F; \mathbf{Z}/3)$, where $A_F = V[[x, y]]/(3 + x^2 - y^3)$ and V is a complete mixed-characteristic $(0, 3)$ discrete valuation

ring with uniformizer 3. In that example, an integral lift acquires divisibility by three over the fraction field while the integral K_2 restriction is injective [24, Corollary 5.1].

A.5. An explicit number-field source when the prime is three. Let $u^2 = -3$ and $\zeta = (1 + u)/2$. Then $\zeta = e^{i\pi/3}$ at the chosen embedding and

$$1 - \zeta = \zeta^{-1}.$$

Thus the pre-Bloch symbol $[\zeta]$ has zero wedge boundary. Its Bloch–Wigner dilogarithm is positive, since

$$D(e^{i\pi/3}) = - \int_0^{\pi/3} \log(2 \sin(t/2)) dt > 0.$$

The rational Bloch-group description of $K_3^{\text{ind}}(E)$ [50, VI, Theorem 5.2] supplies an explicit possible rational source class. The main proof does not require fixing this comparison or its integral normalization: Borel’s theorem and Lemma 12.4 already produce an integral anti-invariant source with nonzero regulator.

APPENDIX B. A MIXED-SQUARE OBSTRUCTION TO A SIMPLICIAL-FUNCTOR CONSTRUCTION

Theorem 5 of [37] asserts that, for a noetherian local ring B and a non-zero-divisor g , the functor $\mathcal{P}_{B/gB} \rightarrow \mathcal{M}_B(1)$ induces the zero map on K -theory. Its application to $B = A$ and $g = x$ would kill the supported class used above. We therefore examine a specific step in its proof. The following calculation concerns Lemma 13(III) as written; it does not, by itself, establish that Theorem 5 is false.

Let B be a local ring, let g be a non-zero-divisor in its maximal ideal, and put $t = (1 + g)^{-1}$. Write

$$G = \text{Typ}(g) = [B \xrightarrow{g} B], \quad I = \text{Typ}(1) = [B \xrightarrow{1} B], \quad M = G \oplus I,$$

with complexes in homological degrees 1, 0. These are respectively the objects $(1, 0)_B$, $(0, 1)_B$, and $(1, 1)_B$ of the category $\mathcal{C}$ in [37, Definition 9]. For $b \in B$ define chain maps $\alpha_b : I \rightarrow M$ and $\beta_b : M \rightarrow G$ by

$$\begin{aligned} (\alpha_b)_1 &= \begin{pmatrix} b \\ 1 \end{pmatrix}, & (\alpha_b)_0 &= \begin{pmatrix} gb \\ 1 \end{pmatrix}, \\ (\beta_b)_1 &= (1 \quad -b), & (\beta_b)_0 &= (1 \quad -gb). \end{aligned}$$

They give a split short exact sequence of complexes

$$\mathcal{S}_b : \quad 0 \longrightarrow I \xrightarrow{\alpha_b} M \xrightarrow{\beta_b} G \longrightarrow 0.$$

Indeed $s : G \rightarrow M$, given by $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ in both degrees, is a chain section of β_b , while $r : M \rightarrow I$, given by $(0 \quad 1)$ in both degrees, satisfies $r\alpha_b = 1$ and $\alpha_b r + s\beta_b = 1$. In the block notation of [37, Conventions 11], the arrows are $\begin{pmatrix} b \\ 1 \end{pmatrix}$ and $(1 \quad -b)$, respectively. Both are upper triangular. Thus $\mathcal{S}_b$ is an object of $S_2^\Delta \mathcal{C}$ in that paper’s notation.

There is an isomorphism of short exact sequences $\mathcal{S}_1 \rightarrow \mathcal{S}_t$ whose components on I, M, G are, respectively,

$$\ell = (1 + g) \text{id}_I, \quad v_1 = \begin{pmatrix} 1 & 0 \\ g & 1 \end{pmatrix}, \quad v_0 = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad q = t \text{id}_G.$$

The relation $\text{diag}(g, 1)v_1 = v_0 \text{diag}(g, 1)$ verifies that v is a chain map. All three components are isomorphisms. The four naturality equations can be checked without the block convention:

$$\begin{aligned} v_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= \begin{pmatrix} 1 \\ 1+g \end{pmatrix} = \begin{pmatrix} t \\ 1 \end{pmatrix} (1 + g), & v_0 \begin{pmatrix} g \\ 1 \end{pmatrix} &= \begin{pmatrix} g \\ 1+g \end{pmatrix} = \begin{pmatrix} gt \\ 1 \end{pmatrix} (1 + g), \\ (1 \quad -t) v_1 &= (t \quad -t), & (1 \quad -gt) v_0 &= (t \quad -gt). \end{aligned}$$

In the paper’s block notation v is $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, so every component is lower triangular. This is therefore a morphism in $i_{\text{lower}} S_2^\Delta \mathcal{C}$.

The assignments μ'_1, μ'_2 preceding Definition 12 take an object $(n, m)_B$ to $\text{Typ}(g)^{\oplus n}$ or $\text{Typ}(1)^{\oplus n}$, and a morphism to its upper-left $n' \times n$ block in both degrees. Each β_b consequently maps to the identity of $\text{Typ}(g)$ or $\text{Typ}(1)$. The component v maps to the identity, whereas q maps to multiplication by t . Naturality of the resulting right-hand square would require $1 = t$. This is false, since $1 - t = g/(1 + g) \neq 0$. The assignments thus fail to send this morphism of S_2 -objects to a morphism, contrary to [37, Lemma 13(III)].

The failure already occurs with $B = \mathbf{Z}_{(3)}$, $g = 3$, and $t = 1/4$. The calculation uses a regular DVR with regular quotient. It identifies a failure of the stated simplicial-functor construction used in the subsequent additivity argument.

AUTHORING AND VERIFICATION

This article consolidates the source manuscripts and the subsequent proof audit with assistance from ChatGPT and Codex. The accompanying source repository records the statements, proofs, and outcomes of independent model-based reviews of the central constructions and their dependencies. A separate Lean 4.34.1 development verifies ten polynomial identities and two inequalities in natural numbers used in the calculations. The Lean development does not formalize the complete algebraic K -theory, regulator, or comparison arguments. The mathematical proofs are given in the article; the repository records the precise scope of the auxiliary checks and preserves the original sources.

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